

Digital Communication

ICT for Internet and Multimedia

Year 2025/2026

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Chapter 1

Week 13-19 October

1.1 Introduction

In most of the course we'll handle wireless communications, as they're much more difficult than wired communications. In this course we'll stay at the physical layer. We'll see:

- **Channel models:** taking radio communications and making a mathematical model for them. We'll find an accurate model and then we'll apply a tradeoff to it, simplifying it so that it is manageable.
- **OFDM:** Orthogonal Frequency Division Multiplexing.
- **MIMO:** we have multiple antennas at the transmitter and multiple antennas at the receiver; we'd like to see how much this improves communications. The book he selected this year handles fundamentals of MIMO communications. MIMO is the latest and greatest achievement, he chose a book which goes from the basics. We might have single user MIMO or even multiple user MIMO: we're communicating to multiple phones through multiple antennas.

We can ask ourselves: what is the best that we can do? This is what our approach will be, so that we can know the boundaries. We'll follow the slides from the author of the book, hopefully they'll be good. We'll also have exercises with solutions: we can look at them from cambridge.org, specifically from the page of the book.

1.2 Primer

The first chapter is a primer in information theory and MSE estimation; we'll go fast as it should summarize stuff we know, as mutual information between two variables.

Note

The signals are always complex-valued and the constellations are always normalized to have unit power. ∞ -QAM: we increase the number of points making them denser inside a specific set.

1.2.1 Complex symmetric random variables, ∞ -modulations

We'll start from chapter 1 of the book: *A primer on information theory and MSE*.

We should already know this stuff, but we'll cover it in order to have a common starting ground.

Prof's backstory moment: In the end, he studies **physical layer** and that's it.

We ask ourselves: "**what is information?** When can we say that someone is giving us information?". ***Professor begins ranting about information***: information is related to **probability**! The more unlikely something is, the more informative it is!

Today we'll talk about **mutual information** between two random variables: the information a variable can supply to another variable.

We'll talk about **channel capacity**: the capacity at which a channel can convey information; it answers to "how much information can I send through this channel, reliably?". By reliably we mean that we can lower the probability of error as much as we want.

Binary coding and **decoding** are near optimum, but there can be other ways, which could be better: we'll show that binary is indeed **optimum**.

LMMSE is optimum when the signal and noise are **Gaussian**.

Let's start from **signals**; in the book, signals are normally **complex**, unless we specifically say otherwise. Normally **complex random variables** are **complexly symmetric**:

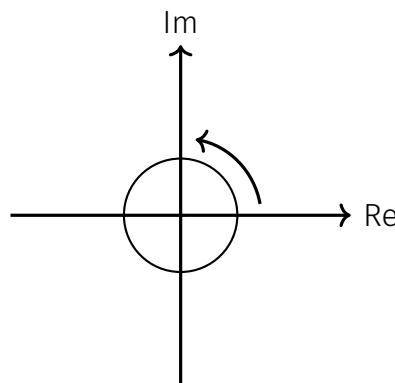


Figure 1.1: Rappresentazione del piano Im-Re con una PFD simmetrica centrale e frequenza di rotazione.

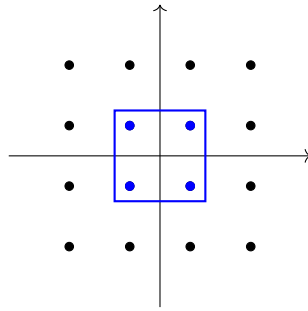
About **M-PSK** and **M-QAM**: M is usually a **power of 2**. Let's see **16-QAM**:

In the book they always keep the **same variance of the constellation**: the book squeezes the values so that the **power stays constant**.

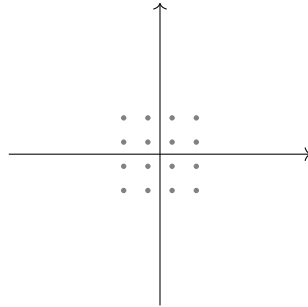
The book also has **inf-PSK** and **inf-QAM**: We take the **normal constellation** and we keep **adding points**.

It's a useful way to see what happens in the **asymptotic regime**. Also, the **PDF of a complex Gaussian variable exists** (see notes).

If σ^2 is the variance, the **Standard deviation** is $\sqrt{\sigma^2} = \sigma$.



The power of the ones in blue is lower than the power of the other signals.



NOT $\pm\sigma^2$!

We'll try and be as precise as possible: there is no **negative standard deviation**! As an example:

$$f(x) = \sqrt{x} \in \mathbb{R} \text{ is def. only for } x \geq 0$$

$$x^2 = 16 \text{ has two sol.}$$

What about information? There have been many results, dating back to **Shannon**! (See pag 7 of the slides btw).

1.3 Discrete entropy, proper signals

1.3.1 Entropy

Def 1.3.1: Entropy of a discrete-time random variable.

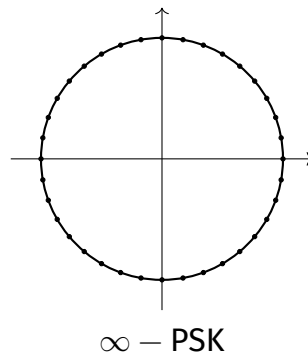
$$\mathcal{H}(x) = - \sum_x p_x(x) \log_2 p_x(x)$$

$$= -\mathbb{E}[\log_2 p_x(x)]$$

Def 1.3.2: Joint entropy.

$$\mathcal{H}(x_0, x_1) = - \sum_{x_0} \sum_{x_1} p_{x_0 x_1}(x_0, x_1) \log_2 p_{x_0 x_1}(x_0, x_1)$$

$$= -\mathbb{E}[\log_2 p_{x_0 x_1}(x_0, x_1)]$$



Nonnegative function that quantifies the amount of uncertainty in discrete valued RV x

Conditional entropy

$$\mathcal{H}(x|y) = - \sum_x \sum_y p_{xy}(x, y) \log_2 p_{x|y}(x|y).$$

Entropy connections

$$\mathcal{H}(x, y) = \mathcal{H}(x) + \mathcal{H}(y|x).$$

The entropy is used in statistical physics.

Differential information

Δ Diff. inform. can be negative.

From this, we can use the Bayes theorem to derive:

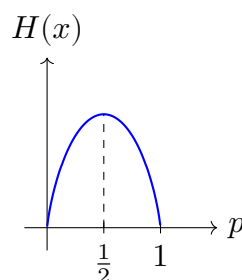
$$\begin{aligned} H(x, y) &= H(x|y) + H(y) \\ &= H(y|x) + H(x) \end{aligned}$$

Example 1.3.1: Bernoulli Variable. *We have a Bernoulli variable:*

$$X = \{0, 1\} \quad P = \{p, 1 - p\}$$

$H(X)$ will be ≤ 1 , where the equality holds if $p = 1/2$.

If we plot it, we get:

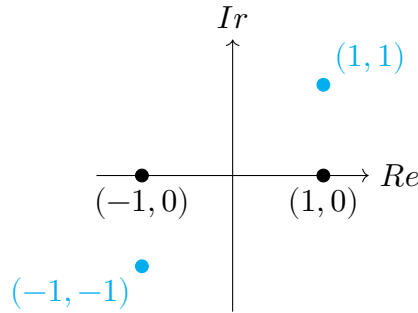


We get maximum information when we flip a ***fair*** coin! If the coin always flips heads or tails, we get no information!

We could say that the messier the information, the most we get out of it.

Prof forgot stuff about signals: we're always talking about "proper" signals; the real and imaginary part of those numbers are uncorrelated!

Properness is preserved by affine transformations. An intuitive reason why we consider only proper complex numbers is that if we have real and imaginary parts of a random variable:



$(-1, -1)$: This is a fake complex variable; we want to avoid these cases.

1.3.2 Proper complex

A random variable (or a random process/random vector) is said to be proper complex if it satisfies **Properness**:

Def 1.3.3: Properness. A complex random variable is said to be **proper** if

$$E[x^2] = E[x]^2$$

holds. Properness requires $\text{Re}[x], \text{Im}[x]$ to be uncorrelated and have the same variance.

A complex random vector is said to be **proper** if

$$E[\mathbf{x}\mathbf{x}^T] = E[\mathbf{x}]E[\mathbf{x}]^T$$

holds, which is equal to asking that $\text{Re}[\mathbf{x}], \text{Im}[\mathbf{x}]$ have identical Covariance matrices and cross-covariance equal to 0.

Rem. Properness is preserved in affine transformations!

1.3.3 Entropy rate

Let's say we have a sequence of discrete random variables indexed by time, $x_0, \dots, x_{N-1}$. If they are **IID**, then the entropy of the whole process grows linearly with N at a rate $\mathcal{H}(x_0)$.

In general, a process grows according to the so-called **entropy rate**

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \frac{1}{N} \mathcal{H}(x_0, \dots, x_{N-1}). \quad (1.32)$$

If the process is **stationary**, we can show that the entropy rate is equal to

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \mathcal{H}(x_N | x_0, \dots, x_{N-1}). \quad (1.33)$$

All of this can be extended for continuous processes to differential entropy.

A process is said to be **nonregular** if its present value is perfectly predictable from noiseless observations of the entire past, while the process is **regular** (which happens if $\mathfrak{h} > -\infty$) if its present value cannot be perfectly predicted from noiseless observations of the entire past.

1.3.4 Differential entropy

Def 1.3.4: Differential entropy.

$$h(x) = \int f_x(x) \log_{1/2} f_x(x) dx = \mathbb{E} [\log_{1/2} f_x(x)]$$

For vectors:

$$h(\underline{x}) = \mathbb{E} [\log_{1/2} f_{\underline{x}}(\underline{x})]$$

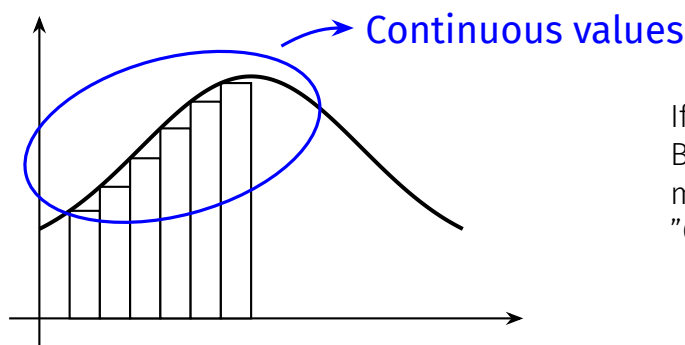
The differential entropy rate does not measure the information contained in a random variable. If we only have a random variable, we only have a shot. In order to consider a **rate**, we consider a **random process**: in this case, we have a random variable every certain amount of time: we can say that we have a certain amount of information per second!

A binary random process with $p = 0.5$ which gives a value every second, gives 1 bit/s. So the rate does not measure the information contained in a random variable. We also need to keep in mind that the differential entropy can be negative.

Th 1.3.1. A **Gaussian distribution maximizes the differential entropy, for a fixed variance**.

The differential entropy of a b bit quantization of a random variable x is $h(x) + b$. (what happens if we quantize more and more finely?)

If we discretize a variable in b bits, then what we obtain is another random variable. What if we want to compute the differential entropy of that random variable? Why do we use differential entropy even though we discretized? We just elongate it:



If we push $b \rightarrow \infty$, it diverges!
Btw if we want a reference for information theory, we can use the book "Cover & Thomas" (p. 247).

Pag 10 of the slides: how to calculate the differential entropy of a Gaussian complex vector.

Example 1.6: Differential entropy of a complex Gaussian vector

Let $\mathbf{x} \sim \mathcal{N}_c(\boldsymbol{\mu}, \mathbf{R})$. From (C.15) and (1.22).

$$h(\mathbf{x}) = -\mathbb{E}[\log_2 f_{\mathbf{x}}(\mathbf{x})] \quad (1.23)$$

$$= \log_2 \det(\pi \mathbf{R}) + \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu})] \log_2 e \quad (1.24)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu})]) \log_2 e \quad (1.25)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbb{E}[\mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu}) (\mathbf{x} - \boldsymbol{\mu})^*]) \log_2 e \quad (1.26)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbf{R}^{-1} \underbrace{\mathbb{E}[(\mathbf{x} - \boldsymbol{\mu}) (\mathbf{x} - \boldsymbol{\mu})^*]}_{\mathbf{R}}) \log_2 e \quad (1.27)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbf{I}) \log_2 e \quad (1.28)$$

$$= \log_2 \det(\pi e \mathbf{R}), \quad (1.29)$$

Where in (1.25) we used the fact that a scalar equals its trace, while in (1.26) we invoked the commutative property in (B.26).

Note: $\det(\exp(\mathbf{A})) = \exp(\text{tr}(\mathbf{A}))$

Example 1.5: Differential entropy of a complex Gaussian scalar

Let $x \sim \mathcal{N}_c(\mu, \sigma^2)$. Invoking the PDF in (C.14),

$$h(x) = \mathbb{E} \left[\frac{|x - \mu|^2}{\sigma^2} \log_2 e + \log_2(\pi \sigma^2) \right] = \log_2(\pi e \sigma^2).$$

Equivalent scalar result

Professor freaks out over the fact that we don't know what the trace function is. The trace of a matrix is the sum of the diagonals. If we want to compute the sum of the eigenvalues, we can just compute the trace of a matrix.

So we have that:

$$h(\underline{x}) = \log_2(\det(\pi e \underline{R}))$$

$$h(x) = \log_2(\pi e \sigma^2)$$

At least not for a Gaussian R.V.!

The differential entropy DOES NOT depend on the mean of the variable! Another very important thing is that the differential entropy is a property of the distribution of a random variable, not of the random variable itself. Two random variables having the same distribution will have the same random variable.

$$\begin{array}{ccc} R.V. \text{ is a map } & \Omega & \longrightarrow \mathcal{S} \\ & \downarrow & \downarrow \\ & \text{events} & \text{probability} \end{array}$$

Elaborating a bit on the computation of the differential entropy: we have that

$$h(\mathbf{x}) = -\mathbb{E}[\log_2 f_{\mathbf{X}}(\mathbf{x})] \quad (1.23)$$

$$\begin{aligned} &= -\mathbb{E} \left[\log_2 \left(\frac{1}{\det(\pi \mathbf{R}_x)} \exp(-(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)) \right) \right] \\ &= \log_2(\det(\pi \mathbf{R}_x)) - \mathbb{E} \left[\frac{\ln(\exp(-(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)))}{\ln 2} \right] \\ &= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)] \log_2 e \end{aligned} \quad (1.24)$$

and now we observe that $(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)$ actually is a scalar. We can then exploit the fact that a scalar is equal to its trace:

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbb{E} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)]] \log_2 e \quad (1.25)$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [\text{Tr} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)]] \log_2 e \quad \text{Linearity}$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [\text{Tr} [\mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x) (\mathbf{X} - \boldsymbol{\mu}_x)^*]] \log_2 e \quad \text{Ciclicity}$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbf{R}_x^{-1} \mathbf{R}_x] \log_2 e$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbf{I}] \log_2 e \quad (1.28)$$

if then we consider $\mathbf{R}_x, \mathbf{I}$ as $N \times N$ matrices, we'll have

$$= \log_2(\det(\pi \mathbf{R}_x)) + N \log_2 e$$

$$= \log_2(e^N \det(\pi \mathbf{R}_x))$$

and, finally, for a matrix $\mathbf{A}$ of dimensionality $N \times N$ it holds that $a^N \det(\mathbf{A}) = \det(a\mathbf{A})$, so:

$$h(\mathbf{x}) = \log_2(\det(\pi e \mathbf{R}_x)). \quad (1.29)$$

Finally, we can simplify this for the scalar case as the determinant of a scalar is the scalar itself, getting that:

$$h(x) = \log_2(\pi e \sigma^2).$$

 **Note:** Btw the B.26 relation that they mentioned is

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}), \quad (\text{B.26})$$

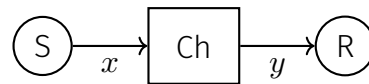
which was here applied to $\mathbf{A} = (\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1}$, $\mathbf{B} = (\mathbf{x} - \boldsymbol{\mu})$.

1.4 Information divergence, mutual information

We have that the Kullback-Leibler divergence is defined as:

$$\mathcal{D}(p||q) = \begin{cases} \sum_x p(x) \log_2 \left(\frac{p(x)}{q(x)} \right) & \text{discrete } x \\ \int p(x) \log_2 \left(\frac{p(x)}{q(x)} \right) & \text{continuous } x \end{cases} \quad (1.34+1.36)$$

this is also called the **relative entropy**. Another way to measure such thing is the Earth-Mover distance.



The mutual information quantifies the reduction in uncertainty between two random variables if one of them is observed.

Discrete:

$$\begin{aligned}
 I(s; y) &= H(s) - H(s|y) = H(y) - H(y|s) \\
 &= D(p_{sy} || p_s p_y) \\
 &= \sum_s \sum_y p_{sy}(s, y) \log_2 \left(\frac{p_{sy}(s, y)}{p_s(s) p_y(y)} \right)
 \end{aligned}$$

Continuous:

$$\begin{aligned}
 &= \mathbb{E}[\text{information density}] \equiv \mathbb{E}[i(s; y)] \\
 I(s; y) &= h(s) - h(s|y) = h(y) - h(y|s) \\
 &= D(f_{sy} || f_s f_y) \\
 &= \iint f_{sy}(s, y) \log_2 \left(\frac{f_{sy}(s, y)}{f_s(s) f_y(y)} \right) ds dy
 \end{aligned}$$

He talked about the **Landau symbols**; he didn't say anything relevant really.

1.4.1 Power series expansions

Useful means to approximate functions especially low or high SNR.

Zero order expansion about 0

$$f(x) = f(0) + \mathcal{O}(x)$$

First order expansion about 0

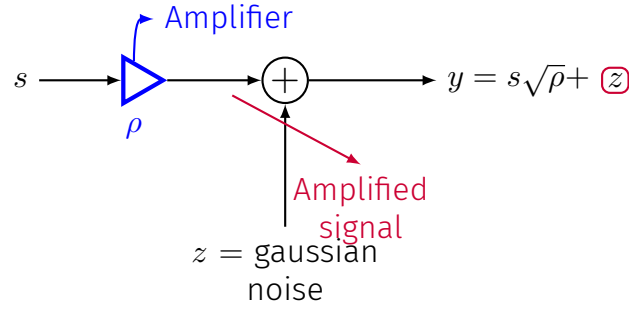
$$f(x) = f(0) + f'(0)x + \mathcal{O}(x^2)$$

First order expansion about Infinity

$$f(1/x) = f(0) + \frac{f'(0)}{x} + \mathcal{O}\left(\frac{1}{x^2}\right)$$

```
In[11]:= Series[f[x], {x, 0, 2}]
Out[11]= f[0] + f'[0]x + 1/2 f''[0]x^2 + O[x]^3
```

Important! → SL. **16**



We can ask ourselves: how much is the gaussian noise corrupting what I'm sending?

We amplify ρ (which is $\sqrt{\rho}$ in power) $\Rightarrow$ we get an ampl. of $\sqrt{\rho}$ on power!

1.4.2 Mutual information of a Gaussian scalar through an AWGN channel

We were calculating the **mutual information** for a Gaussian complex scalar.

We talked about the relation $y = \sqrt{\rho}s + z$. We have that s, z are **independent**. It might seem weird that we're putting noise in a channel, but it will make sense later.

Our constellation input always will have $\sigma^2 = 1$, such that $s \sim \mathcal{N}_c(0, 1)$; we'll also have a noise $z \sim \mathcal{N}_c(0, 1)$. We have that y is made of a sum of Gaussian independent random variables, and will thus be a Gaussian random variable, with a variance given by a sum of variances: we'll have

- $y \sim \mathcal{N}_c(0, 1 + \rho)$
- $y|s \sim \mathcal{N}_c(\sqrt{\rho}s, 1)$: this is y with s being fixed. We now want to calculate the mutual information between the input and the output of the signal.

We know the differential entropy of a Gaussian scalar and we know that:

$$\begin{aligned}
 \mathcal{I}(\rho) &= I(s; \sqrt{\rho}s + z) \\
 &= h(\sqrt{\rho}s + z) - h(\sqrt{\rho}s + z|s) \\
 &= h(\sqrt{\rho}s + z) - h(z) \\
 &= \log_2(\pi e(1 + \rho)) - \log_2(\pi e) \\
 &= \log_2(1 + \rho).
 \end{aligned} \tag{1.51}$$

Relevant comments: In $h(\sqrt{\rho}s + z|s) = h(z)$, we know that we've already **fixed** s ; s multiplied by whichever scalar won't bring any extra information than z , as we already know what s is; for this reason, the differential entropies are the same.

If we then consider $\rho \rightarrow 0$, we have:

$$\log_e(1 + x) \xrightarrow{x \rightarrow 0} x - \frac{1}{2}x^2 + \mathcal{O}(x^3)$$

So we need to **convert to base e** and we'll then be able to use the expansion:

$$\log_2(1 + \rho) = \left(\rho - \frac{1}{2}\rho^2 \right) \log_2 e + \mathcal{O}(\rho^3)$$

And **for large** ρ we'll then have:

$$\log_2(1 + \rho) \xrightarrow{\rho \rightarrow \infty} \log_2(\rho) + \mathcal{O}(1/\rho)$$

 **Note:** This is **crucially important** as it allows to model an AWGN channel! It allows us to define the **capacity** as

$$\frac{1}{2} \log(1 + \text{SNR})$$

(we don't have 1/2 since this is complex; we'd have 1/2 if it wasn't complex!).

Chapter 2

Chapter 2 Week 27 October - 2 November

2.1 MIMO Intuition

We'll use random vectors in order to model MIMO communications Our book is **MIMO**! For a long time we had a setup with a transmitter and a receiver.

People asked: what can we do in order to increase the **spectral efficiency** of our signals? The spectral efficiency is how many **bit/s/Hz** we can squeeze.

The spectral efficiency we've reached is between **1 and 5 bit/s/Hz**; we generally have **4 bit/s/Hz**.

$B \cdot \nu$ is how many bits we can send (B = bandwidth, ν = spectral efficiency).

What if we have **two antennas** at the transmitter and **two at the receiver**? It's been shown that by doing such, **it's like having twice the bandwidth**: this is the whole concept of **MIMO**.

Of course we have to pay **double**.

With two antennas we're transmitting two symbols, aka a **vector**:

$$\begin{bmatrix} T_x \\ S_1 \\ S_2 \end{bmatrix}$$

And the received signal vector is:

$$\begin{bmatrix} L_{RC} \\ y_1 \\ y_2 \end{bmatrix} = A \begin{bmatrix} S_1 \\ S_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

We thus need to use **vectors**, we can't stay with **scalars**!

2.2 Mutual information of a gaussian vector through an AWGN channel (+figure of noise)

Let's say we have a vector $\mathbf{s}$ and a matrix $\mathbf{A}$ (which is needed since every signal is communicating with each antenna!): **we're now in the MIMO setting**.

We have a **Gaussian vector** $\mathbf{s}$, a **matrix** $\mathbf{A}$ and a **noise Gaussian vector** $\mathbf{z}$; we'll have an output

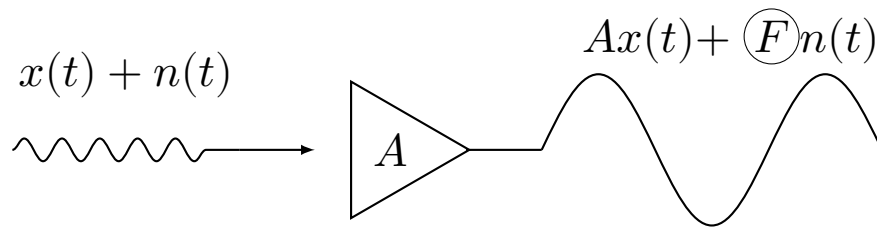
$$\mathbf{y} = \sqrt{P_A} \mathbf{s} + \mathbf{z},$$

where $\mathbf{A}$ is considered **deterministic** (we'll see it's not really true), $\mathbf{z}$ is the **noise**, which originates from **thermal noise: electrons are not still**; the more we increase the temperature, the more **electrons move**; we want to transmit a certain voltage, but we'll measure a certain fluctuating value added to our transmitted signal: the **fluctuating addition comes from thermal noise**!

2.2.1 Noise figure

✎ Noise figure

What about **noise figure**? When we amplify a signal, do we pay a price? We're amplifying the noise as well, but we might amplify the noise more: the amplifier will add its own noise to the noise, which is F , the noise figure:



We now want to calculate the mutual information between source and output; the idea is the same, only with matrices and vectors; we have

$$\mathcal{J}(\rho) = I(\mathbf{s}; \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) \quad (1.74)$$

$$= h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) - h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z} | \mathbf{s}) \quad (1.75)$$

$$= h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) - h(\mathbf{z}) \quad (1.76)$$

$$= \log_2 \det(\pi e(\rho \mathbf{A} \mathbf{R}_s \mathbf{A}^* + \mathbf{R}_z)) - \log_2 \det(\pi e \mathbf{R}_z) \quad (1.77)$$

$$= \log_2 \det(\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^* \mathbf{R}_z^{-1}), \quad (1.78)$$

with (1.77) following from the definition of **differential entropy of a gaussian vector**. The correlation matrix of $\sqrt{\rho} \mathbf{A} \mathbf{s}$ is $\rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*$; we then have a sum, so we'll have $\mathbf{R}_z$. The fact that in the last passage (1.78) we have $\mathbf{R}_z$ at the denominator, means we'll be multiplying by $\mathbf{R}_z^{-1}$; this follows from the fact that

$$\frac{\det(\mathbf{A})}{\det(\mathbf{B})} = \det(\mathbf{A}\mathbf{B}^{-1}),$$

here applied with $\mathbf{A} = \rho\mathbf{A}\mathbf{R}_s\mathbf{A}^* + \mathbf{R}_z$ and $\mathbf{B} = \mathbf{R}_z$.

How do we know that $\sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$ is Gaussian?

I think the idea is that we can consider $\sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$ as a multiplication between the matrix

$$[\sqrt{\rho}\mathbf{A} \quad 1]$$

and the vector

$$\begin{bmatrix} \mathbf{s} \\ \mathbf{z} \end{bmatrix},$$

which will result in a random vector with distribution

$$\sim \mathcal{N}_{\mathbb{C}} \left([\sqrt{\rho}\mathbf{A} \quad 1] \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix}, [\sqrt{\rho}\mathbf{A} \quad 1] \cdot \begin{bmatrix} \mathbf{R}_s & 0 \\ 0 & \mathbf{R}_z \end{bmatrix} \cdot \begin{bmatrix} \sqrt{\rho}\mathbf{A}^* \\ 1 \end{bmatrix} \right),$$

according to

Transformation of a Gaussian vector

If we have:

- A Gaussian vector $\mathbf{x} \sim \mathcal{N}_{\mathbb{C}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ of dimensions $N \times 1$,
- a matrix $\mathbf{A}$ of dimensions $M \times N$
- a vector $\mathbf{b}$ of dimensions $M \times 1$

we have that the vector

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$$

will be distributed according to

$$\mathbf{y} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^*).$$

► Extension of [Uni generale](#).

Derivative of logarithm of determinant

In order to go on, we consider the following relations:

$$\begin{aligned} \left. \frac{\partial}{\partial \rho} \log_e \det(\mathbf{I} + \rho\mathbf{B}) \right|_{\rho=0} &= \text{tr}(\mathbf{B}) \\ \left. \frac{\partial^2}{\partial \rho^2} \log_e \det(\mathbf{I} + \rho\mathbf{B}) \right|_{\rho=0} &= -\text{tr}(\mathbf{B}^2) \end{aligned}$$

We can then apply some linear algebra: we'll get that for $\rho \rightarrow 0$ follows

$$\mathcal{J}(\rho) = \left[\text{tr}(\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1})\rho - \frac{1}{2}\text{tr}((\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1})^2)\rho^2 \right] \log_2 e + o(\rho^2)$$

About this passage: it all depends on the fact that $\log \det$ is a **convex function**! We know the derivative of $\log \det$ and we're applying it to the fact that we're summing the eigenvalues: that's how we get the **trace**!

Page 74 of a linear algebra book he recommended

This is a very useful result!

If N_s, N_y are the dimensions of the vectors, for $\rho \rightarrow +\infty$ we get the result

$$\mathcal{J}(\rho) = \min(N_s, N_y) \log_2 \rho + \log_2 \det(\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1}) + O\left(\frac{1}{\rho}\right)$$

Professor said that this equation has a minor detail which is **wrong**, proceeded into not telling us what it was :D

2.3 Gaussian mutual information for an arbitrary constellation

Let's see the Gaussian mutual information for an arbitrary constellation.

Let $\mathbf{s}$ be a zero-mean unit-variance discrete random variable taking values in $\mathbf{s}_0, \dots, \mathbf{s}_{M-1}$ with probabilities $p_0, \dots, p_{M-1}$. This subsumes **M-PSK**, **M-QAM**, and any other discrete constellation. The PDF of $\mathbf{y} = \sqrt{\rho}\mathbf{s} + \mathbf{z}$ equals

$$f_{\mathbf{y}}(\mathbf{y}) = \frac{1}{\sqrt{2\pi}} \sum_{m=0}^{M-1} p_m e^{-|\mathbf{y} - \sqrt{\rho}\mathbf{s}_m|^2} \quad (1.65v)$$

whereas $y|\mathbf{s} \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}\mathbf{s}, 1)$. Thus,

$$\mathcal{J}^{\text{aw}}(\rho) = I(\mathbf{s}; \sqrt{\rho}\mathbf{s} + \mathbf{z}) \quad (1.66)$$

$$= - \int f_{\mathbf{y}}(\mathbf{y}) \log_2 f_{\mathbf{y}}(\mathbf{y}) d\mathbf{y} - \log_2(\pi e) \quad (1.67)$$

with integration over the complex plane.

$$\mathcal{J}_{\text{aw}}^{\text{aw}}(\rho) = \log_2 M - \epsilon \quad \text{Large a first order expansion} \quad \log_2 e = \frac{d_{\min}^2}{4} \rho + o(\rho)$$

dove

$$d_{\min} = \min_{k \neq l} |\mathbf{s}_k - \mathbf{s}_l|$$

Comments:

- In order to obtain (1.65) we used

$$f_{\mathbf{y}}(y) = \sum_{m=0}^{M-1} f_{\mathbf{y}|\mathbf{s}}(y|\mathbf{s}_m) p_{\mathbf{s}}(\mathbf{s}_m),$$

with $p_{\mathbf{s}}(\mathbf{s}_m) = p_m$.

- (1.67) is just $h(\mathbf{y}) - h(\mathbf{y}|\mathbf{s})$, considering that $\mathbf{y}|\mathbf{s}$ is a **Gaussian RV with variance 1** and thus $h(\mathbf{y}|\mathbf{s}) = h(\mathbf{z}) = \log_2(\pi e)$.
- We don't see σ^2 since it's $\sigma_{\mathbf{z}}^2 = 1$!
- ϵ is first defined here

(1.66) is the definition of mutual information. Solving the integral is pretty difficult.

What professor wants us to remember though is that

$$\mathcal{I}^{\text{aw-ary}}(\rho) \xrightarrow{\rho \rightarrow +\infty} \log_2(M) - \epsilon,$$

with

$$\log_2(\epsilon) = \frac{d_{\min}^2}{4} \rho + o(\rho)$$

intuitively, a constellation with M points asymptotically gives $\log_2(M)$ bits of information per channel use; of course this is not the case and we need to lower that term using ϵ , which brings us from the **best case scenario** to a more realistic case.

2.4 Mutual information of an encoded word

2.4.1 Theoretical treatment: results for generic $\mathbf{s}, \tilde{\mathbf{y}}$ (+for iid values and for a stationary channel)

We have an abstraction of the communication link:

We have an encoder, which maps a signal to a codebook with $2^{N_{\text{bits}}}$ words. We send a codeword of N bits $x[n], n \in [0, N-1]$. We then send our signal through the channel, which is characterized by **noise, amplification, ...**; we'll receive a whole word $\mathbf{y}[n], n \in [0, N-1]$. If we want to calculate the **mutual information between input words and output words**, we need to use the **conditioned information function**, not the joint one: we know the probability of receiving a certain message $\mathbf{y} = [y[0], \dots, y[N-1]]$ (as it is $f_{\mathbf{y}}$) when sending a certain $\mathbf{s} = [s[0], \dots, s[N-1]]$ and we need to work with that!

If we assume that the various symbols are **equiprobable**, we find that

$$f_{\mathbf{y}}(\mathbf{Y}) = \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{Y}|\mathbf{s}_m) p_{\mathbf{s}}(\mathbf{s}_m) \stackrel{\text{equiprobable } \mathbf{s}_m}{=} \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{Y}|\mathbf{s}_m) \frac{1}{2^{N_{\text{bits}}}}$$

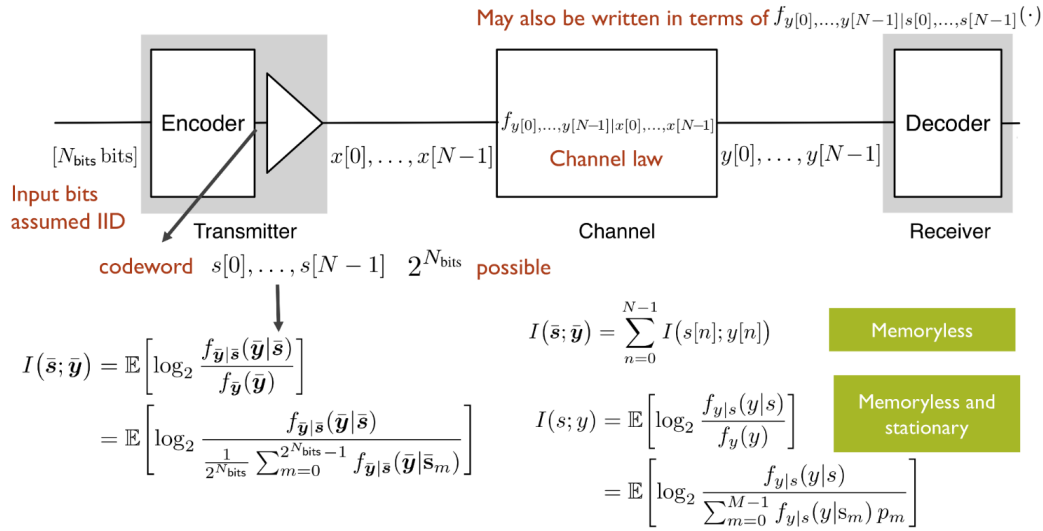


Figure 2.1: Abstraction of the communication link

and this enables us to write

$$I(\mathbf{s}; \mathbf{y}) = \mathcal{H}(\mathbf{y}) - \mathcal{H}(\mathbf{y} | \mathbf{s}) \quad (1.88)$$

$$= \mathbb{E} \left[\log_2 \left(\frac{f_{\mathbf{y} | \mathbf{s}}(\mathbf{y} | \mathbf{s})}{f_{\mathbf{y}}(\mathbf{y})} \right) \right]$$

$$= \mathbb{E} \left[\log_2 \left(\frac{f_{\mathbf{y} | \mathbf{s}}(\mathbf{y} | \mathbf{s})}{\frac{1}{2^{N_{\text{bits}}}} \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y} | \mathbf{s}}(\mathbf{y} | \mathbf{s}_m)} \right) \right]. \quad (1.89)$$

If the channel is **memory-less** (which might not be true: $y[1]$ might depend on $y[0]$ etc etc), the mutual information of input and output is the **sum of mutual information of every couple of input and output**: we don't have vectors anymore (and we like this!).

If this is the case, the channel law will factor out as

$$f_{\mathbf{y} | \mathbf{s}}(\mathbf{y} | \mathbf{s}) = \prod_{n=0}^{N-1} f_{y[n] | s[n]}(y[n] | s[n]).$$

with

$$I(s[n]; y[n]) = \mathbb{E} \left[\log_2 \frac{f_{y[n] | s[n]}(y[n] | s[n])}{f_{y[n]}(y[n])} \right]. \quad (1.93)$$

Hidden message

Looks like one hidden message is that if the conditioned probability factorizes, we can easily rewrite the mutual information as a summation. This follows from the fact that if the single elements are iid, then the expected value factorizes as well.

If the channel is also **stationary**, we can simplify further, as all the symbols will have the same statistics: $s \equiv s[n] \forall n$ and $y \equiv y[n] \forall n$ will hold, and can write (1.93) as

$$I(s; y) = \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)} \right] \quad (1.94)$$

$$= \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{\sum_{m=0}^{M-1} f_{y|s}(y|s_m) p_m} \right], \quad (1.95)$$

with p_m being the probability that a symbol s is s_m ; we can of course further simplify this writing if we assume equiprobability, $p_m = 1/M$.

This comes from the fact that any output does not store any information about any input other than the one which corresponds to it:

$$I(s[k]; y[n]) = \begin{cases} 0 & k \neq n \\ \text{Something} & \text{otherwise} \end{cases}.$$

2.5 Memoryless channel law with Gaussian noise

Let's see an example; if our output is defined as

$$y[n] = \sqrt{\rho} s[n] + z[n] \quad n = 0, \dots, N-1 \quad (1.96)$$

and we have that $z[0], \dots, z[N-1]$ are IID with $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ the channel law will be the same as z , now centered in mean in the same value as the transmitted s :

$$f_{y|s}(y|s) = \frac{1}{\pi} e^{-|y - \sqrt{\rho} s|^2}. \quad (1.97)$$

The channel law is the probability that we receive y when we're transmitting s .

2.5.1 Capacity of a channel, information stability, spectral efficiency (+codeword error, information density)

We come to an extra important concept: **capacity**. We can express it in different ways; we'll see **bit/s**; when we'll put the symbols one after the other.

Ideally we must study the book, in order to fully grasp these concepts

The **codeword error** probability for equiprobable words of length N is

$$p_e = \frac{1}{2^N} \sum_{m=0}^{2^N-1} P[\hat{w} \neq m | w = m],$$

where w is the transmitted word and $\hat{w}$ is the received word.

⚙️ Capacity (bits/symbol)

The capacity C in bits per symbol is the highest rate that can be achieved such that for large values of N we have

$$p_e \xrightarrow{N \rightarrow +\infty} 0 .$$

The **information density** between an input word $\bar{s}$ and an output word $\bar{y}$ is defined as

$$i(\bar{s}; \bar{y}) = \log_2 \left(\frac{f_{\bar{s}, \bar{y}}(\bar{S}, \bar{Y})}{f_{\bar{s}}(\bar{S}) f_{\bar{y}}(\bar{Y})} \right) .$$

If the **channel** is **information stable**, then we have that

$$\lim_{N \rightarrow +\infty} \frac{1}{N} i(\bar{s}, \bar{y}) = \lim_{N \rightarrow +\infty} \frac{1}{N} \mathbb{E}[i(\bar{s}, \bar{y})] \stackrel{*}{=} \lim_{N \rightarrow +\infty} \frac{1}{N} I(\bar{s}, \bar{y}) .$$

👤 Professor's definition of information stable

A channel is information stable if the mutual information does not vary if we jump in the future or past; it's not taking the first symbols and conveying less information than the symbols coming in later.

This relation is saying that the information conveyed by $y[0], \dots, y[N-1]$ about $s[0], \dots, s[N-1]$ is invariant, provided that N is large enough

*: note that, by **definition**,

$$\mathbb{E}[i(s; y)] = I(s; y) .$$

⚙️ Sufficient condition for information stability

A sufficient condition for a channel to be information stable is that **the channel is stationary and ergodic** (ref)

✍️ Ergodicity in mean

We have a random process $x(n)$ and we know that $\mathbb{E}[x(n)] = B$.

A signal is **ergodic in mean** if

$$\lim_{N \rightarrow +\infty} \sum_{n=0}^{N-1} \frac{x(n)}{N} = B :$$

the average in time is the same as the average taken in the probability space

If the channel is stationary and ergodic, then

$$C = \max_{\text{signal constraints}} \left(\lim_{N \rightarrow +\infty} \frac{1}{N} I(\bar{s}, \bar{y}) \right) .$$

The maximization is over the joint distribution of the unit-power codeword symbols $s[0], \dots, s[N-1]$.

Zero mean signals

The mean has no influence over the mutual information (and thus the capacity), but it does influence the average power spent for the transmission; for this reason, from now onwards, we'll only consider zero-mean signals.

We want the codeword error probability to go to zero for long words. We define an **information stable channel**; we then have the information density: if it's information stable, the information density must become the mutual information for large N s. The capacity is the **maximum mutual information which we can achieve by changing the distribution of the input signal**. We have very long codewords and we want the probability of error on the codeword to be as low as we want.

If we have a pipe, the capacity is how much water we can push through without losing any water

In order to reach the capacity, what we can do is changing the input distribution of the symbols.

If the channel is **static, memoryless and ergodic**, then we simply have

$$C = \max_{\text{signal constraints}} I(s; y).$$

The bitrate R at which we can reliably communicate with a period T satisfies $R \leq C/T$. From the sampling theorem we have $1/T \leq B$, where B is the passband bandwidth of our transmission. By joining the two conditions, we obtain that

$$\underbrace{\frac{R}{B}}_{\nu, \text{ spectral efficiency}} \leq C \leftrightarrow \nu \leq C,$$

where C uses the alternative measure bit/s/Hz and where the equality is reached asymptotically.

This is telling us that the capacity is not only the maximum bitrate, but also the maximum spectral capacity which we can reach.

The capacity involves searching in all the possible distributions, we'll always stay underneath it, and that's something we always need to take into account!

2.6 BICM architecture (actually: Information loss when assuming bit independence, interleavers, soft decoding)

From the book: In paragraphs **before 1.5.4**, the book showed that using **binary encoding** and **constellation mapping** together yields no loss of optimality. We can ask ourselves whether this also holds when we combine **soft demapping** and **binary decoding**. In order to investigate this, we consider a stationary and memoryless channel as well as an M -point equiprobable constellation.

A primer on information theory and MMSE estimation (pp. 3-56), p. 28

In this setting, codewords with IID entries are optimum and thus bits mapped to distinct symbols can be taken to be independent. However, **the channel does introduce dependencies among same-symbol bits**. Even with the coded bits produced by the binary encoder being statistically independent (as we know from *past paragraphs*), after demapping at the receiver dependencies do exist among the soft values for bits that traveled on the same symbol. Unaware, a binary decoder designed for IID bits ignores these dependencies and regards the channel as being memoryless, not only at a symbol level but further at a bit level [91]. Let us see **how much information can be recovered under this premise**.

First of all, we look at the decoder counterpart of (1.94), (1.95), if the channel is also stationary, we can simplify further, as all the symbols will have the same statistics: $s \equiv s[n] \forall n$ and $y \equiv y[n] \forall n$ will hold, and can write (1.93) as

$$I(s; y) = \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)} \right] \quad (1.94)$$

$$= \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{\sum_{m=0}^{M-1} f_{y|s}(y|s_m) p_m} \right] \quad (1.95)$$

with p_m being the probability that a symbol s is s_m ; we can of course further simplify this writing if we assume equiprobability, $p_m = 1/M$, which is

$$I_{\text{aw}}^{\text{BICM}} = \sum_{k=0}^{m_k-1} I(\mathbf{b}_k; \mathbf{y})$$

after much hard work, we find that it is

$$I_{\text{aw}}^{\text{BICM}} = \mathbb{E}_{\mathbf{y}} \left[\sum_{k=1}^{m_k} \mathbb{E}_{\mathbf{s}} \left[\log_2 \frac{\sum_{i=0}^{2^{m_k}-1} \mathbb{E}_{\mathbf{s}_{\setminus k}} [f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s}_i)]}{\sum_{i=0}^{2^{m_k}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s}_i)} \right] \right] \quad (1.137)$$

where $\mathbf{s}[k]$ are the subsets of constellation points where k -th bit is either 0 or 1; just to give an example:

With constellations such as BPSK and QPSK, where a single coded bit is mapped to the in-phase and quadrature dimensions of the constellation and thus **no dependencies**

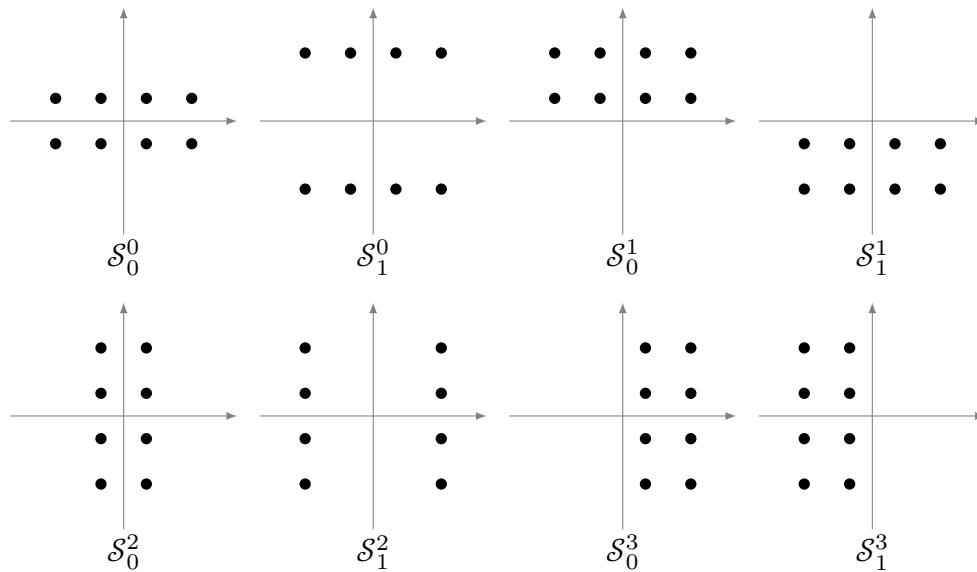


Figure 2.2: Above, 16-QAM constellation with Gray mapping. Below, subsets $\mathcal{S}_0^\ell$ and $\mathcal{S}_1^\ell$ for $\ell = 0, 1, 2, 3$ with the bits ordered from right to left.

among same-symbol coded bits, we have that $\sum_k I(b_k; y) = I(\mathbf{s}; \mathbf{y})$, since the binary decoder is not disregarding information. Instead, **if the decoder disregards information** (acquiring information about other bits in the same symbol), the binary coding yields $\sum_k I(b_k; y) < I(\mathbf{s}; \mathbf{y})$.

Skipped

Several examples have been skipped: 1.20, 1.21, 1.22, 1.23

Anyhow, it's true that the difference between $\sum_k I(b_k; y)$ and $I(\mathbf{s}; \mathbf{y})$ is generally tiny, if the mapping of coded bits to constellation is chosen wisely. An example of wise choice is **Gray mapping**, where neighboring constellation points only differ of one bit.

There's just one missing thing in order to have a full information-theory representation of modern transmission systems, which is **interleaving**, which generally concerns shuffling either symbols or bits in an invertible manner. Although symbol-level interleaving would suffice to break off bursts of poor channel conditions, bit-level interleaving has the added advantage of shuffling also the bits contained in a given symbol; if the interleaving were deep enough to push any bit dependencies beyond the confines of each codeword, then the gap between $\sum_k \mathcal{I}(b_k; y|\mathbf{s})$ would be closed. It's true that interleaving has no effect when $N \rightarrow \infty$ and will thus have no effect on the mutual information calculations, but it's also true that it will improve the performance of actual codes with finite N .

The usage of **binary coding/decoding, bit-level interleaving, constellation mappers/soft demappers** gives the so-called **Bit-Interleaved Coded Modulation (BICM)** architecture. BICM is information-theoretically equivalent to signal-space coding for BPSK and Gray-Mapped QPSK. For higher-order constellations, even if the dependencies among same symbol bits are not fully eradicated by interleaving and the receiver ignores them, it

is only slightly inferior.

In class:

What we usually do is that we have the input bits and we do **channel coding**; normally, channel coding is a **binary coding**.

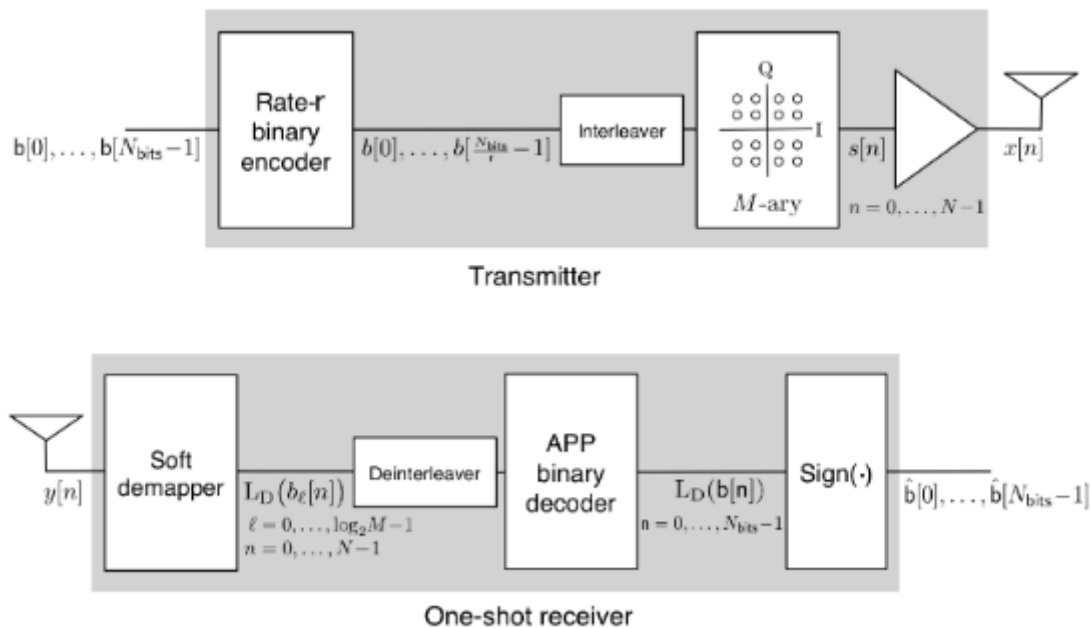


Figure 2.3: Top: BICM Transmitter Chain. Bottom: BICM Receiver Chain with Soft Demapper and Decoder.

Interleaver: sometimes the channel does errors in bursts, (it might be in a bad condition, ...); in that situation, the decoder cannot work as we'd have a long string of errors. What we do is to scramble the bits; if we have a matrix of memory, we write the values in rows

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

We read:

$\longrightarrow$ 1, 2, 3, 4,
 $\longleftarrow$ 8, 7, 6, 5,
 $\longrightarrow$ 9, 10, 11, 12,
 $\longleftarrow$ 16, 15, 14, 13

And then:

↓ 1, 5, 9, 13,
 ↓ 2, 6, 10, 14,
 ↓ 3, 7, 11, 15,
 ↓ 4, 8, 12, 16

This way, bits which were close are now far apart, avoiding many consecutive errors; this is a simple interleaver! Then of course we'll do the reverse in the output.

There are many ways to implement interleavers, of course.

A positive **L-value** indicates that the bit in question is more likely to be a 1 than a 0, and vice versa for a negative value, with the magnitude indicating the confidence of the decision.

For a certain bit b , we define the **L-value** as

$$L(b) = \log \frac{P(b=1)}{P(b=0)}$$

such that if $L(b) > 0$, it's more likely that we have $b = 1$ and if $L(b) < 0$ it's more likely that we have $b = 0$. The bigger the magnitude, the more certain we are about our result.

At page 25 of the book, this concept is applied to an actual message, in the form of a **soft decoder**.

This thing we've just described (together with many other much more complex things handled from page 25 to page 30) is what is called an "A Posteriori Probability" decoder or "**APP**" decoder for short.

page 28

Whichever the type of code, the sign of the L-values generated by an APP decoder for the message bits directly gives the MAP decisions

APP decoders are indeed really cool.

Let's say we transmit bits; we have a system which gives us values between -1 and 1; the closer we are to 1, the more we're sure that the value will be 1; the closer we are to -1, the more we're sure the value will be 0. We then use the sign function and we have back the binary values.

From what I gathered, an **interleaver** is such to scramble bits in order to remove dependency between bits.

The rest: didn't get it at all :D

Chapter 3

Week 3-9 November

3.1 Spectral efficiency for finite-length codewords

We can reach the Shannon bound only by pushing N to infinity; only if we're sending a **really** long codeword can we reach the capacity. We'd need to wait a long time (an **infinite** amount of time, I think) to fill that codeword and then we'd have to wait lots of latency in order to receive such a large signal.

We can show that by using small (non infinite!) N values, we don't reach the capacity and we can indeed show to which spectral efficiency we actually get!

$$\frac{R}{B} = C - \sqrt{\frac{V}{N}} Q^{-1}(p_e) + \mathcal{O}\left(\frac{\log N}{N}\right). \quad (1.143)$$

How below we stay depends on N, V, p_e ; we don't need to remember the capacity, but we do need to remember the fact that **it's lower when compared to the capacity**.

Takeaway points are:

1. If the **block length N increases**, we get closer to the **ideal** capacity
2. If the **probability of error decreases**, we get closer to the **ideal** capacity (since $p_e \downarrow \Rightarrow Q \uparrow \Rightarrow (1/Q) \downarrow$)
3. If the **information density has a lower variance** (totally intuitively, i.e. if input/output are more related), we get closer to the **ideal** capacity

V is the variance of the information density, defined as

$$V = \text{var}[i(s; y)]. \quad (1.144)$$

This is a really cool result, as it allows us to understand how far away we are from ideal capacity.

About the exam

The sections published on the website are the program for the exam.

3.2 MMSE estimation

Ref: [Chapter_1_EE_381S_Spg_2019, 08.1_pp_3_56_A_primer_on_information_theory_and_MMSE_estimation, page 37.](#)

Probably next week we'll receive an email about the 1st of November and about where the lesson has been moved.

18th, 19th of 20th of November (and even more), professor will be in Paris. We'll have 3 extra dates, probably wed 16:30-18:30. We'll be able to find all the information about this on elearning.

3.2.1 Intro to MMSE estimation, scalar estimator (and the properties an estimator should have)

Today we'll do Minimum Mean Squared Error Estimation: MMSE estimation. This is something between telecommunications and control engineering. It goes back to Gauss and friends, but we've had much progress in the 20th century after the world war; the two major achievements are the Wieman and Calman filter. Without the Calman filter, we wouldn't have gone to the moon.

We'll go over different MMSE estimators; usually **estimating means we want to estimate a parameter**; here we have a transmitter and at the receiver we want to estimate the transmitted signal. This is the idea of this. This is not the only estimator though (we've seen MAP, ML, MD, ...). The last part of the chapter shows that the mutual information can be derived, instead of the usual way we've seen till now, as a derivate of what we'll call information MSE.

Our problem is to estimate a signal in noise

$$y = \sqrt{\rho}s + z ;$$

we have the observation y , we have $\sqrt{\rho}$, where ρ is the SNR and then we'll have z , Gaussian noise and s , unknown. We have some statistical information, such as:

- Posterior probability, probability density of s given y , $f_{s|y}(\cdot)$
- We have the likelihood function: we know the probability density of y given s , $f_{y|s}(\cdot)$
- Sometimes we have even the probability density of s , $f_s(\cdot)$!

Estimation problem, solution $\hat{s}(y)$

We want to find the solution $\hat{s}(y)$ which minimizes the MSE, $\mathbb{E}[\|s - \hat{s}(y)\|^2]$.

We can see this as a statement, but we'll do exercise 1.34 about this; the solution is

$$\hat{s}(y) = \mathbb{E}[s|y] . \quad (1.151)$$

When we make an estimator, we first of all need to check that the estimator is **unbiased**, aka that on average it gives me the actual quantity that I want to estimate: "If I estimate many times, I should get my estimate!"

The thermometer is kind of estimating the temperature; the thermometer might be biased, though. The thermometer might give always 1 degree more than the temperature. A good estimator should on average give a temperature of 24 if the temperature actually is 24.

This is **unbiased** in the sense that

$$\mathbb{E}[\hat{s}(y)] = \mathbb{E}[\mathbb{E}[s|y]] \stackrel{*}{=} \mathbb{E}[s] . \quad (1.153)$$

* comes from the **theorem of total expectation**, which, by definition is such that

$$\mathbb{E}_y[\mathbb{E}_s[s|y]] = \mathbb{E}_s[s] .$$

The estimator **must satisfy orthogonality**; if we multiply the error by **any function** of the observation, it must be equal to 0:

$$\mathbb{E}[g(y^*)(s - \hat{s}(y))] = 0 \quad \forall \text{ function } g(\cdot) .$$

We're conjugating y ; at the same time, g can contain a conjugate, so it's whatever: the conjugate is irrelevant. The conjugate will be useful when we'll manipulate the problem later.

Let's give an heuristic interpretation of orthogonality: if we have a plane with two vectors (y, z , at least I think) and a vector going out of the plane (s); we want to give the best estimation of the vector which is going out of the plane:

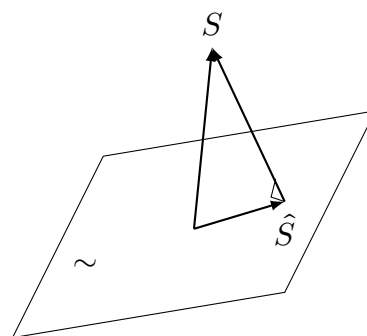


Figure 3.1: Interpretazione geometrica dell'ortogonalità: l'errore di stima è ortogonale al piano delle osservazioni.

We'll estimate the vector which is out of the plane, we'll have that our estimate is the projection of the vector on the plane and the error will be orthogonal to the plane (as can be seen in the figure).

We'll also have that the error is also orthogonal to $\hat{s}$ and to what it's available.

And what actually is the MMSE (until now we've only seen the estimator!)? It's the actual error we're making:

$$\text{MMSE}(\rho) = \mathbb{E}[\|s - \mathbb{E}[s|y]\|^2]. \quad (1.156)$$

In this equation it's important to look at the fact that the MMSE is only dependent on ρ .

3.2.2 MMSE estimator for a Complex Gaussian scalar through a channel

Let's say we have $y = \sqrt{\rho}s + z$ with

$$z \sim \mathcal{N}_{\mathbb{C}}(0, 1) \implies y|s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}s, 1) \implies f_{y|s} = \frac{1}{\pi} e^{-|y - \sqrt{\rho}s|^2}$$

We fix s ; we can try and see what $y|s$ is: it's a complex Gaussian with mean $\sqrt{\rho}s$ and variance 1.

Let's look at the solution:

$$\begin{aligned} \hat{s}(y) &= \mathbb{E}[s|y = y] \\ &= \int s f_{s|y}(s|y) \, ds \\ &= \int \frac{s f_{y|s}(y|s) f_s(s)}{\int f_{y|s}(y|s) f_s(s) \, ds} \, ds \\ &= \frac{\int s f_{y|s}(y|s) f_s(s) \, ds}{\int f_{y|s}(y|s) f_s(s) \, ds} \\ &= \frac{\int s e^{-|y - \sqrt{\rho}s|^2} f_s(s) \, ds}{\int e^{-|y - \sqrt{\rho}s|^2} f_s(s) \, ds} \end{aligned}$$

We go from $f_{s|y}$ to $f_{y|s}$ by using the Bayes theorem.
We can also have f_y as

$$f_y(Y) = \int f_{y,s}(Y, S) \, dS = \int f_{y|s}(Y|S) f_s(S) \, dS,$$

from the marginalization rules.

Then specifically, for Gaussian signals,

$$\hat{s}(y) = \frac{\int s e^{-|y - \sqrt{\rho}s|^2} e^{-|s|^2} \, ds}{\int e^{-|y - \sqrt{\rho}s|^2} e^{-|s|^2} \, ds} \quad (1.167)$$

$$= \frac{\int s e^{-\left|\frac{y}{1+\rho} - \sqrt{1+\rho}s - \frac{\sqrt{\rho}}{1+\rho}y\right|^2} \, ds}{\int e^{-\left|\frac{y}{1+\rho} - \sqrt{1+\rho}s - \frac{\sqrt{\rho}}{1+\rho}y\right|^2} \, ds} \quad (1.168)$$

$$= \frac{\int s e^{-|s - \frac{\sqrt{\rho}}{1+\rho}y|^2 / \frac{1}{1+\rho}} \, ds}{\int e^{-|s - \frac{\sqrt{\rho}}{1+\rho}y|^2 / \frac{1}{1+\rho}} \, ds} \quad (1.169)$$

which can be noticed to be the mean over the PDF of a complex gaussian distribution of

$$s' \sim \mathcal{N}_{\mathbb{C}} \left(\frac{\sqrt{\rho}}{1+\rho} y, \frac{1}{1+\rho} \right),$$

and we can then rewrite the values in (1.169) as

$$= \frac{\mathbb{E}[s']}{1} = \frac{\sqrt{\rho}}{1+\rho} y.$$

About the passages from (1.167)

To simplify the exponent in both numerator and denominator:

$$-|y - \sqrt{\rho}s|^2 - |s|^2$$

We expand:

$$|y - \sqrt{\rho}s|^2 = |y|^2 - 2\Re(y^* \sqrt{\rho}s) + \rho|s|^2$$

So,

$$-|y - \sqrt{\rho}s|^2 - |s|^2 = -|y|^2 + 2\Re(y^* \sqrt{\rho}s) - (1+\rho)|s|^2.$$

Our aim now is to try and isolate s from y ; we then consider that for $a, b \in \mathbb{R}$,

$$|as + by|^2 = a^2|s|^2 + 2ab\Re(y^*s) + b^2|y|^2;$$

if we look at the previous expression (forgetting for now of the sign flip), we can impose

$$a = \sqrt{1+\rho},$$

from which follows that

$$ab = -\sqrt{\rho} \implies b = -\frac{\sqrt{\rho}}{\sqrt{1+\rho}},$$

giving that

$$-|y - \sqrt{\rho}s|^2 - |s|^2 = -a^2|s|^2 + 2ab\Re(y^*s) - |y|^2:$$

we're missing a $-b^2|y|^2$ term! Then we sum and subtract $-b^2|y|^2$, getting

$$= -a^2|s|^2 + 2ab\Re(y^*s) - b^2|y|^2 + b^2|y|^2 - |y|^2,$$

which we can then collect as

$$= -|as + by|^2 + (b^2 - 1)|y|^2.$$

Since the term $(b^2 - 1)|y|^2$ will be in both exponentials, we can just cancel it out and forget about it. We're then left with

$$-|as + by|^2 = -\left|\sqrt{1+\rho}s - \frac{\sqrt{\rho}}{\sqrt{1+\rho}}y\right|^2 = -\left|s - \frac{\sqrt{\rho}}{1+\rho}y\right|^2 \Big/ \frac{1}{1+\rho}:$$

our initial expression will now be

$$\frac{\int_{-\infty}^{+\infty} s \exp\left(-\left|s - \frac{\sqrt{\rho}}{1+\rho}y\right|^2 \Big/ \frac{1}{1+\rho}\right) ds}{\int_{-\infty}^{+\infty} \exp\left(-\left|s - \frac{\sqrt{\rho}}{1+\rho}y\right|^2 \Big/ \frac{1}{1+\rho}\right) ds},$$

where the top is the expectation of $s \sim \mathcal{N}_{\mathbb{C}}\left(\frac{\sqrt{\rho}}{1+\rho}y, \frac{1}{1+\rho}\right)$ and the bottom is the integral of the whole PDF of such RV; we'll then have that this is

$$= \frac{\frac{\sqrt{\rho}}{1+\rho}y}{1} = \frac{\sqrt{\rho}}{1+\rho}y.$$

If we suppose $s \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, using the previous example and some calculus we find that the MMSE estimator for that s is

$$\hat{s}(y) = \frac{\sqrt{\rho}}{1+\rho}y. \quad (1.172)$$

This is good because we'll often suppose that the input is Gaussian. This is a linear transformation of the observation!

Importance of Gaussian results

Whichever type of result which assumes $s \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ is extremely important in that most of the information theory we'll derive for capacity and similar will assume a Gaussian input.

If I have to calculate the best estimation and we have a Gaussian input, we can take the input, multiply it by the quantity which we've found and we'll have our estimator.

The minimum of the mean squared error is actually

$$\text{MMSE}(\rho) = \mathbb{E} \left[\left| s - \frac{\sqrt{\rho}}{1 + \rho} y \right|^2 \right] \quad (1.173)$$

$$= \mathbb{E}[|s|^2] - 2 \frac{\sqrt{\rho}}{1 + \rho} \Re[\mathbb{E}[ys^*]] + \frac{\rho}{(1 + \rho)^2} \mathbb{E}[|y|^2] \quad (1.174)$$

$$= \mathbb{E}[|s - \hat{s}(y)|^2] = \frac{1}{1 + \rho}, \quad (1.175)$$

where (1.175) follows from the fact that $\mathbb{E}[|s|^2] = 1$, $\mathbb{E}[ys^*] = \sqrt{\rho}$, $\mathbb{E}[|y|^2] = 1 + \rho$.

This means that if ρ is big, the error becomes smaller, which is what we expect. If we have low noise, we expect to have no error. With a SNR of 0 (no signal, all error), we have an error of 1.

3.2.3 Vectorial estimator

Let's reformulate for vectors:

$$\mathbf{y} = \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}, \quad (1.186)$$

where we know that $\mathbf{A}$ is a matrix (not necessarily a square matrix though!). We have another assumption, that the SNR is the same for every signal in our vector.

MMSE estimator, MSE matrix for a random vector

The MMSE estimator is

$$\hat{\mathbf{s}}(\mathbf{y}) = \mathbb{E}[\mathbf{s}|\mathbf{y}]. \quad (1.187)$$

The MSE matrix is

$$\mathbf{E} = \mathbb{E}[(\mathbf{s} - \hat{\mathbf{s}}(\mathbf{y}))(\mathbf{s} - \hat{\mathbf{s}}(\mathbf{y}))^*]. \quad (1.188)$$

The matrix $\mathbf{E}$ is a matrix which in the main diagonal will have for a vector $\mathbf{s} = (s_1, s_2, \dots, s_n)$ that the j, j element will give $\mathbf{E}_{jj} = \text{MSE}(s_j)$.

3.2.4 MMSE estimator for a Complex Gaussian vector representing a channel

Example of estimation of a vector Gaussian: we have $\mathbf{s} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}_s)$ and $\mathbf{z} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{I})$, for a channel expressed as $\mathbf{y} = \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}$.

We can find the estimator (by using (1.187)) as

$$\hat{\mathbf{s}}(\mathbf{y}) = \sqrt{\rho} \mathbf{R}_s \mathbf{A}^* (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*)^{-1} \mathbf{y}. \quad (1.189)$$

Just like in the scalar case, the estimator is still linear.

We also have to calculate the error matrix; it will be

$$\begin{aligned} \mathbf{E} &= \mathbb{E}[\mathbf{s} \mathbf{s}^*] - \mathbb{E}[\hat{\mathbf{s}} \hat{\mathbf{s}}^*] - \mathbb{E}[\hat{\mathbf{s}} \mathbf{s}^*] + \mathbb{E}[\hat{\mathbf{s}} \hat{\mathbf{s}}^*] \\ &= \mathbf{R}_s - 2\rho \mathbf{R}_s \mathbf{A}^* (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*)^{-1} \mathbf{A} \mathbf{R}_s \\ &\quad + \rho \mathbf{R}_s \mathbf{A}^* (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*)^{-1} (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*) (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*)^{-1} \mathbf{A} \mathbf{R}_s \\ &= \mathbf{R}_s - \rho \mathbf{R}_s \mathbf{A}^* (\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*)^{-1} \mathbf{A} \mathbf{R}_s. \end{aligned}$$

We can rearrange those matrices in an elegant manner:

$$\mathbf{E} = (\mathbf{R}_s^{-1} + \rho \mathbf{A}^* \mathbf{A})^{-1}.$$

This follows from the matrix inversion lemma (with $\mathbf{A} \rightarrow \mathbf{R}_s^{-1}$, $\mathbf{B} \rightarrow \rho \mathbf{A}^*$, $\mathbf{C} \rightarrow \mathbf{I}$, $\mathbf{D} \rightarrow \mathbf{A}$),

Matrix inversion lemma

We have that

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{B} (\mathbf{C}^{-1} + \mathbf{DA}^{-1} \mathbf{B})^{-1} \mathbf{DA}^{-1}.$$

This means that we only want $\mathbf{A}$, $\mathbf{C}$ to be invertible!

Thanks to the matrix inversion lemma, we find something which is pretty similar to what we had before,

$$\text{MMSE}(\rho) = \mathbb{E} \left[\left| s - \frac{\sqrt{\rho}}{1 + \rho} y \right|^2 \right] \quad (1.173)$$

$$= \mathbb{E}[|s|^2] - 2 \frac{\sqrt{\rho}}{1 + \rho} \Re[\mathbb{E}[ys^*]] + \frac{\rho}{(1 + \rho)^2} \mathbb{E}[|y|^2] \quad (1.174)$$

$$= \mathbb{E}[|s - \hat{s}(y)|^2] = \frac{1}{1 + \rho}, \quad (1.175)$$

Skipped

Wrt the book, we're only missing the expansion of the error matrix for low-SNR.

3.3 I-MMSE relationship in Gaussian noise

A relation has been found around 20 years ago between information theory (I) and estimation theory (MMSE)!

Information-MMSE scalar relationship

Referring to the relation $y = \sqrt{\rho}s + z$ with s, z independent and $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, by using $\mathcal{I}(\rho) = I(s; y)$, we have the equivalence

$$\frac{1}{\log_2 e} \frac{d}{d\rho} \mathcal{I}(\rho) = \text{MMSE}(\rho) \quad (1.196)$$

or

$$\frac{1}{\log_2 e} \mathcal{I}(\rho) = \int_0^\rho \text{MMSE}(\xi) d\xi. \quad (1.197)$$

If we check this relation on the Complex Gaussian scalar case, we get that

$$\mathcal{I}(\rho) = \log_2(1 + \rho) \quad (1.198)$$

$$\text{MMSE}(\rho) = \frac{1}{1 + \rho}, \quad (1.199)$$

which satisfies the equation.

Information-MMSE vector relationship

What about the vector case? We use the gradient operator!

$$\frac{1}{\log_2 e} \nabla_{\mathbf{A}} I(\mathbf{s}; \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) = \rho \mathbf{A} \mathbf{E} .$$

Let's test this relation for complex Gaussian vectors:

Example 1.35 (I-MMSE relationship for a complex Gaussian vector) As established in Example 1.13, when the noise is $\mathbf{z} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{I})$ while $\mathbf{s} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}_{\mathbf{s}})$,

$$\mathcal{J}(\mathbf{s}; \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) = \log_2 \det(\mathbf{I} + \rho \mathbf{A} \mathbf{R}_{\mathbf{s}} \mathbf{A}^*) \quad (1.208)$$

$$= \log_2 \det(\mathbf{I} + \rho \mathbf{A}^* \mathbf{A} \mathbf{R}_{\mathbf{s}}) \quad (1.209)$$

and, applying the expression for the gradient of a log-determinant function given in Appendix D, we obtain

$$\frac{1}{\log_2 e} \nabla_{\mathbf{A}} I(\mathbf{s}; \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) = \rho \mathbf{A} \mathbf{R}_{\mathbf{s}} (\mathbf{I} + \rho \mathbf{A}^* \mathbf{A} \mathbf{R}_{\mathbf{s}})^{-1} \quad (1.210)$$

$$= \rho \mathbf{A} (\mathbf{R}_{\mathbf{s}}^{-1} + \rho \mathbf{A}^* \mathbf{A})^{-1} , \quad (1.211)$$

which indeed equals $\rho \mathbf{A} \mathbf{E}$ with $\mathbf{E} = (\mathbf{R}_{\mathbf{s}}^{-1} + \rho \mathbf{A}^* \mathbf{A})^{-1}$ as determined in Example 1.31 for a complex Gaussian vector.

We can conclude that the expression also works for vectors!

3.4 LMMSE for random variables

Disclaimer

This paragraph is not to be confused with the future paragraph **LMMSE equalization**, which instead discusses how to use the contents of this paragraph in order to equalize the output of a noisy channel, so to recover the original signal.

We can find a different type of estimator which can be used in order to **estimate when quantities which we don't know are involved** (i.e.: transformations made by the channel): it is a linear estimator which will prove to be more robust and versatile, but also inferior, when compared with a conditional-mean estimator (a conditional mean estimator is an estimator between the ones which have been discussed until now), which we'll refer to as the "Linear MMSE estimator" or "LMMSE estimator".

We can use wiener filtering if we do some operations beforehand, but only to stationary filters.

If the process is not stationary, we need to use Calemann filters.

The LMMSE problem sees us finding a vector $\mathbf{b}^{\text{MMSE}}$ and matrix $\mathbf{W}^{\text{MMSE}}$ such that we minimize the mean squared error between $\mathbf{s}$ and our estimate

$$\hat{\mathbf{s}} = \mathbf{W}^{\text{MMSE}} \mathbf{y} + \mathbf{b}^{\text{MMSE}} . \quad (1.223)$$

The term $\mathbf{b}$ is used **in order to ensure that the estimator is unbiased**; it is generally set to $\mathbf{0}$, since we generally assume we're using zero-mean signals.

The estimator is simply

$$\mathbf{W}^{\text{MMSE}} = \mathbf{R}_y^{-1} \mathbf{R}_{ys} . \quad (1.224)$$

In order to find (1.224), we consider that we want to minimize the mean-squared error; **we then write the mean-square error on the estimation of the j th entry of $\mathbf{s}$ via a generic linear filter $\mathbf{W}$ (s.t. $\hat{\mathbf{s}} = \mathbf{W}^* \mathbf{y}$) as**

$$\mathbb{E}[|\mathbf{s} - \hat{\mathbf{s}}|_j|^2] = \mathbb{E}[|\mathbf{s} - \mathbf{W}^* \mathbf{y}|_j|^2] \quad (1.225)$$

$$= \mathbb{E}[|s_j - \mathbf{w}_j^* \mathbf{y}|^2] \quad (1.226)$$

$$= \mathbb{E}[|s_j|^2] - \mathbb{E}[\mathbf{w}_j^* \mathbf{y} s_j^*] - \mathbb{E}[s_j \mathbf{y}^* \mathbf{w}_j] + \mathbb{E}[\mathbf{w}_j^* \mathbf{y} \mathbf{y}^* \mathbf{w}_j] \quad (1.227)$$

where $\mathbf{w}_j \triangleq [\mathbf{W}]_{:,j}$ is the j th column of $\mathbf{W}$, which is the only part interested in estimating the entry $s_j \triangleq [\mathbf{s}]_j$.

Then if we consider the gradient wrt $\mathbf{w}_j$, we see that, since

$$\nabla_{\mathbf{x}}(\mathbf{x}^* \mathbf{r}) = \mathbf{r} \quad (D.5)$$

$$\nabla_{\mathbf{x}}(\mathbf{r}^* \mathbf{x}) = \mathbf{0} \quad (D.6)$$

$$\nabla_{\mathbf{x}}(\mathbf{x}^* \mathbf{R} \mathbf{x}) = \mathbf{R} \mathbf{x} , \quad (D.7)$$

$$\begin{aligned} \nabla_{\mathbf{w}_j} \mathbb{E}[|\mathbf{s} - \hat{\mathbf{s}}|_j|^2] &= -\mathbb{E}[\mathbf{y} s_j^*] + \mathbb{E}[\mathbf{y} \mathbf{y}^* \mathbf{w}_j] \\ &= -\mathbb{E}[\mathbf{y} (s_j - \mathbf{w}_j^* \mathbf{y})^*] \\ &= -\mathbb{E}[\mathbf{y} [\mathbf{s} - \hat{\mathbf{s}}]_j^*] , \end{aligned}$$

which, once equated to zero (in order to minimize it) gives

$$\begin{aligned} -\mathbb{E}[\mathbf{y} (\mathbf{s} - \mathbf{W}^{\text{MMSE}*} \mathbf{y})^*] &= \mathbb{E}[\mathbf{y} \mathbf{y}^*] \mathbf{W}^{\text{MMSE}} - \mathbb{E}[\mathbf{y} \mathbf{s}^*] \\ &= \mathbf{0} \end{aligned}$$

and thus that

$$\mathbf{R}_y \mathbf{W}^{\text{MMSE}} - \mathbf{R}_{ys} = \mathbf{0} , \quad (1.233)$$

which then easily leads to (1.224).

About the concept of unbiased

For the concept of unbiased, refer to

This is **unbiased** in the sense that

$$\mathbb{E}[\hat{s}(y)] = \mathbb{E}[\mathbb{E}[s|y]] \stackrel{*}{=} \mathbb{E}[s] . \quad (1.153)$$

About the inversion

Question: is the -1 over the matrix a problem? An autocorrelation matrix is semidefinite positive. If the vectors are proper (and they are thanks to our assumption), it is semidefinite positive. This means that if we take the matrix and multiply it by two vectors, we get something positive:

$$\mathbf{v}^* [\mathbf{R}] \mathbf{v} > 0$$

From algebra we know that semidefinite positive matrices are always invertible. Multiplying a semidefinite positive matrix by a positive value keeps the property of semidefinite positivity.

We always need to pay attention when we see a -1 over a matrix!

Let's then derive the **MMSE matrix for the LMMSE**:

$$\mathbf{E} = \mathbb{E}[(\mathbf{s} - \hat{\mathbf{s}}(\mathbf{y}))(\mathbf{s} - \hat{\mathbf{s}}(\mathbf{y}))^*] \quad (1.241)$$

$$= \mathbf{R}_s - \mathbf{R}_{s\hat{s}} - \mathbf{R}_{\hat{s}s}^* + \mathbf{R}_{\hat{s}} \quad (1.242)$$

$$= \mathbf{R}_s - \mathbf{R}_{\hat{s}} - \mathbf{R}_{\hat{s}}^* + \mathbf{R}_{\hat{s}} \quad (1.243)$$

$$= \mathbf{R}_s - \mathbf{R}_{ys}^* \mathbf{R}_y^{-1} \mathbf{R}_{ys} ,$$

with (1.243) following from the facts that:

$$\mathbf{R}_{\hat{s}} = \mathbb{E}[\hat{\mathbf{s}}\hat{\mathbf{s}}^*] \quad (1.234)$$

$$= \mathbf{W}^{\text{MMSE}*} \mathbb{E}[\mathbf{y}\mathbf{y}^*] \mathbf{W}^{\text{MMSE}} \quad (1.235)$$

$$= \mathbf{R}_{ys}^* \mathbf{R}_y^{-1} \mathbf{R}_y \mathbf{R}_y^{-1} \mathbf{R}_{ys} \quad \text{def. of } \mathbf{W}^{\text{MMSE}} \quad (1.236)$$

$$= \mathbf{R}_{ys}^* \mathbf{R}_y^{-1} \mathbf{R}_{ys} . \quad (1.237)$$

$$\mathbf{R}_{s\hat{s}} = \mathbb{E}[\mathbf{W}^{\text{MMSE}*} \mathbf{y}\mathbf{s}^*] \quad (1.238)$$

$$= \mathbf{R}_{ys}^* \mathbf{R}_y^{-1} \mathbf{R}_{ys} \quad \text{def. of } \mathbf{W}^{\text{MMSE}} \quad (1.239)$$

$$= \mathbf{R}_{\hat{s}} \quad \text{from (1.237)} \quad (1.240)$$

3.5 Extension (from random variables to random processes)

If we extend this to random processes, the relation in that case is

$$y[n] = \sqrt{\rho} s[n] + z[n] .$$

We'll need to generalize from random variable to random process. In order to find LMMSE, we can use a causal or a noncausal filter:

$$\begin{aligned} \text{MMSE} &= 1 - \int_{-1/2}^{1/2} \frac{\rho S(\nu)}{1 + \rho S(\nu)} d\nu && \text{noncausal} \\ \text{MMSE} &= \left[\exp \left(\int_{-1/2}^{1/2} \log_e(1 + \rho S(\nu)) d\nu \right) - 1 \right] && \text{causal} \end{aligned}$$

Causal means that we can't use the future to compute something now: we can't use future values of x, z, y to compute an estimate. In the real world we can only use causal filters.

About this being asked

I'm pretty positive this will never be asked.

$S(\cdot)$ is the power spectrum of $s[\cdot]$, btw.

Problem 1.34 (proof of the orthogonality principle and of the linear MMSE estimator)

Exercise time! Text:

1.34 Consider the transformation $y = \sqrt{\rho}s + z$.

- Prove that, for any arbitrary function $g(\cdot)$, $\mathbb{E}[g(y)(s - \mathbb{E}[s|y])] = 0$. This is the so-called **orthogonality principle**.
- Taking advantage of the orthogonality principle, prove that the MMSE estimate is given by $\hat{s}(y) = \mathbb{E}[s|y]$.

For what concerns question (a), we can exploit the total probability theorem, thanks to which here we can write

$$\begin{aligned} \mathbb{E}[g(y)(s - \mathbb{E}[s|y])] &= \mathbb{E}_y[\mathbb{E}[g(y)(s - \mathbb{E}[s|y])|y]] \\ &= \mathbb{E}_y[g(y)(\mathbb{E}[s|y] - \mathbb{E}[s|y])] \\ &= \mathbb{E}_y[g(y) \cdot 0] = 0 \end{aligned}$$

where the very first passage was one of the various applications of the total probability theorem.

Instead, for question (b), we know that the MMSE is $\mathbb{E}[|s - \hat{s}(y)|^2]$; we can minimize it through:

The expectation $\mathbb{E}[s|y]$ is a function of y ; to search for the MMSE solution, we need to take this into account. Point **a** of the exercise is telling us about orthogonality between the error and **any** function of y . The intuition about this is the one which has been shown using vectors and planes.

Let's then express the MMSE. We use a trick by adding and subtracting $\mathbb{E}[s|y]$ inside the norm:

$$\mathbb{E}[|s - \hat{s}(y)|^2] = \mathbb{E}\left[\underbrace{|s - \mathbb{E}[s|y]|^2}_a + \underbrace{|\mathbb{E}[s|y] - \hat{s}(y)|^2}_b\right]$$

We know that for $a, b \in \mathbb{C}$, we have an equation for the modulo:

$$|a + b|^2 = |a|^2 + |b|^2 + 2\Re\{ab^*\}$$

Applying this to our expectation:

$$\begin{aligned} &= \mathbb{E}[|s - \mathbb{E}[s|y]|^2] + \mathbb{E}[|\mathbb{E}[s|y] - \hat{s}(y)|^2] \\ &\quad + 2\Re\left\{\mathbb{E}\left[(s - \mathbb{E}[s|y])\underbrace{(\mathbb{E}[s|y] - \hat{s}(y))^*}_{g(y)}\right]\right\} \end{aligned}$$

Let's analyze the cross-term (the $2\Re\{\dots\}$ part). The term $(\mathbb{E}[s|y] - \hat{s}(y))^*$ is a function of y , let's call it $g(y)^*$. From **before** (part a), we know that

$$\mathbb{E}[g(y)^*(s - \mathbb{E}[s|y])] = 0$$

So the entire cross-term cancels out!

We are left with:

$$= \underbrace{\mathbb{E}[|s - \mathbb{E}[s|y]|^2]}_{\text{Cannot change this}} + \underbrace{\mathbb{E}[|\mathbb{E}[s|y] - \hat{s}(y)|^2]}_{\geq 0 \text{ iff } \hat{s}(y) = \mathbb{E}[s|y]}$$

We can play on $\hat{s}(y)$ in order to minimize the MMSE: this happens for $\hat{s}(y) = \mathbb{E}[s|y]$, as this gets us only to be zero for the part in green (the second term).

we've thus proven the orthogonality principle.

3.6 Signal processing

Book reference: [08.2_pp_57_138_A_signal_processing_perspective](#).

3.6.1 Introduction

We'll look at chapter two, which looks at the DSP perspective; until now we've only looked at information theory and MSE theory; we also saw that the two theories can be related: these are the foundations!

Ref: [Chapter_2_EE_381S_Spg_2019](#).

We need to study on the book oc: only there we will find all the details.

Highlights:

- Complex baseband signal model allows representing passband signals regardless of the actual carrier frequency
- Because the transmit signal is bandlimited, a complete system model can be obtained in discrete-time, including the portion of the channel impacted by the transmit signal
- Several different standard MIMO signal models are used, sometimes vectorized depending on the problem at hand
- Precoding is a structured means to create a correlated transmit signal, and can be normalized in different ways depending on the signal constraints
- OFDM and SC-FDE have an equalization structure that is independent of the memory in the channel, at the expense of some overheads
- Channels are estimated using pilots, and may exploit channel statistics

We'll see that channels are estimated using what we'll call *pilots*.

3.7 Passband signal, in-phase/in-quadrature components

We normally transmit waveforms characterized by a signal of voltage $x_p(t)$, applied to the antenna. This signal is generally passband, in the sense that in the frequency domain most of its energy is concentrated around a **carrier frequency** f_c .

Let's suppose that $x_p(t)$ is an ideal passband signal (by ideal we mean that in the frequency domain it will be nonzero only over a bandwidth B centered on f_c): we'll have that $B \ll f_c$, which we'll refer to as the **narrowband condition**.

Under the narrowband condition, we can write

$$x_p(t) = A(t) \cos(2\pi f_c t + \phi(t)) ,$$

where:

- $A(t)$ is a magnitude function
- $\phi(t)$ is a phase function

Applying trigonometric identities to that equation, we can arrange it into

$$x_p(t) = \underbrace{A(t) \cos(\phi(t))}_{x_c(t)} \cos(2\pi f_c t) - \underbrace{A(t) \sin(\phi(t))}_{x_s(t)} \sin(2\pi f_c t) ,$$

where:

- $x_c(t)$ is called the **in-phase component**; it modulates a cosine carrier
- $x_s(t)$ is called the **in-quadrature component**; it modulates a sine carrier

Complex envelope of a passband signal

We'll now define the **complex envelope** or **complex baseband equivalent** of $x_p(t)$ as

$$x(t) = x_c(t) + jx_s(t) ;$$

this will give that the passband signal will be obtained as

$$x_p(t) = \sqrt{2}\Re\{x(t)e^{j2\pi f_c t}\} .$$

$x_p(t)$ is a physical quantity and it will thus be $\in \mathbb{R}$; $x(t)$, instead, will be $\in \mathbb{C}$.

In the frequency domain we'll have

$$x_p(f) = \frac{1}{\sqrt{2}}[x(f - f_c) + x^*(-f - f_c)] . \quad (2.5)$$

If $x_p(f)$ is bandlimited with passband bandwidth of B , then $x(f) = x_c(f) + jx_s(f)$ and its components x_c, x_s will all be bandlimited with baseband bandwidth $B/2$. Moreover, the two components are orthogonal thanks to the phase difference between the cosine and sine carriers.

The process of going from the baseband signal to the passband equivalent is called **upconversion**, while the opposite process is called **downconversion**.

Upconversion can be easily implemented as follows:

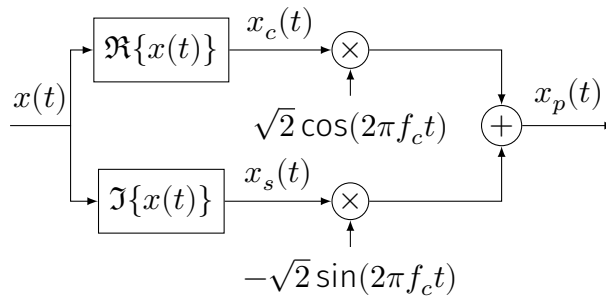


Figure 3.2: Conversion of a baseband signal to a passband signal.

Downconversion is a bit more involved. Let's say $y_p(t)$ is the signal at the receiver. First of all, the signal is bandpass-filtered in order to have it reject transmission on other frequency bands. Being bandpass, $y_p(t)$ will have a baseband equivalent $y(t) = y_c(t) + jy_s(t)$ such that, just like what we said about the sent signal, we'll have

$$y_p(t) = \sqrt{2}[y_c(t) \cos(2\pi f_c t) - y_s(t) \sin(2\pi f_c t)] . \quad (2.6)$$

There are ways to extrapolate $y_c(t)$ and $y_s(t)$ using pertinent low pass filters, which need to be realized while knowing the **carrier frequency** f_c : [08.2_pp_57_138_A_signal_processing_perspective, p.63](#).

f_c is generally not known precisely; estimating it is a matter tackled by "carrier frequency estimation".

The conversion to the baseband is mostly important since most of the processing is done on low frequency signals.

Wireless communications systems send and receive passband signals; those signals are not directly created in passband, though.

Bluetooth is at 5.4GHz; we don't have circuits at those frequencies, as they'd be super intense.

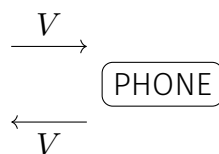
What we generally do is to take signals in baseband and then we scale them.

3.7.1 Random digression on why radio doesn't transmit on baseband and on ADSL

Why are radio signals not transmitted in baseband? ADSL transmits in baseband to avoid attenuation.

Why aren't radio signals transmitted as baseband? We need to go to **high frequencies if we don't want to make gigantic antennas**.

We studied radio; but what about ADSL? We'd have a twisted pair of cables coming to our house. Let's say we have a phone. On a cable we're sending voice, on another we're receiving voice; we do so by changing a certain voltage:

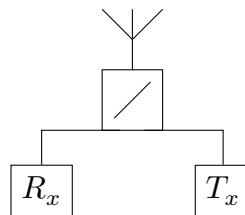


How is it that with just two cables we don't hear a distorted signal (i.e.: how is it that we don't hear our own voice)? We only have one voltage difference! We could implement FDMA, but it would be impractical! In order to synthesize a frequency, we'd need a crystal and a VCO (voltage controlled oscillator). What we actually do is use the *forchetta telefonica* (telephone hybrid); we have a circulator which is able to distinguish the incoming and outgoing signal: it cancels the outgoing voltage from the total, getting the incoming voltage as a result. The telephone hybrid allows us to go from 2 to 4 wires.

With ADSL we have something like FDMA; we have frequencies up to 4KHz and such (this generally applies to voices; instruments and such will have higher frequencies but we don't really care lol).

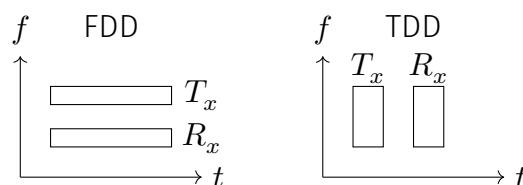
3.7.2 Random digression on FDD (Frequency Division Duplex) and TDD (Time Division Duplex)

FDD and TDD. When we're transmitting with a radio in SISO, we have an antenna transmitting and an antenna receiving. When we're transmitting, we can imagine we have a switch which selects the transmitter; when we want to receive, we'll want to select the receiver:



We also need to be careful about not having contact between receiver and transmitter, as we'd destroy the receiver (which is designed in order to use much smaller voltages)!

FDD: Frequency Division Duplex; we transmit at the same time using different frequencies. **TDD: Time Division Duplex;** we have different time slots for transmitting and receiving.



The problem is that the received signal is much smaller than the transmitted; it's difficult to distinguish them.

Full duplex is now being researched: with full duplex, we use the same frequencies and we transmit in the same time slot.

3.8 Complex baseband channel response

The channel for wireless transmission can be modeled as a LTI channel.

If we have a description of the response of the channel to a passband signal, we denote it as

$$c_p(\tau),$$

such that we'll have

$$y_p(t) = (c_p * x_p)(t)$$

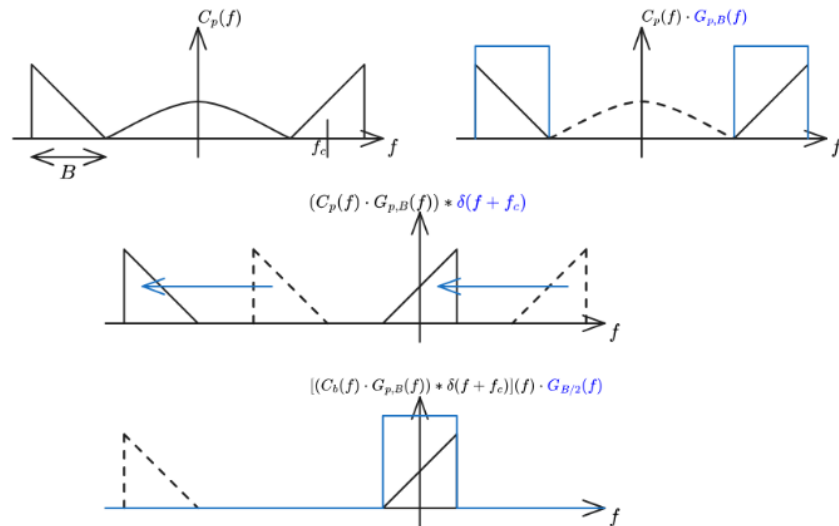


Figure 3.3: Operations to obtain the baseband channel response from the passband channel response.

we want to find another **baseband representation of the channel response**, $c_b(\tau)$, such that

$$y(t) = [c_b * x](t) .$$

About the time invariance: it's more generic to say that the channel is roughly constant over certain intervals of time and frequency.

We thus need to find an expression of c_b starting from c_p . Let's look at the various operations we want our baseband channel to implement:

Let's cover each step:

1. We have our channel response, which in the frequency domain might have extra elements on top of the part which interests us (a section of bandwidth B around the frequency f_c).
2. We select only the part of the response which interests us, by multiplying the values for negative and positive frequencies around f_c with two ideal rect functions

$$G_{p,B}(f) = \text{rect}\left(\frac{f - f_c}{B}\right) + \text{rect}\left(\frac{f + f_c}{B}\right) ,$$

which in the time domain will be

$$g_{p,B}(t) = \underbrace{B \text{sinc}(Bt)}_{\text{env}} \underbrace{2 \cos(2\pi f_c t)}_{\text{shift}}$$

and will be convolved with $c_p(t)$:

$$(c_p(\cdot) * g_{p,B}(\cdot))(t) .$$

3. We shift the selected values towards the left, centering around 0 the part of the response which was around f_c ; this is equal to convoluting with $\delta(f + f_c)$ or, in the time domain, multiplying by the corresponding oscillating signal:

$$e^{-j2\pi f_c t} (c_p(\cdot) * g_{p,B}(\cdot))(t) .$$

4. We only select the central component by multiplying our response with an ideal rect centered around 0:

$$G_{b,B/2}(f) = \text{rect}(f/B) ,$$

which in the time domain will be

$$g_{b,B/2}(t) = B \text{sinc}(Bt) \quad (2.12)$$

and will be convoluted with the result of the previous point:

$$c_b(t) = \left[(e^{-j2\pi f_c(\cdot)} (c_p * g_{p,B})(\cdot)) * g_{b,B/2} \right] (t) .$$

Baseband channel response

Since our signal $x(t)$ will be bandlimited, we can actually skip point 2 and get away with an equation such as

$$c_b(t) = [g_{B/2}(\cdot) * (c_p(\cdot) e^{-j2\pi f_c(\cdot)})](t) .$$

In the equation we have a frequency dependent phase shift: we'll see that this is very nasty.

Pseudo-baseband channel response

We then have that

$$c(t) = c_p(t) e^{-j2\pi f_c t}$$

is the pseudo-baseband channel response, such that if $x(t)$ is bandlimited we can obtain

$$y(t) = [x(\cdot) * c(\cdot)](t)$$

and also such that

$$c_b(\tau) = [g_{B/2}(\cdot) * c(\cdot)](\tau) .$$

This comes from my reasoning after [these statements](#)

The figure shows filtering with passband signals:

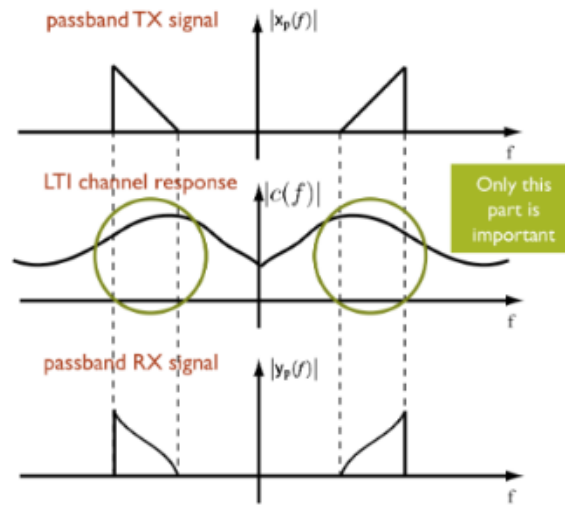
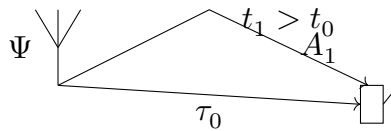


Figure 3.4: This image helps understanding why we simplify the equation of the channel response

3.9 Q paths complex baseband equivalent

Let's look at an example in which we have an antenna and a phone; we'll experience different delays τ_q and different gains A_q for each path followed by the signal:



We can model such a channel as

$$c_p(\tau) = \sum_{q=0}^{Q-1} A_q \delta(\tau - \tau_q) . \quad (2.20)$$

The baseband equivalent (according to "040299") will be

$$c_b(\tau) = g_{B/2}(\tau) * \sum_{q=0}^{Q-1} A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q} . \quad (2.21)$$

which will give an output

$$\begin{aligned} y(t) &= [c_b * x](t) \\ &= \int_{-\infty}^{\infty} \left(g_{B/2}(\tau) * \sum_{q=0}^{Q-1} A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q} \right) x(t - \tau) d\tau \end{aligned} \quad (2.22)$$

$$= \int_{-\infty}^{\infty} \left(\sum_{q=0}^{Q-1} A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q} \right) x(t - \tau) d\tau \quad (2.23)$$

$$= [c * x](\tau) \quad (2.24)$$

with $c(\tau)$ being the **complex pseudo-baseband channel response**, since it's shifted but not bandlimited. $*$ follows from x being bandlimited.

We can obtain the complex baseband channel response from the complex pseudo-baseband channel response as

$$c_b(\tau) = [g_{B/2} * c](\tau) . \quad (2.25)$$

3.10 Time discretization: discrete representation of the channel $c[\ell]$ (+critical sampling)

Since the channels are bandlimited, we can sample with a period $T \leq 1/B$ (equality is what we'll call **critical sampling**) and discretize the input and the channel.

Let's say we have $x(t), y(t)$ represented by their discrete-time samples with $T = 1/B$; we want to find a channel such that

$$y[n] = \sum_{\ell} c[\ell] x[n - \ell] . \quad (2.27)$$

FYI

This is a noiseless **frequency selective, unitary gain channel model**, as will be shown in (2.80).

If we try and discretize the convolution $[c_b * x]$ (with reference to (2.25) and previous equations) with a step $d\tau \rightarrow T$, we get

$$\begin{aligned} y(nT) &= (c_b * x)(nT) \\ &= \int_{-\infty}^{\infty} c_b(\tau) x(nT - \tau) d\tau \end{aligned} \quad (2.28)$$

$$\approx \sum_{\ell} c_b(\ell T) x(nT - \ell T) T \quad (2.29)$$

$$= \sum_{\ell} T c_b(\ell T) x[n - \ell] , \quad (2.30)$$

which, when compared with the equation of $y[n]$ in (2.27), will point us to conclude that $c[\ell] \approx T c_b(\ell T)$.

We can conclude that the discrete time equivalent in (2.27) for any baseband or pseudo-baseband channel is obtained by:

1. Taking the generic channel response $c(\tau)$
2. Filtering it with the low-pass filter $Tg_{B/2}(\tau)$, thus obtaining $c_b(\tau)$
3. Sampling it at $\tau = \ell T$

Discrete time baseband channel equivalent

It will thus be

$$c[\ell] = [Tg_{B/2} * c](\tau)|_{\tau=\ell T}$$

or, if the channel is already bandlimited,

$$c[\ell] = [Tc(\tau)]|_{\tau=\ell T} . \quad (2.31)$$

Where the T is introduced in order to keep the energy constant.

If we want to deal with signal theory, professor suggests to look at "unified signal theory".

What's important here is that we'll forget about the continuous time domain and we continue with the discrete time domain:

[08.2_pp_57_130_A_signal_processing_perspective, page 10](#)

In light of the connection established between passband signals, bandlimited baseband signals, and discrete-time signals, most results in this book are formulated directly in discrete time. Although actual signals are not exactly bandlimited since they are time-limited, and the filters in the front-ends are never ideal, signals are close to bandlimited as they are mandated to have low adjacent channel interference. Commercial systems also feature analog-to-digital and digital-to-analog converters with finite precision. Normally, perfect conversion is assumed in the design stages and then the required tolerances are investigated after the fact via simulation. Quantization and reconstruction errors are not part of the models in this text.

3.11 Idea of pulse shaping: from a digital signal $x[n]$ to an analog signal

Pulse shaping: the book mentions it, but it's not really important.

[08.2_pp_57_130_A_signal_processing_perspective, page 10](#)

The sampling and reconstruction that we have applied to discretize the transmit-receive relationships can be interpreted as basic modulation and demodulation schemes, whereby the complex symbols that make up the codewords are embedded onto $x(t)$ and then extracted from $y(t)$.

The basic concept of pulse shaping is to link the symbols of the message we're transmitting, $x[n]$, with the actual continuous time signal, $x(t)$.

If we know $x[n]$, we might think about using sinc for interpolation; problem is that it is not causal! What we generally do then is to use the **square root raised cosine**.

No matter what functions we'll use for interpolation at the transmitter, we'll call it g_{tx} . At the receiver we'll use a different function, g_{rx} , to filter what we received.

We can generally **reconstruct the signal through interpolation** as

$$x(t) = \sum_n x[n] g_{tx}(t - nT) , \quad (2.41)$$

where $g_{tx}(\cdot)$ is the transmit pulse shape.

At the receiver, we'll then use a specific filter which in reality cannot be the ideal low-pass filter $g_{B/2}(\cdot)$ described in (2.18): it will be a modified filter $g_{rx}(\cdot)$.

The **composite pulse shape** is thus defined by combining the two pulse shapes, as

$$g(\tau) = [g_{tx} * g_{rx}](\tau) . \quad (2.42)$$

Skipped

Possible definitions of $g(\tau)$ can be found from [08.2_pp_57_130_A_signal_processing_perspective, page 13](#) to [08.2_pp_57_130_A_signal_processing_perspective, page 15](#). Professor completely skipped them.

3.12 Channel models (frequency selective, FIR, flat channel); channel normalization and average symbol energy; vector representation of a SISO FIR channel

An important thing is the normalization of the channel; people normally make lots of mistakes about this.

We have the model defined in (2.19). We can model such a channel as

$$c_p(\tau) = \sum_{q=0}^{Q-1} A_q \delta(\tau - \tau_q) . \quad (2.19)$$

If we don't pay attention, this channel could be amplified. We need to pay attention because the energy of the channel is

$$\sum |A_q|^2 ,$$

which might be greater than 1.

What's important is the ratio between the noise and the amplification of the channel (which comes from the antenna itself, its directionality etc etc): the gain is in respect to an isotropic antenna.

Chapter 4

Week 17-23 November

4.1 Channel models (frequency selective, FIR, flat channel); channel normalization and average symbol energy; vector representation of a SISO FIR channel

An important thing is the normalization of the channel; people normally make lots of mistakes about this.

We have the model defined in (2.19). We can model such a channel as

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If we don't pay attention, this channel could be amplified. We need to pay attention because the energy of the channel is

$$\sum |A_q|^2 ,$$

which might be greater than 1.

What's important is the ratio between the noise and the amplification of the channel (which comes from the antenna itself, its directionality etc etc): the gain is in respect to an isotropic antenna.

In short, the problem is that we want to separate the gain from the channel response, which is why we write:

$$c(\tau) = \sqrt{G} h(\tau) \quad (2.64)$$

$$c_b(\tau) = \sqrt{G} h_b(\tau) \quad (2.65)$$

$$c[l] = \sqrt{G} h[l] \quad (2.66)$$

we'll also impose that:

$$\sum_l \mathbb{E}|h[l]|^2 = 1 \quad (4.1)$$

the channel gain should always be one.

Rem: This makes perfect sense. *We know that a generic request on the channel is that*

$$\sum_l \mathbb{E} \|H[l]\|_F^2 = N_t N_r$$

here we're in SISO, thus we have only one receiver and one transmitter; it makes perfect sense to request that the sum of the expectations is 1.

Def 4.1.1: Frequency selective model. The **frequency selective** is a model in which we have $\sqrt{G}$ then a causal and FIR (Finite Impulsive Response) channel and then noise which is IID and $v[n] \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$.

The output of a generic channel, assuming the normalization we mentioned holds, would be:

$$y[n] = \sqrt{G} \sum_l h[l]x[n-l] + v[n] \quad (2.69)$$

If the condition of FIR and of causality hold, though, the output can be simplified as:

$$y[n] = \sqrt{G} \sum_{l=0}^L h[l]x[n-l] + v[n] \quad (2.70)$$

If we then have $L = 0$, the channel is multiplicative and will be called **flat channel**, giving an output of:

$$y[n] = \sqrt{G}h[0]x[n] + v[n] \quad (2.71)$$

all the frequencies are treated the same; we can have this model represent a time variant setup if $h[0]$ actually depends on n .

If $L > 0$ the channel exhibits ISI (Inter-Symbol Interference) and since its response will not be flat in frequency, we say that it is **frequency-selective**.

Symbols are normalized to have an **average symbol energy**:

$$E_s = \frac{1}{N} \sum_{k=0}^{N-1} \mathbb{E} |x[k]|^2 \quad (4.2)$$

We'll see that this specific definition is just the per-block power constraint. We can also have per-symbol or per-antenna power constraints, from which will follow other constraints on the precoding matrix (see [signal power constraints](#)).

This will allow us to write the SNR as:

$$\text{SNR} = \frac{P_r}{N_0/T} = \frac{GE_s}{N_0}, \quad (4.3)$$

where:

- P_r is the power of the received signal, which in turn is equal to $P_r = GP_x = GE_s/T$ (P_x is the power of the transmitted signal);
- N_0 is the covariance of the AWGN.

This last relation will turn out to be useful later on (it is literally (4.2), (4.3) and it will be of use to write (4.6), (4.7)...).

If we have a frequency-selective channel, we can give a vector representation of (2.70). For the channel we'll use a Toeplitz matrix, which is a $N \times (N + L)$ convolution matrix defined as:

$$H_{N,N+L} = \begin{bmatrix} h[L] & \dots & h[0] & 0 & \dots & 0 \\ 0 & h[L] & \dots & h[0] & 0 & \vdots \\ \vdots & & \ddots & & \ddots & 0 \\ 0 & \dots & 0 & h[L] & \dots & h[0] \end{bmatrix} \quad (2.72)$$

the input signal is consequently defined as:

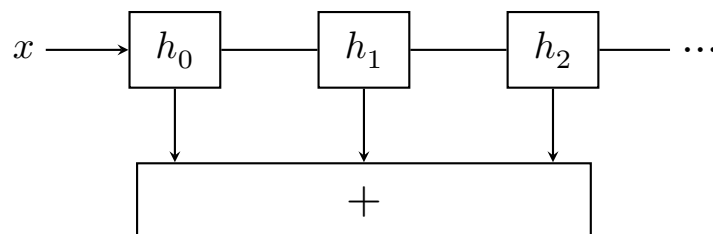
$$x_{N+L} = \begin{bmatrix} x[-L] \\ \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \quad (2.73)$$

giving that if the channel noise is v_N , we'll have:

$$y_N = [y[0] \dots y[N-1]]^T = \sqrt{G} H_{N,N+L} x_{N+L} + v_N \quad (2.74)$$

We can show that whichever n -th row in (2.74) is equivalent to (2.70).

This is like if we had a MIMO system: "Vectorized result makes an equivalence between frequency-selective SISO channels and frequency-flat MIMO channel, and may be useful for deriving estimators or equalizers".



Block diagram of the FIR/Channel model (reconstruction)

This is the same of having a MIMO channel with different gains. We can ask: if we want to estimate a symbol, how can we only get the symbol $x[n]$ we're interested in from $y[n]$?

We'll want to massage the MIMO system to make it similar to our model; we can put together all the matrices and get one only matrix: this is the stacked vector representation.

4.2 Organization detour (+mentions of SVD, Precoding)

Organizational issue: he set up a lecture for next wednesday; this is because on the 1st of November we'll have no lecture. Professor tried to keep the target so that we could finish before christmas. We'll also have some other lectures to look over.

One of the next topics is OFDM; he suggests to read the part of the book dealing with it before going through it with lectures; if we read it beforehand, we'll be in a better position.

Lore of the course moment About MMSE: when we transmit, this is another way to estimate the symbol transmitted.

Blabbering ab MIMO: if we have N tx and M rx, we have NM paths.

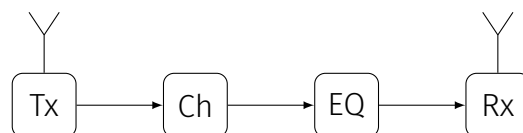
We'll see filters whose coefficients are matrices.

Singular value decomposition: we'll have a specific slide on this.

Precoding: we should try, if possible, to figure what a certain equation means in practice; the antenna may have a main lobe and we can change the main lobe by moving the antenna mechanically. Precoding is changing the type of transmission without moving the antenna mechanically; we do this using a certain matrix which we call precoding. With precoding we can steer where the antennas are transmitting.

We'll move to OFDM (used for 4G, 5G and probably will also be used for 6G); we need to see why we use OFDM! That's the wrong approach, since there are other types of modulation (single carrier modulation, just as an example). When we're transmitting, we have two antennas and a single channel.

We model the channel via spans:



If we have reflection, the previous signal might superpose to the previous one. The channel has a non-linear response: we can equalize it! We have many ways to equalize the channel!

The EQ in MIMO would increase the noise! We thus look at two **different types of equalizations**; we can have a linear or a non-linear equalization, but the latter is more resource-heavy.

We'll then look at the basics of radio propagation and such; there will be many concepts which are really important and will be pointed out. We'll see the Friis equation.

◇ Exam

Example of exam question: we can transmit at 900MHz or at 24GHz; we have a fixed power P and we can decide which frequency.

Thanks to the Friis equation, we know that choosing the lower frequency is better.

The state is getting more money from choosing the higher frequency.

4.3 Problems in discretization: multiple paths, multiple taps (+synchronization, reflection and noise whitening)

We've seen that long time ago people were manipulating analog signals, aka continuous time. Microphones are still in frequency modulation, as an example. Long time ago, we were able to communicate and we supposedly went to the moon.

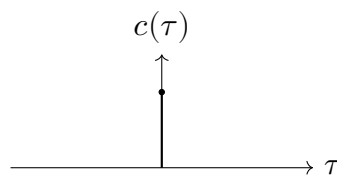
Kalman filtering learned about his filter in Moscow.

We were used at working with analog signals; the leap happened when we began manipulating signals with computers: in order to do this, we need to discretize signals. Analog circuits are different from one another; computers always do the same operations! It's much better for us to manipulate signals in a discrete domain!

What if we have a certain channel with a response which is an impulse?

4.4 Discretized Channel Model

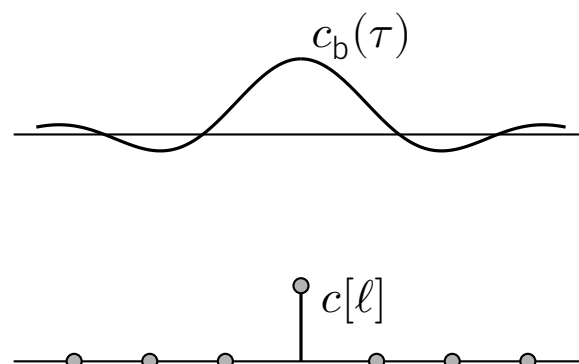
One-tap



Whatever is transmitted, it arrives straight to the receiver. This is true up to a certain point: it might take a while for the signal to travel up to the destination: we might want to move our delta! This is a problem which is called **time synchronization**.

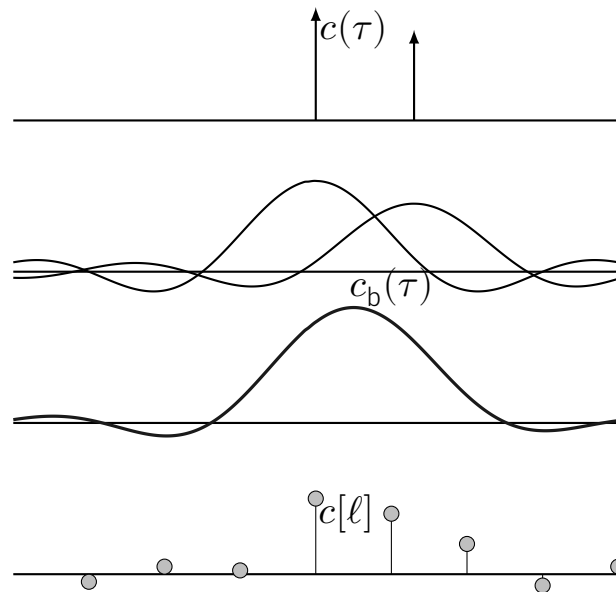
How can the transmitter understand when $t = 0$ is? We might have a form of mismatching if the receiver doesn't understand correctly when the signal started. We might do syncing using autocorrelation: CAZAC sequences are generally used (Constant Amplitude Zero AutoCorrelation waveform).

Here we suppose that our signal is synchronized; let's convolve it with a sinc; we have that sampling it will give a Kronecker delta function.



This only happens if we have synchronization.

Let's look at another channel: we have two paths which will be followed by our signals;



This is a typical example of **reflection**; for each delta, we'll have a sinc; if we sample this, then we'll have four taps (as can be seen in the last figure). In this case what happens is that from two taps, in the discrete domain we got 4.

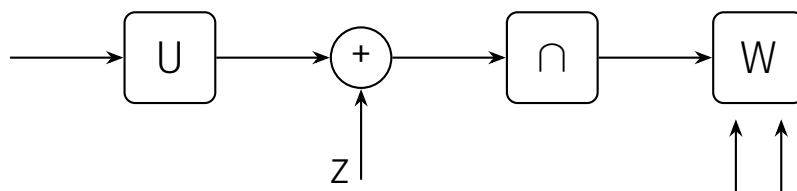
The amount of taps are increasing; we can only sync with one or the other, thus we're always in trouble.

This is a simple problem: if we only have a tap, we'll have a single tap in the frequency domain. If we have multiple taps, stuff might become a nightmare!

Sampling is nice, but we can't do this for free.

Let's make some comments on developing a discrete time model:

- Noise is still IID and the noise samples are still Gaussian with $\sim \mathcal{N}(0, N_0)$
- We can also use root-raised-cosine filter instead of a sinc filter.
- If we want to detect a signal, it's much easier to do so if we have a white spectrum: if the noise goes through certain blocks, though, it will no longer be flat!

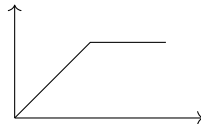


we can add a **block which whitens the noise**; this is good since then all the subsequent equalization and such work much better. Whitening a noise can be much more difficult.

Pulse shaping: if in the frequency domain we have a sinc, this is optimal. This is a sinc tho, which decays slowly in the time domain. This is why we're using a root-

raised-cosine filter: it will have a wider bandwidth but it will be better in the time domain.

Instead of sampling at the Nyquist rate (which allows for perfect recovering), we can say: "what if I sample the signal more frequently? Maybe I can make algorithms which are working better". From the Information theory POV, Nyquist suffices; but if we have to do with practical issues like non linearities, stuff might be different. Amplifiers are generally non-constant: there is a certain range in which the amplifier is linear, but there is a certain point at which whichever signal we put at the input of the amplifier, we'll get the exact same output; this is called clipping:



For synchronization, it helps to have oversampling (not in the book!); if the Nyquist rate is just in between the right points, it might not sync correctly.

4.5 Transitioning to MIMO (frequency selective and FIR channel expressions; channel normalization)

We're now going towards MIMO; we have a certain number N_t of transmitting antennas and N_r of receive antennas; those numbers can def be different.

We have at the input the value $\mathbf{x}$ and at the output $\mathbf{y}$ with noise $\mathbf{v}$.

Let's start from the causal and FIR **frequency flat** channel; we'll have an output of:

$$\mathbf{y}[n] = \sqrt{\rho} \mathbf{H} \mathbf{x}[n] + \mathbf{v}[n]. \quad (2.81)$$

$\mathbf{H}$ is a matrix such that the impulse response between transmit antenna j and receive antenna i will be represented by $\mathbf{H}_{i,j}$.

◇ Frequency flat (also found as frequency-flat)

By frequency flat, the book refers to a channel which has just one tap, aka $L = 0$.

$\mathbf{x}$ is multiplied by $\mathbf{H}$; this means that the **length of $\mathbf{x}$ is N_t** ; the length of $\mathbf{y}$ is N_r .

Skipped

Professor skipped the beginning of **section 2.3.1**, which tackles a continuous IO MIMO relation.

Just like we saw, we're not certain of having a delta; we might have some other paths! We'll model this (a **FIR channel**) just as we did before:

$$\mathbf{y}[n] = \sqrt{\rho} \sum_{l=0}^{L-1} \mathbf{H}[l] \mathbf{x}[n-l] + \mathbf{v}[n]. \quad (2.80)$$

where $[\mathbf{H}[l]]_{i,j}$ represents the response between transmit antenna j and receive antenna i . This means that the **output vector at time n is also depends on previous**

symbols. Each vector is then also weighted by a different matrix, which changes over time.

There is no difference, conceptually, with the scalar case; we only have that we have a matrix instead of a vector.

Before, we had normalized the channel; in this case we sum over all the three indexes; we don't want it to be 1, but instead we'd like to be $1 \cdot \# \text{ of paths} = N_t N_r$.

We can use the definition of the **Frobenius norm** in order to impose the normalization; while we could impose the summation to be equal to 1, this would not allow to include channel models with power asymmetry across antennas; for this reason, we have that

$$\sum_{l=0}^{L-1} \sum_{i=0}^{N_r-1} \sum_{j=0}^{N_t-1} \mathbb{E}[|H[l]_{i,j}|^2] = N_t N_r. \quad (2.82)$$

which can be rewritten more compactly using the definition of Frobenius norm (given in appendix B.5),

$$\sum_{l=0}^{L-1} \mathbb{E}[\|\mathbf{H}[l]\|_F^2] = \sum_{l=0}^{L-1} \mathbb{E}[\text{tr}(\mathbf{H}[l]\mathbf{H}^H[l])] \quad (2.83)$$

$$= \sum_{l=0}^{L-1} \mathbb{E}[\text{tr}(\mathbf{H}^H[l]\mathbf{H}[l])] \quad (2.84)$$

$$= N_t N_r. \quad (2.85)$$

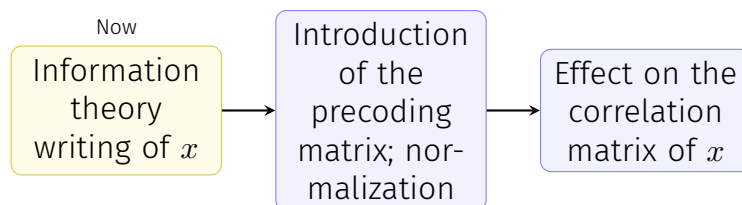
which for the frequency-flat channel simplifies to

$$\begin{aligned} \mathbb{E}[\|\mathbf{H}\|_F^2] &= \mathbb{E}[\text{tr}(\mathbf{H}\mathbf{H}^H)] \\ &= \mathbb{E}[\text{tr}(\mathbf{H}^H\mathbf{H})] \\ &= N_t N_r. \end{aligned} \quad (2.86)$$

The Frobenius norm is taking matrix and substituting it with the trace of the matrix multiplied with the complex conjugate of said matrix.

4.6 Precoding

4.6.1 Introduction (information theory writing of $x[n]$, matrix F ; variation of precoded x)



We have the precoding; we'll start with SISO signals. We'll decompose it in an IT (Information Theory) friendly way:

$$x[n] = \sqrt{E_s P[n]} s[n].$$

where:

- E_s is the average symbol energy
- $P[n]$ is the power allocation at a certain given time
- $s[n]$ are unit-variance symbols (PSK, QAM, Gaussian)

Hello, new writing of $x[n]$! Will you stick around?

Yup. This writing will be reused in chapter 4, when linking OFDM and vector coding (and also when finding the SNR for both).

About units...

From what I gather, if we admit that energy is in $W \cdot s$ and power in W , if x is in units of V , then s must have a unit of $V/\sqrt{J}$.

Sending info using a Gaussian: a **Gaussian** is the base we use for the other ways to send signals; we use it as a **bound** for what we can do. If we don't reach the performance of the Gaussian, there is a reason for that.

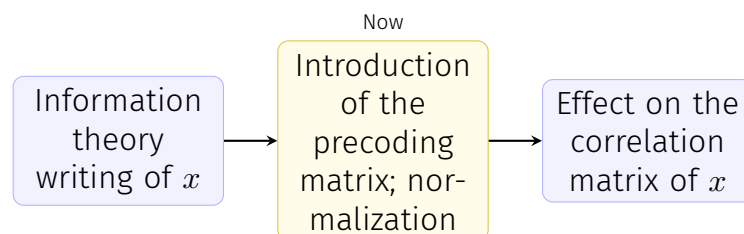
When we're going from 16 to 64 QAM, we're just making the constellation more dense.

$P[n]$ is the power allocation for each signal; about it:

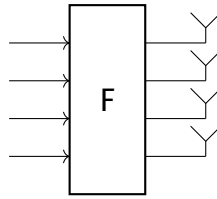
In our phone we have an algorithm called the **power control algorithm**; when we go away from an antenna, we need to raise the power used to transmit; if we're close, we need to have a forward signal!

What is practically done is that we measure the signal from the base station and we'll know how strong our signal should be.

We'll then change the signal when we go farther from a station.



With MIMO we do **transmit precoding**: we suppose we have $N_s \leq N_t$: the number of transmit symbols (N_s) is at most equal to the number of transmit antennas (N_t). If we're not using an antenna, we can think about what to do with it. If we have 5 symbols but just an antenna, we won't be able to transmit that symbol. **Our vector of symbols can only be transmitted if $N_s \leq N_t$.** In the MIMO we can do more than deciding how much power we should use.



we might decide to use a matrix to mix those symbols; it's a degree of freedom which might prove very useful: why not use it?

For any running symbol, we multiply it with a matrix $\mathbf{F}[n]$ (which can vary with n !!):

$$\mathbf{x}[n] = \sqrt{\frac{E_s}{N_t}} \mathbf{F}[n] \mathbf{s}[n]. \quad (2.96)$$

where:

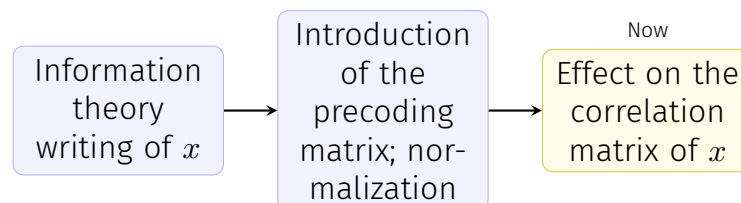
- The $s[n]$ symbols are **IID**.
- E_s is the average energy per symbol (more about this later).
- $\mathbf{F}[n]$ is the precoder matrix; as we'll see, it's normalized in order to respect specific relations. It will have a dimension of $N_t \times N_s$.

We have a restriction on $\mathbf{F}[n]$ which uses the **Frobenius norm**; it grants that the antenna is not amplifying the signal:

$$\frac{1}{N} \sum_{n=0}^{N-1} \|\mathbf{F}[n]\|_F^2 = \frac{1}{N} \sum_{n=0}^{N-1} \text{tr}(\mathbf{F}[n] \mathbf{F}^H[n]) = N_t. \quad (2.97)$$

Later, we'll need the transmit covariance. $s[n]$ are IID codewords, which means that $\mathbb{E}[\mathbf{s}[n] \mathbf{s}^*[n]] = \mathbf{I}$ (this follows from the fact that independence implies uncorrelation and that these vectors are 0-mean; the main diagonal will have the variance of each signal, which is 1), with dimensions $N_s \times N_s$.

We're always transmitting 0-mean and 1-variance signals.



If we take into account $x[n]$, which has multiplication with the matrix, the covariance will be given by $\mathbf{R}_x$. In order to find its value, we'll apply the fact that $(\mathbf{F}\mathbf{s})^* = \mathbf{s}^* \mathbf{F}^*$. We'll thus have

$$\begin{aligned} \mathbf{R}_x &= \frac{E_s}{N_t} \mathbb{E}[\mathbf{F} \mathbf{s} \mathbf{s}^* \mathbf{F}^*] \\ &= \frac{E_s}{N_t} \mathbf{F} \mathbb{E}[\mathbf{s} \mathbf{s}^*] \mathbf{F}^* \\ &\stackrel{(*)}{=} \frac{E_s}{N_t} \mathbf{F} \mathbf{F}^*. \end{aligned}$$

re-introducing the time dependent notation, we have (for ID $\mathbf{s}$)

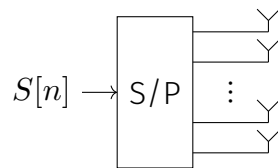
$$\mathbf{R}_x[n] = \frac{E_s}{N_t} \mathbf{F}[n] \mathbf{F}^H[n]. \quad (2.101)$$

Comparison with normal case

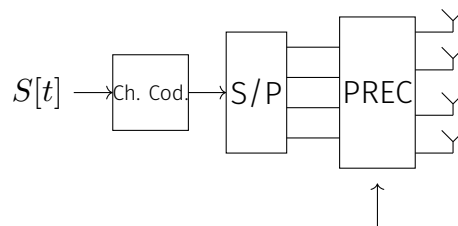
If we were to set $\mathbf{F}[n] = \mathbf{I}$, we'd find that the non-precoded matrix of x is

$$\frac{E_s}{N_t} \mathbf{I}.$$

We have a source of symbols, we pass them through a serial to parallel and then we go through the antennas; usually, we use a code



What we generally do is actually code after the series/parallel transition!



4.7 Recap of singular value decomposition

Let's revisit the concept of singular value decomposition!

This is heavily used in machine learning and such

We have a certain matrix $\mathbf{A}$ of dimensions $N \times M$; we can always decompose it as

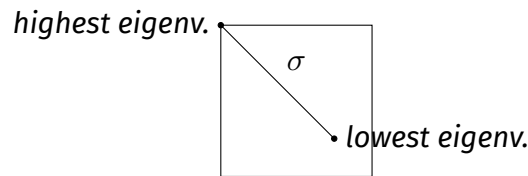
$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*,$$

where:

- $\mathbf{U}$ is unitary $N \times N$.
- $\mathbf{V}^*$ is unitary $M \times M$.
- $\mathbf{\Sigma}$ is $N \times M$ with non negative entries on its diagonal ordered from largest to smallest.

The entries of said matrix **are the singular values of the original matrix, aka the square root of the eigenvalues of $\mathbf{A}$.**

The $\mathbf{\Sigma}$ matrix is generally like



If we choose $\mathbf{u}_j$, a column of $\mathbf{V}$ and $\mathbf{u}_j$, a column of $\mathbf{U}$, we'll have

$$\mathbf{A}\mathbf{v}_j = \sigma_j \mathbf{u}_j$$

$$\mathbf{A}^* \mathbf{u}_j = \sigma_j \mathbf{v}_j$$

where σ_j is the j -th element on the diagonal of Σ .

We just need to remember the dimensions in order to have this work correctly.

Property of unitary matrices

For a unitary matrix, we have that

$$\mathbf{U}^{-1} = \mathbf{U}^*.$$

This is why unitary matrices are very useful.

$\mathbf{U}$ is very easy to invert and same goes for $\mathbf{V}$. Then we might say: it will be easy to invert $\mathbf{A}$.

$\mathbf{A}$ will also have the following properties:

Table B.1 Bases for the column, row, and null spaces of an $N \times M$ matrix $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$

Subspace	Dimension	Basis
Column space of $\mathbf{A}$	$r = \text{rank}(\mathbf{A})$	First r columns of $\mathbf{U}$
Row space of $\mathbf{A}$	r	First r columns of $\mathbf{V}$
Null space of $\mathbf{A}^T$	$(N - r)$	Last $(N - r)$ columns of $\mathbf{U}$
Null space of $\mathbf{A}$	$(M - r)$	Last $(M - r)$ columns of $\mathbf{V}$

About the number of streams

If we have 1 transmitter and 1 receiver, we know many things; but we didn't consider a channel which is dispersive and generates ISI; how do we deal with ISI?? This is handled in [section 2.5](#) of the book.

Let's suppose, at least for now, that we have no ISI; the Shannon bound tells us that we have a maximum amount of bits per symbol per hertz which we can put through the channel.

People said: we want to transmit even more!! In order to do so, we increase the number of transmitter and receiver antennas. Let's say we have 3 transmitters and 5 receivers; the interactions between all of these won't be described by a vector anymore, but rather by a matrix. We'll see that we have **a number of independent streams which is**

$$\# \text{streams} \leq \min\{N_t, N_r\};$$

we use MIMO since we want to be able to transmit even more than what would have been possible while respecting the Shannon bound. It's useless to have 1 transmitter and many receivers: we'd be able to only have 1 stream.

It's difficult to have many antennas in the phone, much more than it is to have many in our station.

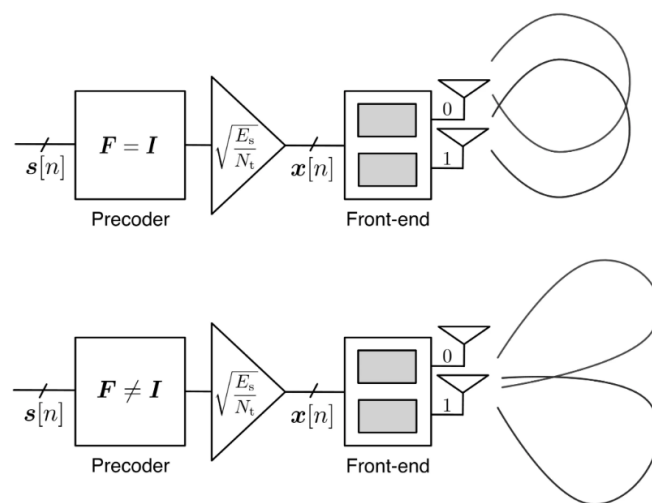
Before looking at this, we want to understand if this works at all, though! We first need to know the theory and that this is possible and only then will we be able to use this.

4.8 Applying precoding: decomposing $\mathbf{F}$ and explaining the various parts of its new form

Let's apply what we've learned to the model

$$\mathbf{x}[n] = \sqrt{\frac{E_s}{N_t}} \mathbf{F}[n] \mathbf{s}[n], \quad (2.98)$$

we know we have $\mathbf{s}[n]$ at the input and let's suppose that $\mathbf{F} = \mathbf{I}$: we go directly in the antennas. If we do this, all the antennas transmit all like a circle (this depends on how they're built), as it comes. If we then have $\mathbf{F} \neq \mathbf{I}$, we can get each antenna to point to a specific user; this allows us to maximize the gain we have from the transmit antenna.



What we want to do is to multiply our symbols with a matrix and do something. What can we achieve? In order to better understand this, we can use **singular value decomposition** to decompose $\mathbf{F}$ as:

$$\mathbf{F}[n] = \mathbf{U}_F[n] \mathbf{\Lambda}_F[n] \mathbf{V}_F^H[n], \quad (2.100)$$

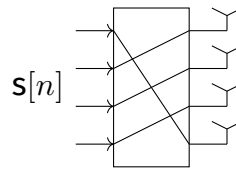
where:

- $\mathbf{V}$ is the **mixing matrix**, which is a $N_s \times N_s$ unitary matrix
- $\mathbf{\Lambda}$ is the **power allocation matrix** and is $N_t \times N_s$ (more about this later)
- $\mathbf{U}$ is the **beam steering matrix**, which is a $N_t \times N_t$ unitary matrix

If we had only

$$\mathbf{V}_F^* \mathbf{s},$$

we'd get N_t symbols; this mixes the various signals towards the various antennas:



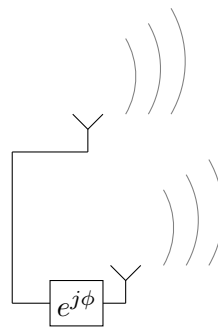
By only doing this we'd create confusion; this is why we also have a power allocation matrix: it has a diagonal which expresses how much power we'll give to each antenna. It will have a shape like

$$\mathbf{P}_F[n] = \begin{bmatrix} \mathbf{P}_2^{\frac{1}{2}} \\ \mathbf{0} \end{bmatrix} \quad (2.101)$$

where $\mathbf{P}$ is a $N_s \times N_s$ matrix which will be defined below.

What's really important is $\mathbf{U}_F$, which allows us to **steer the signal**.

This had been discovered a long time ago! If we have a radar and we transmit with a phase shift on one antenna:



In a certain direction we'll have a specific ϕ ; if $\phi = \pi$, we're not transmitting anything. If we look at another direction, we can tilt an antenna electronically to transmit over a certain direction.

This is used by radars, even though radars transmit an impulse and want to get stuff back. In order to understand where the signal is coming from they use multiple antennas and they use something like a steering matrix. If we take the nose of an airplane, the whole nose of an airplane has a radar in it.

It's nothing new in this sense!

So now we can have different directions from an array of antennas. By using appropriate matrices, we can make spatial streams.

Our tools for designing are $\mathbf{U}_F$ and the amount of power we put in each antenna (the sum of the power cannot be more than N_t), expressed by the matrix

$$\mathbf{P} = \text{diag}(P_1, \dots, P_{N_s-1})$$

such that $\sum_{i=0}^{N_s-1} P_i = N_t$.

The notation P_j is especially interesting since the j th mixed signal (aka $[\mathbf{V}_F^H \mathbf{s}[n]]_j$) will have a power of $\frac{E_s}{N_t} P_j$ allocated to it, where the E_s/N_t term comes from our definition in (2.98).

The j th amplified signal will be launched from all transmit antennas and will be weighted with the coefficients in the j th column of $\mathbf{U}_F[n]$.

If we were to choose a silly (aka useless) setup, we might have both $\mathbf{U}_F, \mathbf{V}_F$ be identity matrices and $\mathbf{P} = \frac{N_t}{N_s} \mathbf{I}$, which entails

$$\mathbf{F}[n] = \sqrt{\frac{N_t}{N_s}} \begin{bmatrix} \mathbf{I}_{N_s} \\ \mathbf{0} \end{bmatrix}. \quad (2.102)$$

With such trivial precoder, each of N_s antennas directly radiates one of the data streams within $\mathbf{s}[n]$ while the remaining $N_t - N_s$ antennas are silent; the transmit signals are independent. If this is the case, the transmission is said to be *isotropic* from a precoding pov or even *unprecoded*.

Example 2.10

Let $N_t = N_s = 2$ and suppose that the two signals within $s[n]$ are QPSK-distributed. Examine the structure of the transmit signal $x[n]$ produced by a time-invariant precoder.

Solution

Dropping the dependence on n , let $\mathbf{P} = \text{diag}(P_1, P_2)$ and

$$\mathbf{U}_F = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix}, \quad \mathbf{V}_F = \begin{bmatrix} V_{00} & V_{01} \\ V_{10} & V_{11} \end{bmatrix}. \quad (2.103)$$

The application of the mixing matrix to $\mathbf{s}$ yields the mixed signal vector

$$\mathbf{V}_F^* \mathbf{s} = \begin{bmatrix} V_{00}^*(\mathbf{s})_0 + V_{10}^*(\mathbf{s})_1 \\ V_{01}^*(\mathbf{s})_0 + V_{11}^*(\mathbf{s})_1 \end{bmatrix} \quad (2.104)$$

whose unit-variance entries conform to 16-ary distributions, in general distinct. Then, after power allocation,

$$\sqrt{\frac{E_s}{2}} \mathbf{P}^{1/2} \mathbf{V}_F^* \mathbf{s} = \begin{bmatrix} \sqrt{\frac{E_s P_1}{2}} (V_{00}^*(\mathbf{s})_0 + V_{10}^*(\mathbf{s})_1) \\ \sqrt{\frac{E_s P_2}{2}} (V_{01}^*(\mathbf{s})_0 + V_{11}^*(\mathbf{s})_1) \end{bmatrix}. \quad (2.105)$$

The top signal is launched from the two antennas, weighted by U_{00} and U_{10} , while the bottom signal is launched weighted by U_{01} and U_{11} . If $\mathbf{F} = \mathbf{I}$, then each signal is radiated from only one antenna. Conversely, if $\mathbf{F} \neq \mathbf{I}$, each signal is transmitted from both antennas and steered in a certain direction. Both possibilities are illustrated in Fig. 2.11.

Comments:

- The $*$ operator on matrix also implies transposition
- A unitary matrix is just rotating the vector: it won't change its magnitude
- If $\mathbf{F} = \mathbf{I}$, each signal is radiated by one antenna
- If $\mathbf{F} \neq \mathbf{I}$, both signals are transmitted by both antennas and we can steer the antennas.

4.9 Signal power constraints

Which power constraint will we enforce? We might have average energy per block or per symbol or even per antenna:

- **Per-block** $\frac{1}{N} \sum_n \|\mathbf{x}[n]\|^2 \leq P$, or equivalently $\frac{1}{N} \sum_n \text{tr}(\mathbf{R}_x[n]) = P_t$.

→ Then the precoder $\frac{1}{N} \sum_n \|\mathbf{F}[n]\|_F^2 = N_t$

- **Per-symbol** $\mathbb{E}[\|\mathbf{x}[n]\|^2] = E_s$ or $\text{tr}(\mathbf{R}_x[n]) = E_s$

→ Then the precoder satisfies $\text{tr}(\mathbf{F}[n] \mathbf{F}^H[n]) = \text{tr}(\mathbf{P}[n]) = N_t$

- **Per-antenna** $(\mathbf{R}_x)_{ii} = \frac{E_s}{N_t}$, which means that

$$\|\mathbf{F}[n]\|_{i,:}^2 = 1 \text{ for } i = 0, \dots, N_t - 1$$

We might even be super strict and say **every symbol should have the same energy**, or more relaxed and say: on average a block should have a certain amount of energy. What we have depends on our situation.

≡ Examples

Regulation says that if we put a sphere around the top of the antenna; the power of the outgoing em field should be below a certain value. If we have that power, we'll need to share it between antennas.

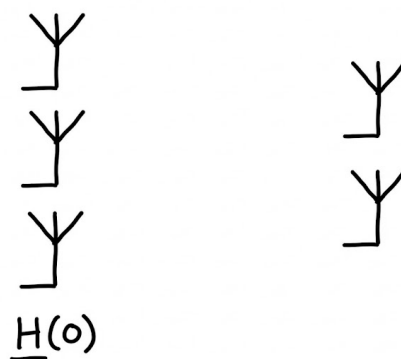
If we have DAS (Distributed Antenna System), which is now used in tunnels, we can put the same power on the various antennas.

The constraint depends on the practical situation which we have.

4.10 Vectorizing the channel: MIMO multipath channel (specifically, stacked notation absorbing convolution, time dependency and MIMO) for receiver i and for the whole system

We were looking at how to vectorize the channel; this paragraph, specifically, investigates how to reach for MIMO a result similar to the (2.74) we reached for SISO. We also include previous values of the signal since the channel has memory. This is like convolution, but in the time domain.

Now, instead of having SISO systems, we can jump to the MIMO case. In summary, instead of having scalars in the matrix H , in the MIMO system we'll have matrices.



$H(o)$ tells how the signal gets from one part to the other. In the SISO case we had $h[k], k = 0, \dots, L$: we had L paths since we weren't in a flat channel. We won't stop at considering only the matrix corresponding to $h[0]$; we'll also look at the matrix for the system's memory.

If we then consider the receiver antenna i , we'll have that the received signal will be the sum of all the values:

$$y_i^{(r)} = \sqrt{\rho} \sum_{j=0}^{N_t-1} [\mathbf{H}]_{i,j} \mathbf{x}_j^{(N+L)} + \mathbf{v}_i^{(r)}. \quad (2.89)$$

where $\mathbf{H}$ is a matrix of matrices and $N + L$ is there since we have L delays. The output does not only depend on one input vector, but it's still a vector of dimension $N + L$ depending on the input of each transmit antenna ($x_j^{(N+L)}$).

What are those $N, N + L$ subscripts?

Those subscripts represent the dimensions of vectors and matrices. Just as an example, $\mathbf{x}$ is a $(N + L) \times 1$ vector and $\mathbf{H}$ is a $N \times (N + L)$ matrix. Specifically, then, each vector contains the time information for signals for $n = 0, \dots, N - 1$, with the relation $\mathbf{H}\mathbf{x}$ representing the convolution operation.

We're considering the antenna i :



We have many antennas transmitting to i ; we need to sum over all the transmitting antennas and over all the vectors transmitted at the j -th antenna. The jump we're doing with this new equation is given by the fact that we have many transmission antennas! y will have a dimension of $N_r N$! We're putting the various vectors one on top of the other:

$$\bar{y}_{N_r N} = \begin{bmatrix} \bar{y}_N^{(0)} \\ \vdots \\ \bar{y}_N^{(N_r-1)} \end{bmatrix}.$$

$$\bar{x}_{N_t(N+L)} = \begin{bmatrix} \bar{x}_{N+L}^{(0)} \\ \vdots \\ \bar{x}_{N+L}^{(N_t-1)} \end{bmatrix}.$$

Before we had $N, N + L$ on the matrix; now we have $N_r N, N_t(N + L)$. The input vector is $N_t(N + L)$: we're stacking the vectors of the input antennas.

All the matrices in $\mathbf{H}$ are $N, N + L$ matrices:

$$\begin{bmatrix} \bar{H}_{N,N+L}^{(0,0)} & \bar{H}_{N,N+L}^{(0,1)} & \cdots & \bar{H}_{N,N+L}^{(0,N_t-1)} \\ \bar{H}_{N,N+L}^{(1,0)} & \bar{H}_{N,N+L}^{(1,1)} & \cdots & \bar{H}_{N,N+L}^{(1,N_t-1)} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{H}_{N,N+L}^{(N_r-1,0)} & \bar{H}_{N,N+L}^{(N_r-1,1)} & \cdots & \bar{H}_{N,N+L}^{(N_r-1,N_t-1)} \end{bmatrix}$$

Each of the matrices is like a SISO matrix; we're considering how the information transmitted from antenna j is arriving at antenna i .

In summary, we can express the output of the MIMO system as

$$\bar{y}_{N_r N} = \sqrt{G} \bar{H}_{N_r N, N_t(N+L)} \bar{x}_{N_t(N+L)} + \bar{v}_{N_r N}. \quad (2.95)$$

By the way, L is the **maximum length of the interference channel**; if we have two antennas with a channel with $L_1 = 3$ and another with $L_2 = 7$, we'll use $L = 7$. We'll just put zeros for all the paths which an antenna does not have:

$$h(0), h(1), h(2), 0, 0, 0, 0.$$

If we want to have a well-formed matrix, we select L as the maximum of all the L s.

The one we got is a compact form, which is the same as the one we had before (with $\bar{y}$), just with bigger dimensions.

By the way, book uses terms *column* and *fat* matrices.

Chapter 5

Lesson 25-11-2025

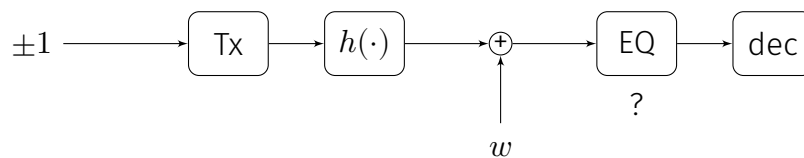
5.1 Linear channel equalization

Ref: *A signal processing perspective* (pp. 57-130), p. 31.

5.1.1 Introduction: super basic block scheme

It's a pretty old topic.

We have a transmitter, then a channel, then a noise which is added **before** anything else.



About magnetic writing; we're writing on a tape, which is noisy; we need to encode before writing.

Fractional sampling: we're sampling at multiples of the Nyquist rate; it's used in order to synchronize.

We don't need to have ± 1 , it can just be whichever input.

5.1.2 Feedback equalizer

If we have that at a certain time the receive signal is

$$y(0) = h(0)x(0) + h(1)x(-1), \quad (5.1)$$

how do we equalize this?

First of all, we'd need to estimate the channel: this means trying to understand the values of $h(0), h(1)$.

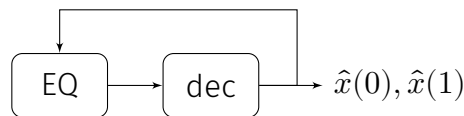
If we suppose we estimated $x(-1)$ as $\hat{x}(-1)$, what we do is implement a feedback equalizer: we'll look at

$$\frac{y(0) - h(1)\hat{x}(-1)}{h(0)} \quad (5.2)$$

which will get us $x(0)$.

We can estimate $\hat{x}(-1)$ since we generally have packets: we know the first symbols, thus we can **bootstrap** this machine.

This is the feedback equalizer:



It's also true that if we fail in decoding a symbol, this could be catastrophic: this would lead to a chain of errors.

We need to play with this type of equalization very carefully.

5.1.3 Linear zero forcing equalization (for both SISO and MIMO; only in the z domain); example 2.17

We'll see just something, but if we want to know more we can look at other books.

What we do is that we go in the z domain (see **z transform**); when we have a sequence, such as $h[0], h[1], h[2], \dots$, we'll write it as

$$\mathbf{h}(z) = h_0 + h_1 z^{-1} + h_2 z^{-2} . \quad (5.3)$$

We can revise this by looking at signals and systems.

A delay of n in the time domain corresponds to a multiplication with z^{-n} in the z domain.

We want to find an equalizer $w^*(z)$ such that the combination with $\mathbf{h}(z)$ is just a delay (this is the Heaviside condition):

$$w^*(z)\sqrt{G}h(z) = z^{-\Delta} . \quad (2.124)$$

The delay is a degree of freedom which can be used when designing.

We however can see that $w^*(z)$ **might not exist or it might be unstable**.

According to the zero-forcing criteria, we'd have

$$w^*(z) = \frac{1}{\mathbf{h}(z)} \cdot \frac{z^{-\Delta}}{\sqrt{G}} ; \quad (5.4)$$

we first of all ask ourselves: **is this stable?** If it's not stable, the noise coming in might make our system explode!

The system is BIBO if the root of the polynomial $w^*(z)$ is inside the unit circle. The noise, even if bounded, might not be bounded after the system.

This is the problem of the zero forcing; the zero forcing means that we want to just have a single delay; point is that we might have just a delay but with a non-stable filter.

We then ask ourselves: **is the response $w^*(z)$ finite or infinite?** → most of the time is **infinite**



We'll then look at the zero forcing **in the MIMO case**. The interesting result is that here for many types of FIR multivariate channel we have FIR zero forcing equalizers, given that we're under certain conditions.

Let's say that the Z-transform of $H[0], \dots, H[L]$ is

$$\mathbf{H}(z) = \sum_{\ell=0}^L \mathbf{H}[\ell] z^{-\ell}; \quad (2.129)$$

this implies that the channel will become a polynomial and **we'll look for an equalizer $\mathbf{W}^*(z)$** such that

$$\mathbf{W}^*(z) \sqrt{G} \mathbf{H}(z) = \mathbf{D}(z), \quad (2.130)$$

where

- $\mathbf{D}(z)$ is a diagonal matrix containing possibly different delay terms in the form of $z^{-\Delta}$.



One thing which should be noted is that **we need to have that a left inverse exists for $\mathbf{H}$ in order for the equalizer to exist**; if we cannot have a left inverse (for various reasons, such as $\mathbf{H}$ not being full-rank), then **we won't have an equalizer**.

One condition which is considered already from this moment is that (quoting from the book):

Existence of a left FIR inverse

A MIMO transfer function $\mathbf{H}(z)$ has a left FIR inverse iff $\mathbf{H}(z)$ has full column rank for every complex $z \neq 0$.

Some other conditions will be tackled later, but just as an anticipation, they'll result in saying that the ZF equalizer exists only if

$$N_t(L_{eq} + L + 1) \leq N_r(L_{eq} + 1) \quad (2.146+)$$

What was $H[\ell]$ again?

We know that $\mathbf{H}[\ell]$ is a matrix such that

$$[H[\ell]]_{i,j} = h_{i,j}[\ell]$$

is the channel's response between receiving antenna i and transmitting antenna j .

From this, it will follow that

$$[\mathbf{H}(z)]_{i,j} = \mathcal{Z}[h_{i,j}[\cdot]] z.$$

A signal processing perspective (pp. 57-130), p. 33

As in the SISO case, it is in practice desirable to restrict the focus to FIR equalizers, meaning that each entry of $\mathbf{w}(z)$ corresponds to an FIR filter of order L_{eq} . Do exact left inverses of $\mathbf{H}(z)$ exist? Yes, as it turns out. More precisely, the answer is found in what is called **perfect recoverability**.

So, if the **perfect recoverability** holds, we'll be sure to find left inverses of the $\mathbf{H}$ matrix; we need the left inverse since it will be used when we'll determine the ZF equalizer.

Here in **MIMO FIR filters may exist!** Before, in SISO, **they might have not!**

About matrices: until now we saw that the inverse of whichever matrix A is such that $AA^{-1} = I$ and $A^{-1}A = I$; this only holds if it's square; generally, the left and right inverses are different.

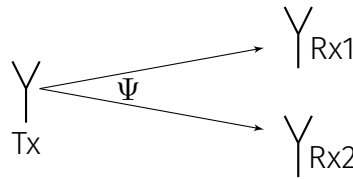
About this paragraph

The notation we wrote here just gives an idea; everything will be treated in the time domain.

Example 2.17

For $N_t = 1$ and $N_r = 2$, find the conditions on $\mathbf{H}(z)$ such that FIR ZF equalizers exist.

If we have 1 transmit antenna and 2 receive antennas; in general we might transmit to both antennas:



! Digital always includes a delay!

We then need to decide what to do; **we might think about selecting the stronger signal**; one might have a reflection and such; this is called **antenna selection**. **It's easy to measure the amplitude of a signal, but we can't know whether the signal is clean.**

Someone came up with something called **MRC (Max Ratio Combining)**: **we weight each signal with the complex conjugate**; this allows us **to have the best SNR**. This is still done, even with analog transmissions.

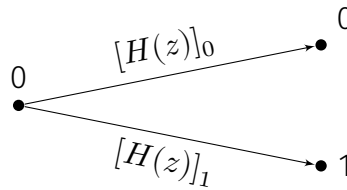
What we have is that we want to satisfy

$$[\mathbf{W}(z)]_0^* \sqrt{G} [\mathbf{H}(z)]_0 + [\mathbf{W}(z)]_1^* \sqrt{G} [\mathbf{H}(z)]_1 = z^{-\Delta}. \quad (5.5)$$

$\mathbf{H}(z)$ is a column vector:

↓ *read example (continued)*

Example 2.17 (continued)



$[H(z)]_0$ might be $= h_{0,0} + h_{0,1}z^{-1} + \dots$
Anyways, we'll have

$$\mathbf{H}(z) = \begin{bmatrix} [H(z)]_0 \\ [H(z)]_1 \end{bmatrix}. \quad (5.6)$$

The only way for this vector to not be full rank is if $\mathbf{H}(z) = \mathbf{0}$ for some z .
If we have that there exists z^* such that $\mathbf{H}(z^*) = \mathbf{0}$ and this happens only if the polynomials have a common root, which is z^* ; we don't have full rank only if those two polynomials share a common root!
If those polynomials are coprime (they don't share a common root!), then it's possible to find an exact FIR inverse.

We care so much about the full rank since a left inverse only exists if $\mathbf{H}(z)$ has full column rank for every $z \neq 0$.

If we have

$$\mathbf{H}(z) = \begin{bmatrix} 1 + z^{-1} \\ 1 + z^{-1} \end{bmatrix},$$

aka the two are equal, we can't find FIR; if we have a half, then it's possible!

$$\mathbf{H}(z) = \begin{bmatrix} 1 + \frac{z^{-1}}{2} \\ 1 + z^{-1} \end{bmatrix}.$$

This is why it's not possible for SISO systems.

5.2 Moore-Penrose pseudo-inverse

In the future, we'll need another version of an inverse; this is the Moore-Penrose pseudo-inverse.

This can be applied to non-square matrices!

The pseudoinverse $A^\dagger$ of an $N \times M$ matrix A is the unique matrix satisfying

$$AA^\dagger A = A \quad (B.30)$$

$$A^\dagger AA^\dagger = A^\dagger \quad (B.31)$$

and also such that $AA^\dagger, A^\dagger A$ are both **Hermitian**.

Taking the already known concept of inversion, we have that:

- If A^*A is invertible, then

$$A^\dagger = (A^*A)^{-1}A^* \quad (\text{B.32})$$

$$A^\dagger A = I; \quad (5.7)$$

this may be the case if $N \geq M$ (tall matrix).

- If AA^* is invertible, then

$$A^\dagger = A^*(AA^*)^{-1} \quad (\text{B.33})$$

$$AA^\dagger = I; \quad (5.8)$$

this may be the case if $N \leq M$ (fat matrix).

- If A is square and invertible, then (B.32), (B.33) coincide and the pseudoinverse reduces to the regular inverse,

$$A^\dagger = A^{-1}.$$

⚠ The right inverse and the left inverse are not the same!

⚠ This only holds if the matrix is **full-rank** (which is then the reason why we ask that the H matrix is full-rank).

We'll use this a lot in the course.

Another point is that if we remember the **singular value decomposition**, then we have

$$A = U\Sigma V^*$$

$$A^\dagger = V\Sigma^{-1}U^*.$$

By the way, we can do this in matlab using the `pinv()` function.

5.3 Actually finding the MIMO zero forcing equalizer

Let's write the problem in the time domain, going from

$$W^*(z)\sqrt{G}H(z) = D(z), \quad (2.130)$$

where $D(z)$ is made of diagonal elements:

$$D(z) = \begin{bmatrix} z^{-\Delta_1} & & 0 \\ & z^{-\Delta_2} & \\ 0 & & \ddots \end{bmatrix}$$

to

$$\sqrt{G} \sum_{\ell=0}^{L_{eq}} W^*[\ell]H[n-\ell] = \text{diag}[\delta[n-\Delta_0], \dots, \delta[n-\Delta_{N_t-1}]]. \quad (2.132)$$

where:

- $\mathbf{W}$ is a $N_r \times N_t$ matrix (I deduced the N_t from the fact that later the columns are indexed with j).
- $\mathbf{H}$ is a $N_r \times N_t$ matrix. (**Note: usually H_{ij} means from j to i**).

This is a diagonal matrix with delays since we designed our equalizer to establish so.

We've allowed ourselves to have different delays ($\Delta_n, n \in [0, N_t - 1]$); finding the zero forcing equalizer means solving the following equation for the stream j :

$$\sqrt{G} \sum_{\ell=0}^{L_{eq}} \mathbf{w}_j^*[\ell] \mathbf{H}[n - \ell] = \mathbf{d}_j[n] \quad n = 0, 1, \dots, \underbrace{L_{eq} + L}_{\rightarrow \text{consequence of convolution}}; \quad (2.133)$$

It makes perfect sense to now ask ourselves what all the parameters are, specifically; let's see them in detail.



$\mathbf{w}_j[\ell]$

$\mathbf{w}_j[\ell]$ is a $N_r \times 1$ column vector such that

$$\mathbf{w}_j[\ell] = \mathbf{W}[\ell]_{:,j} : \begin{bmatrix} w_{1,j}[\ell] \\ \vdots \\ w_{N_r,j}[\ell] \end{bmatrix}.$$

It represents the equalization applied to the signals transmitted by the antenna j towards all the receiving antennas.



$\mathbf{d}_j[n]$

With reference to the equation (2.133) we define $\mathbf{d}_j[n]$ as the **target delays for transmitted signal j** , with $j = 0, \dots, N_t - 1$; it will be a $N_t \times 1$ vector made of **all zeros** and a $\delta[n - \Delta_j]$ **at position j** , where $\Delta_j \in [0, \dots, L_{eq} + L]$ is the target delay for the transmitted signal j .

We can formulate (2.133) in vector notation as

$$\sqrt{G} \bar{\mathbf{w}}_j^* \mathcal{T} = \bar{\mathbf{d}}_j, \quad (2.137)$$

with the various elements being as described in the following:



$\bar{\mathbf{w}}_j$

$\bar{\mathbf{w}}_j$ is a $N_r(L_{eq} - 1) \times 1$ vector obtained by stacking all $\mathbf{w}_j$ vectors on top of one another, as

$$\bar{\mathbf{w}}_j = \begin{bmatrix} \mathbf{w}_j[0] \\ \vdots \\ \mathbf{w}_j[L_{eq}] \end{bmatrix}, \quad (2.134)$$

embedding all the equalizer responses over the delay dimension, always referring to transmit antenna j .



We then define $\bar{\mathbf{d}}_j$ as a $N_t(L_{eq} + L + 1) \times 1$ vector:

$$\bar{\mathbf{d}}_j = \begin{bmatrix} \mathbf{d}_j[0] \\ \vdots \\ \mathbf{d}_j[L_{eq} + L] \end{bmatrix}; \quad (2.135)$$

discussing the positions at which this vector is non-zero deserves its own discussion. Due to how $\mathbf{d}_j[n]$ is defined, we know that it will have a 1 only at position j and only if $n = \Delta_j$. Let's now consider $\bar{\mathbf{d}}_j$ for a simple case such as $N_t = 2, j = 0, L_{eq} + L = 1$; we'll have

$$\bar{\mathbf{d}}_0 = \begin{bmatrix} \mathbf{d}_0[0] \\ \mathbf{d}_0[1] \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \delta[0 - \Delta_0] \\ 0 \end{bmatrix} \\ \begin{bmatrix} \delta[1 - \Delta_1] \\ 0 \end{bmatrix} \end{bmatrix};$$

in this simple example we see that the various subvectors are located at multiples of N_t , specifically at $nN_t, n \in [0, L_{eq} + L]$; in each subvector, the δ is located at the j -th element. We'll thus have the various deltas at positions

$$N_t n + j, \quad n = 0, \dots, L_{eq} + L$$

where j is fixed by the vector $\bar{\mathbf{d}}_j$ we're considering.

It's also true that the δ s will only be 1 if $n = \Delta_j$ and we can thus simplify the previous equation by saying that we'll have 1 in the vector at positions

$$N_t \Delta_j + j,$$

which of course can only be satisfied for those Δ_j which are $\Delta_j \leq L_{eq} + L$.



The Toeplitz matrix $\bar{\mathcal{T}}$ is defined as

$$\bar{\mathcal{T}} = \begin{bmatrix} \mathbf{H}[0] & \dots & \mathbf{H}[L] & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{H}[0] & \dots & \mathbf{H}[L] & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} & \mathbf{H}[0] & \dots \mathbf{H}[L] \end{bmatrix}, \quad (2.136)$$

having dimensions of $N_r(L_{eq} + 1) \times N_t(L_{eq} + L + 1)$.

We reach a linear system of equations $\rightarrow$ under determined means infinite sol, over determined means no-sol.

We then consider that we're trying to minimize the error in (2.137), which is proportional to

$$\|\sqrt{G}\bar{\mathbf{w}}_j^* \bar{\mathcal{T}} - \bar{\mathbf{d}}_j^*\|^2. \quad (\text{RIP Tomba}^\dagger) \quad (2.138\text{-ish})$$

We can show (*A signal processing perspective* (pp. 57-130), p. 35) that (2.138-ish) can be minimized using the left Moore-Penrose pseudo-inverse of the matrix $\tilde{\mathcal{T}}^*$, aka $(\tilde{\mathcal{T}}^*)^\dagger$:

$$\bar{\mathbf{w}}_j^{ZF} = \frac{1}{\sqrt{G}} (\tilde{\mathcal{T}}^*)^\dagger \bar{\mathbf{d}}_j \quad \text{time domain} \quad (2.145)$$

$$\mathbf{W}^{ZF} = \frac{1}{\sqrt{G}} (\tilde{\mathcal{T}}^*)^\dagger \bar{\mathbf{D}} \quad \text{z domain} \quad (2.146)$$

We can see how we can find the zero-forcing solution for MIMO.

I'd like to elaborate a lil bit on this; we essentially have that

$$\sqrt{G} \mathbf{W}^{ZF*} \tilde{\mathcal{T}} = \mathbf{D}^* \implies \mathbf{W}^{ZF*} = \frac{1}{\sqrt{G}} \mathbf{D}^* (\tilde{\mathcal{T}})^\dagger.$$

Now we need to do the complex transpose of that, in order to find the equalizer. Before doing that, it's relevant to know that

$$(\tilde{\mathcal{T}}^\dagger)^* \stackrel{\text{tall}}{=} [(A^* A)^{-1} A^*]^* = A(A^* A)^{-1} = (\tilde{\mathcal{T}}^*)^\dagger;$$

then we see that the complex transpose of the equation up there does indeed give (2.146).

❓ z domain?

I think it was professor to write "z domain"; I'm positive this whole thing goes for the time domain!

For a frequency flat channel, this simplifies to

$$\mathbf{W}^{ZF} = \frac{1}{\sqrt{G}} (\mathbf{H}^\dagger)^*.$$

Just as a sidenote, this equation gives what some exercises refer to as the *perfect equalizer*.

One important thing which was overlooked (until here) is that this comes as a solution to (2.137); thing is that **said solution only exists if**

$$N_t(L_{eq} + L + 1) \leq N_r(L_{eq} + 1) \quad (2.146+)$$

(aka **only exists if $\tilde{\mathcal{T}}$ is either square or tall**).

⚙️ $\mathbf{W}^{ZF}$

The $\mathbf{W}^{ZF}$ matrix is defined as

$$\mathbf{W}^{ZF} = [\bar{\mathbf{w}}_0^{ZF} \quad \dots \quad \bar{\mathbf{w}}_{N_t-1}^{ZF}],$$

in a way such that the j -th column represents the equalizer coefficients for the j th transmitted signal in a way such that the values from kN_r to $(k+1)N_r$, $k \in \mathbb{Z}$ will represent the coefficients used to equalize at the k th receiver.



Matrix $\bar{\mathbf{D}}$ is defined as

$$\bar{\mathbf{D}} = [\bar{\mathbf{d}}_0 \quad \dots \quad \bar{\mathbf{d}}_{N_t-1}],$$

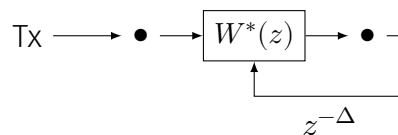
which, due to what has been said before, will be zero except for positions

$$(N_t \Delta_j + j, j),$$

which will be 1, if the position exists in the matrix.

5.4 MIMO Degrees of Freedom and ZF Equalization

Let's conceptually look at a problem which arises only when we look at SISO due to the lack of degrees of freedom. If we have 3 transmitter and 4 receivers, we can introduce delays to make everything look like just one delay. If we just have 1 receiver, we can't make everything look just like one delay, though:



For the first antenna, we modify $W^*(z)$ so that it can transform the first channel response into a delay $z^{-\Delta}$; the other channel response will not be such that the same $W^*(z)$ will be able to enforce the same delay as a result (at least if the response is different, which is generally the case): we've already used the only degree of freedom we had by setting Δ .

Rem. *In the MIMO case, we have more degrees of freedom and that's why we generally have a solution, provided that $\mathbf{H}$ is full rank.*

Example 2.18

Compute the ZF equalizer for a frequency-flat channel with $N_r \geq N_t$.

In a frequency flat channel we can see that the matrix is just a pseudo inverse of $\mathbf{H}$, since we have $T = \mathbf{H} \rightarrow$ delay of L . Then do not $L_{eq} \rightarrow \infty$ (minimize eq. error).

We'll thus have:

$$\mathbf{W}^{ZF} = \frac{1}{\sqrt{C}} \mathbf{H}^+ \quad (5.9)$$

There's a difference between the SISO and the MIMO cases.

- **SISO:** we have the **problem of stability** and even for simple channels we might **not have a causal filter**.
- **MIMO:** The **perfect equalizer** of a FIR channel is FIR.

✓ **Example 2.19**

Verify that the W^{ZF} derived in Example 2.18 effects perfect ZF equalization.

Solution

The application of the equalizer to the channel gives:

$$W^{ZF}\sqrt{C}H = (HH^+)^{-1}H^H H \Rightarrow I \quad \text{Identity} \quad (5.10)$$

If we ignore the noise, we have that the channel behaves like an Identity.

If we have 4 antennas in transmission and reception, it's like if we had 4 parallel channels: this is a big step forward.

5.5 LMMSE equalization



Until now **we ignored noise**: the big problem with zero forcing which does not disappear with MIMO is that **noise generally gets amplified**.

If we multiply the w with y , we get an error on top of the delay. We'll introduce a new concept, aka **trying to minimize the noise**. We're not just inverting the channel: we now have another target; we're trying to invert the channel as much as we can while also limiting the noise as much as we can.

ZF equalization neglects the noise: we instead would like to **take the noise into account**; in order to do so, we use the MMSE metric we already introduced. Our equalizer will then be a MMSE estimator and since we stay in the context of linear equalization, it will be a Linear MMSE estimator (LMMSE).

In this section we'll try to derive a LMMSE estimator using the theoretical concept we defined for MMSE estimation.

For this derivation we consider both $\mathbf{x}[n], \mathbf{s}[n]$ as **wide sense stationary**, using the notations:

$$R_x[l] = E[\mathbf{x}[n]\mathbf{x}^*[n+l]] \quad (5.11)$$

$$R_x[l] = N_0\delta[l] \cdot I \quad (5.12)$$

(Pass on this / get to some)

It can be shown that thanks to the transmit-receive relationships we defined:

$$R_y[l] = G \sum_{i=0}^L \sum_{j=0}^L \mathbf{H}[i] R_x[l+i-j] \mathbf{H}^*[j] + N_0\delta[l]I \quad (5.13)$$

$$R_{yx}[l] = \sqrt{G} \sum_{i=0}^L \mathbf{H}[i] R_x[l+i] \quad (5.14)$$

We want to look at the difference between $\mathbf{x}[n - \Delta]$ (which is the desired signal) and the equalized output (obtained by convolving the output y ($N_r \times 1$) and the equalizer $\mathbf{W}$ ($N_t \times N_r$)). We need to remember that y will have the noise in it:

$$\hat{x}[n] - x[n - \Delta] = \sum_{l=0}^{L_{eq}} \mathbf{W}^*[l] y[n - l] \quad (5.15)$$

⚠ Totally not trivial implication

Differently from what we have in the original treatment, here we're not considering a frequency-flat channel. Rather, this channel has time dependency: for this reason, we're considering the convolution between $\mathbf{W}$ and y .

We want to minimize this difference here! We can also obtain a matricial notation by stacking the various vectors/matrices; we'll obtain:

$$\hat{\mathbf{x}} = \mathbf{x}[n - \Delta] - \mathbf{W}^* \vec{y}[n] \quad (5.16)$$

This is good to find the LMMSE estimator (*in the book are all the details, professor does not really care about them*).

Rem. *The book never states this explicitly, but those matrices are:*

- $\mathbf{W}^* : N_t \times N_r(L_{eq} + 1)$
- $y[n] : N_r(L_{eq} + 1)$

And they're obtained through a simple stacking.

Although neither the book or professor were really clear about this, I'd say that our equalizer is designed so that

$$\mathbf{W}^* y[n] \approx x[n - \Delta] \quad (5.17)$$

The problem is that ZF equalizers amplify the noise; we'd like to have a compromise between equalization of the channel and non-amplification of the noise.

We thus form the $\mathbf{W}^{LMMSE}$ estimator following from **The estimator is simply**

$$\mathbf{W}^{LMMSE} = \mathbf{R}_y^{-1} \mathbf{R}_{yx} \quad (5.18)$$

which in this case (that of **WSS signals/noise**) yields

$$\begin{aligned} \mathbf{W}^{LMMSE} &= \mathbf{R}_{\vec{y}[n]}^{-1} \mathbf{R}_{\vec{y}[n]x[n-\Delta]} \\ &= \begin{bmatrix} R_y[0] & R_y[1] & \dots & R_y[L_{eq}] \\ R_y[1] & R_y[0] & \dots & R_y[L_{eq}-1] \\ \vdots & \vdots & \ddots & \vdots \\ R_y[L_{eq}] & R_y[L_{eq}-1] & \dots & R_y[0] \end{bmatrix}^{-1} \begin{bmatrix} R_{yx}[-\Delta] \\ R_{yx}[1-\Delta] \\ \vdots \\ R_{yx}[L_{eq}-\Delta] \end{bmatrix} \end{aligned} \quad (5.19)$$

$$(5.20)$$

We then have that from

$$E = \mathbb{E}[(\mathbf{x} - \hat{\mathbf{x}}) \mathbf{y}^H (\mathbf{x} - \hat{\mathbf{x}})^H] \quad (5.21)$$

$$= \mathbf{R}_x - \mathbf{R}_{yx} - \mathbf{R}_{yx}^H + \mathbf{R}_y \quad (5.22)$$

$$= \mathbf{R}_x - \mathbf{W}^H \mathbf{R}_y^{-1} \mathbf{R}_{yx} - \dots \quad (5.23)$$

we can find the MMSE matrix by considering the various matrices we've already found, which will specifically give:

$$E = \mathbb{E}[\tilde{x}[n]\tilde{x}^*[n]] \quad (5.24)$$

$$= R_x[0] - R_{yx[n]y[n]}^H R_{y[n]}^{-1} R_{yx[n]x[n-\Delta]} \quad (5.25)$$

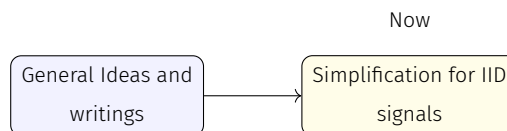
We don't need to remember the manipulations which are involved.
There won't be exercises about this!

! Exam

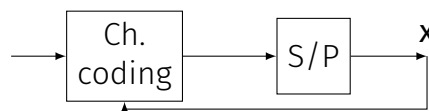
We can bring 1A4 paper, with printings on only one side. We'll be able to use them for everything but not for the quizzes!!

! Warning

NEVER forget the noise, since it's the most important thing here.



If, in order to simplify things, we write as follows:



We have that the values of $\mathbf{x}$ are actually dependent, since the channel coding introduces dependencies and redundancy.

We make an **(incorrect!!)** assumption which is that the vector $\mathbf{x}$ is made of **IID symbols**; this assumption allows us to have a simple writing of $\mathbf{R}_x$:

$$R_x[l] = \frac{E_x}{N_t} \delta[l] I \quad (5.26)$$

If we choose E_x in a specific manner, the R_x matrix becomes an identity, which simplifies things by a lot. This won't work in real systems, tho!

We'll obtain that the MMSE matrix can be written as:

$$\tilde{W}^{MMSE} = \mathbf{R}_{\tilde{y}[n]}^{-1} \mathbf{R}_{\tilde{y}[n]x[n-\Delta]} \quad (5.27)$$

$$= \frac{1}{\sqrt{G}} \left(\mathcal{T}^H \mathcal{T} + \frac{N_t}{\text{SNR}} \mathbf{I} \right)^{-1} \mathcal{T}^H D \quad (5.28)$$

$$= \frac{1}{\sqrt{G}} \mathcal{T}^H \left(\mathcal{T} \mathcal{T}^H + \frac{N_t}{\text{SNR}} \mathbf{I} \right)^{-1} D \quad (5.29)$$

The relevant thing is the SNR: if we push the SNR to $+\infty$, we have that the noise is disappearing and we'll find again the zero forcing equalizer! It makes sense that if the noise is disappearing the noise equalizer and the zero-forcing equalizer converge to be the same.

■ **This will never be asked, but for completeness:**

The $\mathcal{T}$ matrix is

✎ **Toeplitz matrix $\mathcal{T}$**

The Toeplitz matrix $\mathcal{T}$ is defined as

$$\mathcal{T} = \begin{bmatrix} H[0] & \dots & H[L] & 0 & \dots & 0 \\ 0 & H[0] & \dots & H[L] & \dots & 0 \\ \vdots & & \ddots & & \ddots & \vdots \\ 0 & \dots & 0 & H[0] & \dots & H[L] \end{bmatrix} \quad (5.30)$$

having dimensions of $N_r(L_{eq} + 1) \times N_t(L_{eq} + L + 1)$.

and D is all zeros except for

$$[D]_{\Delta N_t, \Delta N_t + N_t - 1, :} = \mathbf{I} \quad (5.31)$$

The alternative form is obtained using the **Matrix inversion lemma**. We know that this is a tool which is very much used.

What's important now is **the error**, which will be:

$$E = \frac{E_x}{N_t} \mathbf{I} - \frac{E_x}{N_t} D^T \mathcal{T}^* \left(\mathcal{T} \mathcal{T}^* + \frac{N_t}{\text{SNR}} \mathbf{I} \right)^{-1} \mathcal{T} D \quad (5.32)$$

✎ **Note: EXAM TYPE OF QUESTION: demonstrate that G goes to zero if n_r goes to infity**

If we **push the SNR to infinity**, we again have that the whole second term becomes $(E_x/N_t)\mathbf{I}$ and thus we'd have $E \rightarrow 0$: we're again in the zero-forcing situation.

We didn't calculate the error in case of zero forcing; we'll see that the error is actually worse than this one.

About the inversion of a matrix of matrices

We can correctly invert a matrix by finding its determinant:

$$A = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix} \quad (5.33)$$

$$\det(A) = d_{11}d_{22} - d_{12}d_{21} \quad (5.34)$$

Instead:

$$\mathcal{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad (5.35)$$

We'll have

$$\det(\mathcal{A}) = \det A_{11} \det A_{22} \quad (5.36)$$

$$- \det A_{12} \det A_{21} \quad (5.37)$$

We can find this as reference 125 in the book.

✗ Recommended read: example 2.21

Example 2.21, which was mentioned by professor many times, uses (2.163) together with some other smart tricks in order to obtain the LMMSE estimator for frequency flat channels, using the precoding writing of the signal,

$$x[n] = \sqrt{\frac{E_s}{N_t}} F[s] s[n] \quad (5.38)$$

, also assuming that the $s[n]$ symbols are IID, strict-sense stationary and unit-variance: this, together with the imposition that $\Delta = 0$, allows us to write that

$$\mathbf{R}_x[0] = \frac{E_s}{N_t} \mathbb{E}[\mathbf{F}[n] \mathbf{s}[n] \mathbf{s}^*[n] \mathbf{F}^*[n]] = \frac{E_s}{N_t} \mathbf{F}[n] \mathbf{F}^*[n]$$

(which is admittedly a bit of a stretch, since the book just said that $\mathbf{R}_x[0] = \mathbf{F} \mathbf{F}^*$, assuming that all symbols are precoded through the same precoder and that $[0]$ expresses a delay of 0).

In turn, this eventually allows us to write

$$\mathbf{W}^{\text{MMSE}} = \frac{1}{\sqrt{G}} \left(\mathbf{H} \mathbf{F} \mathbf{F}^* \mathbf{H}^* + \frac{N_t}{\text{SNR}} \mathbf{I} \right)^{-1} \mathbf{H} \mathbf{F}, \quad (2.182)$$

❓ Normally asked question: MIMO Eq

What is the length of the equalizer?

$L_{eq} = 2$? 1? No.

L_{eq} is **zero** because we don't have memory. **We have only a matrix.**

Convention:

indices: $\boxed{0}, 1, \dots, L_{eq}$
 $\underbrace{\hspace{1.5cm}}_{L_{eq}+1}$

Chapter 6

Digression on Circular Convolution

6.1 Circular Convolution

- **Problem:** in the time domain, **linear convolution** is complicated to perform.

$$y[n] = (x * h)[n] = \underbrace{\sum_{l=0}^L h[l]x[n-l]}_{\text{discrete time}} \quad (6.1)$$

Where:

- h is a **filter**
- x is the **input signal**
- y is the **output signal**

In the frequency domain, convolution becomes a simple **multiplication**!

$$y[n] = x[n] * h[n] \xrightarrow{\text{FFT}} Y[k] = X[k] \cdot H[k] \quad (6.2)$$

So the idea is:

1. Transform the signal (f domain)
2. Equalize it (do stuff)
3. Transform it back (time domain)

Attention

Multiplication holds only for **circular convolution**, not linear.

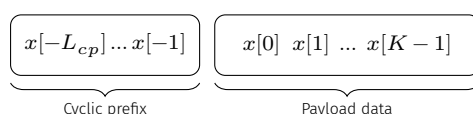
We need a way to “**trick**” the linear convolution channel so that its output corresponds to that of a circular convolution $\Rightarrow$ **CYCLIC PREFIX (CP)**!

Suppose we have a **payload** of length K and a response of the filter h with **memory** or order L .

The **CP** is obtained by taking the last L_{cp} samples of the **payload** and **prepending** them to the start of that block.

6.2 Cyclic Prefix & Fourier Taxonomy

To guarantee that the **linear convolution** is equal to the **circular** one, we must have $L_{cp} \geq L$.



I obtain an **extended signal** $\bar{x}[n]$ of length $K + L_{cp}$. In this way, the **periodic input** necessary for the **circular convolution** is simulated.

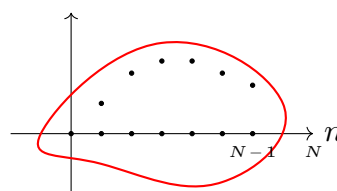
TIME DOMAIN	FOURIER F.T. exists
Periodic $NT = T_p, F = \frac{1}{T_p}$ discrete time T aperiodic	$S(kF) = \sum_{n=0}^{N-1} s(nT) e^{-j2\pi kFnT}$ $S(f) = \sum_{n=-\infty}^{+\infty} s(nT) e^{-j2\pi fnT}$
periodic signals $T_p, F = \frac{1}{T_p}$ continuous time t aperiodic signals	$S(kF) = \int_0^{T_p} s(t) e^{-j2\pi kt} dt$ $S(f) = \int_{-\infty}^{+\infty} s(t) e^{-j2\pi ft} dt$ ↑ this is a generic case

We focus on the behavior at **discrete periodic time**.

discrete time periodic:

$$\sum s(n) e^{-j2\pi \frac{kn}{N}} \quad (6.3)$$

→ Also called **Discrete Fourier Transform (DFT)**.



The **circular convolution** is defined as:

$$c(n) = \sum_{l=0}^{K-1} h[l] x[(n-l)_N] \quad (6.4)$$

Relationship in frequency domain:

$$C(k) = H(k) \cdot X(k) \quad (6.5)$$

6.3 Graphical Representation: SC-FDE

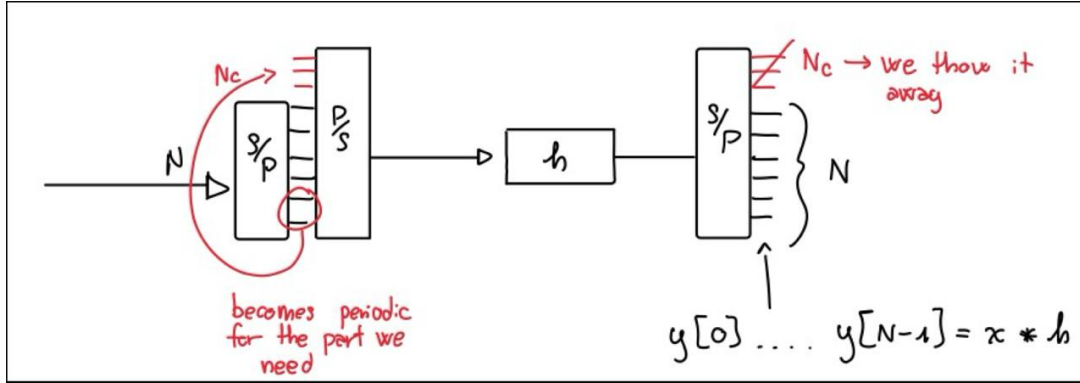


Figure 6.1: SC-FDE

6.3.1 Receiver Structure Notes

- Simple transmitter, complicated receiver.
- We can put the equalizer in the transmitter to make it more balanced.

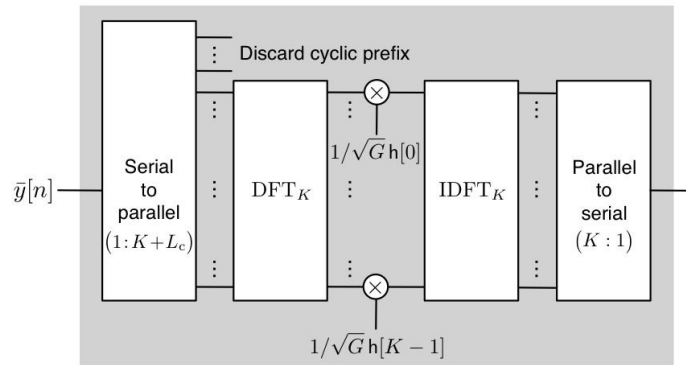


Fig. 2.14 Top: Serial-to-parallel, DFT, Equalization, IDFT

6.3.2 Noise Variance Derivation

$$= \frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|} \dots \frac{1}{|h[K-1]|} \right) \text{diag} \left(\frac{1}{h[0]} \dots \frac{1}{h[K-1]} \right)^* U \quad (2.197)$$

$$= \frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|^2}, \dots, \frac{1}{|h[K-1]|^2} \right) U. \quad (2.198)$$

The above covariance matrix is, in general, not proportional to the identity, indicating that the noise is correlated. The **variance of the noise** on the n -th equalized symbol is:

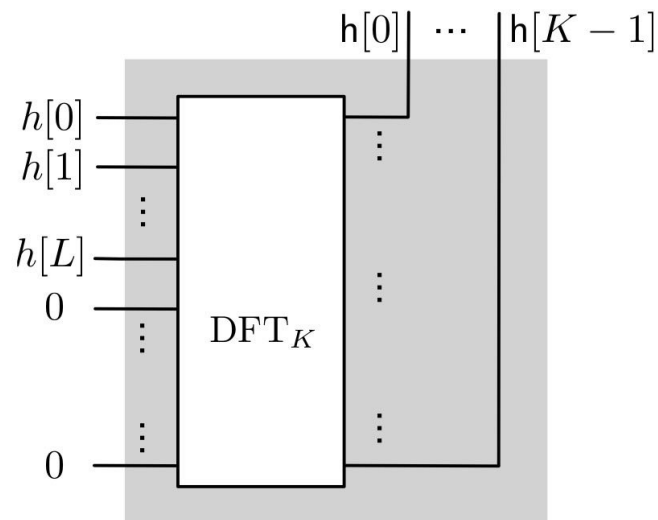


Fig. 2.14 Bottom: Channel taps into DFT

$$\left[\frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|^2}, \dots, \frac{1}{|h[K-1]|^2} \right) U \right]_{n,n} \quad (2.199)$$

$$= \frac{N_0}{GK} \sum_{k=0}^{K-1} e^{-j\frac{2\pi}{K}nk} \frac{1}{|h[k]|^2} e^{j\frac{2\pi}{K}nk} \quad (2.200)$$

$$= \frac{N_0}{GK} \sum_{k=0}^{K-1} \frac{1}{|h[k]|^2} \quad (2.201)$$

which does not depend on n , revealing that the noise variance is constant over the output.

Chapter 7

Lesson 28-11-2025

7.1 Orthogonal Frequency Division Multiplexing (OFDM)

7.1.1 OFDM Transmitter

Notation Change and "Blockization"

In OFDM, we change the standard notation compared to single-carrier systems. Usually, input symbols are indicated as $x[n]$ in the time domain. In OFDM, we denote the input symbols as $x[k]$. The k represents the index in the **frequency domain**, since these symbols will be transmitted on different subcarriers.

The input symbols are grouped ("packed") into blocks of size K .

$$\mathbf{x} = [x[0], x[1], \dots, x[K-1]] \quad (7.1)$$

These symbols may belong to a complex constellation, for example, 256-QAM.

🔗 Student Q&A: Block Stretching

Q: What is meant by "stretching" the blocks?

✓ Answer: Block Stretching

A: During the lecture, a student asked for clarification on "lengthening" the blocks. The professor explained that to transmit a block of K symbols, if one wants to maintain the same input rate, time or bandwidth must be increased. However, in the OFDM perspective, the key concept is that we take K symbols and transmit them **in parallel** over a longer time period compared to a single symbol, occupying a bandwidth $B \approx K/T$ (where T is the block time). Technically, in the context of the explanation, "stretch" referred to the insertion of the Cyclic Prefix, which physically lengthens the duration of the transmitted block from K to $K + L$ samples.

IDFT (Inverse Discrete Fourier Transform)

The fundamental operation at the transmitter is the IDFT applied to each block of K symbols. This step brings the symbols from the frequency domain to the time domain.

The formula to generate the signal in the time domain $x[n]$ is:

$$x[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j\frac{2\pi}{K}nk} \quad \text{for } n = 0, \dots, K-1 \quad (7.2)$$

Cyclic Prefix (CP) Insertion

To mitigate multipath channel effects, a Cyclic Prefix is inserted. If the length of the channel impulse response is L , we insert a prefix of length L (or greater) at the beginning of the block. The extended vector $\bar{x}$ will have dimension $K + L$. The operation consists of copying the last L samples of the block $x[n]$ and placing them at the beginning:

$$\bar{x} = [\underbrace{x[K-L], \dots, x[K-1]}_{\text{Cyclic Prefix (copy)}}, \underbrace{x[0], x[1], \dots, x[K-1]}_{\text{Original Block}}] \quad (7.3)$$

Formally, the negative indices in the prefix correspond to the final indices of the original block: $x[-1] \leftarrow x[K-1]$, $x[-2] \leftarrow x[K-2]$, and so on.

Cyclic Property: The insertion of the CP makes the signal "circular" with respect to the block length K . Mathematically, thanks to the periodicity of the complex exponential, the following relationship holds:

$$x[-1] = x[K-1] \quad (7.4)$$

In fact, $e^{j\frac{2\pi}{K}(-1)k} = e^{j\frac{2\pi}{K}(K-1)k}$.

7.2 The Channel

The transmitted signal passes through a channel modeled as a Finite Impulse Response (FIR) filter with coefficients $h[l]$ of length $L+1$ ($l = 0 \dots L$). The channel output $y[n]$ is given by the linear convolution between the transmitted signal (including the CP) and the channel response:

$$y[n] = \sqrt{G} \sum_{l=0}^L h[l] \bar{x}[n-l] \quad (7.5)$$

Inter-Block Interference (IBI): For indices corresponding to the cyclic prefix (e.g., $n = -1, \dots, -L$), the convolution "picks up" symbols from the previous transmitted block. For example, for $y[-1]$, the summation involves $\bar{x}[-1-L]$, which belongs to the previous block.

🔍 Student Q&A: CP Constraints

Is there a constraint on the length of the Cyclic Prefix?

✓ Answer: CP Constraints

Yes. The length of the Cyclic Prefix (L) must be greater than or equal to the length of the channel impulse response. This ensures that all Inter-Block Interference (IBI) falls within the cyclic prefix, which is then discarded at the receiver, leaving the data block "clean".

7.3 The OFDM Receiver

The receiver must recover the original symbols $x[k]$. The operations are specular to those of the transmitter.

7.3.1 Cyclic Prefix Removal

The first L received samples ($y[-L], \dots, y[-1]$) are contaminated by interference from the previous block. The solution is simple: discard the cyclic prefix. We consider only the vector y of length K :

$$y = [y[0], y[1], \dots, y[K-1]] \quad (7.6)$$

7.3.2 DFT (Discrete Fourier Transform)

On the vector cleaned of the cyclic prefix, we apply the DFT:

$$\tilde{y}[n] = \sum_{p=0}^{K-1} y[p] e^{-j \frac{2\pi}{K} np} \quad (7.7)$$

Here n represents the subcarrier index at reception.

7.4 Mathematical Proof [The "Heart" of OFDM]

Let us see why OFDM works by substituting the channel expression into the DFT formula.

7.4.1 Step 1: Expansion of $\tilde{y}[n]$

Substitute $y[p]$ with the channel convolution:

$$\tilde{y}[n] = \sum_{p=0}^{K-1} \left(\sqrt{G} \sum_{l=0}^L h[l] \bar{x}[p-l] \right) e^{-j \frac{2\pi}{K} np} \quad (7.8)$$

7.4.2 Step 2: Substitution of x (IDFT)

Substitute $\bar{x}[p-l]$ with its IDFT definition (the original input $x[k]$):

$$\tilde{y}[n] = \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^L h[l] \left(\frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j \frac{2\pi}{K} (p-l)k} \right) e^{-j \frac{2\pi}{K} np} \quad (7.9)$$

Note: Thanks to the cyclic prefix, the time shift $p-l$ in the linear convolution acts as a circular shift, allowing the use of the IDFT complex exponentials.

7.4.3 Step 3: Rearrangement of Summations

We bring out the constants and group the terms:

$$\tilde{y}[n] = \frac{\sqrt{G}}{K} \sum_{k=0}^{K-1} x[k] \left(\sum_{l=0}^L h[l] e^{-j\frac{2\pi}{K}lk} \right) \left(\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)} \right) \quad (7.10)$$

Let us analyze the two terms in parentheses:

1. **Channel DFT ($\tilde{h}[k]$):** The term $\sum_{l=0}^L h[l] e^{-j\frac{2\pi}{K}lk}$ is exactly the frequency response of the channel at frequency k , which we call $\tilde{h}[k]$. We assume $h[l] = 0$ for $l > L$.
2. **Orthogonality:** The term $\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)}$ sums complex exponentials. This sum is non-zero only if $k = n$.

$$\sum_{p=0}^{K-1} e^{j\frac{2\pi}{K}p(k-n)} = K \cdot \delta(k - n) \quad (7.11)$$

where $\delta(x) = 1$ if $x = 0$, otherwise 0.

7.4.4 Final Result

Thanks to the Dirac delta $\delta(k - n)$, the summation over k "survives" only for the term where $k = n$. The factor K in the numerator cancels with $1/K$. We obtain:

$$\tilde{y}[n] = \sqrt{G} \cdot \tilde{h}[n] \cdot x[n] \quad (7.12)$$

🔍 Student Q&A: Notation Confusion

Confusion regarding n vs k notation in the final proof.

✓ Answer: Notation Confusion

Some ambiguity was noted in the use of indices. In the final formula, n is used as the output subcarrier index ($\tilde{y}(n)$). The formula proves that the output at index n depends **only** on the input at the same index n ($x[n]$), scaled by the channel $\tilde{h}[n]$. This confirms the absence of Inter-Carrier Interference (ICI).

7.4.5 Interpretation and Equalization

The result $\tilde{y}[n] = \sqrt{G} \tilde{h}[n] x[n]$ is fundamental. It means that, in the frequency domain, there is no interference between symbols or subcarriers. The symbol transmitted on frequency n ($x[n]$) is merely scaled by a complex coefficient $\tilde{h}[n]$ (the channel response at that frequency).

One-Tap Equalization: To recover the transmitted symbol $\hat{x}[n]$, a simple division (Zero-Forcing) is sufficient, assuming the channel is known:

$$\hat{x}[n] = \frac{\tilde{y}[n]}{\sqrt{G} \cdot \tilde{h}[n]} \quad (7.13)$$

Chapter 8

Lesson 2-12-2025

8.1 Mathematical representation of OFDM, noise of a sub-carrier; terminology used by the book; pulse shaping for OFDM

Given that the symbols we're generating and want to transmit are $x[k]$, we have that after IDFT and cyclic prefix, the OFDM waves will be

$$\bar{x}[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j\frac{2\pi}{K}nk} \quad n = -L, \dots, K-1 \quad (2.211)$$

■ Transmitted samples after the IDFT and cyclic prefix addition

$$\bar{x}[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j\frac{2\pi}{K}nk} \quad n = -L, \dots, K-1$$

■ Received signal after discarding the cyclic prefix (first L samples)

$$y[n] = \sqrt{G} \sum_{\ell=0}^L h[\ell] \bar{x}[n - \ell]$$

$$= \sqrt{G} \sum_{\ell=0}^L h[\ell] x[(n - \ell)_K] \quad n = 0, \dots, K-1$$

(now including noise)

$$\Rightarrow y[k] = \sqrt{G} h[k] x[k] + v[k]$$

Simple equalization
✓ Noise remains IID and $\sim \mathcal{N}_c(0, N_0)$

Looks like a circular convolution
between length K signals
(channel zero padded)

Per-subcarrier noise

$$\frac{N_0}{G} \frac{1}{|h[k]|^2}$$

Professor never used the *circular convolution*. We sum the noise which is IID and Gaussian.

Note:

Vedi: ex 2.22, 2.23 (esercizi esame molto simili a quelli del libro + vedi moodle) We can see that each **subcarrier** has a noise variance of

$$\frac{N_0}{G} \frac{1}{|h[k]|^2}. \quad (2.221)$$

This follows from the type of equalization we're performing (specifically by using (2.219) to find the noise covariance by considering $\hat{x}[k]\hat{x}^*[k]$ for symbols $x_k \equiv x[k] = 0$, so to only consider the noise),

My explanation

At a certain point we get that (with reference to A signal processing perspective (pp. 57-130), p. 50):

$$y[n] = \sum_{k=0}^{K-1} x_k h[k] e^{j\frac{2\pi}{K}kn},$$

which is the IDFT of $x_k h[k]$! We know that a zero forcing equalizer W is such that in the frequency domain, we'll have

$$W^*(z) \cdot H(z) = z^{-\Delta}.$$

What we see here is that we have a **dummy frequency domain** which sees the multiplication of x_k and $h[k]$: this will allow us to say that if we choose

$$x_k = \frac{1}{Kh[k]} \text{DFT}[y[n] | k],$$

the relation will be satisfied with $\Delta = 0$: in this sense, what we previously found can be considered a ZF equalizer.

If we then were to also add the channel gain $\sqrt{G}$ into the mix and we defined $\hat{x}[k] \equiv x_k$ and $y[k] = \text{DFT}[y | k]$, we have that the previous equation can be written just like in the book,

$$\hat{x}[k] = \frac{y[k]}{\sqrt{G}h[k]} \quad k = 0, \dots, K-1. \quad (2.219)$$

If the channel is small for a certain k , the noise is increasing in the subcarriers. If the frequency response is small, we're paying this in increasing noise.

This is somehow tied to ~d45a40.

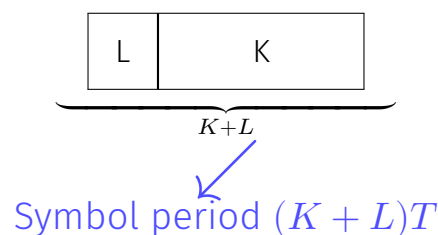
Terminology:

- Input to the IDFT are symbols
- Output of the IDFT are OFDM chips
- Interval between chips is T (would be the symbol period)
- Number of subcarriers is K , the dimension of the IDFT / DFT
- Collection of chips $\bar{x}[-L], \dots, \bar{x}[K-1]$ forms an OFDM symbol
- OFDM symbol period is $(K+L)T$
- Subcarrier spacing is the distance between adjacent subcarriers $\frac{1}{KT}$
- Effective pulse shape is usually a rectangular
- Edge subcarriers may be zero, leading to a spectral efficiency penalty

About chips: professor does not agree to this; **chip** has to do with slicing.

We have that $x[k]$ are symbols, $z[n]$ are chips.

The **interval between chips** is T . The **number of subcarriers** is K ; this is also the dimension of the IDFT/DFT. The collection of chips $\bar{x}[-L], \dots, \bar{x}[K-1]$ forms an **OFDM symbol**: an OFDM symbol is $K + L$.



It's tricky to have to do with this: we do S/P, then DFT, then we add the cyclic prefix. We're adding some symbols: we'll need to go faster in order to keep up with the previous frequency.

One way of choosing the pulse shape is to have a rectangular pulse of duration KT , giving that subcarriers will bring a signal of $KT\text{sinc}(fKT)$. In discrete frequency, the subcarriers are spaced of $1/K$, which, with a sampling period of T , gives that the K subcarriers will **span a total bandwidth of $1/T$** .

The actual (non-ideal) signal will surely have a bandwidth larger than $1/T$, forcing us to set some of the outer carriers to 0. This can be done either by **windowing**, aka by putting specific subcarriers to θ or by just applying a filter before transmission.

One thing which is typically done to improve the overall situation is to use raised cosines or similar signals.

$K \uparrow$ implies that the subcarriers are narrower and decay faster in frequencies. This term directly influences the containment of the signal in $1/T$. Also, the higher this value, the more the subcarriers are susceptible to little changes. (e.g.: if a subcarrier has a small phase shift, this causes more trouble if the symbols are tightly close). As it's clear, K also influences the resolution of the channel's frequency selectivity.

Comparisons

The book makes some comparisons between OFDM and single-carrier techniques. I just skipped over that. If interested, see **A signal processing perspective (pp. 57-130), p. 53.**

One last thing is that whenever the book (or this document) mentions **OFDM resource elements**, we're referring to a specific OFDM symbol in time and a specific subcarrier in frequency, according to the figure:

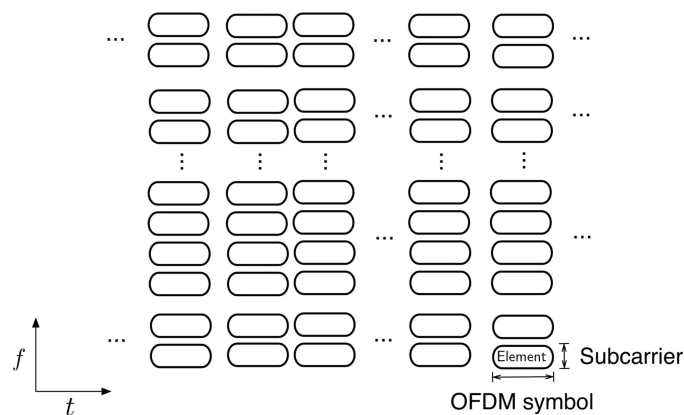


Figure 8.1: OFDM resource elements representation

8.2 Subcarrier spacing (+about the rectangular pulse shape of the signal), spectral efficiency penalty

Subcarrier spacing between adjacent carriers is $1/KT$; this comes from the writing

$$\exp\left(j\frac{2\pi}{K}kn\right) = \exp\left(j2\pi\frac{k}{KT}nT\right).$$

in which k/KT is the frequency term and nT is the time term. The spacing in frequency between each subcarrier will thus be $1/KT$.

The pulse shape is usually rectangular. We'll eventually need to go back to the continuous time domain; if we go back, we'll see that we have $x[n]$ (the transmission will happen in continuous time, of course):

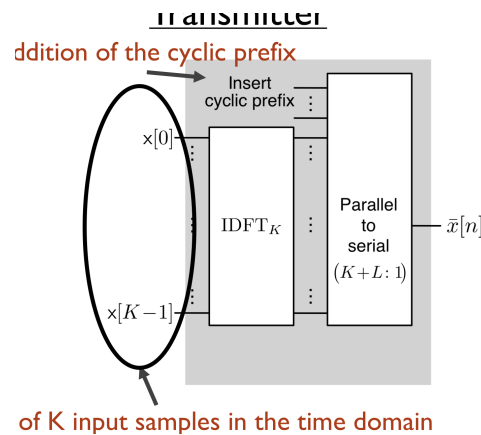


Figure 8.2: Transmitter: addition of the cyclic prefix

We'll actually need to use a rect:

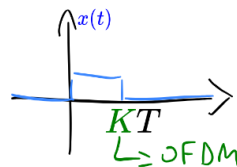


Figure 8.3: Rectangular pulse of duration KT

This is pretty brutal, but we'll have a sinc in the frequency domain; all the multiples of $1/KT$ will be 0:

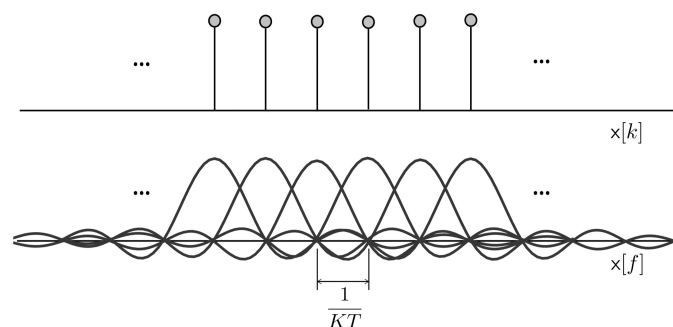


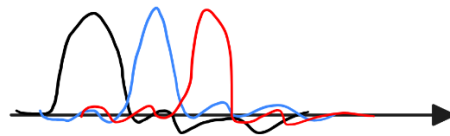
Figure 8.4: Sinc functions in frequency domain spaced by $1/KT$

This will allow for a perfect representation of our signal!

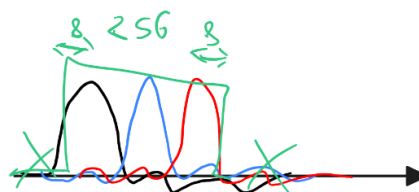
Between OFDM symbols we have KT seconds in time; if we consider them, the rect is KT and the sinc in frequency will go to zero on the multiples of $1/KT$: this means that the carriers have no influence over each other whatsoever.

Edge subcarriers might be 0 (we'll see why shortly), leading to a **spectral efficiency penalty**.

Let's say we have a certain number of subcarriers:



We know that if our filter is a rect, it might interfere, cutting away a portion of the signals in the frequency domain:



We'd like to filter beforehand, but we can't generally do this; what we thus do is to **switch off the edge carriers** (in the example up here we switch off the first and last 8 edge carriers). This means that the filter won't cut that sinc and it won't interfere with neighboring subcarriers.

In the following paragraph a more formal definition of penalty factor is given:

Penalty factor

A completely separate but equally important definition is that of **penalty factor**: it is given by how much the usable bandwidth is in relation to the used bandwidth; the point is that in reality we'll always have some excess bandwidth which will go over our constraints: just to give an example, in **example 2.23** the subcarrier spacing is 15KHz but it turns out that the actual subcarrier bandwidth is $\frac{500\text{MHz}}{300} = 16.66\text{KHz}$, more than what would be available. The penalty factor is thus calculated as

$$\text{Penalty factor} = \frac{\text{Usable bandwidth}}{\text{Used bandwidth}};$$

this value is useful as whenever we have a channel of known spectral efficiency R/B but we also know that we're exceeding the channel's bandwidth, we can correct the spectral efficiency as

$$(\text{Penalty factor}) \cdot \frac{R}{B}.$$

Just as an extra piece of information, another way to see this is

$$\text{Penalty factor} = \frac{\text{Theoretically used bandwidth}}{\text{Actually used bandwidth}}.$$

8.3 Examples of small exercises; many definitions of small values related to OFDM

In the example we turn off the first and last 8 subcarriers for a system for which we have $K = 256$: we'll have an **efficiency** which is limited:

$$\frac{240}{256} < 1.$$

We're thus having a total efficiency of

$$\frac{256}{L + 256} \cdot \frac{240}{256} < 1.$$

We have some examples which **can also be conceived as some exam exercises** (as can be seen, they were indeed asked in the multiple-choices part of the exam).

Example 2.22

Let $T_{\text{active}} = 3.2 \mu\text{s}$ with $K = 256$ subcarriers and a cyclic prefix of length $L = 64$. Find the chip period T , the passband bandwidth $1/T$, the subcarrier spacing, and the guard interval spanned by the cyclic prefix.

Solution

The chip period T satisfies $(256)T = 3.2 \mu\text{s}$ and thus $T = 12.5 \text{ ns}$. Then, $1/T = 80 \text{ MHz}$ and the subcarrier spacing is $\frac{1}{KT} = 312.5 \text{ kHz}$. The guard interval is $LT = 0.8 \mu\text{s}$ and therefore channels with impulse responses of this duration can be handled.

[Alternative reading based on visual shape of blurry numbers]:

The chip period T satisfies $(256 + 64)T = 3.2 \mu\text{s}$ and thus $T = 10 \text{ ns}$. Then, $1/T = 100 \text{ MHz}$ and the subcarrier spacing is $\frac{1}{KT} = 390.6 \text{ kHz}$. The guard interval is $LT = 0.64 \mu\text{s}$ and therefore channels with impulse responses of this duration can be handled.

Example 2.23

In LTE, the subcarrier spacing is 15 kHz and every 5 MHz of bandwidth accommodates $K = 300$ subcarriers. What penalty factor should be applied to the spectral efficiency computed with $B = 1/T$?

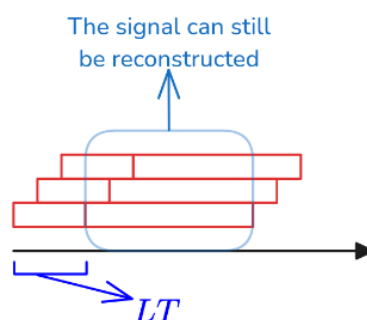
Solution

Since $300 \cdot 15 \cdot 10^3 \text{ Hz}$ out of $5 \cdot 10^6 \text{ Hz}$ are used, the penalty factor equals $\frac{300 \cdot 15 \cdot 10^3}{5 \cdot 10^6} = 0.9$. For NR, this tightens up to 0.96 .

When the solution of the first says **of this duration**, the actual correct thing is **of this duration or less**! In the second problem, the actual amount of carriers would be 333 , but we use less so that we can avoid collision between external carriers.

💀Guard interval

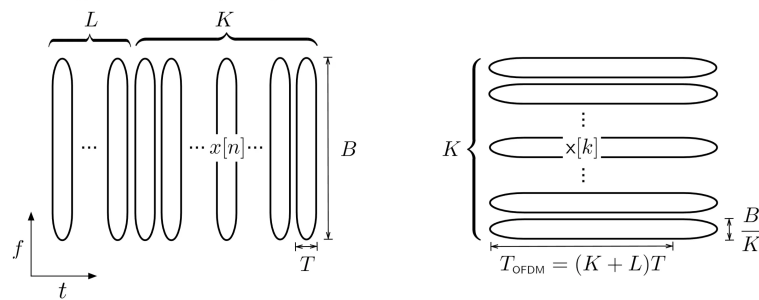
The guard interval is the amount of time occupied by the cyclic prefix; it's called like this because it's an upper bound to how much our signal can be shifted while still being recoverable:



The signal can still be reconstructed (blue arrows indicating LT shift)

 Several definitions: $T, T_{OFDM}, \dots$

Although we never did so in class, it seems necessary to introduce some values, with reference to the following figure (fig. 2.16):



- T is the **chip period**.
- LT is the **guard interval**, as described above.
- T_{OFDM} is the **period of a whole OFDM signal**, given by

$$T_{OFDM} = (K + L)T.$$

- $B = 1/T$ is the **bandwidth** occupied by the **whole amount of subcarriers** collectively.
- $B/K = 1/KT$ is the **bandwidth occupied by a single subcarrier**; it also is the **subcarrier spacing**.

Penalty factor

A completely separate but equally important definition is that of **penalty factor**: it is given by how much the usable bandwidth is in relation to the used bandwidth; the point is that in reality we'll always have some excess bandwidth which will go over our constraints: just to give an example, in **example 2.23** the subcarrier spacing is 15kHz but it turns out that the actual subcarrier bandwidth is $\frac{500\text{MHz}}{300} = 16.66\text{kHz}$, more than what would be available. The penalty factor is thus calculated as

$$\text{Penalty factor} = \frac{\text{Usable bandwidth}}{\text{Used bandwidth}};$$

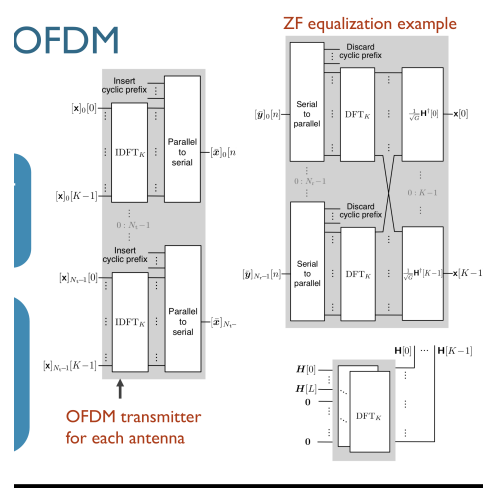
this value is useful as whenever we have a channel of known spectral efficiency R/B but we also know that we're exceeding the channel's bandwidth, we can correct the spectral efficiency as

$$(\text{Penalty factor}) \cdot \frac{R}{B}.$$

Just as an extra piece of information, another way to see this is

$$\text{Penalty factor} = \frac{\text{Theoretically used bandwidth}}{\text{Actually used bandwidth}}.$$

The same structure is found when extending this to MIMO-OFDM:



All the IDFTs at the inputs of the different antennas are of the same sizes; we need the responses to be of the same sizes; L must be the greatest of all channels and K must be calculated knowing the penalty.

If we want to use as few resources as possible, we should customize L, K for every

antenna; we won't actually do this.

Chapter 9

Lesson 3-12-2025

9.1 Channel estimation

3 Dec 2025

General idea (pilots) and formal writing of $\bar{Y}$, $\text{vec}(\bar{Y})$

Ref: *A signal processing perspective* (pp. 57-130), p. 54

We've seen OFDM and single carrier equalization. Today professor wants to finish channel estimation. If we want to transmit with OFDM, we need to estimate the channel.

$$y[k] = \sqrt{G}h[k]p[k] + v[k]$$

↑ *pilot component*

$\xrightarrow{\text{IDFT}}$ of length K

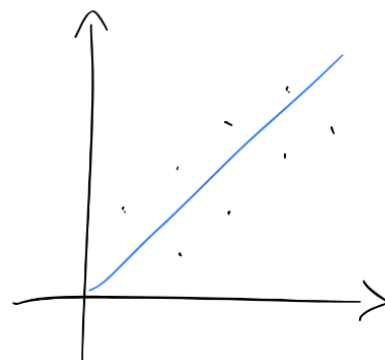
How can we do channel estimation? There are two approaches:

- Maximum likelihood
- Least squares $\rightarrow$ *like MD (Minimum Distance)*

In the book the difference between least square and MMSE is not made clear.

If we have something like this and we're asked to draw a line which minimizes the error we're doing, that is Least Squares (LS).

How do we do channel estimation? We use what we'll call a **pilot**: this is a symbol at a certain frequency and time that we know.



LS minimization visualization

Since we know the symbol we've sent and we know its power,

we can learn the response of the channel accordingly.

Formal writing:

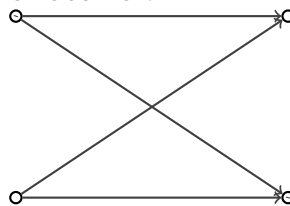
We can model the received signal as

$$y[n] = \sqrt{G} \sum_{\ell=0}^L H[\ell] x[n - \ell] + v[n].$$

$$y[k] = \sqrt{G} \underset{\substack{\uparrow \\ \text{useful}}}{h[k]} p[k] + \underset{\substack{\uparrow \\ \text{noise}}}{v[k]}$$

How the H matrices represent the delay

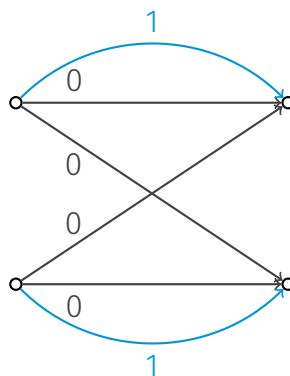
Let's suppose we have 2 transmit and 2 receive antennas and all the signals arrive without any delay at the receiver:



We'll use a matrix such as

$$H = \begin{bmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{bmatrix}$$

to characterize the channel. Even in this case, there could be other paths, with different delays!



Here the channel won't be flat anymore (the arrows in blue arrive at time 1): the one before will be

$$H(0) = \begin{bmatrix} h_{00}(0) & h_{01}(0) \\ h_{10}(0) & h_{11}(0) \end{bmatrix}$$

and we'd also have

$$H(1) = \begin{bmatrix} h_{00}(1) & 0 \\ 0 & h_{11}(1) \end{bmatrix};$$

we have some form of reflection which makes the signal arrive later at the receiver.

Of course if we were to have reflections between 0,1 and 1,0, we'd have

$$H(1) = \begin{bmatrix} h_{00}(1) & h_{01}(1) \\ h_{10}(1) & h_{11}(1) \end{bmatrix}.$$

It's similar to what we had in the SISO case:

$$y[n] = \sum_{\ell=0}^L h[\ell]x[n-\ell]; \quad (9.1)$$

this is why we wrote the channel as

$$y[n] = \sqrt{G} \sum_{\ell=0}^L H[\ell]x[n-\ell] + v[n]. \quad (9.2)$$

Previous symbols interfere with current symbols!

This is the IO relationship. If we want to estimate the channel, what do we want to do? We don't transmit an unknown symbol x , but rather some **known symbols**; this is indicated with p , which stands for **pilots**.

Before diving further, let's address the values which they'll use. We are indeed sending N_p pilots, but since the channel has L taps, all received symbols up until the $L - 1$ th will have **contributions which we don't want**. In order to remove such contributions **we only consider the received signals from L to $N_p - 1$** .

Specifically, we have the synthesized form

$$\begin{aligned} [y[L] \quad \dots \quad y[N_p - 1]] &= \\ &= \sqrt{G} [H[0] \quad H[1] \quad \dots \quad H[L]] \begin{bmatrix} p[L] & p[L+1] & \dots & p[N_p - 1] \\ p[L-1] & p[L] & \dots & p[N_p - 2] \\ \vdots & \ddots & \ddots & \vdots \\ p[0] & p[1] & \dots & p[N_p - L - 1] \end{bmatrix} + \bar{V} \\ &\Leftrightarrow \bar{Y} = \sqrt{G} \bar{H} \bar{P} + \bar{V}. \end{aligned} \quad (9.3)$$

Where:

- since each pilot is a $N_t \times 1$ vector, the $\bar{P}$ matrix will be a $N_t(L+1) \times (N_p - L)$ matrix.
- Since each $H[\ell]$ matrix is a $N_r \times N_t$ matrix, the $\bar{H}$ matrix will be a $N_r \times N_t(L+1)$ matrix.
- Both $\bar{Y}$, $\bar{V}$ will be $N_r \times (N_p - L)$ as a result of the product $\bar{H}\bar{P}$.

N_p is the number of pilots we transmit and it's up to us to decide its value. If we transmit too many pilots, we can't transmit the information symbols. We need to

transmit some pilots (generally a vector of all pilots) and then some pilots scattered around the time and frequency domains.

The book, in order to make the notation more compact, uses the **Kronecker product** of two matrices.

| Of course in order to actually understand this, we need to explode this notation.

We also have the **vec operator**. We have a specific property we'll use:

Relation with the Kronecker product

We have that

$$\text{vec}(ABC) = (C^T \otimes A)\text{vec}(B), \quad (9.4)$$

$$\text{vec} : \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a \\ c \\ b \\ d \end{bmatrix}$$

where $\otimes$ represents the **Kronecker product**.

Even for SISO, we used to make a compact notation by using matrices which imply time dependance.

| This is similar to how with OFDM we use a matrix for representing the transform

| If we look at exercise 20, we'll end up with 2 matrices representing the channel

After going to our matrix notation, we further progress to a vector notation, knowing that if we were to stack one on the other all the values of $\bar{Y}$, we'd get

$$\begin{aligned} \text{vec}(\bar{Y}) &= \sqrt{G}\text{vec}(\bar{H}\bar{P}) + \text{vec}(\bar{V}) \\ &= \sqrt{G}(\bar{P}^T \otimes I_{N_r})\text{vec}(\bar{H}) + \text{vec}(\bar{V}) \\ &= \sqrt{G}P_{\otimes}\text{vec}(H) + \text{vec}(\bar{V}). \end{aligned} \quad (2.229)$$

We can apply the forementioned property; in applying it, we use $A = I_{N_r}$, we can develop like it's written above here. Question is: what's $P_{\otimes}$? It's just $\bar{P}^T \otimes I_{N_t}$.

9.2 Linear estimation for a MIMO AWGN channel

Now we estimate, either with ML or LS.

In this estimation, **we assume that we know the conditional distribution of the received signal given the channel**. In other words, we know that our transmit receive relation will follow (2.229) and that our noise $\bar{V}$ will be AWGN. Thanks to the fact that we can choose our pilots to be **IID**, we know that the **ML estimator will be equivalent to the MAP estimator**.

Considering (2.229), we have that $P_{\otimes}$ is known and that the observation given by a certain channel realization $\mathbf{h}$ (which realizes the distribution of $\text{vec}(\bar{H})$) is complex Gaussian with mean

$$\mathbb{E}[\text{vec}(\bar{Y})] = \mathbb{E}[\sqrt{G}\bar{P}_{\otimes} \underset{\text{realization}}{\mathbf{h}} + \text{vec}(\bar{V})] = \sqrt{G}\bar{P}_{\otimes}\mathbf{h} + 0 \quad (9.5)$$

and covariance

$$\begin{aligned} \mathbb{E}[(\text{vec}(\bar{Y}) - \mu_{\text{vec}(\bar{Y})})(\text{vec}(\bar{Y}) - \mu_{\text{vec}(\bar{Y})})^*] &= \mathbb{E}[\text{vec}(\bar{V})\text{vec}^*(\bar{V})] \\ &= N_0 I. \end{aligned} \quad (9.6)$$

We'll thus have that, given a certain realization of the channel, $\text{vec}(\bar{Y})$ is Gaussian with PDF (according to [Uni Generale/Digital Communications/Complex Gaussian distribution > 57a95e](#))

$$f_{\text{vec}(\bar{Y})|\text{vec}(\bar{H})}(y|\mathbf{h}) = \frac{1}{(\pi N_0)^{(N_p-L)N_r}} \exp\left(-\frac{\|y - \sqrt{G}\bar{P}_{\otimes}\mathbf{h}\|^2}{N_0}\right), \quad (2.230)$$

which, thanks to the monotonicity of the exponential, is equivalent to minimizing $-\log f_{\text{vec}(\bar{Y})|\text{vec}(\bar{H})}(\cdot)$, ultimately giving

$$\text{vec}(\hat{\bar{H}})(\mathbf{y}) = \arg \min_{\mathbf{h}} \|\mathbf{y} - \sqrt{G}\bar{P}_{\otimes}\mathbf{h}\|^2. \quad (2.232)$$

This is identical to the deduction of the zero forcing equalizer [we already discussed](#) (equations (2.145), (2.146)), ultimately giving

$$\text{vec}(\hat{\bar{H}})(\mathbf{y}) = \underbrace{\frac{1}{\sqrt{G}}(\bar{P}_{\otimes}^* \bar{P}_{\otimes})^{-1} \bar{P}_{\otimes}^*}_{W^{\text{LS}*}} \mathbf{y}, \quad (2.233)$$

and thus the linear **least squares estimator**

$$W^{\text{LS}} = \frac{1}{\sqrt{G}} \bar{P}_{\otimes} (\bar{P}_{\otimes}^* \bar{P}_{\otimes})^{-1} \quad (2.233+)$$

to the observation (we have $^{-1}$ and not † since we know that $\bar{P}_{\otimes}^* \bar{P}_{\otimes}$ is a square matrix).

🧑 How does this tie together with the ZF filters?

For ZF, we were trying to minimize

$$\|\sqrt{G}\bar{w}_j^* \bar{T} - \bar{d}_j^*\|^2. \quad (2.138\text{-ish})$$

and we got to

$$\bar{w}_j^{\text{ZF}} = \frac{1}{\sqrt{G}}(\bar{T}^*)^\dagger \bar{d}_j \quad \text{time domain} \quad (2.145)$$

$$\bar{W}^{\text{ZF}} = \frac{1}{\sqrt{G}}(\bar{T}^*)^\dagger \bar{D}. \quad \text{z domain} \quad (2.146)$$

we know that the Moore-Penrose pseudoinverse for a tall matrix A is

$$A^\dagger = (A^* A)^{-1} A^*$$

if we then consider the complex conjugate of the argument of (2.232), we get

$$\|y^* - \sqrt{G}\mathbf{h}^* \bar{P}_\otimes^*\|,$$

which fits neatly in (2.138-ish), to give (from 2.145)

$$\mathbf{h} = \frac{1}{\sqrt{G}}(\bar{P}_\otimes)^\dagger y,$$

which finally leads to (2.233).

| Professor would like of us to remember the reasoning and that we introduced $\otimes$ and vec only for the reasons of having vectors, since with vectors we know the joint probability distribution of a Gaussian vector.

🧑 Sorry... which type of estimator are we talking about?

We have that this estimator is both the ML, MAP and LS estimator!

| Note though that this is NOT the LMMSE estimator!!

Due to the discussion we made about zero forcing equalizers, **we need to ensure that $\bar{P}_\otimes$ is square or tall**, requiring to check that

$$N_p \geq (L + 1)N_t + L \quad (2.234)$$

and also that $\bar{P}_\otimes$ is full rank (which requires $\bar{P}$ to be full rank).

📖 About the monotonicity

While the book explains this passage using the monotonicity of the exponential, this is nothing else than putting into play the equivalence between ML and MD criteria when we have AWGN noise.

Through the derivation which can be found from A signal processing perspective (pp. 57-130), page 57 to A signal processing perspective (pp. 57-130), page 58, we find that

through the linear estimator we found we'll have

$$E^{\text{LS}} = \frac{N_0}{G} (\bar{P}_{\otimes}^* \bar{P}_{\otimes})^{-1} \quad (2.239)$$

$$\hat{H}^{\text{LS}} = \frac{1}{\sqrt{G}} \bar{Y} \bar{P}^* (\bar{P} \bar{P}^*)^{-1}, \quad (2.247)$$

where (2.239) expresses the Mean-Square Error Matrix while (2.247) expresses the best estimation of the channel. The advantage of this equation with respect to (2.233) is that it entails less operations and also that since $\bar{P}$ is known, we can consider $\bar{P}^* (\bar{P} \bar{P}^*)^{-1}$ as already known.

Hmmm

I'm sure there are more complicated and accurate reasons behind the procedure which gets us to (2.247), but for the time being I'd just observe that it seems as if we just switch $\bar{P}_{\otimes}$ and y and then replace $\bar{P}_{\otimes} \rightarrow \bar{P}^*$

A signal processing perspective (pp. 57-130), page 58

Also notice that the i th row of $\bar{H}$ corresponds to the channel coefficients between the N_t transmit and the i th receive antenna, while the i th row of $\bar{Y}$ corresponds to the observation at the i th receive antenna

Due to how we defined $\bar{H}$, it doesn't look like this is true...we have that the i th row will contain coefficients such as $h_{00}[0], h_{01}[0], h_{00}[1], h_{01}[1]$, aka the coefficients of transmission from antenna 0 to all other antennas.

Skip

From page A signal processing perspective (pp. 57-130), page 58 to page A signal processing perspective (pp. 57-130), page 59 we find a brief discussion of sensible values for the pilots.

We can find the discussion of a LMMSE estimator at A signal processing perspective (pp. 57-130), page 61.

ML is one of the criteria! I know I received the joint probability distribution of all elements in the vector. What we want to do is to find the minimum of the probability density distribution while **playing** with our values. This is a pretty well known problem and has the solution we see in the slides. If the noise is Gaussian, at the end of the day we forget everything and minimize the distance we see. MAP requires to know the distribution of the input; we know that **if the input are equiprobable and independent, we'll have $\text{MAP} \equiv \text{ML}$. If the noise is Gaussian, we have $\text{ML} \equiv \text{MD}$** . This is why we're using the MD criterium. If we're doing minimum distance we can use LS and the solution is the one in the slides.

It's invertible if it's tall and we chose the pilots in an appropriate manner. There are indeed recipes to select the pilots; we should remember the line of reasoning; it **should not be taken for granted that stuff is invertible**. If we have to estimate at least four numbers and we're in the OFDM setting and we have $K = 256$, we need to make 256 estimates; the minimum is that we only need 8 numbers of H [numbers].

The numbers in time domain are much less! We don't need to have 256 pilots, but we can't have less than 8 pilots. With less than 8 known variables we cannot estimate the 8 numbers in the matrices.

How do we distribute the pilots? We're going to distribute them evenly, otherwise we won't be able to get all the numbers.

 In class

Derivation to simplify the LS estimator:

- Exploiting the fact that

$$\begin{aligned} (\bar{P}^* \bar{P})^{-1} \bar{P}^* &= ((\bar{P}^* \otimes I)^* (\bar{P}^T \otimes I))^{-1} (\bar{P}^* \otimes I)^* \\ &= ((P \otimes I) (\bar{P}^T \otimes I))^{-1} (\bar{P} \otimes I) \\ &= ((P \bar{P}^T) \otimes I)^{-1} (\bar{P} \otimes I) \\ &= (P \bar{P}^T)^{-1} \bar{P} \otimes I, \end{aligned}$$

- Gives

$$\text{vec}(\hat{H}) = \frac{1}{\sqrt{G}} ((\bar{P} \bar{P}^*)^{-1} \bar{P}^* \otimes I) \text{vec}(\bar{Y})$$

- Exploiting $\text{vec}(ABC) = (C^T \otimes A) \text{vec}(B)$ in reverse

$$\begin{aligned} \hat{H} &= \frac{1}{\sqrt{G}} \bar{Y} ((\bar{P}^* \bar{P}^T)^{-1} \bar{P}^*)^T \\ &= \frac{1}{\sqrt{G}} \bar{Y} \bar{P}^* (\bar{P} \bar{P}^*)^{-1}, \end{aligned}$$

We're not required to remember all of this, but it's recommended to go and check that.

Suppose we made all of our calculations and we have $\hat{H}$ and we're using OFDM; we know OFDM works with MIMO. If we know that the prefix is L , it means that we won't have more than L numbers in the input's response. A brutal technique is: suppose we are in the SISO setting and we have the $\hat{h}$ with 256 carriers. If $L = 16$, what should I expect? We should expect that the vector $\hat{H}$ should be nonzero from the element 0 to element 15.

From this we do IDFT and we have 256 elements:

$$h = \begin{bmatrix} \text{stuff}_0 \\ \vdots \\ \text{stuff}_{15} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \updownarrow 256$$

If we don't have zero, it means we have noise; we then just replace everything with zeros and we do the DFT: we have H^* , a better version of $\hat{H}$.

Assuming we haven't made a wrong decision on L (we thus must have $L < K$), we can cut everything after L , which will be noise. This is what we mean by saying **we can work in the time domain**.

$$\hat{H} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \xrightarrow{\text{IDFT}} \hat{h} = \begin{bmatrix} \downarrow L \\ \sim \\ \text{noise} \end{bmatrix} \xrightarrow{\text{cut}} \begin{bmatrix} \downarrow L \\ 0 \\ \vdots \\ 0 \end{bmatrix} \xrightarrow{\text{DFT}} H^*$$

The figure is showing that we can go in the time domain, cut everything after L (since it's noise) and then go back to the frequency domain, having a better signal.

9.3 Zero forcing equalizer for MIMO OFDM

We'd like to somehow adapt

to the observation (we have $^{-1}$ and not † since we know that $\bar{P}_{\otimes}^* \bar{P}_{\otimes}$ is a square matrix).

so to estimate an OFDM transmission.

We can indeed extend what we saw to OFDM, considering that our objective is to distribute the various pilots in an even manner across the K subcarriers (A signal processing perspective (pp. 57-130), page 65), also considering that now our channel is defined as

$$\mathbf{y}[k] = \sqrt{G} \mathbf{H}[k] \mathbf{p}[k] + \mathbf{v}[k] \quad (2.288)$$

$$= \sqrt{G} \left(\sum_{\ell=0}^L H[\ell] e^{-j \frac{2\pi}{K} \ell k} \right) \mathbf{p}[k] + \mathbf{v}[k] \quad k = k_0, \dots, k_{N_p-1}, \quad (2.289)$$

which, in terms of $\bar{H} = [H[0] \dots H[L]]$, gives

$$\mathbf{y}[k] = \sqrt{G} \bar{H} \begin{bmatrix} I_{N_t} \\ e^{-j \frac{2\pi}{K} k} I_{N_t} \\ \vdots \\ e^{-j \frac{2\pi}{K} k L} I_{N_t} \end{bmatrix} \mathbf{p}[k] + \mathbf{v}[k] \quad (2.290)$$

$$= \sqrt{G} \bar{H} \begin{bmatrix} \mathbf{p}[k] \\ e^{-j \frac{2\pi}{K} k} \mathbf{p}[k] \\ \vdots \\ e^{-j \frac{2\pi}{K} k L} \mathbf{p}[k] \end{bmatrix} + \mathbf{v}[k]. \quad (2.291)$$

By defining

$$\mathbf{u}[k] = \begin{bmatrix} 1 & e^{-j \frac{2\pi}{K} k} & e^{-j \frac{2\pi}{K} k 2} & \dots & e^{-j \frac{2\pi}{K} k L} \end{bmatrix}^T \quad (2.292)$$

9.4 A signal processing perspective

and through the derivation at A signal processing perspective (pp. 57-130), page 65 we see that

$$\mathbf{y}[k] = \sqrt{G} (\mathbf{u}[k] \otimes \mathbf{p}[k]) + \mathbf{v}[k] \quad (2.293)$$

and we eventually get to

$$\underbrace{\begin{bmatrix} \mathbf{y}[k_0] \\ \mathbf{y}[k_1] \\ \vdots \\ \mathbf{y}[k_{N_p-1}] \end{bmatrix}}_{\bar{\mathbf{y}}} = \sqrt{G} \underbrace{\begin{bmatrix} \mathbf{u}[k_0]^T \otimes \mathbf{p}[k_0]^T \otimes \mathbf{I}_{N_r} \\ \mathbf{u}[k_1]^T \otimes \mathbf{p}[k_1]^T \otimes \mathbf{I}_{N_r} \\ \vdots \\ \mathbf{u}[k_{N_p-1}]^T \otimes \mathbf{p}[k_{N_p-1}]^T \otimes \mathbf{I}_{N_r} \end{bmatrix}}_{\bar{\mathbf{P}}_{\otimes}} \text{vec}(\bar{H}) + \underbrace{\begin{bmatrix} \mathbf{v}[k_0] \\ \mathbf{v}[k_1] \\ \vdots \\ \mathbf{v}[k_{N_p-1}] \end{bmatrix}}_{\bar{\mathbf{v}}} \quad (2.296)$$

We need to ensure that $\bar{\mathbf{P}}_{\otimes}$ is tall, which requires

$$N_p \geq (L + 1)N_t$$

📖 Elaborating a lil bit...

- $\mathbf{u}$ is $(L + 1) \times 1$, $\mathbf{p}$ is $N_t \times 1$;
- $\mathbf{u} \otimes \mathbf{p}$ is thus $(L + 1)N_t \times 1$;
- Then we must have that $\mathbf{u}^T \otimes \mathbf{p}^T \otimes \mathbf{I}_{N_r}$ is $N_r \times N_r N_t (L + 1)$;
- Finally, we stack a bunch of those (N_p) on top of one another, getting $N_r N_p \times N_r N_t (L + 1)$.

We can now see that

$$N_r N_p \geq N_r N_t (L + 1) \implies N_p \geq N_t (L + 1).$$

The number of pilots is equal to the one of frequency flat channels and is lower of L when compared with frequency selective channels.

We could also have frequency domain estimation, see brief discussion at A signal processing perspective (pp. 57-130), page 66.

Examples (2.24, 2.30, 2.31)

Example 2.24: Least-square estimator for a frequency-flat SISO channel

Normally our estimator is

Through the derivation which can be found from A signal processing perspective (pp. 57-130), page 57 to page 58, we find that through the linear estimator we found we'll have

$$E^{LS} = \frac{N_0}{G} (\bar{P}_{\otimes}^* \bar{P}_{\otimes})^{-1} \quad (2.239)$$

$$\hat{H}^{LS} = \frac{1}{\sqrt{G}} \bar{Y} \bar{P}^* (\bar{P} \bar{P}^*)^{-1}, \quad (2.247)$$

where (2.239) expresses the Mean-Square Error Matrix while (2.247) expresses the best estimation of the channel.

The advantage of this equation with respect to (2.233) is that it entails less operations and also that since $\bar{P}$ is known, we can consider $\bar{P}^* (\bar{P} \bar{P}^*)^{-1}$ as already known.

Here though we have both that $L = 0$ (meaning that $\bar{H} \rightarrow H$) and that $N_t = N_r = 1$, meaning that we're in a SISO case, such that $H \rightarrow h$ and that our $\bar{P}$ matrix will reduce to a horizontal vector containing input pilots for $p[0], \dots, p[N_p - 1]$.

In order to have

Due to the discussion we made about zero forcing equalizers, **we need to ensure that $\bar{P}_\otimes$ is square or tall**, requiring to check that

$$N_p \geq (L + 1)N_t + L \quad (2.234)$$

and also that $\bar{P}_\otimes$ is full rank (which requires $\bar{P}$ to be full rank).

To be satisfied, we require that $N_p \geq 1$ so we just choose $N_p = 1$, implying that $\bar{P} \rightarrow p[0] = \sqrt{E_s} e^{j\phi}$.

The rest follows quite easily:

Example 2.24 (Least-squares estimator for a frequency-flat SISO channel)

Let $L = 0$ and $N_t = N_r = 1$. For this simplest of settings, a single scalar pilot suffices. With such a pilot being $\sqrt{E_s} e^{j\phi}$ where ϕ is an arbitrary phase, (2.247) reduces to

$$\hat{h} = \frac{e^{-j\phi}}{\sqrt{GE_s}} y \quad (2.249)$$

while E in (2.239) reduces to

$$\text{MMSE} = \frac{N_0}{GE_s} \quad (2.250)$$

$$= \frac{1}{\text{SNR}}, \quad (2.251)$$

which does not depend on ϕ ; the pilot's phase can thus be set to zero.

For $N_p > 1$, $\bar{P}$ becomes a row vector and the pilot sequence requires no special structure: in this setting, $\bar{P} \bar{P}^*$ is always a scalar, even if the same pilot symbol is simply repeated. It is easy to verify that

$$\text{MMSE} = \frac{1}{N_p \text{SNR}}, \quad (2.252)$$

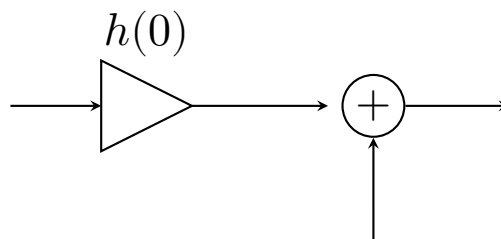
indicating that the accrual of energy over multiple pilots has the same effect as increasing the energy of a single pilot.

Professor's comments (Hz 616 ex 2.29)

We have $h[\ell] \neq 0$ only for $\ell = 0$: this is what *frequency flat* means.

Approximation Logic: Consider the term $\frac{1}{1+x}$:

- For $x \gg 0 \Rightarrow \approx \frac{1}{x}$
- For $x > 0$ (small) $\Rightarrow \approx 1 - x$



We only need $h(0)$ in order to have an estimate of the channel. We just take the output and we divide it by what we had transmitted. The error we get is 1 over the SNR. If the noise goes to zero, we have infinite error and infinite SNR $\rightarrow$ no error.

What we can do is to send several symbols one after the other; of course we're paying in the fact that we're sending multiple symbols.

If we use N_p pilots, the power of the noise is divided by N_p : **we can gain much in this sense but of course nothing comes for free and while we get this, we're also not sending information**: we loose in the bitrate but we gain in the accuracy detection.

 Example 2.30 (LMMSE estimator for a vec channel)

Here we're instead asked to find the LMMSE estimator, which we know to be defined as

The estimator is simply

$$W^{\text{MMSE}} = R_y^{-1} R_{ys}. \quad (1.224)$$

; since we have IID and unit-vector values in $\bar{H}$, we have that

$$\begin{aligned} R_{\text{vec}(Y)} &= \mathbb{E}[G\bar{P}_{\otimes} \text{vec}(\bar{H}) \text{vec}^*(\bar{H}) \bar{P}_{\otimes}^* + \underbrace{0 + 0}_{\text{Indep.}} + \text{vec}(\bar{H}) \text{vec}^*(\bar{H})] \\ &= G\bar{P}_{\otimes} \bar{P}_{\otimes}^* + N_0 I \end{aligned}$$

and also that

$$R_{\text{vec}(\bar{Y})\text{vec}(\bar{H})} = \mathbb{E}[\sqrt{G}\bar{P}_{\otimes} \text{vec}(\bar{H}) \text{vec}^*(\bar{H})] = \sqrt{G}\bar{P}_{\otimes},$$

in which btw we used $\bar{H}$ instead of the supposed signal because here we're trying to estimate $\bar{H}$, not the channel. From this, it follows that

Example 2.30 (LMMSE estimation for an IID MIMO channel)

Let the entries of $\bar{H}$ be IID and, by normalization, of unit variance. Then, $R_{\text{vec}(H)} = I$ and

$$W^{\text{MMSE}} = \sqrt{G}\bar{P}_{\otimes}(N_0 + G\bar{P}_{\otimes}^*\bar{P}_{\otimes})^{-1}, \quad (2.270)$$

from which

$$\text{vec}(\hat{H}) = \sqrt{G} \left(G\bar{P}_{\otimes}^*\bar{P}_{\otimes} + N_0 I_{(L+1)N_t N_r} \right)^{-1} \bar{P}_{\otimes}^* \text{vec}(\bar{Y}), \quad (2.271)$$

while the MMSE matrix in (2.262) specializes to

$$E = I - G\bar{P}_{\otimes}^*(G\bar{P}_{\otimes}\bar{P}_{\otimes}^* + N_0 I)^{-1}\bar{P}_{\otimes}. \quad (2.272)$$

In the foregoing expression for $\text{vec}(\hat{H})$, the dimensionality of the identity matrix has been noted explicitly to set the stage for the decomposition $N_0 I_{(L+1)N_t N_r} = N_0 I_{(L+1)N_t} \otimes I_{N_r}$. Recalling that $\bar{P}_{\otimes} = \bar{P}^T \otimes I_{N_r}$, we can then write

$$\text{vec}(\hat{H}) = \sqrt{G} \left(G\bar{P}^c \bar{P}^T \otimes I_{N_r} + N_0 I_{(L+1)N_t} \otimes I_{N_r} \right)^{-1} (\bar{P}^c \otimes I_{N_r}) \text{vec}(\bar{Y}) \quad (2.273)$$

$$= \left[\sqrt{G}(G\bar{P}^c \bar{P}^T + N_0 I_{(L+1)N_t})^{-1} \bar{P}^c \otimes I_{N_r} \right] \text{vec}(\bar{Y}) \quad (2.274)$$

and, resorting once more to the identity $\text{vec}(ABC) = (C^T \otimes A)\text{vec}(B)$, what ensues is

$$\hat{H} = \sqrt{G}\bar{Y}\bar{P}^*(G\bar{P}\bar{P}^* + N_0 I)^{-1}, \quad (2.275)$$

which coincides with its least-squares counterpart in (2.247), save for the argument of the inverse being regularized in proportion to the noise strength. This similarity is consistent with the fact that the channel statistics, which differentiate the least-squares and LMMSE estimators, are trivial when the entries of $\bar{H}$ are IID; an SNR-dependent regularization suffices to optimally bias the estimate of each realization. The possibility of unvectorizing the estimate in the form (2.275) is also a consequence of the receive antennas being uncorrelated, as then the LMMSE estimate of each row of $\bar{H}$ can be produced separately, based only on the corresponding row of $\bar{Y}$.

I Professor's comments.

Entries are random: we're dealing with a stochastic channel! We want to estimate this channel. If we want to estimate a random variable, we want to estimate the probability density function of the random variable or some other complete description of that variable. We expect the channel is random but we need to deal with the realization.

** Realization**

If I have a Bernoulli, before throwing the coin we have a random variable; when we're looking at the result, we have the realization of that random variable. If we take many realization we expect to have half heads and half tails (This of course is the frequentistic approach). We can only estimate a realization of random variable!

We have the realization, we do the calculation and again we have a noise term to the minus one. It's what is called a regularization term, which ensures the matrix is always invertible (refer to [pag 46 slide](#)).

There's a nice example.

We have

$$\begin{bmatrix} 1 & -1 \\ \rho & -1 \end{bmatrix}$$

for which we see what's up when ρ changes between -1 and 1 . **The matrix will become singular when $\rho \rightarrow 1$. If we use the same noise and we have $\rho = 1$, we'll see that the noise of zero forcing will escalate while the MS will stay still** (the zero forcing tries to invert a singular matrix!).

[color=ForestGreen!50!black] Random: we need to check for consistency when we look at our results.

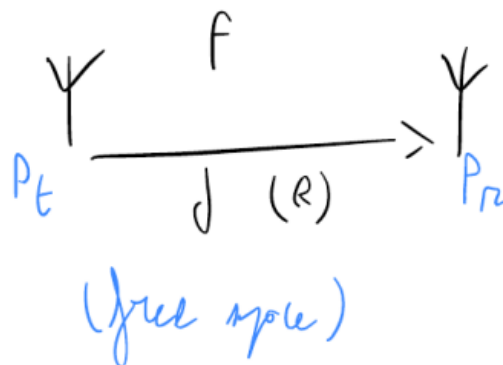
Chapter 10

Lesson 5-12-2025

10.1 Channel modeling

10.1.1 Friis' equation and discussion on the choice of the transmission frequency

Let's say we have



Which parameters do we need? The gains of the transmit and receive antennas: G_T, G_R .

Friis' equation

What's the equation which binds the two?

$$P_r = P_t G_t G_r \frac{1}{d^2 f^2} \left(\frac{c}{4\pi} \right)^2 = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 ;$$

this is Friis' equation; what does this tell us?

We can't do anything on the distance! **We can do something on the frequency** though; if we double the frequency, the received power will be divided by 4. The lower frequencies will be the best ones. Of course we can't go baseband: the size of the antenna is

somehow proportional to λ ; if we have $f = 0$, we'll need to have $\lambda = \infty$ and thus we'd need an extremely large antenna.

In Europe the unlicensed band is around **868MHz**. We then have other ones at **2.4, 5GHz**: those are used by the wifi. If we have a router and home, let's imagine we have a switch between the two; normally it transmits at both. When we transmit, we can decide which to use; we might think we'll certainly choose 2.4GHz, since it will get 4 times the power the other gives! There are other considerations; speed is something we want: the bitrate does not only depend on the power, but also on the bandwidth. Here the bandwidth is larger. At 5GHz we have **higher bitrate but more attenuation**, while at 2.4GHz we don't have as much attenuation but we also have a lower bandwidth.

If we consider the channel itself, it would prefer lower frequencies. It's also true that **sometimes we want attenuation** (voice is attenuated: we can talk without disturbing the neighbour by lowering the voice).

We don't want the router of our neighbour to enter our house, as that would create interference. The fact that the waves are attenuating is good, if we look at that kind of principle.

Here the exponent is 2. We know that now wifi is converging towards mmwaves; these are frequencies for which the λ is comparable with **1mm**, meaning that the frequency is $f > 6\text{GHz}$. Why so? mmwaves have some other problems; if we are in the air (not in the free space anymore), we have oxygen, water, ...If we take a frequency which is resonating certain molecules, then we have peaks of absorption at certain frequencies. The frequencies at mmwaves experience the phenomenon of **blockage**. If the wave just meets a leaf, it's immediately blocked. Such high frequencies (like mmwaves) are blocked by many types of obstacles. Point is that higher frequencies have very large and non-crowded bandwidth: if we're in line of sight, we can upload gigabits per second!

Of course the more bandwidth we have, the larger the bitrate we can achieve (this follows easily from $R/B = C$: for a fixed capacity, increasing the bandwidth gives us a better bitrate). Let's suppose we can get the line of sight.

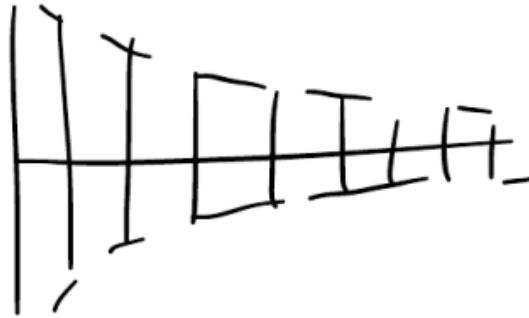
If we consider that each cellular phone has a bandwidth of 2MHz, this means that

$$\frac{60\text{GHz}}{2\text{MHz}} = 3 \cdot 10^4$$

(where 60GHz is the frequency of one of the more recent standards) this is a really big attenuation factors; how can we do this, then? 5G is using mmwaves; 6G is thinking of THz! We solve the problem of the high attenuation by working on the gain of antennas; amplifiers are not the way, as they introduce noise; the antennas are passive elements.

Since we have mmwaves, we can make an antenna which is just of the size of a millimeter; we can thus have multiple small antennas in a very small area. Going back again: we can set the gains really high. This is like directing the power specifically in a certain direction.

The gain comes from the fact that the signal is propagating specifically in one direction.



“ In-class stuff

THz is really horrible; generally telecommunications have always been like: “we make this, somebody will use it”; now telecommunications has a market pull: they say ***we need something*** and we’re the ones finding such things. The use case for THz waves is to replace the cables in data centers. This would allow for great simplifications in situations with many cables.

Shows a picture with attenuation depending on the transmission frequency. The spots of minima are candidates for 5G; wifi instead uses the spots where attenuation is higher; this is because wifi knows it does not need to go far in distance. The maximas are used depending on the application; if we just need low distance transmission, we can use the maxima as well.

Now we should have an idea of how the frequencies are selected. There is a lot of discussion about the human damages which according to certain people are being cause by electromagnetic waves.

Controversy of people not wanting 5G; we have a parameter called SAR (specific absorption rate): we can’t commercialize stuff which has wrong values of this. In principle, high frequencies on molecules should not be more dangerous than the lower ones; actually, the low frequencies are even worse.

10.2 Modeling approaches

10.2.1 Modeling of radio propagation (LOS, NLOS, effects on the signals)

Handwritten note: Two approaches → deterministic → ray tracing (difficult) (GPU). Stochastic → lights... used for underwater communications.

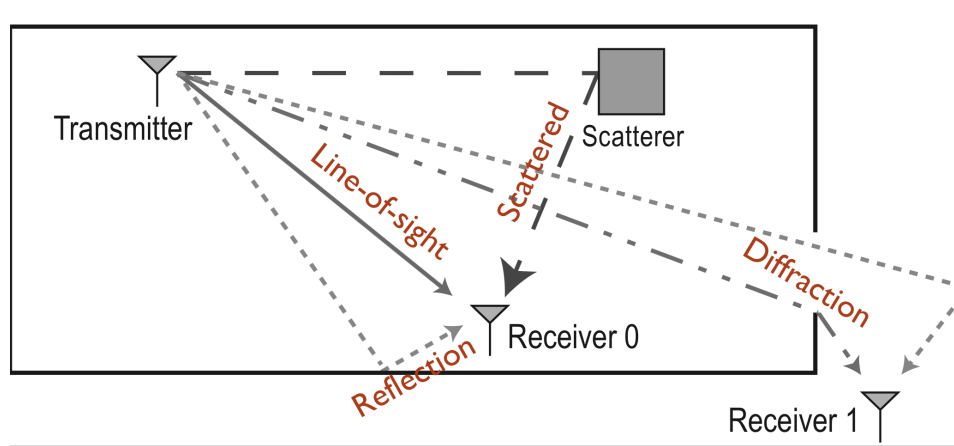
Highlights; we’ll skip them.

Propagation modeling; there are different effects that we should take into account. The aim is to develop a mathematical model of what happens physically!

The best that can happen is that our signal directly reaches the receiver, through a line of sight (LOS); generally, what happens is that the signal actually reaches the receiver

through some different type of path, aka a non-line of sight (NLOS). We have different possibilities, when it comes to NLOS:

1. A wave might bounce off a smooth object of size exceedingly large relative to the wavelength of the signal: we have **reflection**; each incoming wave is mapped onto a single reflected direction.
2. **Scattering** is said to have occurred when a wave travels through a medium that contains objects whose dimensions are smaller than or comparable to the wavelength, and where the number of such objects is large. Then, each incoming wave is mapped onto many scattered ones.
3. Radio waves also bend around sharp edges, a phenomenon that is referred to as **diffraction** and that plays a pivotal role in urban settings. Indeed, a major contributor to the propagation of radio signals in such settings is rooftop diffraction.



Ray tracing example: Reflection, Scattering, Diffraction

The Maxwell equations are not enough to describe the various effects we can have, such as diffraction, reflection, scattering. Only if we're lucky do we have a line of sight. In order to solve this we'd need to solve differential equations which are exceedingly more complicated: we could not get out of that, not even with super computers.

Clutter

In an urban scenario, we refer to the **propagation clutter** as the set of objects which cause the NLOS effects; we can have either an elevated transceiver (if it's elevated with respect to the clutter) or a **within clutter** transceiver.

We'll then derive an abstraction simple enough for us to be able to handle the model, but also complicated enough to be able to capture the major effects which are happening. Such a model is key of information engineering. We always need to be careful about the conditions which allowed us to form the model.

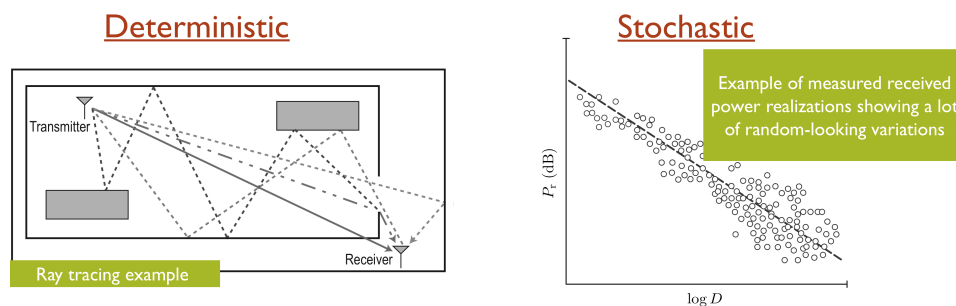
It's difficult (impossible!) that things go better than what is predicted by the Friis equations, which only work in line of sight. We'll then develop a mathematical description of multi-path propagation channels inspired by propagation channel measurements.

Sensitivity of the receiver

Sensitivity of the receiver: the minimum amount of power needed to correctly receive a signal with a certain probability. Sensitivity of -100dBm; if we receive any greater power, we'll receive the signal and interpret it correctly with a good probability. We have very expensive softwares able to map a whole area this way. It's also true that "high power signal" does not mean that we have a clean signal.

10.2.2 Modeling approaches (deterministic, stochastic); elements of the modelization

We know that in the end we won't generally develop a deterministic model, as that would be extremely complicated (involving ray tracing and such); we only do this sometimes, like with millimeter waves (since they generally require ray tracing of indoor sites). More often, we do stochastic models!



Deterministic vs Stochastic approaches

Channel modeling (pp. 131-208), page 4

In general, a stochastic model is embodied by the **conditional distribution of the channel output given its input**. When the channel is linear, as is the case in the wireless medium, this is tantamount (*equivalente*) to a stochastic impulse response.

A critical distinction which needs to be made when modeling is that between large-scale and small-scale phenomena.

Channel modeling (pp. 131-208), page 4

Specifically, what is **assumed** is that the **distribution of the channel response is locally stationary within certain neighborhoods around transmitter and receiver**.

We'll just assume wide-sense stationarity, but we might as well assume strict-sense stationarity. The neighborhoods in which the signal can be considered to be stationary is generally in the order of tens to hundreds of wavelengths.

Channel modeling (pp. 131-208), page 4

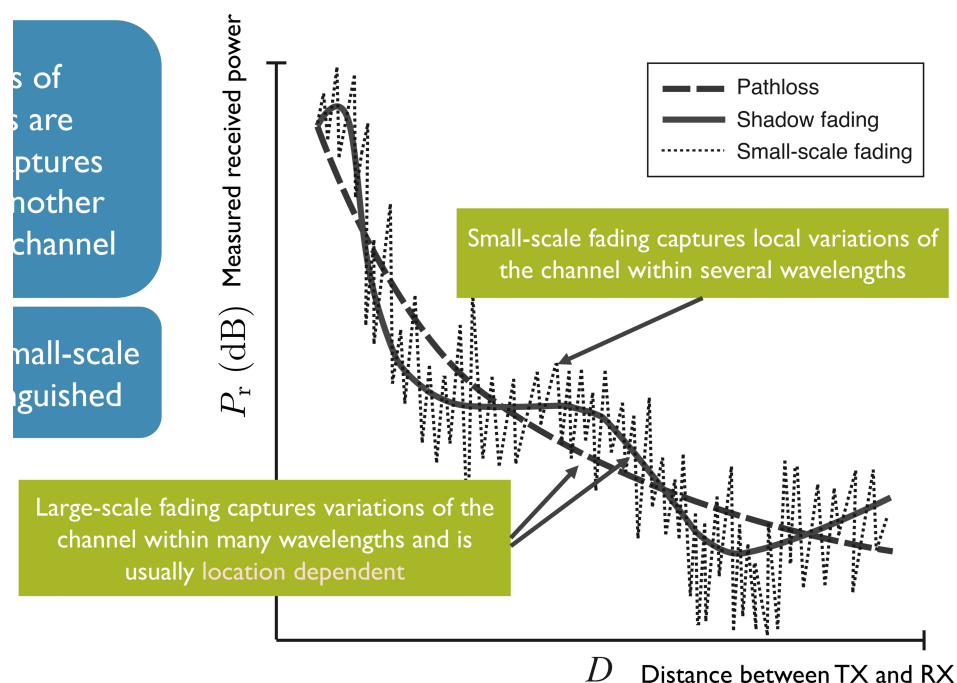
Large-scale phenomena impact the local distribution to which the small-scale phenomena conform, **generally the local-average power**.

Channel modeling (pp. 131-208), page 5

Because of the importance of the decoupling of large- and small-scale phenomena, we use distinct variables to model them and carry those throughout the derivations in the entire book. While this takes a small toll in terms of the compactness of some expressions, it reinforces the separation between quantities that change at very different scales in time and frequency, and which have markedly different effects on the communication.

Just an insight on the approach we'll further discuss later: we generally draw a regression line trying to minimize the distance between the line and the various values. We make great campaigns of measurements, in order to obtain sufficient data to model this. We then decompose everything in two (+1) components:

1. **Large-scale fading:** if we go around the corner, we have another propagation, changing abruptly; this depends heavily on where we are! This change is not smooth.
2. **Small-scale fading:** depends on how the various waves come to the receiver and how they arrive at the receiver; if we move our phone up and down, we'll have variations in the signal.
3. **Line loss:** this changes depending on what we have in-between. There are parameters to tune if the loss is due to walls, ceilings, iron, ...; this needs to be managed carefully.



Decoupling of fading types over distance

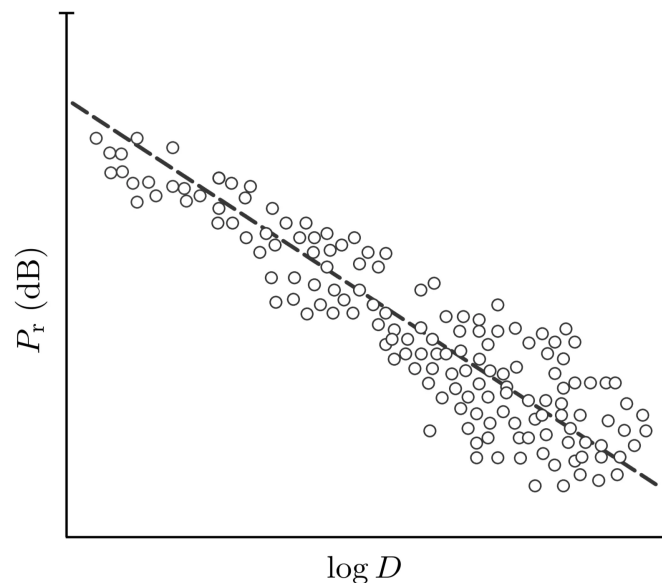
In MIMO, we want to have full-rank matrices; small scale fading can help to make the various matrices full rank!

10.3 Large scale phenomena

10.3.1 Pathloss and shadow fading (their separation, distribution of the shadow fading), pathloss; large-scale channel gain G and partial values P_t, P_r, G_t, G_r

Channel modeling (pp. 131-208), page 5

Under the separation argued in the previous paragraph, large-scale phenomena are those that determine the local distribution of the stochastic channel response. The prime feature of that distribution is, of course, the power. If one plots, for a given transmit power, the local-average received power (in log-scale) obtained from many empirical measurements in otherwise identical conditions as a function of the distance (also in log-scale) between transmitter and receiver, the result is invariably a cloud of scattered points, such as the one in the figure below



The line we see in the figure above is a linear regression line which relates the local-average received power to the distance. Such a law represents what we call the **pathloss**, which we'll denote as L_p ; the scattered measurements around the law is what we call **shadow fading**, denoted by χ . The pathloss and the shadow fading put together give the large-scale description of virtually every existing channel model.

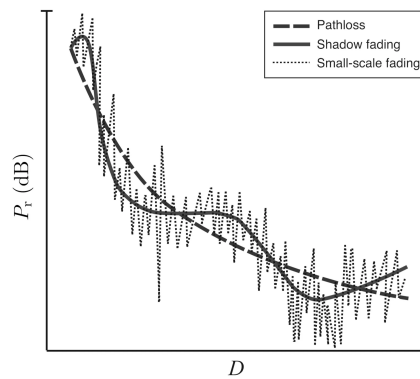
Channel modeling (pp. 131-208), page 6

Although recent work indicates, on solid analytical footings based on random walks and diffusion theory, that nonlinear regressions may sometimes be superior [203, 204], the **combination of pathloss and shadow fading suffices for our purposes**.

** Channel modeling (pp. 131-208), page 6**

The pathloss can be interpreted as the expected attenuation over an ensemble of realizations of a given propagation scenario, whereas the shadow fading individualizes, through randomization, the attenuation of each such potential realization.

Another common interpretation of the large-scale phenomena is that the **pathloss quantifies the local-average attenuation** tied exclusively to distance whereas the **shadow fading accounts for the presence of obstacles** that cause the local-average attenuation to deviate from the value predicted by the pathloss. This is illustrated in Fig. 3.4 (below here), which depicts how both pathloss and shadow fading affect the evolution of the received signal power as the distance between transmitter and receiver increases over a generic trajectory; the pathloss is monotonic, whereas the shadow fading can oscillate as obstacles come in and out of the way. It was suggested in [205] that **the shadow fading attenuation, χ , could follow a lognormal distribution** and empirical measurements have repeatedly confirmed the excellent fit to data offered by such distribution [206, 207]. The most widely accepted justification for this log-normal nature is that shadow fading represents the product of the losses introduced by the several obstacles traversed by the signal and, in log-scale, such a product becomes a sum of independent terms. Despite the fact that those terms can hardly be assumed to conform to the same distribution, and that the number of such terms may not be particularly large, the robustness of the central limit theorem (see Appendix C.1.10) renders the sum surprisingly well modeled by a Gaussian random variable.



Received signal power as a function of distance, illustrating the large-scale (pathloss and shadow fading) and the small-scale propagation phenomena.

Figure 10.1: Received signal power as a function of distance, illustrating the large-scale (pathloss and shadow fading) and the small-scale propagation phenomena. Fig. 3.4

We thus have that under the log-normal model, $\chi|_{\text{dB}}$ is Gaussian with PDF

$$f_{\chi|_{\text{dB}}}(\xi) = \frac{1}{\sqrt{2\pi}\sigma_{\text{dB}}} \exp\left(-\frac{\xi^2}{2\sigma_{\text{dB}}^2}\right), \quad (3.1)$$

where σ_{dB} is the standard deviation which is generally assumed to be independent of the distance for the sake of simplicity.

Channel modeling (pp. 131-208), page 7

Outdoors, σ_{dB} tends to range between 8 and 12 dB. Indoors, it is usually smaller.

Log-normal random variable is like a Gaussian in the log domain! We use a Gaussian because it is given by the summing of various different random variables.

The signal was represented as zero-mean: we can easily include a mean in the log-scale Pathloss. We can also rewrite the distribution in linear scale, see [08.3_pp_131_208_Channel_modeling](#) page 7.

📖 Channel modeling (pp. 131-208), page 7

A relevant property confirmed through empirical observations is that, as intuition would have it, shadow fading **can be broken up into distinct pieces: one** that depends on the **location of the transmitter** and **one** that depends on the **location of the receiver** [207]. Two mutually distant receivers communicating with the same transmitter (or two mutually distant transmitters communicating with the same receiver) experience the same shadow fading for one piece but a completely different shadow fading for the other. These two pieces, whose values add up in log-scale, tend to have similar strengths. Furthermore, measurements reported in [208] point to an exponential model for the spatial autocorrelation of each piece, with a correlation distance that is on the order of tens of meters outdoors, depending on the environment, and substantially shorter indoors [209]. For more refinements on the modeling of shadow fading, the reader is referred to [210, 211] and references therein.

We can express the local-average received power P_r as a balance of gains and losses according to the **link budget equation** :

$$P_r|_{\text{dB}} = P_t|_{\text{dB}} + G_t|_{\text{dB}} + G_r|_{\text{dB}} - \underbrace{(L_p|_{\text{dB}} + \chi|_{\text{dB}})}_{\text{Pathloss function}}, \quad \xrightarrow{\text{better to measure it in dBm}} \quad (3.4)$$

It's somewhat like the Friis equation; we have the plus operation because we're in the log domain.

We can also rewrite it **more compactly as**

$$P_r|_{\text{dB}} = P_t|_{\text{dB}} + G|_{\text{dB}}, \quad (3.5)$$

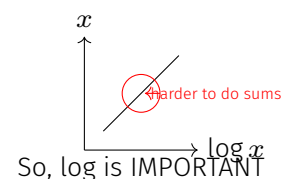
where **the gain**

$$G = \frac{G_t G_r}{L_p \chi} \quad (3.6)$$

Why they created logs?

In math, to "convert" products to sums.

$$\log(a \cdot b) = \log(a) + \log(b)$$



is the **large-scale channel gain**, a quantity which was already introduced in the signal processing section to normalize the channel response; only now it is acquiring its full significance. Of course if we have a log, the exponent can change.

10.4 Two-exponents model

We then have another **path loss modeling**; it gives the loss depending on the distance, which only applies after a certain reference distance D_{ref} . We have an exponent

η , which is normally higher than 2 and is called the pathloss exponent. We can also write this in dB:

$$L_p(D) = K_{\text{ref}} \left(\frac{D}{D_{\text{ref}}} \right)^\eta \quad (3.7)$$

$$L_p(D)|_{\text{dB}} = K_{\text{ref}}|_{\text{dB}} + 10\eta \log_{10} \left(\frac{D}{D_{\text{ref}}} \right) \quad D > D_{\text{ref}} \quad (3.8)$$

with D_{ref} being the reference distance and K_{ref} being the pathloss at D_{ref} .

This is a very rudimental model; we're only looking at the path loss.

Values of K_{ref}

If no value is specified for K_{ref} , it's safe to assume we can use the freespace value, as described in (3.9).

Channel modeling (pp. 131-208), page 8

The exponent η is the most consequential parameter, with typical values ranging between 3.5 and 4 if either the transmitter or the receiver are elevated. If both are within the propagation clutter, higher values can be encountered. Values below 2 are rare but possible in settings such as indoor corridors, outdoor urban canyons, or tunnels.

Channel modeling (pp. 131-208), page 8

The parameter K_{ref} is sometimes referred to as the intercept at D_{ref} because it is the value at which a plot of $L_p(D)$ intercepts a vertical axis drawn at D_{ref} . Given a reference distance, K_{ref} can be measured empirically. Alternatively, if the reference distance is chosen properly, K_{ref} can be computed under the assumption that the propagation from the antenna up to that point follows the free-space law described next. In this case, the representation in (3.8) becomes the two-slope model in Fig. 3.5 (**see below here**), namely **free-space decay** up to D_{ref} and decay **with exponent η thereafter**. This two-slope behavior can be justified analytically [51, section 2.4.1] and on the basis of experimental observations [213]. Besides the representation in (3.8), suitably complemented by log-normal shadow fading, several pathloss modeling alternatives are available. These alternatives are the subject of the remainder of this section. While the focus here is exclusively on how the large-scale propagation phenomena affect the local-average received power, models do exist for how other features of the local distribution (introduced later in this chapter) are affected, e.g., the Rice factor, the power delay profile, or the angle spread [214, 215].

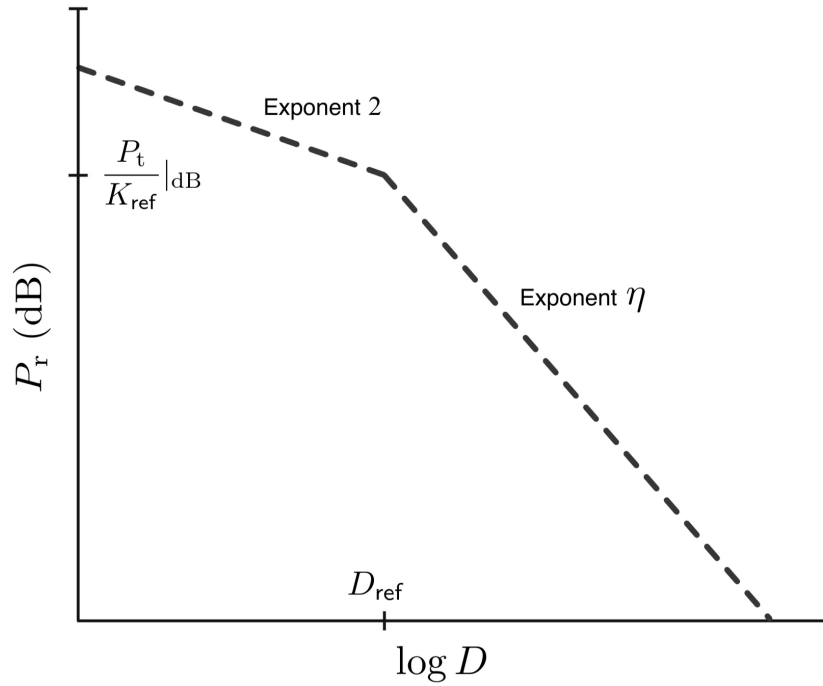


Fig 3.5

The first slope in the figure 3.5 is the **pathloss in free space**, given by

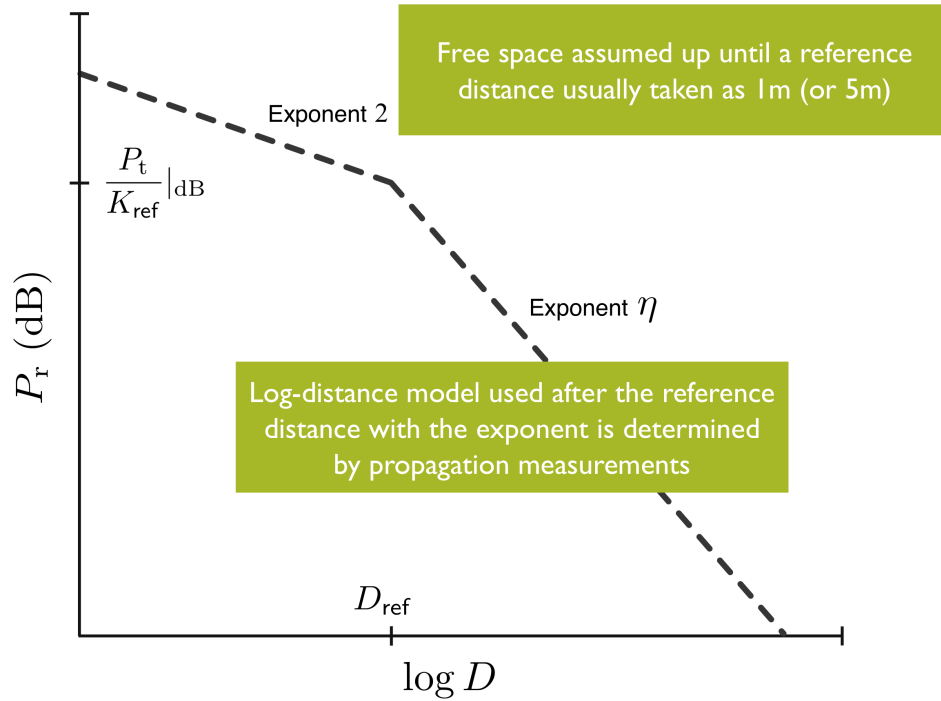
$$L_p(D)|_{\text{dB}} = 20 \log_{10} \left(\frac{4\pi D}{\lambda_c} \right), \quad (3.9)$$

which is none other than (3.8) adapted with $K_{\text{ref}} = 1$ and $\eta = 2$; λ_c is the carrier wavelength.

It **holds only when D lies in the far-field region of the antennas**, the rule of thumb being $D \geq 2d_a^2/\lambda_c$ where d_a is the largest physical dimension of either antenna. By definition, this model applies **only when there are no obstructions between transmitter and receiver or in the vicinity** of the trajectory linking them; this restricts its applicability to very specific situations, e.g., the computation of the intercept at a nearby point. There is no accompanying shadow fading. The presence of objects of any kind, even simply the ground, alters the square-law distance decay in (3.9), yielding a more general exponent η .

In class

We also have a two-slope model: up to a certain distance we have an exponent; then we have another exponent η :



If we look at the book, we see we have various models from [08.3_pp_131_208_Channel_modeling, page 8](#) to [08.3_pp_131_208_Channel_modeling, page 12](#).

10.5 Macrocell models

These are models with radius exceeding 1km.

10.5.1 Okumura-Hata model

This is a model derived by empirical measurements; it is

$$L_p|_{\text{dB}} = 69.55 + 26.16 \log_{10} f_c - 13.82 \log_{10} h_b - b(h_m) + (49.9 - 6.55 \log_{10} h_b) \log_{10} D \quad (3.10)$$

where:

- $f_c \in [150, 1500]$ is the carrier frequency in MHz
- $h_b \in [30, 200]$ is the base station antenna height in meters
- $h_m \in [1, 10]$ is the mobile user antenna height in meters
- D is in kilometers

and where

$$b(h_m) = \begin{cases} (1.1 \log_{10} f_c - 0.7)h_m - 1.56 \log_{10} f_c + 0.8 & \text{small to medium cities} \\ 8.29[\log_{10}(1.54h_m)]^2 - 1.1 & \text{large cities, } f_c \leq 300\text{MHz} \\ 3.2[\log_{10}(11.75h_m)]^2 - 4.97 & \text{large cities, } f_c \geq 300\text{MHz} \end{cases} \quad (3.11)$$

For **suburban areas** we should add to (3.10) the term:

$$-2[\log_{10}(f_c/28)]^2 - 5.4; \quad (3.12)$$

for **rural areas**, we should add to (3.10) the term:

$$-4.78(\log_{10} f_c)^2 + 18.33 \log_{10} f_c - 40.94. \quad (3.13)$$

This does not adapt to microcell; the COST-231 model adapts up to 2GHz.

10.5.2 COST-231 Hata model

This model works for frequencies from 150MHz to 2GHz. In urban areas, we have

$$L_p(D)|_{\text{dB}} = 46.3 + 33.9 \log_{10} f_c - 13.82 \log_{10} h_b - b(h_m) \\ + (44.9 - 6.55 \log_{10} h_b) \log_{10} D \quad (3.14)$$

with $b(h_m)$ just like in (3.11). In **dense metropolitan areas**, an extra 3dB must be added.

10.6 SUI model

This model works for fixed wireless antennas, like rooftop antennas, elevated of 15-40m over the clutter.

We have 3 types of terrains:

- A. Hilly terrains with heavy tree density
- B. Mostly flat terrains with moderate-to-heavy tree densities or hilly terrains with light tree densities.
- C. Flat terrains with light tree densities.

The main equation is a refinement of (3.8), aka

$$L_p(D)|_{\text{dB}} = K_{\text{ref}}|_{\text{dB}} + 10\eta \log_{10} \left(\frac{D}{D_{\text{ref}}} \right) + X(f_c) + X(h_m) \quad D > D_{\text{ref}}, \quad (10.1)$$

with $D_{\text{ref}} = 100\text{m}$ and $K_{\text{ref}}|_{\text{dB}}$ being the intercept at D_{ref} (assuming free space up to D_{ref}). We then have

$$\eta = a - bh_b + \frac{c}{h_b}, \quad (3.16)$$

with the constants being given according to the terrain type,

Table 3.1 Values of a, b, and c in (3.16)			
Parameter	Terrain A	Terrain B	Terrain C
a	4.6	4.0	3.6
b (m ⁻¹)	0.0075	0.0065	0.005
c (m)	12.6	17.1	20

h_m is the user height, which influences the model together with λ_c through:

$$X(f_c) = 6.0 \log_{10} \left(\frac{f_c}{2000} \right) \quad (3.17)$$

$$X(h_m) = \begin{cases} -10.8 \log_{10}(h_m/2000) & \text{Terrains A and B} \\ -20 \log_{10}(h_m/2000) & \text{Terrain C} \end{cases}. \quad (3.18)$$

10.7 Microcell models

Models with radius between 100m and 1km.

10.7.1 COST-231 Walfisch-Ikegami model

This model is defined for frequencies between 800MHz and 2GHz.

We define the model under the **3GPP parameters** (base station height $h_b = 12.5\text{m}$, building height 12m, building-to-building distance of 50m, street width of 25m, user height $h_m = 1.5\text{m}$).

We have that for **NLOS propagation**

$$L_p|_{\text{dB}} = -55.9 + 38 \log_{10} D + \left(24.5 + 1.5 \frac{f_c}{925}\right) \log_{10} f_c, \quad (3.19)$$

with f_c being in MHz, while D is in meters and must be $D \geq 20$.

Under **LOS propagation**,

$$L_p|_{\text{dB}} = -35.4 + 26 \log_{10} D + 20 \log_{10} f_c. \quad (3.20)$$

Here, the recommended standard deviation for shadow fading is $\sigma_{\text{dB}} = 4\text{dB}$. **(?) End**
6/12/2025

Chapter 11

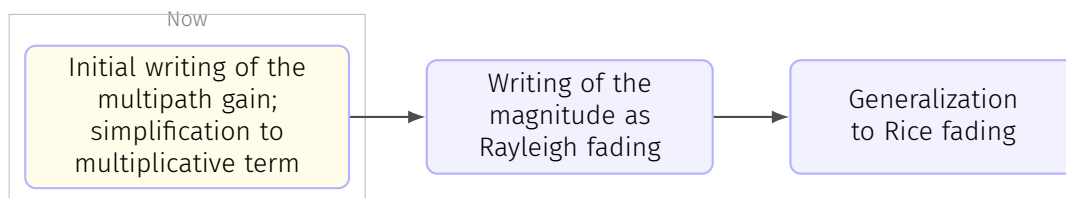
Lesson 9-12-2025

11.1 Picocell and indoor models

Generally we have a path loss like (3.8), with $\eta = 3.5$. The shadow fading is usually ranging between 3 and 6dB.

11.2 Small-scale fading

11.2.1 Multipath propagation (+Rayleigh and Rice fading)



Channel modeling (pp. 131-208), page 13

Let us now zoom in and enter the small-scale realm. When dealing with continuous-time representations, we use the complex **pseudo-baseband channel response**. We also recall from Section 2.2.3 that, **to discretize this representation**, lowpass filtering is in general required before sampling, although in those cases when the discretization returns a singletap response and the effect is only multiplicative, the complex gain of the pseudo-baseband channel can be borrowed directly.

Due to the various NLOS effects, what the receiver observes is a superposition of a number of replicas of the transmitted signal; the overall channel response is a sum of all these various contributions.

Each path can be described using the attenuation A_q and the delay τ_q , under the form

$$A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q}.$$

Channel modeling (pp. 131-208), page 13

Since they stem from objects within the local neighborhoods, all the relevant paths can be assumed to be subject to similar pathloss and shadow fading, suitably modified by the local reflection/diffraction/scattering, and thus **the complex gains $\{A_q\}$ may not exhibit major differences**. However, because at carrier frequencies of

interest the phase $2\pi f_c \tau_q$ shifts radically with the slightest variation in τ_q , the differences in path delays are always sufficient to regard the path phases as independent random variables.

Channel modeling (pp. 131-208), page 13

Suppose for now that the transmit signal $x(t)$ corresponds to a passband sinusoid at frequency f_c . **Since the effect of a delay on such a signal is merely a phase shift, we can absorb each delay τ_q as a shift $2\pi f_c \tau_q$ in the phase of A_q .** Then, the overall channel impulse response can be written as

$$\underbrace{\left(\sum_q A_q e^{-j2\pi f_c \tau_q} \right)}_{\text{Gain}=\sqrt{G} \cdot h} \delta(\tau), \quad (3.21)$$

which is only possible if we eliminate the delay by transforming

$$\delta(\tau - \tau_q) \rightarrow \delta(\tau),$$

while actually considering only the phase shift caused by the complex exponential.

In class

If we're in the narrowband case (we have a narrow band signal), this means that the symbol duration is much larger than the inverse of the bandwidth. This means that we have a large symbol, then a delay, the symbols are more or less aligned:

$$\begin{array}{c} e^{j\varphi_1} \\ \hline e^{j\varphi_2} \end{array}$$

We can use the same coefficient but they might have a different phase (blue). We thus assume we have just one delta and that's where the second summation (the one on the right) comes from. In this case we have a one tap channel with a gain of $\sqrt{G} \cdot h$. We know h is random but must be either modulus one or variance 1. This is modeled due to the Central Limit Theorem like a complex Gaussian with $\sigma^2 = 1$. We have that such a random variable is irrotational. The phase varies randomly between 0 and 2π .

The signal reaching the receiver will thus be

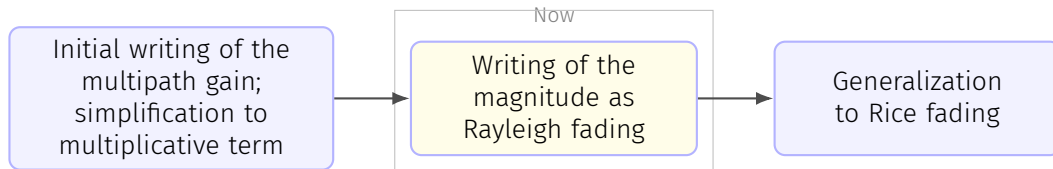
$$y(t) = \sqrt{G}h \cdot x(t).$$

☰ **Channel modeling (pp. 131-208), page 14**

Thus, upon a sinusoid the channel does have **only a multiplicative effect**, i.e., a complex gain. Furthermore, the preceding consideration on the path phases suggests modeling this gain as a sum of IID random variables; if the number of paths is large enough and some mild conditions are satisfied [221], application of the central limit theorem leads to a complex Gaussian distribution. The result of the sum in (3.21) changes rapidly with the position of both transmitter and receiver: a mere displacement on the order of a wavelength alters the path lengths, and thus the delays thereon, by an amount that shifts the path phases drastically, causing major swings in the sum. Put differently, the paths can add constructively or destructively depending on their relative phases, which are completely modified by small changes in position. The same effect transpires if, instead of a change in position, there is a change in frequency that shifts the phases substantially.

☰ **Channel modeling (pp. 131-208), page 14**

The picture that emerges is thus as follows: the large-scale channel features (pathloss and shadow fading) determine the average channel conditions within the local neighborhood while the small-scale fading determines, subject to that local-average, the instantaneous channel conditions at each position and frequency. This invites decoupling the complex gain in (3.21), namely $\sqrt{G} \cdot h$ where $\sqrt{G}$ determines the local-average received power, P_r , as per (3.5), whereas the small-scale fading h is normalized to have unit power. This largely achieves the desired separation between large-scale and small-scale features, yet a mild coupling remains in that certain other parameters of the local small-scale fading, e.g., the Rice factor, may depend on (or be correlated with) the large-scale gain. Ideally, the value of the large-scale features should be taken into account when setting those smallscale parameters, or else there is a danger that the hugely convenient decoupling between large- and small-scale gains entails a loss of relevant information.



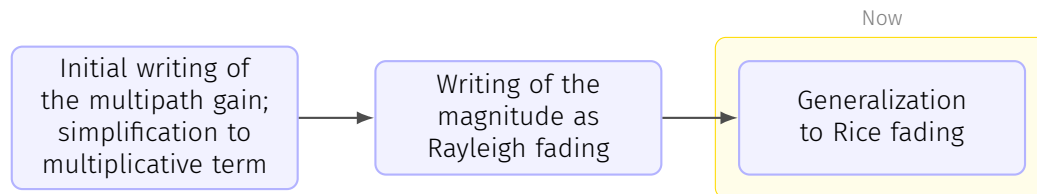
The **small-scale fading is locally modeled as a stationary random process**, in principle zero-mean and complex Gaussian, normalized to have unit variance. At a given position and frequency, then, it is $h \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ with **uniformly distributed phase and with a magnitude that abides by the Rayleigh distribution** (see Appendix C.1.9), hence the popular designation of small-scale fading as simply **Rayleigh fading**. Precisely,

$$f_{|h|}(\xi) = \xi e^{-\frac{1}{2}\xi^2}, \quad (3.22)$$

while the power $|h|^2$ follows the exponential distribution

$$f_{|h|^2}(\xi) = e^{-\xi}, \quad (3.23)$$

which of course is only defined for $\xi \geq 0$.



If a multipath term dominates over the rest, e.g., because it propagates through LOS and it is subject to a distinct pathloss, then the receiver can lock onto that signal component and render it **unfaded**. It follows that $h \sim \mathcal{N}_{\mathbb{C}}(\mu_h, \sigma_h^2)$ with mean $\mu_h \neq 0$ and with variance $\sigma_h^2 = 1 - |\mu_h|^2$. For this **so-called Rice fading**, we have

$$f_{|h|}(\xi) = 2(K+1)\xi e^{-(K+1)\xi^2 - K} I_0(2\sqrt{K(K+1)}\xi), \quad (3.24)$$

where $K = |\mu_h|^2 / \sigma_h^2$ is the **Rice factor** while $I_0(\cdot)$ is the zero-order modified Bessel function of the first kind.

In class

The K factor is the ratio between the mean signal strength and the variance of a random component.

For $K = 0$, Rice fading reverts to Rayleigh fading; for $K \rightarrow \infty$, the fading vanishes as the value of h becomes deterministic. Thus, the Rice distribution bridges these extremes and allows, through K , to regulate the severity of the fading. Analytically speaking, however, the PDF in (3.24) is not particularly friendly because of the presence of $I_0(\cdot)$.

We can't calculate the Bessel functions!

Mynote

As it is clear, the distribution of the modulus of the channel $|h|$ is described by (3.24).

A more tractable distribution is the Nakagami-m distribution, see [08.3_pp_131_208_Channel_modeling page 15](#).

There are other magnitude distribution; generally Ricean is used when we don't have a direct line of sight.

11.3 Digression on the importance of channel modeling, standardization institutions

We were talking about the channel models; a few words before the actual lecture: tomorrow we'll have a regular lecture.

About this chapter: in general there's not a well-shaped channel modeling theory; if we feel we're confused, it's normal at the first approach. At the beginning we said we're trying to make a mathematical model for a really complicated environment: to make a deterministic model for such a situation would be too difficult by far. We resorted to models which should mimic reality. Unfortunately, we don't have an agreed model; for the noise, we know that we have WGN. Here we have a confusing situation: what we want to design depends on our purpose; we have models which are useful in certain situations, other models which are useful in certain others. Specifically, we're looking for the limits of our systems; when we know what the ultimate limit is, we know how far we are from that ultimate limit. If we are at 3dB from the ultimate performance, there's room for improvement: our effort can pay off. If we're 0.01dB from the limit, our work is done from an engineering perspective; even if we improve, we're really close to the theoretical limit.

When we approach this chapter, we know we have many different models; some are useful (e.g.: a model has gaussian variables as an input of the system). In specific cases we don't have gaussian values, though.

Important point is that we have very different models depending on what we're doing. In this chapter we also find practical stuff btw. When we talk about LTE and 5G, who defines what those are? **Standardization** is really complicated; internet standards is handled by the IETF (Internet Engineering Task Force): it establishes protocols. Other things standardized are how the wireless network is done and how different networks can communicate; what most interests us is the communication interface: there must be a precise standard to allow communication between different towers etc etc; this is called **radio interface**. There must be a document saying how we transmit every single bit; we want to have a competitive market, in which multiple brands can improve on the communication.

In the past, mobile phones were chosen depending on reception capabilities; now we have a field so high that it's very easy to receive signal.

We need to have specifications about sending bits from the central tower to a mobile device and vice versa. There is not a specification on how the mobile phone should be made, but rather only on how the phone should communicate.

If we know how bits are sent downlink, we might think to use ZF or even better, LMMSE; it must be specified how bits are sending data to the base stations. We can make base stations however we want: we're the ones deciding the compromise between cost and performance.

For cellular standards we have 3GPP, which is third generation ?? program. How can we participate in specification of 3GPP? We can't do that easily: it's made by several regional standard organizations; here in Europe we have ETSI (European Telecommunications Standard Institute). In Japan we have ARIB, ...Who is member of those partners can send standards to 3GPP; we can be members if we pay a fee, which depends on turnover of company, ...; in the 3GPP is dealing with radio access networks. How do they decide if we use OFDM or whatever?

3: why? 1: analog, 2: GSM, 3: after GSM, 4: LTE, 5G, 6G, ...

2G was specified by ETSI and adopted by other countries. For 3G they wanted to have global roaming; that's how 3GPP was formed. It was decided that 3G would use WCDMA (Wideband CDMA). Let's suppose we have different technologies. Today we have OTFS, which is a way to transmit bits; how do we decide in a group which of the various options we have is better? What happens is that we do theoretical simulations; in all of this exercise there is a critical point: which channel model do we use? Depending on the model, it might happen that a technology is better than another; an initial debate was in fact on which channel model should have been used. The first thing we need to do when developing standards is agreeing on the channel model.

In section 3.8 we'll find MIMO in standards.

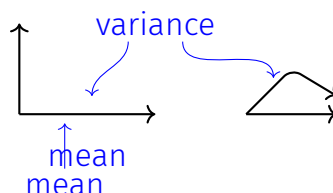
Different groups adapt different models; there are researches being made on identifying good and representative channel models.

Professor advises us to go through the chapter and then go back and redo it, since the picture will be much clearer. We cannot forget reality; at the same time, we're doing something inspired by reality.

11.4 Space selectivity

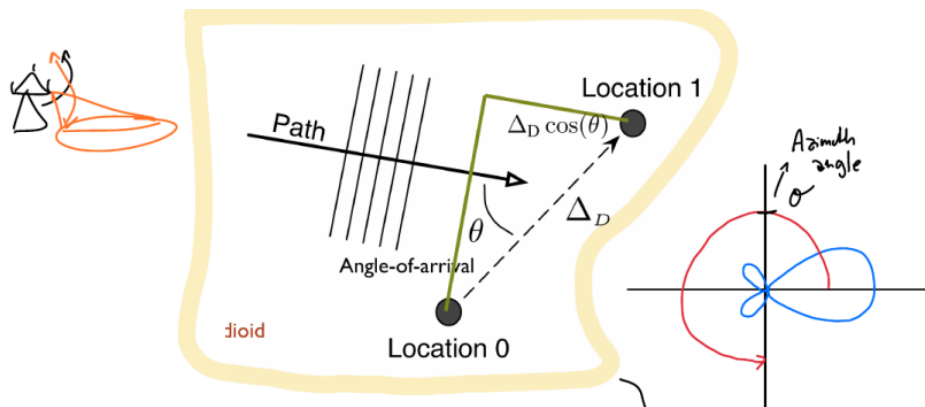
In class

We have a factor modulating the straight path and the other paths; the average of the sum is given by the straight, while the average of the power is given by considering all.

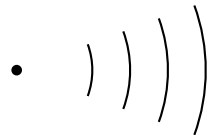


Space selectivity; we need somehow to have this. the two locations could be the

locations of the two antennas; let's say we have a wave which is hitting the two antennas:



We know that if we go far the curvature is almost null.



There is a difference of length at various points: this will also mean we have a difference in phase. This difference matters a lot; this means that **selectivity stays not in frequency, but in space**. In the previous drawing we might have a phase perfect and a non-in-phase signal: we can't just sum them, otherwise we'd have a problem. We might have scattering clusters, if we don't do anything. There are many reasons to have space selectivity. We also have a PAS, but we'll see it next week. We know the spectrum of the signal tells us how the power is distributed on the various frequencies. If we talk about Power Angle, we talk about distribution depending on the angulation of transmission. This is what PAS means.

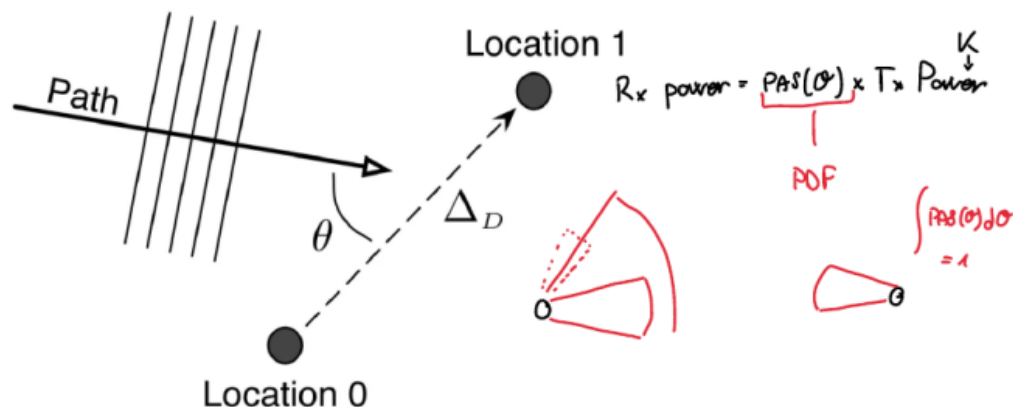
In the previous drawing we were in a 2-d setting; we live in a 3-d setting; we should consider not only the azimuth but also the elevation. Generally we don't have influence in the elevation, while we have a lot of influence if we change the azimuth.

11.5 PAS → Power Angle Spectrum

What's the PAS?

$$\begin{aligned} \text{PAS}(\theta) &\rightarrow \text{function} \\ \text{TX func} &\rightarrow \text{RX func} \\ \text{RX power} &= \text{PAS}(\theta) \times \text{Tx power} \end{aligned}$$

From the book: The **phase shifts** between different locations we referred to in equations such as (3.21), are governed, at least **in the local neighborhood**, by the angle between the propagation paths and the segment connecting those locations, as can be seen in the figure (which shows the segment connecting two locations with a propagation path at an angle θ):



The change of path length between the two is simple given by $\Delta_D \cos(\theta)$, aka by projecting the segment on the path direction.

Power Angle Spectrum (PAS)

Since the projection depends on θ , it is crucial to have a function which establishes the angular distribution of paths; this function is the PAS (Power Angle Spectrum), which expresses the received signal power as a function of angle when acting as a receiver; instead, when we're acting as a transmitter, instead, the PA expresses the transmitted signal power that reaches the receiver at the other end of the link as a function of the angle of departure.

Channel modeling (pp. 131-208), page 16

The PAS is a continuous function of angle, **normalized so it can be interpreted as a PDF**, which is a definition consistent with the unit-power normalization of the small-scale fading.

The standard deviation of the PAS, viewed as a distribution, gives the root-mean-square (RMS) **angle spread**.

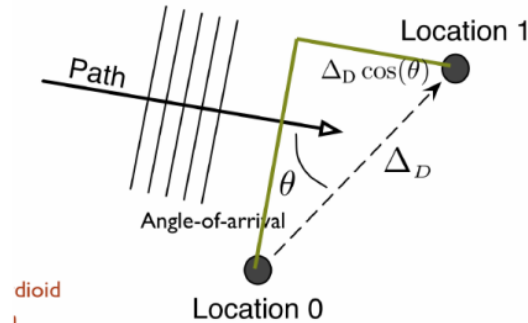
Channel modeling (pp. 131-208), page 16

Although, in principle, the definition of the PAS as a continuous function of angle models a diffuse propagation scenario, it does not preclude modeling **situations where**

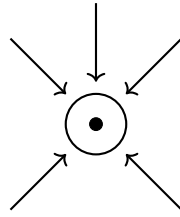
power is received only from a finite number of discrete angles; these situations can be accommodated by constructing the PAS on the basis of discrete delta functions. Formally, the PAS is a function of both azimuth and elevation. However, empirical measurements have indicated that in most outdoor wireless systems the dependence on elevation plays only a secondary role [224]. Historically then, the focus has been on characterizing the marginal PAS on the azimuth plane—in fact, PAS can also be taken to stand for power azimuth spectrum—yet the elevation dimension should not be carelessly dismissed in other types of deployment, e.g., indoors or in vertical urban picocells. We return to this aspect at the end of the chapter. Several functions have been proposed and applied to model the marginal PAS in azimuth, which in this text is denoted by $\mathcal{P}(\cdot)$.

In class:

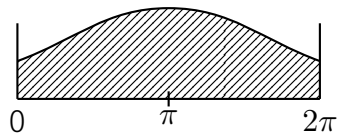
Last lesson, we were talking about space selectivity. We talk about it because we may have 2 (or more) antennas: a part of the signal might be traveling a larger distance; in our case, it is $\Delta_D \cos(\theta)$.



Normally, this is not so big that the strength of signal goes down, but the phase may change. If we apply a phase shift on location 1, it's like as if we made the signal **start late**. We can steer the wave with different angles by changing the phase of a certain amount. This is the intuition behind what happens when a certain wave is hitting the two antennas. If we have multiple antennas, we just apply the same principle. We can thus make the antenna sensitive to a certain direction: even if we have an omnidirectional antenna, the Power Angle Spectrum is modeling the fact that if we have an antenna, we might have waves coming from multiple directions



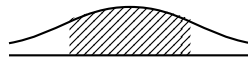
What do we do in such a situation? The PAS is a probability distribution which tells where it's most likely that the signal is arriving.



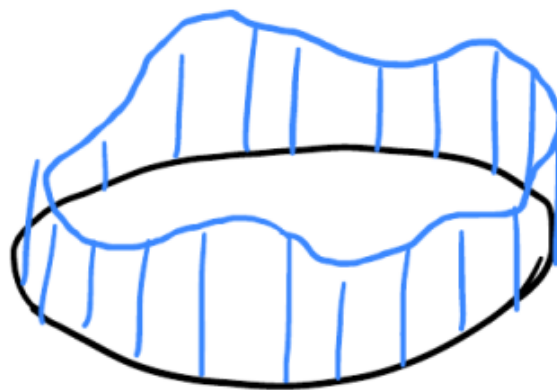
We don't need to confuse the word **spectrum** being used here: we're not talking about frequencies!

$$x(t) \rightarrow g(\cdot) \rightarrow y(t)$$

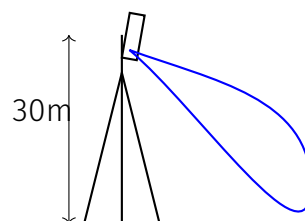
$$P_y(f) = |G(f)|^2 P_x(f)$$



This allows us to understand where we can capture more energy. The PAS tells where we have more power depending on the angle.

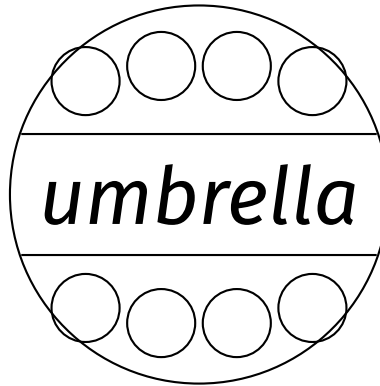


Antennas are generally 30m high and have a tilting on their top of a few degrees:



we need to decide the tilting depending on the application we want to have.

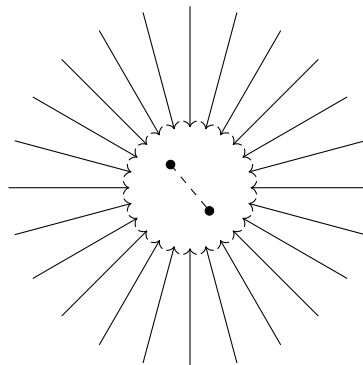
If we have a motor wave we have a central base station and then smaller others. If a network sees we're jumping, it will assign us to an umbrella base station: the probability that the calls drop is lower.



Clarke-Jakes PAS → take it as example.

We have various examples of PAS; Clarke-Jakes' is a uniform one:

Possible question: what is a PAS?



With

$$\mathcal{P}(\theta) = \frac{1}{2\pi} \quad \theta \in [-\pi, \pi) \quad (3.27)$$

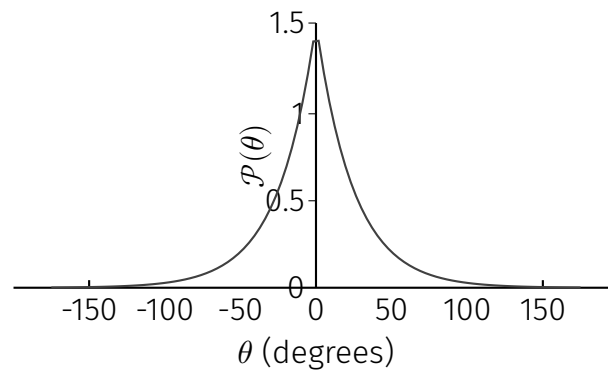
Channel modeling (pp. 131-208), page 16

It fits surprisingly well the behavior observed at mobile users, capturing the essence of multipath propagation in a manner that is friendly to analysis. Such friendliness, however, drops markedly as soon as the angular range is reduced below 2π and more convenient alternatives exist if directional preference (particularly important in transceivers located above the clutter, e.g., elevated base stations) is to be introduced.

The Clarke-Jakes PAS is widely used. There are other spectrums related to this one, but it shall not be confused with stuff about the delay.

11.6 Truncated Laplacian

We may use another $\mathcal{P}(\theta)$, the Truncated Laplacian.



With

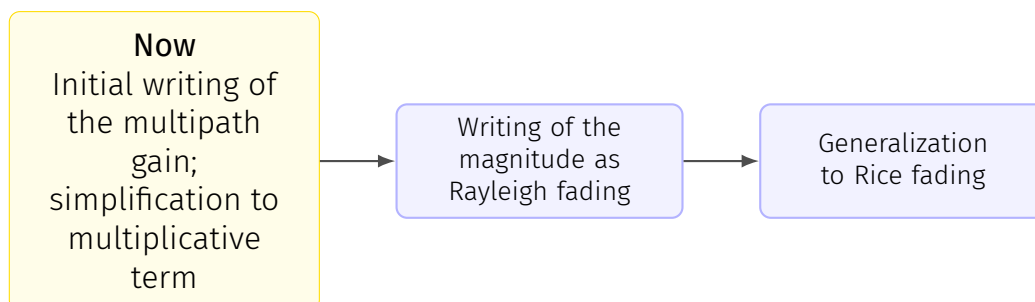
$$\mathcal{P}(\theta) = \frac{K_L}{\sqrt{2}\sigma_\theta} \exp\left(-\left|\frac{\sqrt{2}(\theta - \mu_\theta)}{\sigma_\theta}\right|\right), \quad \theta \in [\mu_\theta - \pi, \mu_\theta + \pi) \quad (3.32)$$

where K_L is a factor which makes the integral be 1, thus defined as

$$K_L = \frac{1}{1 - \exp(-\sqrt{2}\pi/\sigma_\theta)}. \quad (3.33)$$

11.7 Small-scale fading

11.7.1 Multipath propagation (+Rayleigh and Rice fading)



Channel modeling (pp. 131-208), page 13

Let us now zoom in and enter the small-scale realm. When dealing with

continuous-time representations, we use the complex **pseudo-baseband channel response**. We also recall from Section 2.2.3 that, **to discretize this representation**, lowpass filtering is in general required before sampling, although in those cases when the discretization returns a singletap response and the effect is only multiplicative, the complex gain of the pseudo-baseband channel can be borrowed directly.

🔍 My questions

Question: why we like to work in pseudo baseband?

Refers to: *pseudo-complex baseband* → (what is this?)

Due to the various NLOS effects, what the receiver observes is a superposition of a number of replicas of the transmitted signal; the overall channel response is a sum of all these various contributions.

Each path can be described using the attenuation A_q and the delay τ_q , under the form

$$A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q} .$$

Chapter 12

Lesson 10-12-2025

12.1 Space correlation

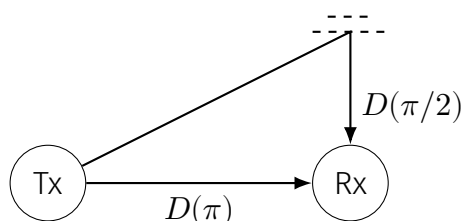
12.1.1 About space correlation (dependance of y on θ , arrival angle); specialization for when x is sinusoidal

Let's suppose the **transmitter is fixed**. Let's study the fading dynamics as a function of the position of the receiver: in other words, let's see what happens if the receiver moves during the transmission.

 **Channel modeling (pp. 131-208), page 19**

The simultaneous dependence on the positions of both transmitter and receiver, essential for MIMO channel modeling, is deferred to later in the chapter.

We transmit a signal x which experiences a delay depending on $D(\theta)$, difference in path length depending on the angle: the experienced delay will thus be $D(\theta)/c$.



Simple illustration showing what we mean by $D(\theta)$.

In this context in which the angle is important, we don't have an omnidirectional pattern: it's more sensitive in certain angles and less in others: this is characterized through $\mathcal{P}(\theta)$, the **PAS**; $G_r(\theta)$ is the directional gain.

We'll thus have that the **baseband noiseless signal at a given location can be expressed as**

$$y(t) = \sqrt{G} \int_{-\pi}^{\pi} \sqrt{\mathcal{P}(\theta) G_r(\theta)} x(t - D(\theta)/c) d\theta. \quad (3.34)$$

 Handwritten notes on (3.34):

- On $G_r(\theta)$: antenna gain is included in G , the pattern is normalized.
- On $D(\theta)/c$: difference in path length as a function of angle.

If we have that $x(t)$ is a sinusoid (expressed as $xe^{-j\phi(t)}$ with x fixed in time), in base-band it will be constant (i.e.: we know that the baseband representation of a signal is obtained by centering its oscillation frequency around zero; if we do so with $e^{-j\omega t}$, we're sure to obtain a constant term), of phase given by the delay $\phi(\theta) = 2\pi D(\theta)/\lambda_c$ (more about this below):

$$y = \sqrt{G} \int_{-\pi}^{\pi} \sqrt{\mathcal{P}(\theta)G_r(\theta)} x e^{-j\phi(\theta)} d\theta \quad (3.35)$$

$$= \sqrt{G} x \underbrace{\int_{-\pi}^{\pi} \sqrt{\mathcal{P}(\theta)G_r(\theta)} e^{-j\phi(\theta)} d\theta}_{h_0} \quad (3.36)$$

$$= \sqrt{G} x h_0.$$

it is a constant, 1! why?

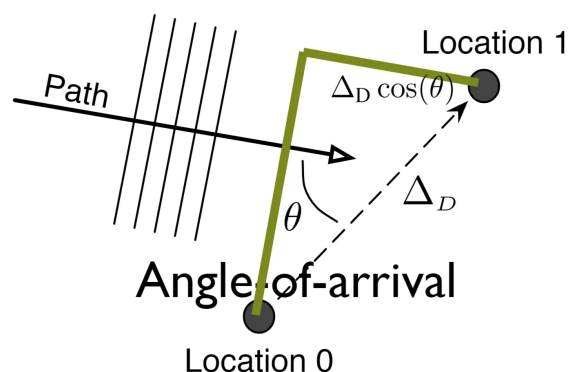
Note: $e^{-j2\pi f_c t}$ baseband equivalent

All this turns out to be a constant, hence why we call it h_0 ; this h_0 depends on many things. We don't know $\mathcal{P}(\theta)$, but it is reasonable to say that:

- if we have enough paths which are independent in phase
- and if the PAS can be modeled as uniform

this enables us to treat h_0 as a complex Gaussian variable.

Let's say we're now not at location 0, but at location 1 (a setting of distance Δ_D from location 0 we considered from (3.34) to (3.36)), with reference to the image below.



We want to find the correlation between the two received signals; we'll have (for a sinusoidal signal)

$$h_1 = \int_{-\pi}^{\pi} \sqrt{\mathcal{P}(\theta)G_r(\theta)} e^{-j[\phi(\theta)+2\pi\Delta_D \cos(\theta)/\lambda_c]} d\theta. \quad (3.38)$$

💡 About the phase shift

We know the space difference between two arrival points from the point of view of a plane wave is of $\Delta_D \cos(\theta)$; we also know that the (baseband) em waves have a dependance on time through a term $e^{-j\omega t}$.

Our delay will be equal to $\Delta_D \cos(\theta)/c$, rendering the exponent as

$$\exp\left(-j\frac{\omega_c \Delta_D \cos(\theta)}{c}\right) = \exp\left(-j\frac{2\pi c \Delta_D \cos(\theta)}{c\lambda_c}\right),$$

ultimately giving what we have in (3.38), aka a phase shift of

$$2\pi\Delta_D \cos(\theta)/\lambda_c.$$

We can make our calculations to **find the correlation** (see 08.3 Channel modeling (pp. 131-208), page 19): we want to find it because we don't want to collect replicas of the signal; we want to have signals being collected independently so that we can process them and enjoy their independence. If they are completely correlated, if one goes down the other will as well. If they're completely uncorrelated, if one goes down we can receive the signal at another receiver.

In the end we find that **if the phase at each angle is uniformly distributed** ($\theta \sim \mathcal{U}[-\pi, \pi)$) s.t. the expectation is nonzero only for $\theta_0 = \theta_1$), the correlation will be

$$R_h(\Delta_D) = \mathbb{E}[h_0 h_1^*] = \int_{-\pi}^{\pi} \underbrace{\mathcal{P}(\theta)G_r(\theta)}_{\text{to be known}} e^{j2\pi\Delta_D \cos(\theta)/\lambda_c} d\theta. \quad (3.41)$$

remember the result...

The antenna pattern $G_r(\cdot)$ can in principle be designed to modify the environment's PAS, and thus $R_h(\cdot)$, at the expense of failing to capture all the incoming power [239, 240]. However, the **habitual design principle** is to apply an **antenna pattern that is broader than the PAS, and approximately flat** thereupon, so as to essentially capture all of the power. If the PAS is unknown or the antenna cannot be properly pointed, as is often the case, a uniform pattern is then preferable and (3.41) simplifies to

$$R_h(\Delta_D) = \int_{-\pi}^{\pi} \mathcal{P}(\theta) e^{j2\pi\Delta_D \cos(\theta)/\lambda_c} d\theta. \quad (3.42)$$

💀 How did we get rid of the double integral?

I think that in the end we got rid of the integral mainly by assuming that h_0, h_1 are independent of one another if their respective signals were received from different angles; for this reason, they were only considered for $\theta_0 = \theta_1 = \theta$.

Btw this last paragraph was from the book

12.2 Space correlation function for Clarke-Jakes PAS

For Clarke-Jakes PAS

With

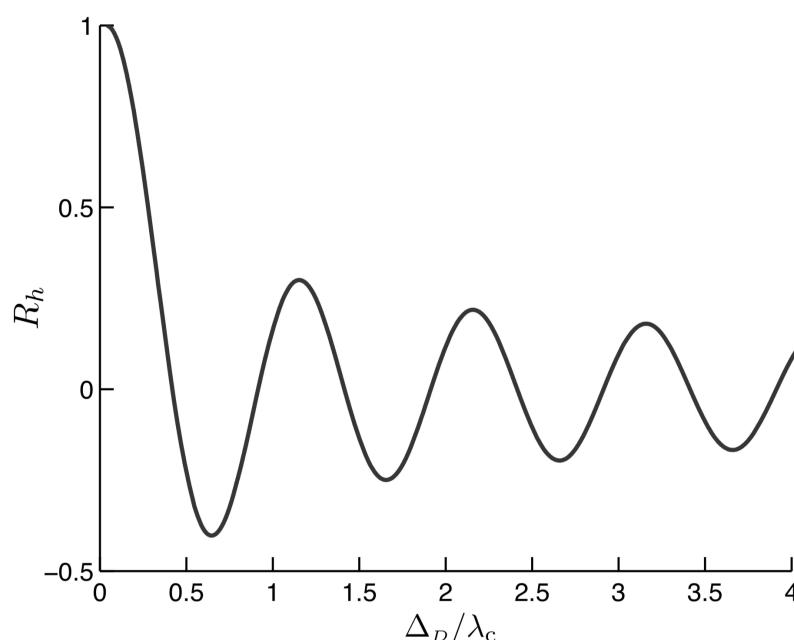
$$\mathcal{P}(\theta) = \frac{1}{2\pi} \quad \theta \in [-\pi, \pi) \quad (3.27)$$

, integral (3.42) gives

$$R_h(\Delta_D) = J_0(2\pi\Delta_D/\lambda_c), \quad (3.43)$$

where $J_0(\cdot)$ is the zero-order Bessel function of the first kind.

The graph of such space correlation is



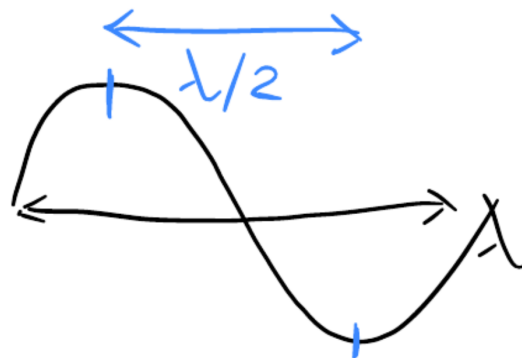
We see we cross the 0 multiple times and also that at a certain point we'll converge to 0, aka uncorrelation; this is a Bessel function of order 0. The fact that **the first crossing of the zero happens for $D_c = 0.38\lambda_c$ is important**: it tells us how far we need to space the antennas in order to receive zero-correlation.

Coherence distance

The mentioned value D_c is the so-called **coherence distance**: it is a distance Δ_D such that $R_h(\Delta_D) = 0$; generally the value of the first crossing of 0 is considered. This is of crucial importance when evaluating a channel, as it expresses how spaced apart two antennas should be in order for the received signals to be uncorrelated.

For the Clarke-Jakes, we have $D_c = 0.38\lambda_c$.

This is an idealized case, though. The rule of thumb is that we need to be **at least half a wavelength far away from one another**. If we want to use MIMO, we should try to have this situation; this translates to the fact that we want to have a full-rank matrix. If we want to have independent antennas, we need to space them of half a wavelength. Really rough intuition:



If one antenna receives the bottom, the other will receive the top.

Other space correlations

There are other already-calculated space correlations, see [Channel modeling \(pp. 131-208\)](#), page 21.

12.3 Time selectivity

12.3.1 Time correlation, coherence time

What about time selectivity? We consider the case in which we have an antenna and we're moving towards the antenna or in some other direction. We're moving and we pick various samples; we ask ourselves: are the two samples we pick in different moments correlated? Of course this depends on the velocity: depending on how fast we're going, how correlated are the signals?

Of course we have the Doppler effect. The delay between antennas is $\Delta_t = \Delta_D/v$. We find

$$R_h(\Delta_t) = \int_{-\pi}^{\pi} \mathcal{P}(\theta) e^{j2\pi f \Delta_t \cos(\theta)/c} d\theta, \quad (3.46)$$

with slight abuse of the notation $R_h(\cdot)$.

Relation with space correlation

If we consider

$$R_h(\Delta_D) = \int_{-\pi}^{\pi} \mathcal{P}(\theta) e^{j2\pi \Delta_D \cos(\theta)/\lambda_c} d\theta. \quad (3.42)$$

we clearly see that (3.46) is the same exact equation, just considering that:

- $\Delta_D = \Delta_t v$
- $1/\lambda_c = f/c$.

Coherence time

Just like we defined $\Delta_t = \Delta_D/v$, with reference to the [definition of coherence distance \$D_c\$](#) we define the coherence time (the delay which should occur in order for the two replicas of a signal to be uncorrelated) as

$$T_c = \frac{D_c}{v}. \quad (3.47+)$$

From what can be gathered from exercise [3.13](#), it looks like **in a time and frequency selective channel**, we can consider the coherence time as scaled by

$$T'_c = T_c B_c,$$

with B_c being the coherence bandwidth.

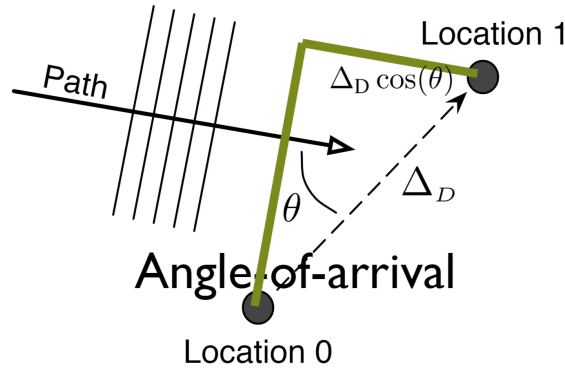
It's clear that this makes no sense from a unit pov, but this does not stop the book from doing it!

12.4 Doppler spectrum

As the receiver travels a distance Δ_D , the length of a path at an angle θ changes by $\Delta_D \cos(\theta)$,

This rotates the phase of a signal at frequency f_c by

$$2\pi \Delta_D \cos(\theta)/\lambda_c \stackrel{\lambda_c=c/f_c}{\stackrel{\Delta_D=\Delta_t v}{=}} 2\pi f_c \Delta_t v \cos(\theta)/c, \quad (*)$$



which is not different from **what we already considered** (with the exception that now we have a moving receiver).

Since the time incurred to travel Δ_D is precisely Δ_t , this phase rotation corresponds to a frequency $f_c v \cos(\theta)/c$. In other words, a signal of frequency f_c arriving over a path at angle θ relative to the direction of motion **suffers a frequency increase** (which can be negative, hence a decrease) **equal to** $\nu = f_c v \cos(\theta)/c$. This frequency shift, a mere manifestation of the Doppler effect, is within the range $[-\nu_M, \nu_M]$, where

$$\nu_M = \frac{v}{c} f_c \quad (3.48)$$

is aptly termed the **maximum Doppler shift**. Since every pair of paths at angles $\pm\theta$ map to a unique Doppler shift, (3.46) can be rewritten as an integral over Doppler shifts rather than angles.

💡 What are f_c, v ?

v is the speed at which the antenna is moving, while f_c is the carrier frequency.

When we have a random process, we have correlation. Let's say we have a stationary random process; by looking at its Fourier transform, we know where the power of the process will be located in the frequency domain; here we do the same. We use ν ; this is because this is the discrete time Fourier transform! This means we're looking at the correlation in the frequency domain. What is happening is that we have a **zero outside of a certain domain** (no power outside from a certain domain): this comes from the fact that we have a bound on the speed (the speed bounds the frequency shift caused by the doppler effect). Specifically, we'll have that the correlation $R_h(\Delta_t)$ will be the Fourier antitransform of

$$S_h(\nu) = \begin{cases} \frac{\mathcal{P}(-\arccos(\nu/\nu_M)) + \mathcal{P}(\arccos(\nu/\nu_M))}{\sqrt{\nu_M^2 - \nu^2}} & \nu \in [-\nu_M, \nu_M] \\ 0 & \nu \notin [-\nu_M, \nu_M] \end{cases} \quad (3.53)$$

💡 Insight!

The idea is that ν/ν_M should quantify the shift in a scale $\in [0, 1]$, with 1 being the 2π shift; we then find the radians representation of this by using $\arccos$ and then compute $\mathcal{P}$ for the obtained value. We finally normalize over $\sqrt{\nu_M^2 - \nu^2}$.

✎ Doppler spectrum

The doppler spectrum $S_h(\nu)$ for $h(\tau)$ is defined as

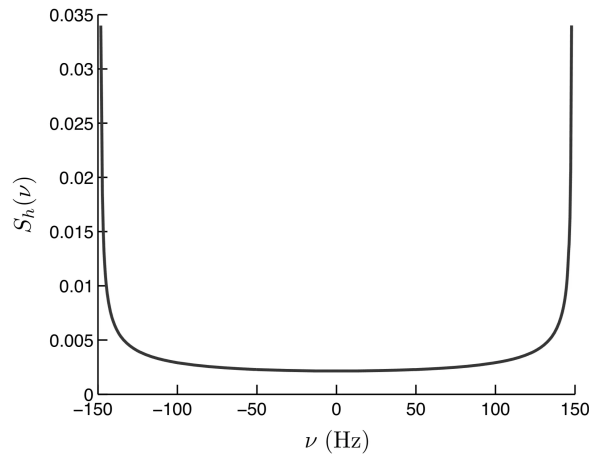
$$S_h(\nu) = \mathcal{F}[R_h(\Delta_t)] = \int_{-\infty}^{+\infty} R_h(\Delta_t) e^{-j2\pi\nu\Delta_t} dx.$$

This is a sort of re-interpretation of (3.51)

Doppler spectrum can be interpreted as follows: a transmit sinusoid of frequency f_c gets smeared into received components at every frequency within $[f_c - \nu_M, f_c + \nu_M]$ as indicated by $S_h(\nu)$. As can be verified, we have

$$\int_{-\pi}^{\pi} \mathcal{P}(\theta) d\theta = \int_{-\nu_M}^{\nu_M} S_h(\nu) d\nu. \quad (3.54)$$

Professor does not want us to remember (3.53). Instead, we need to remember the shape of this value, which is the **Clarke-Jakes Doppler spectrum**:



This is for $\nu_M = 148\text{GHz}$, which (for example) corresponds to $v = 80\text{km/h}$ at $f_c = 2\text{GHz}$.

It goes to ∞ at the extremis. If we're at 0, we have little energy; this means that the process is low energy in that part: this means that it's stable, it does not move. We have more energy towards higher spectrum, higher Doppler. **That's where we have more chance to have higher independence.**

The higher the frequency, the lower the correlation. If we don't move, the signal will be the same when we start and when we end. If we're really fast, the signal will be

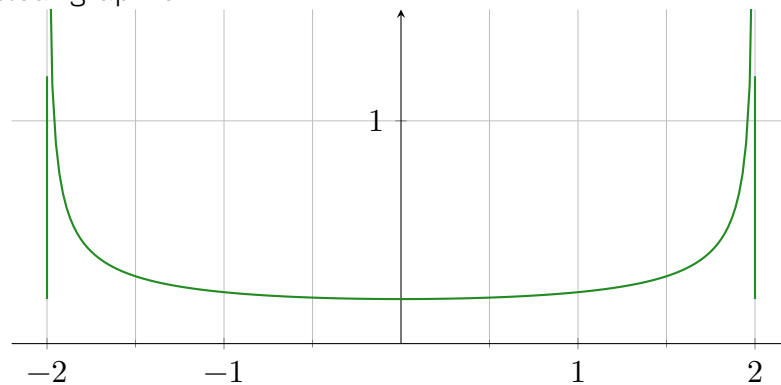
really different.

☠ Clarke-Jakes PAS

It's really not that hard to find the Doppler Spectrum for the Clarke-Jakes PAS. From (3.53), we just have

$$S_h(\nu) = \begin{cases} \frac{\frac{1}{2\pi} + \frac{1}{2\pi}}{\sqrt{\nu_M^2 - \nu^2}} & \nu \in [-\nu_M, \nu_M] \\ 0 & \text{Otherwise} \end{cases}.$$

This has a clear graph of



Which here is plotted for $\nu_M = 2$.

This depends on signal and carrier; when we have a low carrier, if we move 1m nothing changes, because the carrier is having a long wavelength.

Here we're considering **nonregular fading**, which:

- Comes from the assumption that a signal can bounce only once from an object
- Makes it so that $S_h(\cdot)$ has a bandlimited nature which allows us to predict the present value in a deterministic manner, given that noiseless observations of the whole past are available.

If we get rid of the **one-bounce** assumption, then we get doppler shifts larger than ν_M and that the present value cannot be predicted from observations of past values. This is called **regular fading**.

Unless otherwise stated, we consider nonregular fading bandlimited to ν_M . While Doppler shifts exceeding this value are possible, the corresponding signal components are likely to be very weak relative to those within $[-\nu_M, \nu_M]$ and hence, barring exceedingly high SNRs, those components beyond $\pm\nu_M$ are likely to be buried well below the noise level.

12.5 Time discretization: time selectivity applied to discrete transmission

For fading conforming to $h(t, \tau) = h(t)\delta(\tau)$ as we are considering thus far, the channel has the multiplicative effect $y(t) = \sqrt{G}h(t)x(t)$ and, as we learned in Example 2.3, the sampled pseudo-baseband gain $h[n] = h(nT)$ suffices for a discrete-time representation; we can directly write $y[n] = \sqrt{G}h[n]x[n]$. Strictly speaking, the time-discretization of $y(t)$ should entail a sampling period $\frac{1}{B+\nu_M}$ because the bandwidth B of the transmit signal increases to $B + \nu_M$ as **the signal undergoes time-varying fading**.

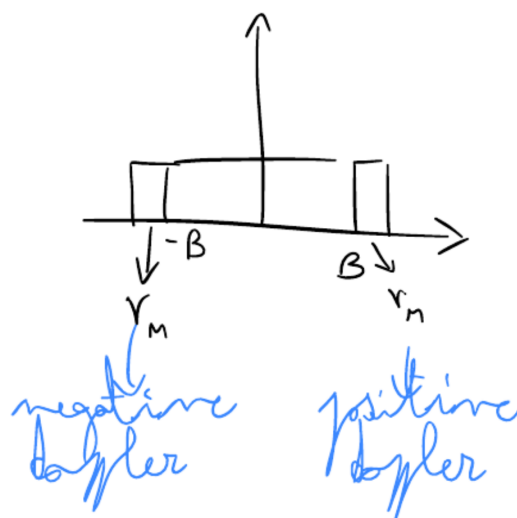
💀 About the enlargement of bandwidth

From what I understand, the bandwidth of $[-\nu_M, \nu_M]$ only holds for a complex exponential; if we have a real signal of bandwidth B , we'll need to have ν_M before and after that bandwidth, as we can consider each slice of band as a complex exponential.

Up to now, we've been in the continuous time domain. If we want to go to the discrete time domain, we want to sample the signal. We have a signal which no longer could very well not be an exponential at this point; it has a certain bandwidth which we'll sample with the inverse of that bandwidth. This is not mathematically complete. We might actually have a bandwidth which goes

from $B + \nu_M$ to $-B - \nu_M$, with

$$\nu_M = \frac{v}{c} f_c \quad (3.48)$$



We can have another approximation, aka that $\nu_M \ll B$: if they were comparable, we should sample at a higher sampling rate. Even if ν_M is very small, we still violate the Nyquist principle; point is that if we actually have $\nu_M \ll B$, we'll have a small loss

even if we sample with $B = 1/T$. Also, normally signals don't really have a specific spectrum; if we have a specific spectrum, we have something more in the lines of a rcos and such. We should not forget $\nu_M \ll B$ because of the concept; **what's really important is to have a Time-varying flat-fading model**. If we take a large B but the channel has many taps (delays), then we'll have a frequency-selective channel.

Since $h[n] = h(nT)$, we'll equivalently have that **the autocorrelation of $h[n]$ equals $R_h(\ell T)$, where $R_h(\tau)$ is the autocorrelation of $h(t)$** .

Given $S_h(\nu)$ (aka the Doppler spectrum of $h(t)$), the Doppler spectrum of $h[n]$ equals

$$\frac{1}{T} S_h \left(\frac{\nu'}{T} \right), \quad \nu' \in [-\nu_M T, \nu_M T] \quad (3.60)$$

with the factor $1/T$ ensuring that the scaling of the frequency axis ν to $\nu' = \nu T$ does not affect the power.

The book uses a different notation, but I can't seem to find the specific font setting to have that ν .

The book is looking at the different aspects one at a time; now it looks at how flat channels are correlated in time and then correlation in frequency. At the end of the day, all of this converges together. When we send $x[n]$, we have a different $h[n]$: this is a problem if we need to estimate it!

If every symbol has a different channel, it's impossible to communicate; or this is not the case, otherwise we wouldn't be able to transmit.

There are models doing this using AI or neural networks.

It might also be relevant to know that for a frequency-flat channel (which might very well hold also for just a complex exponential in whichever channel), we have

$$T_c \approx \frac{1}{2\nu_M} \quad (3.59)$$

12.6 Simplified models

12.6.1 Why do we need simplified models?

As mentioned at the outset of the chapter, the art of channel modeling must seek a **compromise between realism and tractability**, distilling what is essential for each purpose. When it comes to fading dynamics, it is sometimes desirable to have models that capture the essence of time selectivity but are otherwise stripped down to the bones. We next introduce two such models, formulated directly for the discrete-time complex channel

12.7 Gauss-Markov model

A first possibility is to resort to a first-order Gauss-Markov process, or first-order autoregressive process, whereby the fading at symbol n satisfies

$$h[n] = \sqrt{1 - \epsilon} h[n - 1] + \sqrt{\epsilon} w[n], \quad (3.61)$$

where $\{w[n]\}$ is a sequence of IID random variables with $w \sim \mathcal{N}_{\mathbb{C}}(0, 1)$.

In turn, ϵ is the one-step prediction error, i.e. the variance of the error to which $h[n]$ can be predicted from a noiseless observation of $h[n - 1]$, something that in a first-order Markov process is as informative as a noiseless observation of the entire past. By definition, a Gauss-Markov process is regular and its degree of decorrelation after ℓ symbols is determined by ϵ via

$$R_h[\ell] = \mathbb{E}[h[n] h^*[n + \ell]] \quad (3.62)$$

$$= (1 - \epsilon)^{\ell/2}. \quad (3.63)$$

What people sometimes do is to say *the next channel is the previous channel with a factor $\sqrt{1 - \epsilon}$ and since we want to make it random, we add a bit of noise, $w[n]$* . This is a random path: the channel is changing but not too much, because in the ingredients we have a predictable part and also some noise. This is called Gauss-Markov model. The correlation here is $(1 - \epsilon)^{\ell/2}$, where ℓ is the distance between the sampling of the channel. If we have $\epsilon \rightarrow 0$, the channel is completely stable.

12.8 Block-fading model

Although professor is a madlad and might ask this no matter what, this is not in the course programme!

A second possibility, illustrated in Fig. 3.12, is to resort to a block-fading structure, whereby the **fading remains constant over blocks of duration T_c** while changing values across blocks. It is worth noting that this model is not stationary but rather **cyclostationary with period T_c** . The fading values at consecutive blocks may be correlated, but are most often assumed IID.

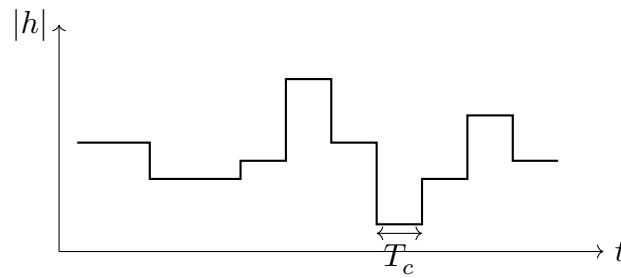


Figure 3.12

If N_c is the number of single-carrier symbols (where the single-carrier bit is specified only to allow us to apply the results which we saw for single carrier cases) we have per block, we have that

$$N_c = \frac{T_c}{T} \quad (3.65)$$

$$= BT_c \quad (3.66)$$

and, if the blocks are IID, then it makes sense to set T_c to coincide with the **coherence time of a stationary process**. Recalling (3.59), which says that $T_c = 1/2\nu_M$, this leads to

$$N_c = \frac{1}{2\nu_M T}. \quad (3.67)$$

However, care must be exercised with both (3.66) and (3.67) because they **hold only if the bandwidth B is "not too large"** where the precise meaning of "not too large" is to become clear in the next section.

They are most likely referring to the coherence bandwidth concept.

The block-fading model can be broadened to allow for continuous variability within each block, with transitions to independent values across blocks. This more general blockfading model can sometimes serve as a convenient bridge with a true continuous fading model [245, 246].

We have jumps in time: we know that the signal has infinite bandwidth. We now go and look at how the frequency response of the channel is changing. This is called **block fading model**: at every symbol it looks like we have fading, a jump. We know this does not actually happen, in fact this is a model.

This is not a coherent theory, it's just an excursus of various channel models.

Yesterday we stopped at the block fading channel. We've seen this is a simplified model which is quite used. The problem is setting ϵ . Clearly deciding what to set it to is a problem.

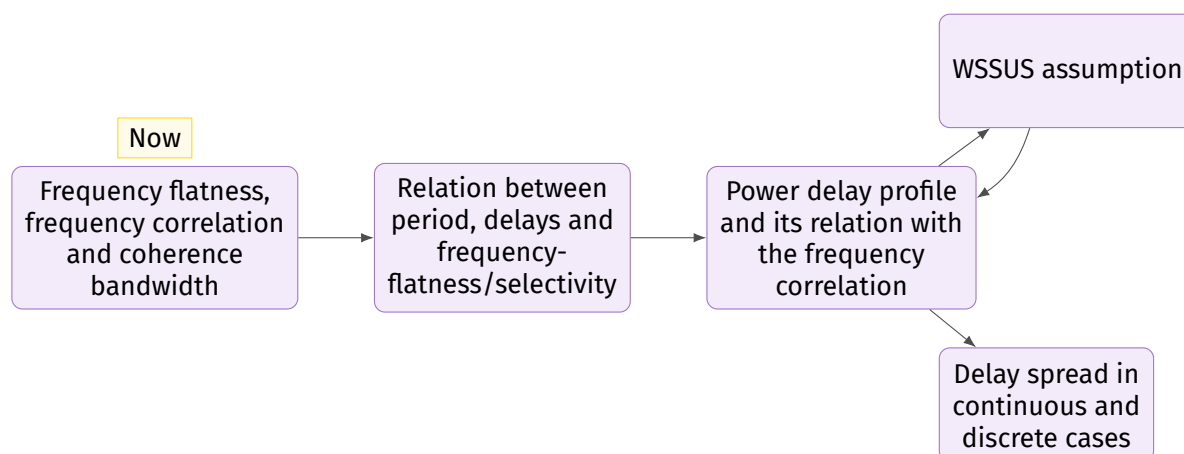
Chapter 13

Lesson 12-12-2025

13.1 Frequency selectivity

12/12/2025

13.1.1 Important parameters (R_h , T_d , PDP) and WSSUS assumption {3.4.2}



Now we'll talk about the coherence bandwidth B_c ; in order to do so, we define R_h which looks kinda like a correlation.

In our characterization of space and time selectivity, we have focused on the fading experienced by a sinusoid (both for what concerns the influence of **space correlation** and, although implicitly, the influence of **time correlation**) at frequency f_c . The derivations hold approximately for frequencies close enough to f_c , and thus the fading goes (roughly) unchanged over a certain frequency span. **If the entire signal fits within a swath of frequencies that experience (roughly) the same fading, the fading is frequency-flat**; the channel is then multiplicative, i.e., it merely applies a complex gain. The signal can also be said to be narrowband, although such narrowness must be understood only in reference to the fading.

If the signal bandwidth is large enough, however, it is bound to exceed the range of frequencies experiencing (roughly) the same fading. At a given location then, in static conditions, the noiseless received signal is $y(f) = \sqrt{G}h(f)x(f)$ where $h(f)$ is no longer constant over the bandwidth B . **The signal undergoes frequency-selective fading**, and it can be said to be fading-wise wideband; the channel is not merely multiplicative, but rather it effects a nontrivial convolution that modifies the shape of the symbol pulses. The degree of selectivity over a frequency range Δ_f is quantified by

$$R_h(\Delta_f) = \mathbb{E}[h(f)h^*(f + \Delta_f)] \quad (13.1)$$

Note: what is implied in this formula? There is a dependency only in Δf , but it should be on f and Δf (depends only on Δf).

whose chief measure is the **coherence bandwidth** B_c , the natural counterpart to the coherence time and the coherence distance. **The fading is frequency-flat if $B \ll B_c$.**

Handwritten Note

For most channels this is true in general, $e^{j2\pi f_0 t^2} \xrightarrow{\text{der}} \frac{1}{2\pi} \frac{d}{dt}(2\pi f_0 t^2) \Rightarrow f_0 t \rightarrow f$

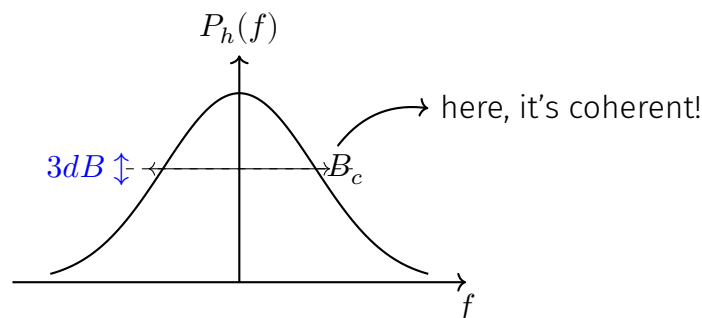
We ask ourselves the question: if we take a frequency and another frequency separated of Δ_f from the first one, which is the difference in terms of the channel? We can see that **the function R_h does not depend on the frequency** (in fact, it's expressed as $R_h(\Delta_f)$ and not as $R_h(\Delta_f, f)$);

this is not always true: it's like when we show that the correlation of the stationary process only depends on the difference over time.

Depending on what the shape of $h(f)$ is, we can define what the coherence bandwidth is.

Possible definition of Coherence Bandwidth B_c

The coherence bandwidth is sloppy as a concept; just to give an example of definition:



we might decide it's 3dB from the top.

The **coherence bandwidth** is that bandwidth in which the channel stays almost constant; this is extremely important when we do OFDM, since that tells us where we can position a pilot without having a frequency-dependent distortion.

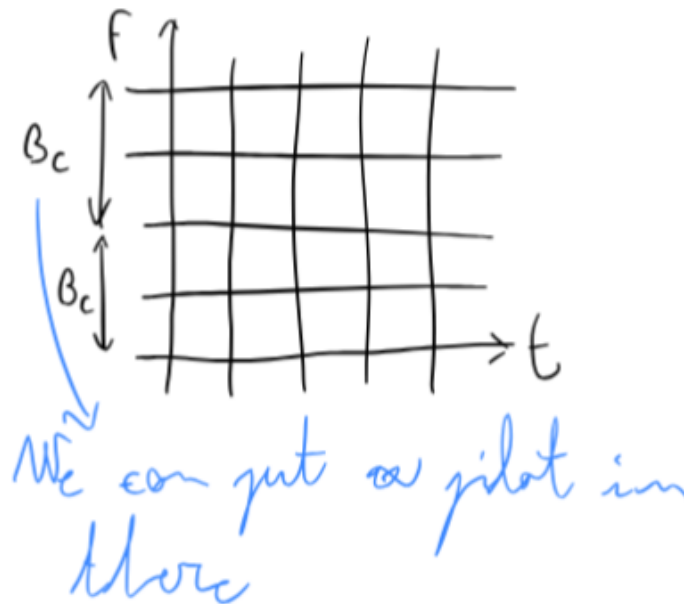
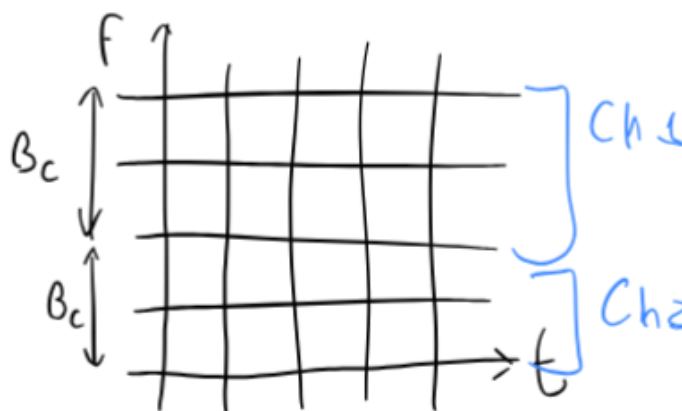


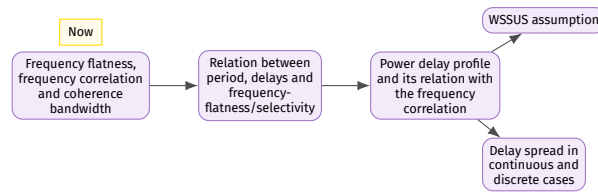
Figure 13.1: Coherence bandwidth

We can put the pilots spaced of not more than the coherence bandwidth; if we do more, the channel in the middle can change. This is definitely important; citing the slides: "Coherence bandwidth B_c determines the bandwidths where the channel should be considered frequency selective".

This means that somehow in the previous example we have 2 different channels which are **frequency selective**:



Normally, the **time-invariant impulse response corresponding to $h(f)$ is $h(\tau) = h\delta(\tau)$, with h being a complex gain at a given position.** Of course, if we were to introduce motion, it'd become time-dependent, giving $h(t, \tau) = h(t)\delta(\tau)$.

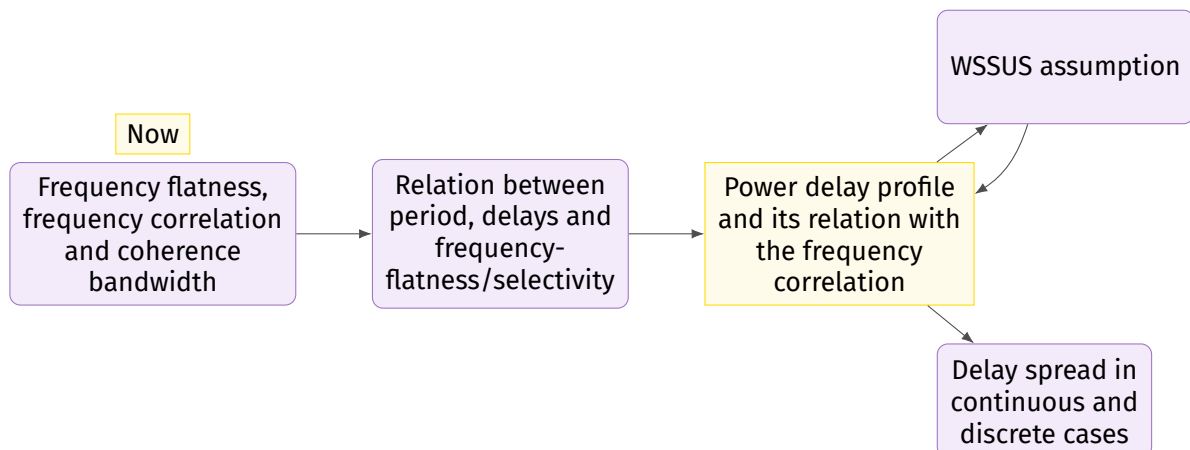


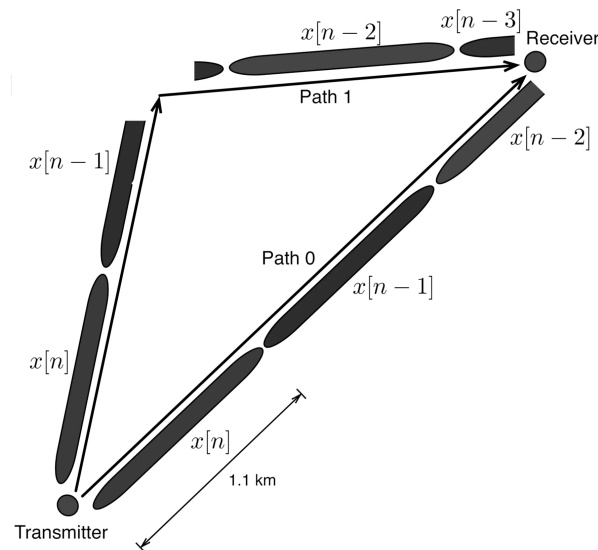
If we consider a **motionless channel** for which the minimum and maximum delays are $\tau_{\min}, \tau_{\max}$, we'd then have that if $\tau_{\max} - \tau_{\min} \ll T$ (with T being the symbol period), then we'd have that the channel can still be modeled as $h(\tau) = h\delta(\tau)$ with little error. If instead the condition is not satisfied, there is collision between the symbol n from a certain path and the symbols $\neq n$ from other paths, causing not only fading but also ISI.

To be sure, any form of fading is the result of self-interference among multi-path signal replicas of each symbol; the difference between frequency-flat and frequency-selective fading is that, in the former, every symbol essentially interferes exclusively with replicas of itself, whereas **in the latter it experiences significant interference with adjacent symbols as well**. The same exact channel can exhibit either frequency-flat or frequency-selective fading depending on T or, equivalently, depending on the bandwidth through which it is observed.

Ultimately, the delay-dispersive nature of fading channels, and thus the frequency selectivity, is rooted in the fact that the speed of light is finite. To appreciate this, the reader is referred to Fig. 3.14, which depicts a transmit sequence of symbols traveling over two distinct propagation paths. The finite value of c bestows upon the symbol separation a finite length in space and opens the door to distinct symbols colliding at the receiver if the difference in length between the paths is sufficiently large. In contrast, if c were infinite, then the symbol sequences received over the multiple paths would always align. The foregoing interpretation can be further stretched to motivate the use of OFDM: since ISI arises because of the finite length of the symbols, **the antidote to ISI is to make the symbols longer**.

Symbol sequence traveling over two distinct propagation paths with lengths differing by substantially more than the symbol period. The sequences received over the two paths are staggered by about half a period, which for $T = 3.7 \mu\text{s}$ would happen if the path lengths differed by about 500 m.





Symbol sequence traveling over two distinct propagation paths with lengths differing by substantially more than the symbol period. The sequences received over the two paths are staggered by about half a period, which for $T = 3.7 \mu\text{s}$ would happen if the path lengths differed by about 500 m.

We can write down the **power delay profile (PDP)**, which **quantifies the energy distributed as a function of delay in the time domain**; it is normalized so that it can be interpreted as a PDF (describing how likely we are to have a certain amount of power at a certain delay) and it is denoted by

$$S_h(\tau) = \mathbb{E}[|h(\tau)|^2]. \quad (3.68)$$

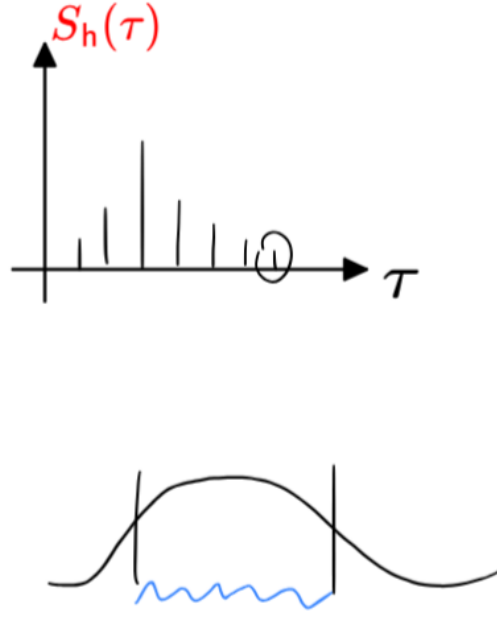
Although in general a continuous function of delay, discrete PDP descriptions are also possible, and in fact fairly common. In our formulation, these are readily accommodated by means of delay-shifted delta functions, each modeling a group of paths—a ray—with similar delays. Because of the plurality of paths per ray, each ray is still subject to fading.

For a certain τ , we measure the expectation of the amplitude of the channel; if we have a **ray-based channel** (we have group of paths with similar delays), we'll get a **channel given by delayed deltas**:

💡 What does that graph mean?

I think the intuition is that a certain delay between two signals defines a certain bandwidth; if we know the average of the signal in that certain bandwidth, given by $S_h(\tau)$, we can draw some conclusions about the bandwidths at which we should transmit.

This might allow us to put the various signals only in a certain part of the channel.



The two quantities R_h and S_h are related through a transform. First we have that

$$R_h(\Delta_f) = \mathbb{E}[h(f)h^*(f + \Delta_f)] \quad (3.73)$$

$$= \mathbb{E} \left[\left(\int_{-\infty}^{+\infty} h(\tau_0) e^{-j2\pi f \tau_0} d\tau_0 \right) \left(\int_{-\infty}^{+\infty} h^*(\tau_1) e^{j2\pi (f + \Delta_f) \tau_1} d\tau_1 \right) \right] \quad (3.74)$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mathbb{E}[h(\tau_0)h^*(\tau_1)] e^{j2\pi [(f + \Delta_f) \tau_1 - f \tau_0]} d\tau_0 d\tau_1 \quad (3.75)$$

but if we now consider a peculiar case (which is the **WSSUS case**, see below), aka that for which

$$\mathbb{E}[h(\tau_0)h^*(\tau_1)] = 0 \quad \text{if } \tau_0 \neq \tau_1,$$

we can now consider the integral only for $\tau = \tau_0 = \tau_1$, seeing that it reduces to

$$R_h(\Delta_f) = \int_{-\infty}^{+\infty} \mathbb{E}[|h(\tau)|^2] e^{j2\pi \tau \Delta_f} d\tau \quad (3.76)$$

$$= \int_{-\infty}^{+\infty} S_h(\tau) e^{j2\pi \tau \Delta_f} d\tau. \quad (3.77)$$

The two are related with one another (the correlation over frequency differences is the **IFT of the power delay profile**)!

🧠 Elaborating further...

The idea is that there is a tight relation between how our signal is distributed across delays (S_h) and how it is distributed in frequency (R_h): (3.77) allows us to draw a relation between the two values, furthermore allowing us to understand whether **certain frequency difference spans** experience highly correlating signals (implying that the signal is really redundant) or perhaps mostly uncorrelated

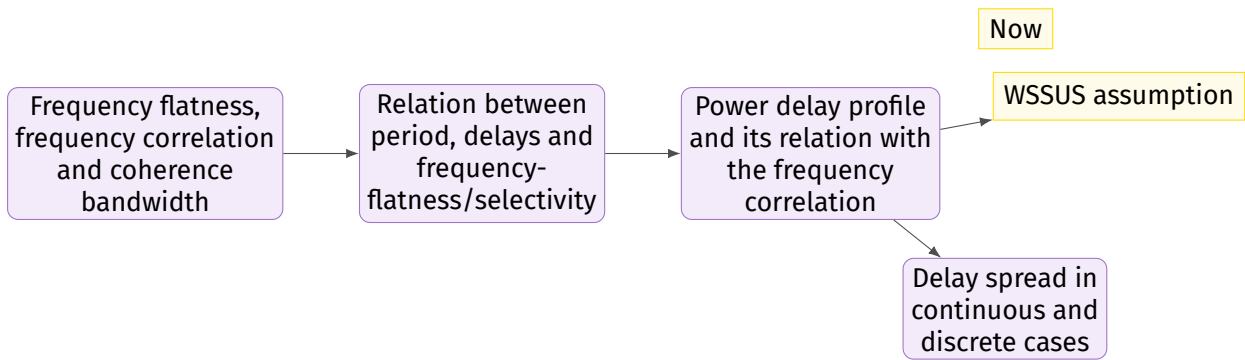
signals (implying that if said delay is achieved, the received signals across delays will contain little to no information about each other). If we then consider $S_h(\tau)$ as a PDF, expressing the relative amount of power which is given to a delay, we can interpret (3.77) as an average of $e^{j2\pi\tau\Delta_f}$ across τ .

Coherence bandwidth

Thanks to the relation between $R_h(\cdot)$ and $S_h(\cdot)$, we can write that

$$B_c \propto \frac{1}{T_d} : \quad (3.78)$$

this is the closest thing we have to a definition for the coherence bandwidth!!



If we have a response such as:



we'll have a completely flat retrospective (the correlation will be identical no matter the frequency we choose, since $R_h(\Delta_f) = \text{IFT}[S_h(\cdot)|\Delta_f] = \text{constant}$). There are many technicalities, of course! In order to be able to write (3.77), we assume that the different resolvable delays are independent.

Uncorrelated Scattering (US) and WSSUS

- By **resolvable delays** we mean delays $\tau_0 \neq \tau_1$ such that the channel's frequency response can be considered different between τ_0 and τ_1
- By independent, we imply that $\mathbb{E}[h(\tau_0)h^*(\tau_1)] = 0$ (which follows from their **uncorrelation**)

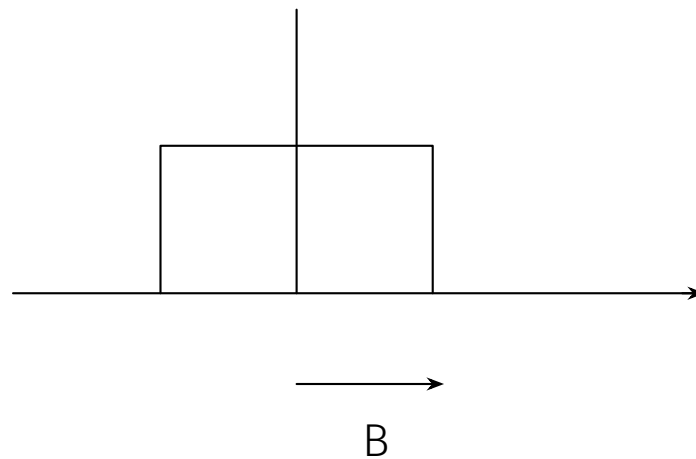
The fact that $\mathbb{E}[h(\tau_0)h^*(\tau_1)] = 0$ for $\tau_0 \neq \tau_1$ is called the **uncorrelated scattering condition**, which, according to the book:

is grounded in the reasonable premise that the propagation mechanisms giving rise to paths having resolvable delays are essentially independent. Together with the assumption of local stationarity, the uncorrelated scattering condition under-

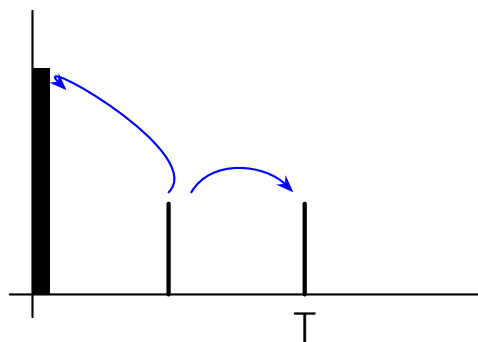
pins the traditional modeling of small-scale fading **conforming the wide-sense stationary uncorrelated scattering (WSSUS) framework** introduced by Bello in 1963 [250]

Further note is that from (3.77) the uncorrelated scattering condition ensured that $R_h(\cdot)$ only depends on Δ_f , not on f : this implies **the signal is wide sense stationary in the frequency domain as well.**

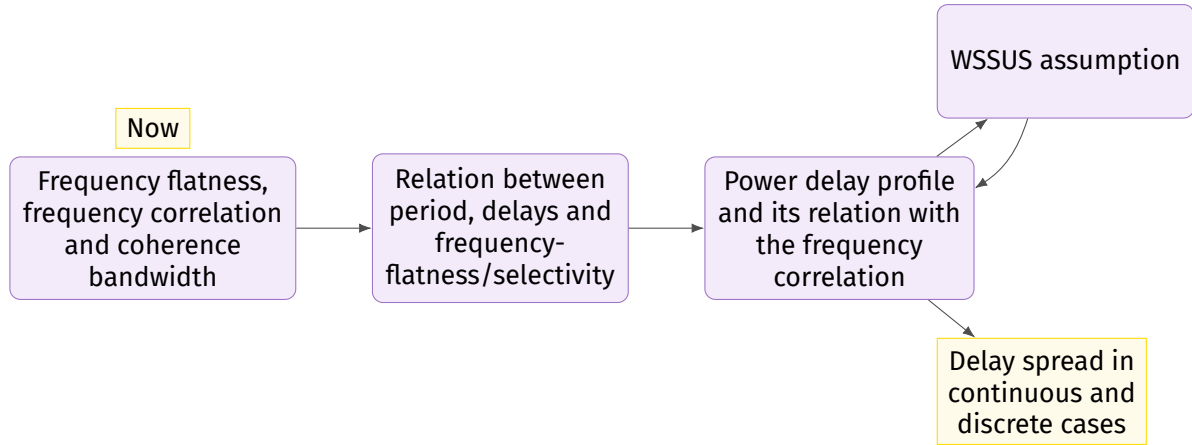
We then ask: when is a delay resolvable? We always use a pass band filter. If we're in baseband domain, we have a filter such as



which allows for perfect reconstruction if we sample at $T = 1/2B$. What if we have another delay (meaning an actually wider bandwidth)? This means that a path will partly get its energy to neighboring paths:



This only works if the various paths are independent. In the drawing up here they are dependent, since they come from the same real delay; if one goes down, the other will as well.



Usually we have some parameters to describe the frequency fading; one is the **delay spread**

Handwritten Note:

$$T_d = \sqrt{\int_{-\infty}^{+\infty} (\tau - \mu_\tau)^2 S_h(\tau) d\tau}, \quad (3.70)$$

channel is flat

if $B \ll B_c$

$$\frac{1}{T} \ll \frac{1}{T_d}$$

with μ_τ being the average delay, expressed as

$$\mu_\tau = \int_{-\infty}^{+\infty} \tau S_h(\tau) d\tau. \quad (3.69)$$

T_d is somewhat like measuring the variance, especially if we consider $S_h(\tau)$ as a PDF. If the PDP happens to be discrete, than the delay spread is redefined as

$$T_d = \sqrt{\sum_{q=0}^{Q-1} (\tau_q - \mu_\tau)^2 P_q} \quad (3.72)$$

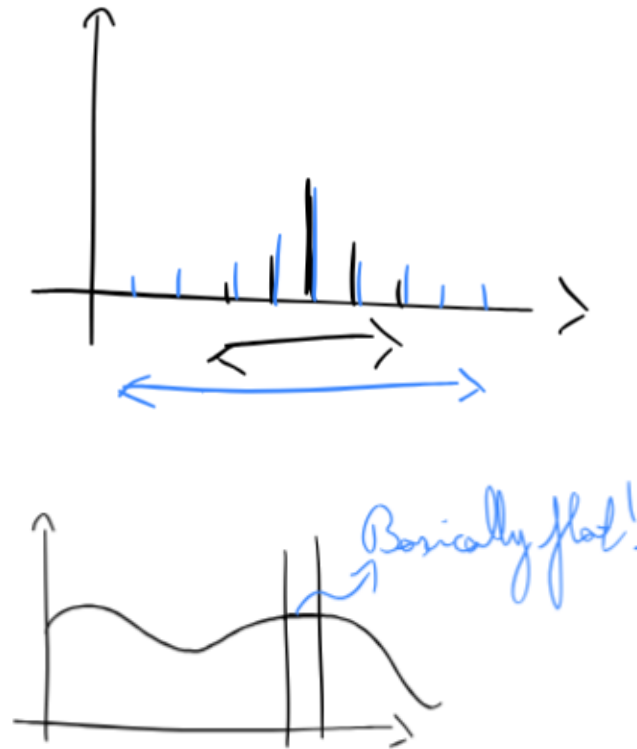
$$\mu_\tau = \sum_{q=0}^{Q-1} \tau_q P_q,$$

which is nothing but an adaptation of (3.69), (3.70) with S_h being a train of deltas weighted with P_q .

The blue channel is much more spread in time! An ideal channel of course has just a path, thus RMS=0. The smallest the RMS, the (generally) better. The principle of only having direct path is not something we like so much: by having only direct paths, sometimes the matrix might not be full rank anymore.

The channel is frequency flat if $T_d \ll T$. This means that **the symbol is much bigger than the delay, thus the ISI is so small that we can basically neglect it.**

This is like saying that the bandwidth d is much less than the coherence bandwidth. We thus have that the inequality in frequencies will be $B \ll B_c$:



Of it's really important to decide whether the channel is flat or not, as this allows to decide whether we need equalization or not.

⚡ Exam

This would certainly be one question at the exam, if something needs to be asked about this chapter.

The slides then state: "characterizing channels in terms of time, frequency and space coherence may be done under the unified framework of wide sense stationary uncorrelated scattering". This is found under the acronym of **WSSUS**; this is the assumption we did until 15/20 years ago; nowadays the new channels can't have that being true anymore. WSS means that R_h only depends on Δf (and not on f) or in other words that S_h only depends on τ (and not on t).

When we use the WSSUS framework, everything becomes easier; we won't be requested to discriminate between the two, at least not in this course.

13.1.2 Example 3.25: delay spread calculation

Just as a reference, we have that the PDP is defined as

We can write down the **power delay profile (PDP)**, which **quantifies the energy distributed as a function of delay in the time domain**; it is normalized so that it can be interpreted as a PDF (describing how likely we are to have a certain amount of

power at a certain delay) and it is denoted by

$$S_h(\tau) = \mathbb{E}[|h(\tau)|^2]. \quad (3.68)$$

and its parameters instead are

If the PDP happens to be discrete, than the delay spread is redefined as

$$T_d = \sqrt{\sum_{q=0}^{Q-1} (\tau_q - \mu_\tau)^2 P_q} \quad (3.72)$$

$$\mu_\tau = \sum_{q=0}^{Q-1} \tau_q P_q,$$

which is nothing but an adaptation of (3.69), (3.70) with S_h being a train of deltas weighted with P_q .

The question of the example is "Compute the delay spread for the vehicular PDPs in Table 3.3"; table 3.3. is as follows:

Table 13.1: ITU vehicular PDP

Vehicular A		Vehicular B	
Delay (μs)	Power share	Delay (μs)	Power share
0	0.485	0	0.322
0.31	0.385	0.3	0.574
0.71	0.061	8.9	0.03
1.09	0.048	12.9	0.057
1.73	0.015	17.1	0.002
2.51	0.005	20.0	0.014

Handwritten Notes:

- ITU: International Telecommunication Union (United Agency of UN)
- IMT-2020 $\rightarrow$ 5G

And describes the power share $S_h(\tau)$ for various delays in two setups, **Vehicular A** and **Vehicular B**.

By simply putting the numbers in the calculator, we'll find that $T_{d,A} = 0.37\mu s$, $T_{d,B} = 4\mu s$.

In vehicular channel, the **power share will eventually sum to 1** (so it can be considered just as if it was a probability distribution!). The delays have very different values for the various channels. If we try to plot them, we can see that they are pretty different!

💬 In-class blabbering

When people are evaluating systems, they first of all establish the channel. The table shows an example for the ITU; 3GPP sends its standards to ITU; rn ITU is working on a standard called IMT-2030. Other organizations as well might be sending standards for evaluation at ITU, which have other channel models which might as

well be different from the ones of 3GPP (3GPP might send 6G standard to ITU for evaluation). 5G has various techs, one of which is DECT-NR (NR: New Radio), which has been recognized by ITU as a 5G standard.

We don't know which channel models reality the best; when we select a certain technology, we want the tech to work well in both (or all) the channels we may be considering! When it comes to models, there's a lot of uncertainty.

The power density profile is different; one goes down, the other goes up and down.

13.2 Single-user SISO

13.2.1 Introduction: capacity of AWGN channel, plotting p_e , various writings of C , total channel gain

This chapter is revolving around the formula of capacity, trying to understand the relation between the various fundamental values.

We know we have that an AWGN channel has a capacity of

$$C(\text{SNR}) = \log_2(1 + \text{SNR})$$

achieved by using Gaussian distributed codewords. In this chapter we'll see waterfilling and about the frequency selective channels.

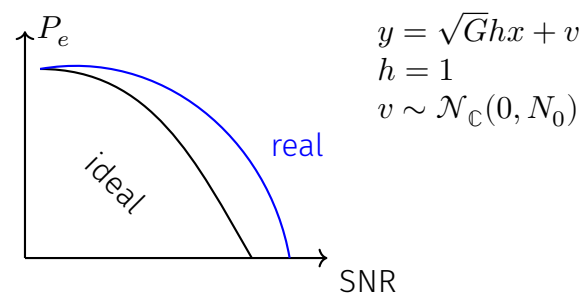
We'll look at the capacity if the transmitter knows the channel information or if only receiver does or if both do.

We may be using **OFDM and not know the channel**: we'll use equal power for every carrier; of course if we use OFDM we at least need to know the length of the channel, as we'll need to set the cyclic prefix length accordingly. The energy we spend for the cyclic prefix is kind of wasted: we use it to simplify our relations, but it is wasted. On top of that, if we know the channel we can distribute the power across the subcarriers in an optimum way. This is why we use **pilots**: using them, we can estimate the channel. This as well can be seen as a **waste**, since we could be transmitting information; point is that then the receiver gets to know the channel; if the receiver feeds the info of the channel back to the transmitter, then the transmitter can optimize itself.

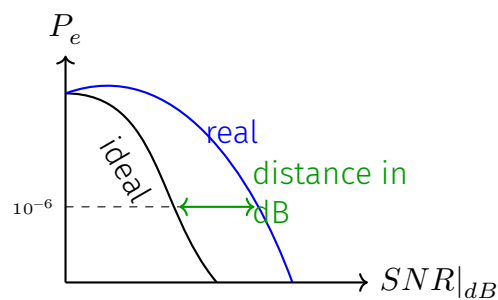
From the last chapter we know the RMS of the channel; we generally interpret that to be Gaussian. We can then make some assumptions and verify if they're true (starting with equal power) and then we'll have a **return channel** used to gather feedback from the receiver. This feedback loop is crucial if we want to get the most out of OFDM. Another thing is that the feedback channel is such that we send some information related to the channel itself. It's also important to take into account the fact that we'll need to quantize our channel: how many bits should we use to quantize the channel? This is an open problem!

There are many papers discussing this; the amount of bits used are really variable.

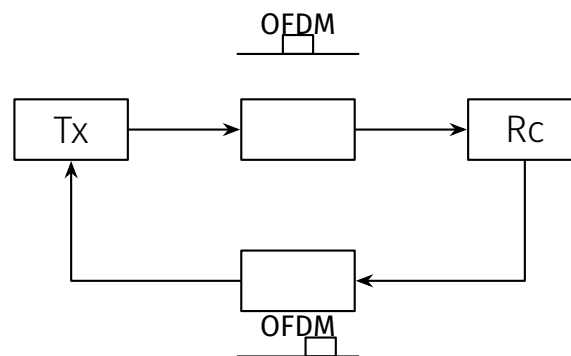
Usually we have an idea, we implement and measure the performance: the difference is called **implementation gap**. We'll be left with a high probability of error, which will be in dB. The simple idea used in practical system in order to measure the error in dB is to have a curve such as:



We select a certain value and then we measure a distance, which is in dB:



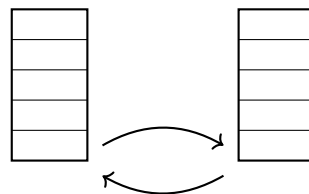
And then how do we implement the return channel?



We can't assume that the return channel is error-free, of course.

We need to have a reliable link; having probability of error equal to 0 means we'd need to have a code with a block length which is very large (actually, infinite). Point is that such a code would give a pretty large delay, while our problem is also that we'd like to have a small delay: if we have too much, by the time we deliver info about the channel, it has already changed.

Normally in the books this is forgotten, but it's important: we need to change this so that we're bringing **not-too-much service information around**; if we don't, we're not bringing any user information.



We're mostly interested about the exchange of billing information.

We could see **many formulations of the formula of capacity**; we can find

$$C = \frac{1}{2} \log_2(1 + \text{SNR}) \quad [\text{bits/use} \leftrightarrow \text{bits/s/Hz}]; \quad (13.2)$$

the $\frac{1}{2}$ depends on whether we're considering the channel as complex (no $1/2$) or not (has $1/2$). We have no B in front of the equation before; without B is the capacity per channel use; we suppose we use the channel with an interval $T = 1/B$, meaning that the rate at which we use the channel is B .

This is why we might also find

$$C = B \log_2(1 + \text{SNR}) \quad [\text{bits/s}]. \quad (13.3)$$

Let's then consider

$$\log_2(1 + \text{SNR}) :$$

which is the range for SNR? Normally it is $\in [0, \infty)$. We might as well express it as dB: the SNR can get really close to 0, meaning that for $\text{SNR} = 0$, we get a capacity of 0.

We of course have

$$\text{SNR}|_{\text{dB}} = 10 \log_{10} \text{SNR}.$$

Of course if in dB we can have a negative value, meaning that the signal has less power than the noise.

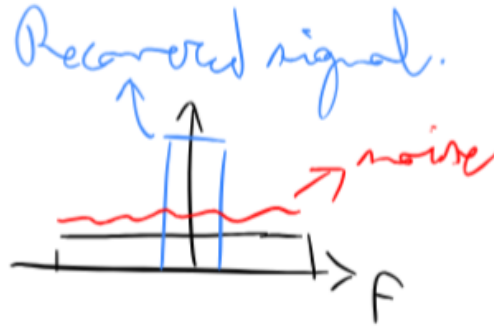
We need to measure the SNR at the point of decision; there are systems working at **negative SNR in dB**, meaning that the power of the noise is much higher than the power of the signal; what is the use case? Military use! In the army we don't want the enemy to get the message, even if it is encrypted; if my transmission is detected, the radar can detect where I am (we just have 2 receivers which can understand the location by looking at the power while they rotate). We don't even want the enemy to know we're transmitting. If we stay below a certain transmission noise, we can accomplish this! LORA is such that if we transmit very far, the power goes below the level of the noise. If we've seen a different kind of modulation, we see that we can also work on the other side; the way to do this is to use DSSS ([Direct Sequence Spread Spectrum](#)): we have a +1, -1 signal and we multiply this with a pseudo-random sequence



We're spreading this signal to a larger spectrum, eventually going below the level of noise. How do I recover the signal? I multiply again by the +1, -1 we had before, getting back the original signal. We get back to a small bandwidth: we have again sufficient SNR for reliable transmission.

This is just a silly PAM, but we could apply it to more elaborate transmissions.

We know the SNR at which a certain model/protocol will work! We look at the SNR at the point in which we'll transmit; we don't care about what happens in the rest of the channel.



The crucial point for this lecture was understanding whether we could put the $\frac{1}{2}$ or B in the equations.

The key SISO quantities are the **bitrate, transmit power and bandwidth**, R, P_t, B . Normally the bandwidth is given; sometimes, we can decide to enlarge the bandwidth, switch it etc etc; now GPS is not reliable because when it was conceived, the bandwidth was large enough that it was impossible to jam the signal in there. Now it's actually really easy. We know the frequency at which GPS is transmitting and we can make the receiver/transmitter blind.

The rate here would be referenced to the channel capacity; this means that we always compare the rate (R) to the channel capacity; we need to know if we're far from the channel capacity or if we're close to it. If we're far, we need to work.

Of course the channel capacity and the channel gain play a central role in the formulation of the tradeoffs between R, P_t and B . The **total channel gain** is, by definition, the ratio between the received and the transmit signal powers, aka

$$\frac{\mathbb{E}[|\sqrt{G}hx|^2]}{\mathbb{E}[|x|^2]} = G \frac{\mathbb{E}[|hx|^2]}{E_s}, \quad (4.1)$$

which will normally consist in attenuation; we'll only have gain if we include antennas. When we want to calculate $\mathbb{E}[|hx|^2]$, is it safe to say it is $= \mathbb{E}[|h|^2]\mathbb{E}[|x|^2] = 1 \cdot E_s$? **Only if they're uncorrelated!** This happens for fading-independent transmissions, in which x is independent of the unit-power small-scale channel coefficient h . If the signal and channel are uncorrelated, then we have that the total channel gain is equal to G .

To illustrate the various definitions and relationships put forth in this section, we have the running example of the channel $y = \sqrt{G}hx + v$ with $h = 1$ and $v \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$. This basic channel, which permeates many of the derivations in Chapter 1, is termed the AWGN channel because it is only impaired by AWGN; its capacity function, borrowed here and then formally derived in Section 4.3, is $C(\text{SNR}) = \log_2(1 + \text{SNR})$. Delving into the tradeoffs between R, P_t , and B , we begin by observing that, rather than relate these quantities directly, it is preferable to relate the ratios $R/B, P_t/B$, and P_t/R , which happen to have an operational importance of their own.

⚡ Exam

We should have in mind the order of magnitude of N_0 . Let's say at the exam professor says like $N_0 = ???$; what's the order of magnitude of that? In the book we find it is $10^{-20} \text{W}/\sqrt{\text{Hz}}$.

13.3 POTS and telephone system detour

POTS (telephone system; we then had ISDN (Integrated System Digital Network); nowadays we only have the IP network); even in the IP network we have control traffic: we need to exchange info about the routing tables.

About fixed phones

An example: we pick up the phone; we hear the tone, which means that the line is available. Some time ago, it could happen to not hear the tone of three line, but the busy line tone. We're using a protocol in this exchange of information. Once we get the line, we start making the number; the network will understand when we're finished writing. Then there's a dialogue between various nodes to understand where the person we're calling is. If the other person is found, their phone starts ringing. Only when the other person picks up, the user data starts. Once the phone is put on hook, there's another signal saying **the communication has been closed**. Mobile phones use a different protocols: they don't know when we've finished putting the number. This is why we have the green number.

At the time we had a phone sending impulses to the local switch; when the phone which turned sent an impulse at every tick while coming back. If we pressed 9, for example, there was just 1 tick. The one with the buttons understood the numbers in another way; it still was connected to the central exchange; info was covered using the DTMF (Dual Tone MultiFrequency); we have various different frequencies: depending on what we press, we'll get a different transmitted signal. The tones (=frequencies) are such to give the coordinates of the number that we want to call. The prefix code is what's used for the communication, actually. If we have a double 0 at the start, the network understands we want to go outside of Italy; by pressing 00 we express the fact that we want to go somewhere else. We then have country codes: 3, 4 mean **Europe**. Italy is 039: of course we can't have a country with a prefix of 0391, because otherwise Italy wouldn't be valid anymore. If we have 039049 this means we want to contact Padova. There must be a worldwide agreement on the numbers. The CEPT is the one assigning the prefixes.

That of the phone was the **first example of standardization**; we cannot do this with the mobile phone since mobile phones are digital; it would be a nightmare to find codes related to mobile networks: they are ISDN numbers. In the mobile network there's another **hidden** number which allows us to (for example) change operator. If we send a message, sending a number one by one would definitely congest the network. It was preferred to have the green button and send the message all at once.

Random story on the africa linking between the satellitar and *terrestrial* network; terrible fiasco rip.

We need to have a function which should inform the network about the location of a certain phone; if an italian phone is in egypt, network contacts the number; HRR is contacted and that register knows where you are. We can then transfer our request to connect to the register network. The operator does not know where we are exactly, as this would be too much information to hold; the cell is too much informaiton as well; they did a location area, which is a set of cells such that if our cell is in there and not engaged in a call: how many cells there are depends on our network design; the location area is written in the HRR: when a call reaches us, the cell broadcasts information: it tells which people are requested to connect every certain amount of milliseconds; the phone understands that message and if it is requested to connect to a call, it starts the transmission. The resource is so scarce that we only allocate it when we know that the transmission is going well.

The network keeps track of what's happening in order to send the bill; we then need to read the bills and understand if the person was at a certain location or not.

13.4 Infineon detour

Infineon detour; he talked about what infineon does and what they sell to who.

Aurix: architecture designed for cars; this is needed since the interrupts need to be served really quickly. Their processors range in various power values. The ones who made Aurix are mostly focused in analog, they develop the peripherals for the car. They are developing telecom systems: one example is the mandatory system which detects how many people are in the car when we close it. More than that, we also need to sense the environment and whether the driver is fully awake. Another example are the sensors which detect whether we're drunk.

Infineon is a very good place to work: they take care of people for their growth; salaries are aligned with the international market; open positions are first broadcast to internal people. We then have a lot of project management: when working in automotive it's needed to design things really carefully.

They also need to comply with many regulations. Infineon uses machine learning in order to ensure a really good production yield (e.g.: they use ML to have a frequency as close as possible to the desired one).

We can also decide whether we want to continue with a technical career or a managerial career.

⚡ What is the structure of the exam?

First of all, we might be checked on the background. The first part is closed book. We can have 1A4 paper which must have our name and student id; we can only write on the front!! At the start of the exam, all our cheat sheets will be put on the desk and we won't be able to look at them. We'll then need to answer to ten questions in a **crocette** format. We then have a theoretical exercise: example "prove that the MMSE estimator is ...": there are no numbers here! The first exercise is a theoretical exercise. We'll have 2 sets of papers; when we decide we have finished, we can deposit the theoretical part and pick up the cheat sheet: we'll then have a numerical exercise, in which there **will** be numbers. We of course need to put together our sequence number. We can also take a calculator with us, but it shall not be connected to the internet.

For tomorrow's tutor lecture, professor selected some exercises.

13.5 SNR (+several other definitions: spectral efficiency, transmit energy, energy per bit); expressions of capacity $C, \mathbb{C}$

We know that the **spectral efficiency** is how many bits we can send per Hz: it's R/B ; the unit is

$$\frac{\text{b/s}}{\text{Hz}} :$$

If I have a spectral efficiency of 100 [b/s/Hz] and we have 4 Hz, we can send 400 [bit/s]: if we enlarge our B , we're improving the channel; of course we can't do this without any limit: we know that the capacity is an upper bound to the amount of bps we can send. If we know the receive power, the number of bits per hertz has a limit which is the Shannon bound.

We have as a rule of thumb that we transmit with period $T = 1/B$; what actually happens is that we use a little bit more than B , as we know (since we can't really use *rect* signals in frequency, as they'd give an endless signal in time).

The **transmit energy** is

$$P_t/B = P_t T = E_s = \mathbb{E}[|x|^2]. \quad (13.4)$$

The SNR must be measured **at the decision point**; we see that

$$\text{SNR} = \frac{GP_t/B}{N_0} \quad (4.2)$$

$$= \frac{GE_s}{N_0} : \quad (4.3)$$

this is measured at the receiver, at the decision point (this means that we're doing this at the receiver). Why is B there? It can be put below, as BN_0 is the power of the noise.

Instead, P_t/R is the **energy spent per bit** (this is because R is in bits/s); this is at the receiver (we can tell this as we have G):

$$E_b = G \frac{P_t}{R} \quad (4.4)$$

$$= \frac{GE_s}{R/B} : \quad (4.5)$$

this is an especially useful quantity when referred to N_0 (we'll see it's often used as a reference values for our graphs); specifically, it gives

$$\frac{E_b}{N_0} = \frac{1}{N_0} \frac{GE_s}{R/B} \quad (4.6)$$

$$= \frac{\text{SNR}}{R/B}. \quad (4.7)$$

This is just a game of substitution.

E_b is the energy per bit; the value E_b/N_0 is generally put on the x axis: we can see (as it is clear from (4.7)) that it is linked to the SNR.

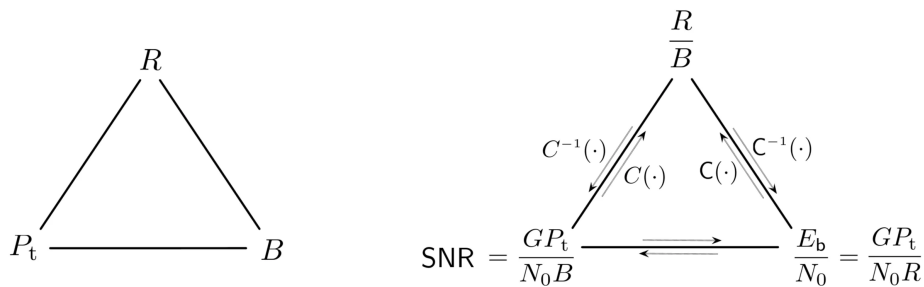
Sometimes it's not so obvious where the **decision point** is.

We've already seen the capacity bounds; C is the capacity we have seen quite some time ago; we also have the power P_t , though. They give:

$$R = B \cdot C \left(\frac{GP_t}{N_0 B} \right) = B \cdot C(\text{SNR}) \quad (4.12)$$

$$P_t = \frac{N_0 B}{G} C^{-1} \left(\frac{R}{B} \right). \quad \xrightarrow{\text{how much power to transmit to get at the capacity}} \quad (4.13)$$

Depending on the reference framework we choose to adopt, we can move across one another following the scheme below:



Problem here is in the notation: we have two different notations of capacity, $C \neq \mathcal{C}$.

We have that

$$C(\text{SNR}) = \log(1 + \text{SNR})$$

but we can also write this using

$$\text{SNR} = \frac{B}{R} \frac{E_b}{N_0}$$

giving

$$\log \left(1 + \frac{B}{R} \frac{E_b}{N_0} \right),$$

which is a different function from $C(\text{SNR})$: we have $\mathfrak{C} \left(\frac{E_b}{N_0} \right)$. We might want to plot the capacity as a function of either the SNR or as a function of E_b/N_0 . From the triangle we see that on the left we have C and on the right we have $\mathfrak{C}$.

⚡ Exam

In the exam, normally C is reserved for the capacity formula: we need to make it clear what the formula we're using is.

From the fact that $C = \mathfrak{C} = R/B$, we can write that

$$\frac{E_b}{N_0} = \frac{\text{SNR}}{C(\text{SNR})} \quad (4.8)$$

and

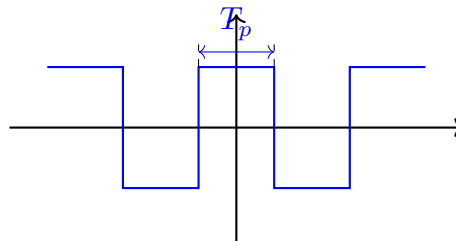
$$\text{SNR} = \frac{E_b}{N_0} \mathfrak{C} \left(\frac{E_b}{N_0} \right). \quad (4.9)$$

The reason why the link between P_t, B has **no function** is that in those situations we don't know what R is, thus it's not easy to go from one to the other. *Per i figli, fino a 4.3*

Appendix A: Transforms

DigiCom/Notes

This appendix reviews several Fourier transforms, instrumental in the transition between the time and the frequency domains, as well as the Z-transform.



1.1 Fourier transforms

1.1.1 Continuous-time Fourier transform

A well-behaved continuous-time function $x(t)$ and its Fourier transform $x(f)$ are related by the analysis and synthesis equations

$$\text{Analysis} \quad x(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt \quad (\text{A.1})$$

$$\text{Synthesis} \quad x(t) = \int_{-\infty}^{\infty} x(f) e^{j2\pi ft} df. \quad (\text{A.2})$$

A detailed account of the technical conditions that are necessary for a Fourier transform to exist is beyond the scope of this book, and interested readers are referred for instance to [1053]. It suffices to say that all physically realizable signals do have Fourier transforms. For our purposes, therefore, we can take "well-behaved" to simply mean a signal for which the integral in (A.1) exists.

Those properties of the continuous-time Fourier transform that are invoked somewhere in the text are listed in Table A.1. Likewise, relevant Fourier transform pairs are presented in Table A.2.

1.1.2 Continuous-time Fourier series

Periodic functions do not fall under the umbrella of well-behaved functions and yet they are very important in the analysis of communication signals. This obstacle can be side-stepped through the definition of the Fourier series and by invoking the Dirac delta function, $\delta(\cdot)$.

Consider a periodic signal $x(t)$ whose period, $T > 0$, is the smallest real number such that $x(t) = x(t + T)$. The continuous-time Fourier series of such signal is defined as

$$\text{Analysis} \quad x[n] = \frac{1}{T} \int_0^T x(t) e^{-j\frac{2\pi}{T}nt} dt \quad (\text{A.3})$$

$$\text{Synthesis} \quad x(t) = \sum_n x[n] e^{j\frac{2\pi}{T}nt}. \quad (\text{A.4})$$

Table 1.2: Continuous-time Fourier transform properties

Property	Aperiodic signal $x(t)$	Fourier transform $x(f)$
Linearity	$ax(t) + b y(t)$	$ax(f) + by(f)$
Time shift	$x(t - t_0)$	$e^{-j2\pi f t_0} x(f)$
Conjugation	$x^*(t)$	$x^*(-f)$
Time reversal	$x(-t)$	$x(-f)$
Time scaling	$x(at)$	$\frac{1}{ a } x(\frac{f}{a})$
Convolution	$x(t) * y(t) = \int x(\tau) y(t - \tau) d\tau$	$x(f)y(f)$ (CONV into PRODUCTS)
Autocorrelation	$x(t) * x^*(-t)$	$ x(f) ^2$
Modulation	$x(t) e^{j2\pi f_0 t}$	$x(f - f_0)$
Conjugate symmetry	$x(t)$ is real	$x(f) = x^*(-f)$
Duality	$x(t) \leftrightarrow x(f)$	$X(t) \leftrightarrow x(-f)$
Parseval's theorem	$\int x(t) y^*(t) dt = \int x(f) y^*(f) df$	$\int x(t) ^2 dt = \int x(f) ^2 df$

Table 1.3: Continuous-time Fourier transform pairs

Time-domain	Frequency-domain
Impulse $\delta(t)$	1
Constant function 1	$\delta(f)$
Complex exponential $e^{j2\pi f_0 t}$	$\delta(f - f_0)$
Cosine $\cos(2\pi f_0 t + \theta)$	$\frac{1}{2} [e^{j\theta} \delta(f - f_0) + e^{-j\theta} \delta(f + f_0)]$
Sine $\sin(2\pi f_0 t + \theta)$	$\frac{1}{2j} [e^{j\theta} \delta(f - f_0) - e^{-j\theta} \delta(f + f_0)]$
Impulse train $\sum_k \delta(t - kT)$	$\frac{1}{T} \sum_n \delta(f - \frac{n}{T})$
Rectangular pulse $rect(\frac{t}{T}) = \begin{cases} 1 & t \leq \frac{T}{2} \\ 0 & \text{else} \end{cases}$	$T \text{sinc}(fT) = \frac{\sin(\pi fT)}{\pi f}$
Sinc pulse $\text{sinc}(Wt) = \frac{\sin(\pi Wt)}{\pi Wt}$	$\frac{1}{W} \text{rect}(\frac{f}{W})$

The continuous-time Fourier series creates as an output a weighting of the fundamental frequency of the signal $e^{-j2\pi/T}$ and its harmonics. From the series it is possible to express the Fourier transform of a periodic signal as

$$x(f) = \sum_n x[n] \delta(f - \frac{n}{T}) \quad (\text{A.5})$$

$$x(t) = \int_{-\infty}^{\infty} x(f) e^{j2\pi ft} df, \quad (\text{A.6})$$

where the unboundedness of (A.1) has been circumvented by means of $\delta(\cdot)$.

1.1.3 Discrete-time Fourier transform

Consider now a well-behaved discrete-time signal $x[n]$. Its discrete-time Fourier transform analysis and synthesis relationships are

$$\text{Analysis} \quad x(v) = \sum_n x[n] e^{-j2\pi nv} \quad (\text{A.7})$$

$$\text{Synthesis} \quad x[n] = \int_{-1/2}^{1/2} x(v) e^{j2\pi nv} dv, \quad (\text{A.8})$$

where the frequency v is defined on any finite interval of unit length, typically $[-1/2, 1/2]$.

1.1.4 Discrete Fourier transform

While the discrete-time Fourier transform offers a sound analytical framework for discrete-time signals, it is inconvenient due to its continuous frequency. An alternative is the discrete Fourier transform (DFT) and its easy-to-implement cousin, the fast Fourier transform (FFT). The DFT is extremely important in digital signal processing and communications, and instrumental in the context of OFDM. It applies to finite-length discrete-time signals and, since any finite-length signal can be repeated to form a periodic discrete-time signal, the DFT can also be interpreted as applying to periodic signals. The DFT synthesis and analysis equations are

$$\text{Analysis} \quad x[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \quad k = 0, \dots, N-1 \quad (\text{A.9})$$

$$\text{Synthesis} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j\frac{2\pi}{N}kn} \quad n = 0, \dots, N-1 \quad (\text{A.10})$$

where the discrete and finite-range nature of both time and frequency are noteworthy. To compactly denote the direct (analysis) and inverse (synthesis) N -point DFT transforms, we introduce the terminology

$$x[k] = DFT_N\{x[n]\} \quad (\text{A.11})$$

$$x[n] = IDFT_N\{x[k]\}. \quad (\text{A.12})$$

Some relevant DFT properties are listed in Table A.3, where $(())_N$ denotes modulo- N operation.

Circular convolution

The circular convolution between two periodic discrete signals of length N is similar to linear convolution, but with an important difference:

- the indices are considered "modulo N ", meaning the ends of the signal "wrap around" on themselves, as if they lived on a circle.
- In practice, it is like performing linear convolution but assuming the signal is periodic with period N .

Table 1.4: Table A.3 DFT properties

Length- N sequence $x[n]$	N -point DFT $x[k]$
$a x[n] + b y[n]$	$a x[k] + b y[k]$
$x[((n - m))_N]$	$e^{-j2\pi km/N} x[k]$
$\sum x[m] y[((n - m))_N]$	$x[k] y[k]$
$x^*[n]$	$x^*[((-k))_N]$
$x[n]$ is real	$x[k] = x^*[((-k))_N]$

1.2 Z-transform

The Z-transform converts a function of a discrete real variable (say the discrete time, n) to a function of a complex variable z . This converts difference equations into algebraic equations and convolution into product. The Z-transform of a causal function $x[n]$ is

$$x(z) = \sum_{n=0}^{\infty} x[n] z^{-n}, \quad (\text{A.13})$$

while the inversion of $x(z)$ back onto $x[n]$ requires an integration on the complex plane [133].

Example A.1

Obtain the Z-transform of $x[n] = \delta[n - \Delta]$

Solution

$$x(z) = z^{-\Delta}$$

The foregoing example generalizes into a time-shifting property according to which, if the Z-transform of $x[n]$ is $x(z)$ then the Z-transform of $x[n - \Delta]$ is $z^{-\Delta} x(z)$.

Appendix B: Matrix algebra

DigiCom/Notes

Fat Matrix: A matrix where the number of columns (M) is greater than the number of rows (N). "Lessgoo!"

2.1 Column space, row space, null spaces

The column space of an $N \times M$ matrix $A = [a_0 \cdots a_{M-1}]$ is the set of all linear combinations of its column vectors $a_0, \dots, a_{M-1}$. It is therefore a subspace (whose dimension is at most M) of the N -dimensional vector space. Since a linear combination with arbitrary coefficients $x_0, \dots, x_{M-1}$ of the columns of A can be written as the product of A with the vector

$$x = [x_0 \dots x_{M-1}]^T \quad (\text{Column vector}) \quad (2.5)$$

i.e.,

$$x_0 a_0 + \cdots + x_{M-1} a_{M-1} = A \begin{bmatrix} x_0 \\ \vdots \\ x_{M-1} \end{bmatrix}, \quad (B.1)$$

the column space of A consists of all possible vectors $y = Ax$.

The row space of A , in turn, equals the column space of A^T or, equivalently, of A^* . Thus, it is a subspace (whose dimension is at most N) of the M -dimensional vector space.

Example B.1

The column space of

$$A = \begin{bmatrix} 0 & 3 \\ 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_2 \\ 2x_1 \\ x_2 \end{bmatrix} \quad (B.2)$$

is the set of vectors $y = [y_0, y_1, y_2]^T$ having the form

$$y = Ax = \begin{bmatrix} 3x_2 \\ 2x_1 \\ x_2 \end{bmatrix} \quad (\text{B.3})$$

These vectors satisfy $y_0 = 3y_2$, which defines a subspace of dimension $M = 2$ (that is, a plane) on a vector space of dimension $N = 3$.

Example B.2

The row space of A in (B.2), in turn, is the set of vectors y having the form

$$y = A^T x = \begin{bmatrix} 2x_1 \\ 3x_0 + x_2 \end{bmatrix} \quad (\text{B.5})$$

which defines the entire vector space of dimension $M = 2$.

The column rank and row rank of A are the dimensions of its column space and row space, respectively. For the matrix in Examples B.1 and B.2, for instance, both equal 2. The fact that the column and row ranks coincide in this case is not a coincidence. Indeed, the row and column ranks always coincide, giving the rank of the matrix. A matrix is said to be full-rank if its rank equals the largest possible, which is $\min(N, M)$, and it is said to be rank-deficient otherwise.

In addition to its column and row spaces, a matrix A elicits two additional subspaces, respectively the orthogonal complements of such column and row spaces.

- The orthogonal complement of the row space, termed the **null space** of A , is the collection of those vectors that are orthogonal to every row of A , i.e., of those vectors x satisfying $Ax = 0$. The null space of A has dimension $M - \text{rank}(A)$.
- The orthogonal complement of the column space of A equals the null space of A^T , with dimension $N - \text{rank}(A)$.

Example B.3

For the matrix A in Examples B.1 and B.2, the null space is empty while the null space of A^T contains all the colinear vectors $y = [y_0, y_1, y_2]^T$ that are orthogonal to the plane defined by $y_0 = 3y_2$.

2.2 Special matrices

2.2.1 Hermitian matrices

A complex matrix A is said to be Hermitian if $A^* = A$.

2.2.2 Unitary matrices

A complex matrix U is said to be unitary if $U^*U = UU^* = I$. In addition:

- U is nonsingular and $U^* = U^{-1}$.
- The columns of U form an orthonormal set, as do the rows of U .
- For any complex vector x , the vector $y = Ux$ satisfies $|y| = |x|$. Thus, y is a rotated version of x and U embodies that rotation.

2.2.3 Fourier matrices

An $N \times N$ Fourier matrix U is a unitary matrix whose (i, j) th entry equals $e^{j2\pi ij/N}$. It follows that the j th column, for $j = 0, \dots, N-1$, is given by

$$u_j = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ e^{j2\pi j/N} \\ \vdots \\ e^{j2\pi(N-1)j/N} \end{bmatrix}. \quad (\text{B.7})$$

The DFT of a vector x is

$$x = \sqrt{N}U^*x \quad (\text{B.8})$$

whereas the IDFT is

$$x = \frac{1}{\sqrt{N}}Ux. \quad (\text{B.9})$$

Indeed, by interpreting the entries of x and vectors as sequences, (B.8) and (B.9) are scaled versions of the $\text{DFT}_N\{\cdot\}$ and $\text{IDFT}_N\{\cdot\}$ transforms in (A.9) and (A.10).

2.2.4 Toeplitz and circulant matrices

A matrix is **Toeplitz** if it is constant along each of its diagonals. A Toeplitz matrix is further **circulant** if it is completely described by any of its rows, of which the other rows are just circular shifts with offsets equal to the row indices. Alternatively, a circulant matrix is described by any of its columns with the other columns just circular shifts thereof.

Example B.4

The real matrices A and B below are Toeplitz and circulant, respectively.

$$A = \begin{bmatrix} 2 & 5 & 1 \\ 4 & 2 & 5 \\ 3 & 4 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 5 & 1 \\ 1 & 2 & 5 \\ 5 & 1 & 2 \end{bmatrix} \quad (\text{B.10})$$

If A is an $N \times N$ circulant matrix, then the following holds:

- The eigenvectors of A equal the columns of the Fourier matrix U in (B.7).
- The eigenvalues of A equal the entries of U^*a where a is any column of A .

Hence, the eigenvalues of a circulant matrix are directly the DFT of any of the columns (or rows) of that matrix [139].

2.2.5 Hankel matrices

A Hankel matrix is an upside-down Toeplitz matrix, i.e., a matrix in which each rightward-ascending diagonal is constant.

Example B.5

The real matrix A below is a Hankel matrix.

$$A = \begin{bmatrix} 3 & 4 & 2 \\ 4 & 2 & 5 \\ 2 & 5 & 1 \end{bmatrix} \quad (2.6)$$

2.3 Matrix decompositions

2.3.1 Eigenvalue decomposition

Any $N \times N$ Hermitian matrix A can be factored into the canonical form

$$A = U\Lambda U^* \quad (B.11)$$

$$= \sum_{i=0}^{N-1} \lambda_i(A) u_i u_i^*, \quad (B.12)$$

where $U = [u_0 \cdots u_{N-1}]$ is a unitary matrix whose columns are the eigenvectors of A whereas $\Lambda = \text{diag}(\lambda_0(A), \dots, \lambda_{N-1}(A))$ is a diagonal matrix whose diagonal entries are the eigenvalues of A . The eigenvectors are the vectors (normalized to have unit norm) that the linear transformation embodied by A does not rotate but merely stretches, and the eigenvalues are the factors by which they are stretched. Thus,

$$A u_i = \lambda_i(A) u_i \quad i = 0, \dots, N-1. \quad (B.14)$$

The eigenvalues of a Hermitian matrix are always real. Furthermore, a Hermitian matrix is **positive-semidefinite** if, for every nonzero complex vector x ,

$$x^* A x \geq 0. \quad (B.15)$$

If the equality in (B.15) is strict (i.e., > 0), then A is **positive-definite**. The eigenvalues of a positive-definite matrix are strictly positive whereas those of a positive-semidefinite matrix may be zero.

The rank of A equals the number of nonzero eigenvalues and A is invertible if it is full rank, i.e., if $\text{rank}(A) = N$. Then, the eigenvectors of A and A^{-1} coincide and

$$A^{-1} = U\Lambda^{-1}U^* = U \text{diag} \left(\frac{1}{\lambda_0(A)}, \dots, \frac{1}{\lambda_{N-1}(A)} \right) U^*. \quad (B.17)$$

Table 2.5: Bases for the column, row, and null spaces of an $N \times M$ matrix $A = U\Sigma V^*$

Subspace	Dimension	Basis
Column space of A	$r = \text{rank}(A)$	First r columns of U
Row space of A	r	First r columns of V
Null space of A^T	$(N - r)$	Last $(N - r)$ columns of U
Null space of A	$(M - r)$	Last $(M - r)$ columns of V

2.3.2 Singular-value decomposition

By means of the singular-value decomposition (SVD), any $N \times M$ matrix A can be factored as

$$A = U\Sigma V^* \quad (\text{B.18})$$

where U and V are unitary, respectively $N \times N$ and $M \times M$ while Σ is an $N \times M$ matrix with real entries on its main diagonal and zero elsewhere. The entries on the main diagonal of Σ are termed the singular values of A , whereas the leading $\min(M, N)$ columns of U and of V are the left and the right singular vectors of A , respectively, such that

$$v_j = \sigma_j u_j \quad j = 0, \dots, \min(M, N) - 1 \quad (\text{B.19})$$

$$A^* u_j = \sigma_j v_j \quad j = 0, \dots, \min(M, N) - 1 \quad (\text{B.20})$$

where σ_j is the j th singular value whereas $u_j = [U]_{:,j}$ is the j th left singular vector and $v_j = [V]_{:,j}$ is the j th right singular vector.

The SVD of A has an intimate relationship with the eigenvalue decompositions of AA^* and A^*A . The squared singular values of A are the nonzero eigenvalues of AA^* and A^*A . The left singular vectors of A are the eigenvectors of AA^* . Conversely, the right singular vectors of A are the eigenvectors of A^*A .

2.3.3 QR decomposition

Given $N \geq M$, any $N \times M$ matrix A can be factored as

$$A = QR, \quad (\text{B.21})$$

where Q is an $N \times N$ unitary matrix while R is an $N \times M$ upper-diagonal matrix whose bottom $(N - M)$ rows are all-zero.

2.4 Trace and determinant

The **trace** of a square matrix equals the sum of the entries on its main diagonal and, also, the sum of its eigenvalues. It is a linear operation that is invariant to a change of basis and, therefore, to pre- or post-multiplication by unitary matrices.

The **determinant** of a square matrix equals the product of its eigenvalues, counted with their corresponding multiplicities. If the determinant is nonzero, the matrix is

invertible and

$$\det(A^{-1}) = \frac{1}{\det(A)}. \quad (\text{B.22})$$

If the determinant is zero, the matrix is said to be singular and it is not invertible.

Useful commutative properties involving the determinant and trace of products of matrices (for properly dimensioned matrices A and B):

$$\det(I + AB) = \det(I + BA) \quad (\text{B.23})$$

$$\det(AB) = \det(BA) = \det(A)\det(B) \quad (\text{B.24})$$

$$\text{tr}(AB) = \text{tr}(BA) \quad (\text{B.26})$$

2.5 Frobenius norm

The Frobenius norm of an $N \times M$ matrix A equals

$$\|A\|_F = \sqrt{\text{tr}(AA^*)} = \sqrt{\text{tr}(A^*A)}. \quad (\text{B.27})$$

Expanding the argument of the square root, what results is

$$\|A\|_F = \sqrt{\sum_{i=0}^{N-1} \sum_{j=0}^{M-1} |[A]_{i,j}|^2}, \quad (\text{B.29})$$

which, in the special case that A is a vector, reduces to the standard Euclidean norm.

2.6 Moore-Penrose pseudoinverse

The pseudoinverse generalizes the concept of an inverse. In particular, the Moore-Penrose pseudoinverse of an $N \times M$ rectangular matrix A is the unique matrix $A^\dagger$ satisfying $AA^\dagger A = A$ and $A^\dagger AA^\dagger = A^\dagger$, and such that $AA^\dagger$ and $A^\dagger A$ are both Hermitian.

- If A^*A is invertible (typically if $N \geq M$):

$$A^\dagger = (A^*A)^{-1}A^* \quad (\text{B.32})$$

- If AA^* is invertible (typically if $N \leq M$):

$$A^\dagger = A^*(AA^*)^{-1} \quad (\text{B.33})$$

2.7 Matrix inversion lemma

Also termed the Woodbury matrix identity:

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}. \quad (\text{B.34})$$

2.8 Kronecker product

The Kronecker product extends to matrices the outer product vector operator. Given an $N_A \times M_A$ matrix A and an $N_B \times M_B$ matrix B , their Kronecker product yields an $N_A N_B \times M_A M_B$ matrix

$$A \otimes B = \begin{bmatrix} [A]_{0,0}B & \cdots & [A]_{0,M_A-1}B \\ \vdots & \ddots & \vdots \\ [A]_{N_A-1,0}B & \cdots & [A]_{N_A-1,M_A-1}B \end{bmatrix}. \quad (\text{B.35})$$

Like the regular matrix product, the Kronecker product is noncommutative, linear, and associative.

$$(A \otimes B)^* = A^* \otimes B^* \quad (\text{B.36})$$

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1} \quad (\text{B.37})$$

Expanded View

Here is the visual expansion of the Kronecker product $A \otimes B$ where A is $m \times n$ and B is $p \times q$:

$$A \otimes B = \left[\begin{array}{ccc|ccc} a_{11}b_{11} & \cdots & a_{11}b_{1q} & & a_{1n}b_{11} & \cdots & a_{1n}b_{1q} \\ \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ a_{11}b_{p1} & \cdots & a_{11}b_{pq} & & a_{1n}b_{p1} & \cdots & a_{1n}b_{pq} \\ \hline & & \vdots & \ddots & & & \vdots \\ a_{m1}b_{11} & \cdots & a_{m1}b_{1q} & & a_{mn}b_{11} & \cdots & a_{mn}b_{1q} \\ \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ a_{m1}b_{p1} & \cdots & a_{m1}b_{pq} & & a_{mn}b_{p1} & \cdots & a_{mn}b_{pq} \end{array} \right]$$

Appendix C: Random variables and processes

DigiCom/Notes

This appendix collects a host of relevant results on random variables (scalars, vectors, and matrices), presents the distributions used throughout the book, and provides a classification of the principal forms of convergence of random sequences. Special treatment is given to the behavior of certain random matrices as their size grows without bound.

3.1 Random variables

The random variables in this section are regarded as continuous unless otherwise indicated, yet most of the concepts extend to discrete distributions with a probability mass function (PMF) in place of the probability density function (PDF).

3.1.1 Bayes' theorem

Given two random variables x and y with joint PDF $f_{xy}(\cdot, \cdot)$ and with marginals $f_x(\cdot)$ and $f_y(\cdot)$, the respective conditional distributions are obtained as:

$$f_{y|x}(y|x) = \frac{f_{xy}(x, y)}{f_x(x)} \quad (\text{C.1})$$

$$f_{x|y}(x|y) = \frac{f_{xy}(x, y)}{f_y(y)} \quad (\text{C.2})$$

respectively for $f_x(x) > 0$ and $f_y(y) > 0$. Bayes' theorem states that:

$$f_{x|y}(x|y) = \frac{f_{y|x}(y|x)f_x(x)}{f_y(y)} \quad (\text{C.3})$$

Bayes' theorem has been said to play a role similar to that of Pythagoras' theorem in geometry, a comparison that seems certainly appropriate in light of its importance and of the triangular relationship it establishes between the joint distribution and the two conditionals.

3.1.2 Expectation

Given a real scalar x with PDF $f_x(\cdot)$, the expected value of x (also called average) can be obtained directly from $f_x(\cdot)$ via:

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} x f_x(x) dx \quad (\text{C.4})$$

and also from the corresponding cumulative distribution function (CDF), $F(\cdot)$, through the relationship:

$$\mathbb{E}[x] = \int_0^{\infty} (1 - F_x(x)) dx - \int_{-\infty}^0 F_x(x) dx \quad (\text{C.5})$$

Note on Expectation

Equation (C.5) works without making derivatives ($f_x(x) = \frac{dF(x)}{dx}$).

Handwritten Note: Mean vs. Median

Context from Page 3: The note illustrates the difference between mean and median using a grading example.

- **Scenario:** 3 out of 4 students think the professor deserves a grade of 6. 1 student thinks the professor deserves a 4.
- **Average:** $\frac{6+6+6+4}{4} = 5.5$.
- **Median:** The median is 6.
- **Interpretation:** The median suggests the professor is sufficient (6), but the average (5.5) drags the score down. The note implies: "Review this because it's confusing... it's like saying the average is 5.5 but the median says he has a grade of 6".

3.1.3 Correlation

The covariance between two random scalars x and y , with respective means $\mu_x = \mathbb{E}[x]$ and $\mu_y = \mathbb{E}[y]$, is given by:

$$R_{xy} = \mathbb{E}[(x - \mu_x)(y - \mu_y)^*], \quad (\text{C.6})$$

which, if $x = y$, reduces to the variance $\text{var}[x] = \sigma_x^2$. Similarly, for two random vectors x and y we can define the covariance/correlation matrix:

$$R_{xy} = \mathbb{E}[(x - \mu_x)(y - \mu_y)^*]. \quad (\text{C.7})$$

Two random variables are **uncorrelated** if $R_{xy} = 0$. If two random variables are independent ($f_{xy}(x, y) = f_x(x)f_y(y)$), then they are also uncorrelated. However, two uncorrelated variables need not be independent.

3.1.4 Properness

A complex random scalar x is said to be **proper complex** if $\mathbb{E}[x^2] = \mathbb{E}[x]^2$. Properness requires that the real and imaginary parts of x be uncorrelated and have the same variance. The concept generalizes to vectors in a straightforward manner: x is a proper complex random vector if $\mathbb{E}[xx^T] = \mathbb{E}[x]\mathbb{E}[x^T]$.

3.1.5 Circular symmetry

A random scalar x is circularly symmetric if its distribution remains unchanged when x is rotated around its mean by an arbitrary angle, i.e., if the distribution of $(x - \mathbb{E}[x])e^{j\phi}$ is identical to that of $(x - \mathbb{E}[x])$.

3.1.6 Unitary invariance

A random matrix X is **unitarily invariant** if its distribution equals that of UX for any unitary matrix U independent of X . Alternatively, X is right unitarily invariant if its distribution equals that of XV^* for any unitary matrix V independent of X .

3.1.7 Linear transformations

If x is a complex random vector with PDF $f_x(\cdot)$ and A is a nonsingular matrix, then the PDF of $y = Ax$ is:

$$f_y(y) = \frac{f_x(A^{-1}y)}{|\det(A)|^2}. \quad (\text{C.11})$$

3.1.8 Kurtosis

Kurtosis informs of the shape of the PDF. A large kurtosis indicates that, for a given power, a random variable exhibits infrequent but extreme deviations (heavy tails). The definition for a real scalar x is:

$$\kappa(x) = \frac{\mathbb{E}[x^4]}{\mathbb{E}^2[x^2]}. \quad (\text{C.12})$$

3.1.9 Relevant distributions

The Gaussian distribution The PDF of a Gaussian scalar x with mean μ_x and variance σ_x^2 is:

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{1}{2}(x-\mu_x)^2/\sigma_x^2} \quad (\text{C.13})$$

denoted as $\mathcal{N}(\mu_x, \sigma_x^2)$.

The complex Gaussian distribution A complex Gaussian scalar x has Gaussian real and imaginary parts. Unless otherwise stated, properness is always assumed. With mean μ_x and variance σ_x^2 , the PDF of x is:

$$f_x(x) = \frac{1}{\pi\sigma_x^2} e^{-|x-\mu_x|^2/\sigma_x^2}. \quad (\text{C.14})$$

For a vector x with mean μ_x and covariance matrix R_x :

$$f_x(x) = \frac{1}{\det(\pi R_x)} e^{-(x-\mu_x)^* R_x^{-1} (x-\mu_x)}. \quad (\text{C.15})$$

The Wishart distribution Let X be an $N \times M$ matrix with entries distributed as zero-mean complex Gaussian. If $M \geq N$, then $W = XX^*$ is an $N \times N$ **central Wishart matrix** with PDF:

$$f_W(W) = \frac{\pi^{-N(N-1)/2}}{(\det R)^M \prod_{i=1}^N (M-i)!} e^{-\text{tr}(R^{-1}W)} (\det W)^{N-M}. \quad (\text{C.22})$$

The chi-square distribution If $x_i \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, then $w = \sum_{i=0}^{M-1} |x_i|^2$ is a **chi-square** random variable with $2M$ degrees of freedom:

$$f_w(\xi) = \frac{\xi^{M-1} e^{-\xi}}{(M-1)!}. \quad (\text{C.33})$$

The Rayleigh distribution For $M = 1$, the chi distribution reduces to the Rayleigh distribution. If $x \sim \mathcal{N}_{\mathbb{C}}(0, \sigma^2)$, then the magnitude $|x|$ has PDF:

$$f_{|x|}(\xi) = \frac{\xi}{\sigma^2} e^{-\frac{1}{2}\xi^2/\sigma^2}. \quad (\text{C.43})$$

3.2 Convergence of sequences

We classify the most relevant forms of convergence for a sequence $\{x_n\}$.

Convergence in distribution: $\lim_{n \rightarrow \infty} F_{x_n}(x) = F_x(x)$ at every x where $F_x(x)$ is continuous. *Example: Central Limit Theorem.*

Convergence in probability (weakly): For every $\epsilon > 0$, $\lim_{n \rightarrow \infty} \mathbb{P}[|x_n - x| > \epsilon] = 0$. *Example: Weak law of large numbers.*

Convergence almost surely (strongly): $\mathbb{P}[\lim_{n \rightarrow \infty} x_n = x] = 1$. *Example: Strong law of large numbers.*

Convergence in the mean-square sense: $\lim_{n \rightarrow \infty} \mathbb{E}[|x_n - x|^2] = 0$.

3.3 Large random matrices

Many performance metrics in MIMO are functions of eigenvalues of random matrices. As $N \rightarrow \infty$, the empirical eigenvalue distribution often converges to a deterministic limit.

- **Wigner's Semicircle Law:** For an $N \times N$ symmetric matrix A with IID entries, the eigenvalues of $A/\sqrt{N}$ converge to:

$$f(\xi) = \frac{1}{2\pi} \sqrt{4 - \xi^2}, \quad \xi \in [0, 2]. \quad (\text{C.52})$$

- **Marčenko-Pastur Law:** For $\frac{1}{M}HH^*$ where H is $N \times M$ with IID unit-variance entries, with $\beta = M/N$:

$$f(\xi) = [1 - \beta]^+ \delta(\xi) + \beta \frac{\sqrt{(\xi - a)(b - \xi)}}{2\pi\xi}, \quad \xi \in [a, b]. \quad (\text{C.53})$$

3.4 Random processes

A random process $x(t)$ is a collection of random variables indexed by time. The autocovariance/autocorrelation is:

$$R_x(t, \tau) = \mathbb{E}[(x(t) - \mu_x(t))(x(t + \tau) - \mu_x(t + \tau))^*]. \quad (\text{C.56})$$

3.4.1 Stationarity

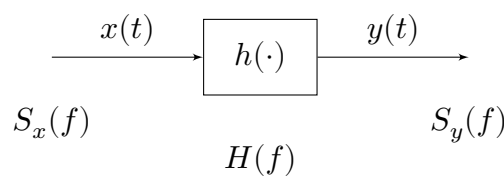
A process is **stationary** if time shifts do not affect its distribution. It is **wide-sense stationary (WSS)** if the mean is constant and autocorrelation depends only on the lag τ . The **Power Spectral Density (PSD)** is the Fourier transform of the autocorrelation:

$$S_x(\nu) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi\nu\tau} d\tau. \quad (\text{C.57})$$

3.4.2 Ergodicity

A process is ergodic if its distribution can be deduced from a single, sufficiently long, realization (time averages equal ensemble averages).

3.4.3 System Analysis (Handwritten Notes)



For a linear time-invariant system:

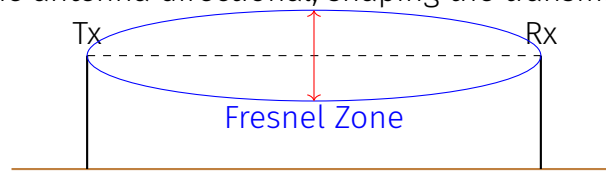
$$S_y(f) = S_x(f) \cdot |H(f)|^2 \quad (\text{3.7})$$

Link Budget and Antenna Gains

Fresnel Zones and Link Budget: The diagram on page 17 illustrates a transmission between two antennas.

- The Fresnel zone is the ellipsoid region between the transmitter and receiver.
- The signal can be disrupted if the Fresnel zone is obstructed (e.g., by the ground or objects).

- G_T and G_R represent the gains of the transmitting and receiving antennas.
- An antenna has gain only with respect to an omnidirectional antenna. We can make the antenna directional, shaping the transmission.



Appendix D: Gradient operator

The gradient of a scalar-valued function $f(x)$ of vector argument x , denoted by $\nabla_x f(x)$ yields a vector-valued function that, at each point x , identifies the direction of greatest increase of $f(\cdot)$, with a magnitude that equals the corresponding rate of increase. It is nothing but the generalization to multiple dimensions of the standard one-dimensional derivative. In rectangular coordinates, where $x = [x_0, x_1, x_2]^T$,

$$\nabla f(x_0, x_1, x_2) = \frac{\partial f(x_0, x_1, x_2)}{\partial x_0} e_0 + \frac{\partial f(x_0, x_1, x_2)}{\partial x_1} e_1 + \frac{\partial f(x_0, x_1, x_2)}{\partial x_2} e_2, \quad (\text{D.1})$$

where e_0 , e_1 , and e_2 are unit vectors along the corresponding coordinates. More generally, for vectors with an arbitrary number of dimensions, and with the allowance of them being complex, we can compactly write

$$[\nabla_x f(x)]_j = \frac{\partial f(x)}{\partial [x^*]_j}, \quad (\text{D.2})$$

which can be further extended to express the gradient of a scalar-valued function $f(X)$ of a complex matrix argument X as

$$[\nabla_X f(X)]_{i,j} = \frac{\partial f(X)}{\partial [X^*]_{i,j}}. \quad (\text{D.3})$$

Example D.1

For the linear function of complex matrix argument $f(X) = \text{tr}(R_0 X^* R_1)$,

$$\nabla_X \text{tr}(R_0 X^* R_1) = R_1 R_0 \quad (\text{D.4})$$

and, as a corollary

$$\nabla_x (x^* r) = r. \quad (\text{D.5})$$

Furthermore, because a vector and its conjugate can be regarded as independent for the purpose of gradient computations,

$$\nabla_x (r^* x) = 0. \quad (\text{D.6})$$

Example D.2

For the quadratic form $f(x) = x^* R x$, the gradient equals

$$\nabla_x (x^* R x) = R x \quad (\text{D.7})$$

Example D.3

While, for $f(X) = \text{tr}(R_0 X R_1 X^*)$

$$\nabla_X \text{tr}(R_0 X R_1 X^*) = R_0 X R_1 \quad (\text{D.8})$$

A scalar function of matrix argument that appears often in this text is $f(X) = \log_e \det(X)$, with gradient

$$\nabla_X \log_e \det(X) = X^{-1}. \quad (\text{D.9})$$

From this expression, in turn, (D.8) and the chain rule of differentiation lead to

$$\nabla_X \log_e \det(I + X^* R_0 X R_1) = R_0 X R_1 (I + X^* R_0 X R_1)^{-1}. \quad (\text{D.10})$$

For definitions of the gradient operator in cylindrical and spherical coordinates, the reader is referred to specialized literature.

Appendix E: Special functions

In order to maximize the generality, elegance, and meaning of their analysis, researchers strive for closed-form results, meaning a combination and composition of elementary functions via the four basic operations. These elementary functions include algebraic functions (solutions of a polynomial equation with integer coefficients) and transcendental functions (including exponentials, logarithms, trigonometric functions, and their inverses).

There are other functions that, while not elementary, appear frequently enough and display sufficiently important properties to have their own names and to be readily tabulated or computable. It is common to relax the interpretation of "closed-form" to also encompass expressions involving these special functions. In this appendix, we survey the special functions that appear throughout this textbook.

5.1 Gamma function

Perhaps the most common special function, and therefore the least special of them all, is the gamma function. For arguments whose real part is positive,

$$\Gamma(z) = \int_0^{\infty} \xi^{z-1} e^{-\xi} d\xi, \quad (\text{E.1})$$

which, for positive integers arguments, reduces to $\Gamma(n) = (n-1)!$ indicating that the gamma function can be interpreted as an interpolator for the factorial function. For noninteger arguments, the best known value is $\Gamma(1/2) = \sqrt{\pi}$.

Partial integrals with the same integrand as (E.1) are referred to as incomplete gamma functions, of which two varieties exist: the upper incomplete gamma function

$$\Gamma(z, s) = \int_s^{\infty} \xi^{z-1} e^{-\xi} d\xi, \quad (\text{E.2})$$

and the lower incomplete gamma function

$$\gamma(z, s) = \int_0^s \xi^{z-1} e^{-\xi} d\xi. \quad (\text{E.3})$$

5.2 Digamma function

The digamma function is the logarithmic derivative of the gamma function, i.e.,

$$\psi(z) = \frac{d}{dz} \log_e \Gamma(z) = \frac{\dot{\Gamma}(z)}{\Gamma(z)} \quad (\text{E.4})$$

which, for integer arguments, can be expressed as

$$\psi(N) = -\gamma_{\text{EM}} + \sum_{l=1}^{N-1} \frac{1}{l} \quad (\text{E.5})$$

given the Euler-Mascheroni constant

$$\gamma_{\text{EM}} = \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{1}{n} - \log_e N \right) \approx 0.5772 \quad (\text{E.6})$$

The digamma function satisfies the recursion

$$\psi(N+1) = 1 + \frac{1}{N} \sum_{n=1}^N \psi(n). \quad (\text{E.7})$$

5.3 Exponential integrals

For real nonzero arguments, the exponential integral function is

$$\mathcal{E}_i(z) = \int_{-\infty}^z \frac{e^{\xi}}{\xi} d\xi, \quad (\text{E.10})$$

from which a class of functions, parameterized by an order n , is defined as

$$\mathcal{E}_n(z) = \int_1^{\infty} \frac{e^{-z\xi}}{\xi^n} d\xi. \quad (\text{E.11})$$

These functions, which we loosely refer to as exponential integrals, are related with the gamma function via

$$\mathcal{E}_n(z) = z^{n-1} \Gamma(1-n, z), \quad (\text{E.12})$$

with the most commonly encountered exponential integral being

$$\mathcal{E}_1(z) = -\mathcal{E}_i(-z) = \Gamma(0, z). \quad (\text{E.13})$$

5.4 Bessel functions

Bessel functions are solutions to a famed differential equation, Bessel's equation. These solutions are parameterized by an order, the most important such orders being integers or half-integers. Bessel functions of the first kind, denoted by $J_n(x)$, are

solutions that are finite at the origin for integer n . They can be expressed as an infinite series or in the integral forms

$$J_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j(n\xi - x \sin \xi)} d\xi = \frac{1}{\pi} \int_0^{\pi} \cos(n\xi - x \sin \xi) d\xi. \quad (\text{E.15})$$

From $J_n(\cdot)$, one can further define the modified Bessel functions of the first kind as

$$I_n(x) = j^{-n} J_n(jx) \quad (\text{E.17})$$

and, in due course, the modified Bessel functions of the second kind as

$$K_n(x) = \frac{\pi}{2} \frac{I_{-n}(x) - I_n(x)}{\sin(n\pi)}. \quad (\text{E.18})$$

5.5 Q-function

The Q-function is the tail probability of a standard Gaussian distribution. If $x \sim \mathcal{N}(0, 1)$, then $Q(x) = \mathbb{P}[x > x]$ and thus the CDF of x satisfies $F_x(x) = 1 - Q(x)$. The usual form for $Q(\cdot)$ is

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\xi^2/2} d\xi, \quad (\text{E.19})$$

which can be rewritten as the finite-range integral

$$Q(x) = \frac{1}{\pi} \int_0^{\pi/2} \exp\left(-\frac{x^2}{2 \sin^2 \phi}\right) d\phi. \quad (\text{E.20})$$

For positive arguments, the Q-function can be bounded as

$$\frac{x}{1+x^2} \frac{e^{-x^2/2}}{\sqrt{2\pi}} < Q(x) < \frac{1}{x} \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad (\text{E.21})$$

where the upper bound can be relaxed into the popular Chernoff bound

$$Q(x) < \frac{1}{2} e^{-x^2/2}. \quad (\text{E.22})$$

Other upper and lower bounds are available in the literature. Alternatively, a widely valid approximation is

$$Q(x) \approx \frac{1}{\frac{\pi-1}{\pi}x + \frac{1}{\pi}\sqrt{x^2 + 2\pi}} \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad (\text{E.23})$$

which, like the bounds, evidences that the tail decays exponentially fast. One can also relate the Q-function with the complementary error function via

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right), \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-\xi^2} d\xi. \quad (\text{E.24})$$

5.6 Hypergeometric functions

A hypergeometric function ${}_pF_q(a_0, \dots, a_{p-1}; b_0, \dots, b_{q-1}; x)$ is defined by its series being hypergeometric, meaning that the ratio of consecutive terms is a rational function of the summation index. If c_n and c_{n+1} are the n th and $(n+1)$ th terms in the series, then

$$\frac{c_{n+1}}{c_n} = \frac{P(n)}{Q(n)}, \quad (\text{E.26})$$

where $P(n)$ and $Q(n)$ are polynomials. We identify two hypergeometric functions in particular.

- The Kummer confluent hypergeometric function is

$${}_1F_1(a; b; x) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} \frac{x^n}{n!}. \quad (\text{E.27})$$

where $(a)_n = x \cdot (x+1) \cdots (x+n-1)$ with $(a)_0 = 1$. In integral form,

$${}_1F_1(a; b; x) = \frac{\Gamma(b)}{\Gamma(b-a)\Gamma(a)} \int_0^1 e^{xt} t^{a-1} (1-t)^{b-a-1} dt. \quad (\text{E.28})$$

- The Gauss hypergeometric function is

$${}_2F_1(a_0, a_1; b; x) = \sum_{n=0}^{\infty} \frac{(a_0)_n (a_1)_n}{(b)_n} \frac{x^n}{n!}. \quad (\text{E.29})$$

Appendix F: Landau symbols

The Landau symbols $\mathcal{O}(\cdot)$ and $o(\cdot)$ allow describing, in terms of simpler functions, the limiting behavior of a function as its argument approaches a particular value or tends to infinity. We present these two symbols in the simplest possible way that suffices for their usage in this book.

A function $f(x)$ is said to be $\mathcal{O}(g(x))$, with $g(\cdot)$ being another—ideally simpler—function, if

$$|f(x)| \leq b|g(x)| \tag{F.1}$$

for some constant b and all values of x .

In turn, $f(x)$ is said to be $o(g(x))$ if

$$\frac{f(x)}{g(x)} \rightarrow 0 \tag{F.2}$$

for x approaching a certain value or tending to infinity, depending on the behavior being described.

Appendix G: Convex optimization

7.1 Convex sets

A set is convex if it contains all segments between any two of its points, meaning that it is free of indentations. An example of convex set is the one depicted in Fig. 5.4.

7.2 Convex and concave functions

A real-valued function $f(x)$ is convex if it satisfies

$$f(\theta x_0 + (1 - \theta)x_1) \leq \theta f(x_0) + (1 - \theta)f(x_1) \quad (\text{G.1})$$

for any real vectors x_0 and x_1 on a certain domain and for every real scalar $\theta \in [0, 1]$. This implies that the graph of the function lies below any chord, i.e., below any segment connecting two points of that graph. If the inequality in (G.1) is strict, then the function is strictly convex. Conversely, if the inequality is a strict equality, then the function is linear.

A function $f(\cdot)$ is concave (respectively strictly concave) if $-f(\cdot)$ is convex (respectively strictly convex), meaning that its graph lies above any chord. Examples of concave functions are shown in Fig. G.1.

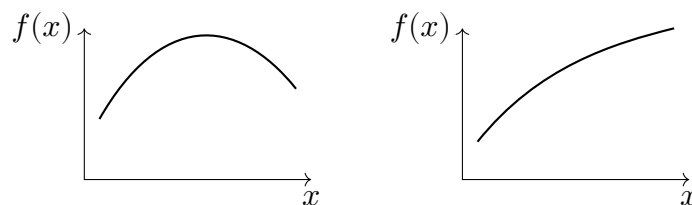


Figure 7.2: Fig G.1: Left, nonmonotonic concave function of a real scalar argument. Right, monotonically increasing concave function.

If $f(\cdot)$ is concave, then the problem of finding its maximum over a convex set readily maps to (G.2) with the objective function given by $-f(\cdot)$.

7.3 Convex optimization problems

An optimization problem has the form

$$\begin{aligned} & \text{minimize } f(x) \\ & \text{subject to } g_i(x) \leq d_i \quad i = 0, \dots, N-1 \end{aligned} \quad (\text{G.2})$$

where $f(\cdot)$ is the objective or cost function whereas $g_i(\cdot)$ are the constraint functions. The solution x^* satisfies $f(x^*) \leq f(x)$ for every vector x meeting the N constraints.

If $f(\cdot)$ as well as $g_0(\cdot), \dots, g_{N-1}(\cdot)$ are convex, then the optimization problem is said to be convex. This means that the set defined by the constraint functions, over which the optimum is to be found, is convex and the objective function thereon is also convex. Convex problems can be interpreted as a generalization of linear problems, which are those where $f(\cdot)$ and $g_0(\cdot), \dots, g_{N-1}(\cdot)$ are linear functions.

While, in general, the solution of generic optimizations may pose considerable difficulty, convex problems can be solved reliably and efficiently even when they involve vectors with many dimensions. For an overview of convex optimization procedures and algorithms, the reader is referred to specialized textbooks. Convex optimization problems arise frequently in communications and, in particular, in MIMO.

A key advantage of convex problems over nonconvex ones is that, if a local minimum exists, then it is a global minimum. As a result, any set of necessary conditions characterizing a minimum are also sufficient for global optimality.

7.4 KKT optimality conditions

The Karush-Kuhn-Tucker (KKT) conditions are necessary and sufficient conditions characterizing the solution to a convex problem. In order to present them, it is useful to rewrite (G.2) distinguishing between those constraints that are inequalities and those that are strict equalities. Also, by suitably modifying the constraint functions, we can set all the constraints to zero, obtaining

$$\begin{aligned} & \min f(x) \\ & \text{s.t. } g_i(x) \leq 0 \quad i = 0, \dots, M' - 1 \\ & \quad h_j(x) = 0 \quad j = 0, \dots, M - 1 \end{aligned} \quad (\text{G.3})$$

Then, the KKT conditions characterizing x^* are

$$\begin{aligned} \nabla f(x^*) + \sum_{i=0}^{M'-1} \lambda'_i \nabla g_i(x^*) + \sum_{j=0}^{M-1} \lambda_j \nabla h_j(x^*) &= 0 \\ g_i(x^*) &\leq 0 \quad i = 0, \dots, M' - 1 \\ h_j(x^*) &= 0 \quad j = 0, \dots, M - 1 \\ \lambda'_i &\geq 0 \quad i = 0, \dots, M' - 1 \\ \lambda'_i g_i(x^*) &= 0 \quad i = 0, \dots, M' - 1 \end{aligned} \quad (\text{G.4})$$

where $\lambda'_0, \dots, \lambda'_{M'-1}$ and $\lambda_0, \dots, \lambda_{M-1}$ are the so-called KKT multipliers.

The KKT conditions play an important role in convex optimization, often serving as targets for the algorithms. Moreover, sometimes the KKT conditions can be analytically resolved, yielding expressions that characterize the solution directly.

Note

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7.5 Lagrange multipliers

If there are no inequality constraints, but only equality ones, the solution of the KKT conditions reverts to the method of Lagrange multipliers. Consider the problem

$$\begin{aligned} \min f(x) \\ \text{s.t. } h_j(x) = 0 \quad j = 0, \dots, M-1 \end{aligned} \quad (\text{G.5})$$

If we augment the objective function with a weighted sum of the constraint functions we obtain the so-called Lagrangian function

$$L(x, \lambda_0, \dots, \lambda_{M-1}) = f(x) + \sum_{j=0}^{M-1} \lambda_j h_j(x), \quad (\text{G.6})$$

where $\lambda_0, \dots, \lambda_{M-1}$ are termed Lagrange multipliers.

Any point x^* where all partial derivatives of $L(\cdot)$ are zero corresponds to an extreme point (minimum or maximum) that satisfies the equality constraints. If $f(\cdot)$ is convex or concave, then this extreme point is guaranteed to be the global one. The conditions satisfied by x^* can be written as

$$\begin{aligned} \nabla_x L(x^*) &= \nabla f(x^*) + \sum_{j=0}^{M-1} \lambda_j \nabla h_j(x^*) = 0 \\ h_j(x^*) &= 0 \quad j = 0, \dots, M-1 \end{aligned} \quad (\text{G.7})$$

which, indeed, can be seen to be a special case of (G.4).

Often, the objective functions encountered in communications are not only convex or concave, but further monotonic in quantities of interest (see Fig. G.1). Taking advantage of that, inequality constraints can be tightened into strict equalities, paving the way for the application of Lagrange multipliers directly, rather than the more general KKT conditions. Examples include the spectral efficiency or the error probability, both of which improve monotonically with the signal power in most settings; then, any applicable power constraints can be considered directly as equalities.

7.6 Jensen's inequality

As stated earlier, the inequality in (G.1) reflects that the graph of a convex function lies below any chord. This generalizes into the widely utilized Jensen's inequality. Given a

convex function $f(x)$ and the vectors $x_0, \dots, x_{N-1}$ in its domain,

$$f\left(\frac{\sum_{i=0}^{N-1} a_i x_i}{\sum_{i=0}^{N-1} a_i}\right) \leq \frac{\sum_{i=0}^{N-1} a_i f(x_i)}{\sum_{i=0}^{N-1} a_i} \quad (\text{G.8})$$

for any positive coefficients $a_0, \dots, a_{N-1}$. As a special case, if $a_i = 1$ for $i = 0, \dots, N-1$, the weighted sum gives the average and thus, if the vectors are random variables, we can write

$$f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]. \quad (\text{G.9})$$

If $f(\cdot)$ is concave, rather than convex, the above inequalities are simply reversed.